ANATOMY

Workbook

A Little progress each day
adds up to BIG RESULTS!
TRUST YOURSELF !!
STAY POSITIVE !!
MAKE IT HAPPEN!!

Dr. Shilpi Agarwal

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
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DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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Contents

1. Upper Limb 10

1. Bones of Upper Limb 10
2. Radiological Anatomy of Upper Limb 15
3. Brachial Plexus & Its lesions 16
4. Pectoral Region (Muscles, Winging of scapula) 21
5. Shoulder Region (Muscle, Deltopectoral groove) 22
6. Back Region (Muscles, Triangle of Auscultation) 24
7. Muscles of Scapular Region 26
8. Rotator Cuff 26
9. Intermuscular Spaces 26
10. Arm (Anterior & Posterior comp.) 27
11. Forearm (Anterior & Posterior compartment) 28

12. Muscles of Hand 31
13. Anatomical Snuff Box 33
14. Flexor Retinaculum (Including Carpal Tunnel Syndrome) 34
15. Cubital Fossa 35
16. Median Nerve - Course, Branches & Lesions 36
17. Ulnar Nerve - Course, Branches & Lesions 38
18. Radial Nerve - Course, Branches & Lesions 40
19. Arteries of Upper Limb (with Axillary & Subclavian A, Anastomosis around scapula) 41
20. Breast 45
21. Axillary Lymph Nodes 45
22. Dermatomes of Upper Limb 46

2. General Anatomy 49

1. Joints 49
2. Epiphysis 52
3. Hybrid/Composite Muscle 53

3. Thorax 55

1. Bones of Thoracic Region 55
2. Intercostal Space - Muscles & Pleural Tap 56
3. Mediastinum 57
4. Heart (Parts, Interior, Blood supply) 58
5. Trachea 62
6. Lung (Including Bronchopulmonary segments) 63
7. Esophagus 67

8. Diaphragm 68

9. Thoracic Duct 70

10. Venous Drainage (SVC, Azygos venous system) 70

4. Head & Neck 75

1. Skull (Including Foramens, Sphenoid, Bony orbit) 76

2. Mandible 80

3. Cervical Vertebra 81

4. Hyoid Bone 82

5. Pneumatic Bones 82

6. Scalp - 82

7. Face (Nerve supply) 83

8. Temporomandibular Joint 84

9. Dangerous area of face 85

10. Cavernous Sinus 86

11. Arterial Supply of Head & Neck (Ext. Carotid A) 86

12. Parasympathetic Ganglia 88

13. Salivary Glands (Including Frey' syndrome) 88

14. Facial Nerve & its Lesions 89

15. Tongue & Cranial Nerve 12 91

16. Extraocular Muscles & Cranial Nerves 3, 4 & 6 92

17. Saggital Section of H & N 94

18. Lymphatic Drainage of Head & Neck (Including Waldeyer Ring) 94

19. Triangles of Neck 95

20. Carotid Sheath 96
21. Ansa Cervicalis 96
21. Thyroid Gland 97
22. Pharynx 97
23. Soft Palate 98
24. Larynx 98
25. Nose (Nasal septum & Lateral wall) 99

5. Neuroanatomy 103

1. Organisation of Nervous System & Meninges & spaces & lumbar puncture 103
2. Cerebrum (with White matter fibres, Functional area) 105
3. Diencephalon 107
4. Basal Ganglia 108
5. Internal Capsule 108
6. Transverse Section of Cerebrum 108
7. Coronal Section of Cerebrum 109
8. Brainstem 109
9. Cerebellum 110
10. Ventricles, CSF & Hydrocephalus 111
11. Arterial Supply of Brain (with Lateral & Medial Medullary Syndromes) 113
12. Spinal Cord (Tracts, Brown Sequard Syndrome) 116
13. Limbic System 120

6. Lower Limb 122

1. Bones of Lower Limb 122
2. Radiological Anatomy of Lower Limb 124
3. Greater & Lesser Sciatic Foramen 125
4. Anterior Compartment of Thigh 126
6. Medial Compartment of Thigh 127
7. Posterior Compartment of Thigh - Muscles 127
8. Iliotibial Tract 130
9. Gluteal Region 130
10. Popliteal Fossa 132
11. Knee Joint 133
12. Sciatic Nerve, Tibial Nerve & Common Peroneal Nerve 133
13. Muscles of Anterior Compartment of Leg 134
14. Muscles of Posterior Compartment of Leg 135
15. Dorsiflexion & Plantar Flexion of Foot 135
16. Muscles of Lateral Compartment of Leg 136
17. Inversion & Eversion of Foot 136
18. Foot Drop 136
19. Arteries of Lower Limb 137
20. Venous Drainage of Lower Limb 138
21. Dermatomes of Lower Limb 139

7. Abdomen & Pelvis 141

1. Abdomen - Planes & Quadrants 141

2. Anterior Abdominal Wall (Layers, Rectus sheath) 142
3. Inguinal Region (with Hesselbach triangle, Hernia) 143
4. Peritoneal Cavity (Sacs & Epiploic foramen) 145
5. Arteries of Abdomen 145
(Abdominal aorta, Celiac trunk, SMA, IMA, Marginal A, Watershed area, Internal & Iliac A)
6. Portal Vein & Portocaval Anastomosis 149
7. Stomach 150
8. Spleen 151
9. Large Intestine 152
10. Small Intestine 152
11. Liver 153
12. Extra Biliary Apparatus 154
13. Pancreas 154
14. Kidney 155
15. Male Urethra & Urethral Injuries 155
16. Female Reproductive System 156
17. Male Reproductive System 158
18. Perineum (Pelvic diaphragm, Perineal body) 159
19. Thoracolumbar Fascia 160
21. Transverse Section of Abdomen 160

8. General Embryology 163

1. Phases of Development 163

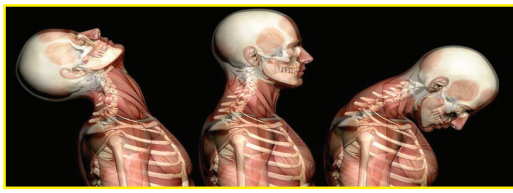
2. Spermatogenesis 163
3. Oogenesis 164
4. Fertilization 166
5. Developmental Period - Week 1 (Morula, Blastocyst, Implantation) 167
6. Developmental Period - Week 2 167
7. Developmental Period - Week 3 167
8. Placenta Development (Decidua, Chorion, Villi, Placental barrier) 168
9. Umbilical Cord 170

9. Systemic Embryology 171

1. Development of CNS 171
2. Development of GIT (Foregut, Midgut, Hindgut, Mesentery) 173
3. Development of Genitourinary System 178
4. Development of Urinary System 180
5. Development of Diaphragm 180
6. Bochdalek Congenital Diaphragmatic Hernia 181
7. Development of Head & Neck (Pharyngeal Apparatus, Tongue, Face) 181

10. Histology 188

1. Epithelium 188
2. Cartilage 192



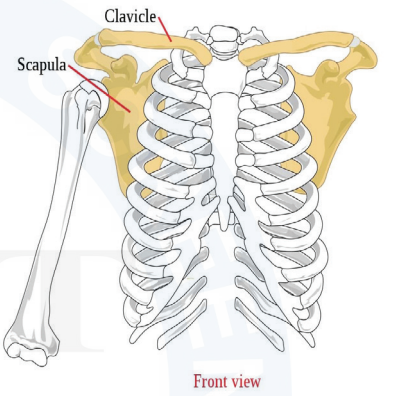
UPPER LIMB

BONES OF UPPER LIMB

1. Clavicle

It is the first to begin ossifying (5^{th}-6^{th} week IUL) & It is last to complete ossification (25 yrs)

❑ **Most common site of fracture of clavicle is** .....



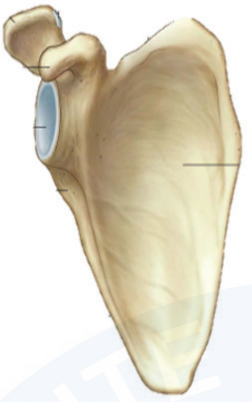
◀**FIG.** Clavicle & scapula (Anterior view)

✦ CLINICAL INTEGRATION

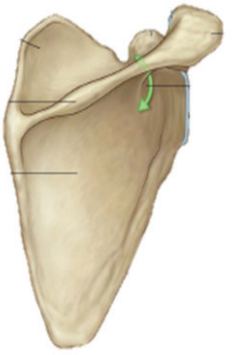
A man had a fall from height & landed on outstretched hand. The patient was found to have fracture of clavicle. What is the most common location for this fracture/where can this fracture be ?

(A) Junction of medial 1/3rd & lateral 2/3rd
(B) Junction of medial 2/3rd & lateral 1/3rd
(C) Junction of medial 1/3rd & lateral 1/3rd
(D) Junction of medial 3/4 & lateral 1/4

2. Scapula



◀FIG. Scapula (Anterior view)



◀FIG. Scapula (Posterior view)

A. Coracoid process

1. Short head of biceps
2. Coracobrachialis
3. Pectoralis Minor

} Tip of coracoid process

✦ Note : Tip of coracoid process is felt 2.5cm below clavicle (on lateral aspect)

3. Humerus

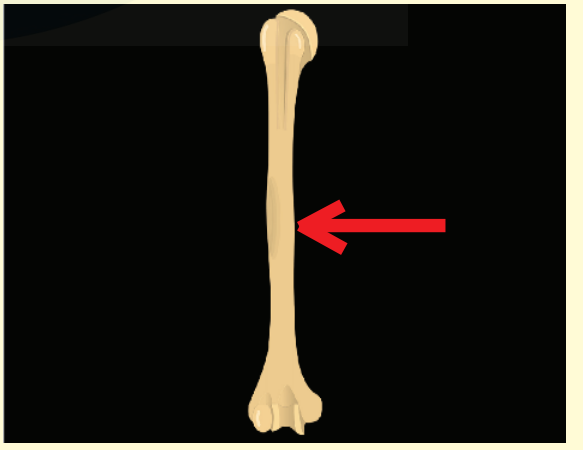
◀FIG. Humerus (Anterior view)

- **Surgical Neck** Related to Axillary Nerve & Posterior Circumflex Humeral Vessels (Artery & Vein)
- **Fracture of Surgical Neck of Humerus:** .....
- **Anterior or Inferior dislocation of Shoulder Joint:** .....
- **Spiral groove** related to : Radial Nerve & Profunda Brachii Vessels (Artery & Vein)
- **Fracture of shaft of Humerus:** .....
- **Behind medial epicondyle** related to : Ulnar Nerve
- **Bicipital Groove** Muscles attached to it :
  1. Pectoralis Major (Lateral Lip)
  2. Teres Major (Medial Lip)
  3. Latissimus Dorsi (Floor)

✦ **CLINICAL INTEGRATION**

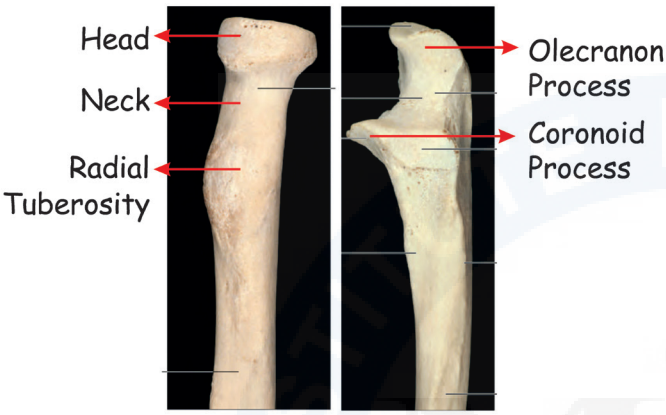
A 35 yr old male presents in emergency after a road traffic accident with fracture at the arrow marked location. He presents with inability to extend his wrist. On examination, there is loss of sensation of dorsum of lateral aspect of hand and fingers. Which nerve will be injured in this case ?

A. Axillary Nerve
B. Radial Nerve
C. Ulnar Nerve
D. Median Nerve

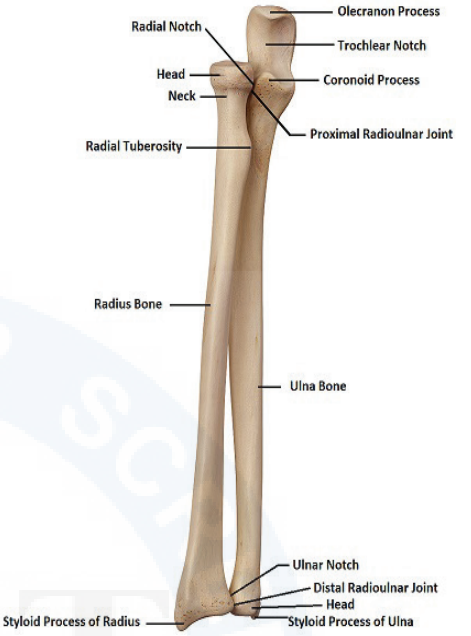


4. Radius & Ulna

- Radius - Radial Tuberosity gives attachment to Biceps Brachii
- Ulna - Olecranon Process gives attachment Triceps.



◀FIG. Radius ◀FIG. Ulna (Lateral View)



◀FIG. Radius & Ulna (Anterior View)

5. Carpal Bones & Phalanges

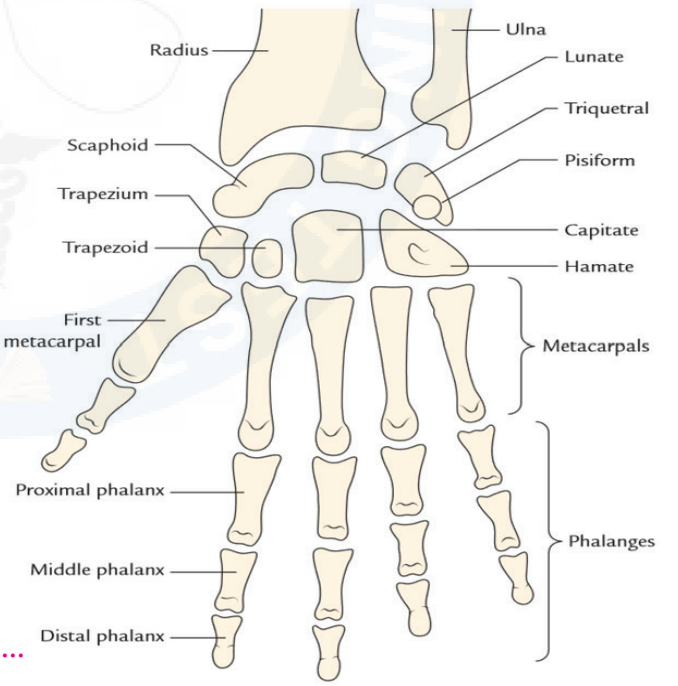
❑ Proximal Row (4) (Lateral to medial)

1. Scaphoid — She
2. Lunate — Looks
3. Triquetral — Too
4. Pisiform — Pretty

❑ Distal Row (4) (Lateral to medial)

1. Trapezium — Try
2. Trapezoid — To
3. Capitate — Catch
4. Hammate — Her

- Largest carpal bone → .....
- Smallest carpal bone → .....
- 1st carpal bone to ossify → .....
- Last carpal bone to ossify → .....

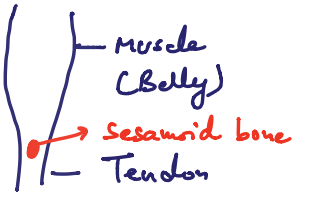


◀FIG. Carpals, Metacarpals & Phalanges

- Scaphoid bone → Most Common carpal bone to Undergo fracture
  (Tenderness at anatomical snuff box)
- Scaphoid bone →Avascular Necrosis
- Lunate dislocation → Median nerve compression

✦ Note - Most common carpal bone to undergo dislocation → Lunate

- Hammate → It has Hook like projection
- Pisiform → Sesamoid bone
  → Articulates with Triquetral



◀FIG. Sesamoid Bone

✦ Note - Sesamoid bone

1. Pisiform → present in tendon of flexor carpi ulnaris
2. Patella → Quadriceps femoris
3. Fabella → Lateral Head of Gastrocnemius

6. Metacarpals

❑ **Normal Epiphysis (only one epiphysis is present in metacarpals)**

Located at Base of 1st metacarpal &
Head of 2nd - 5th metacarpals.

✦ **CLINICAL INTEGRATION**

A person fell on the outstrechted hand & got injured at the anatomical snuff. Which bone is most commonly fractured at anatomical snuff box ?

A) Lunate

B) Scaphoid

C) Trapezium

D) Pisiform

RADIOLOGICAL ANATOMY OF UPPER LIMB

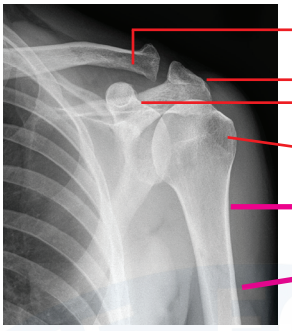


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FIG. X-ray of Shoulder Joint

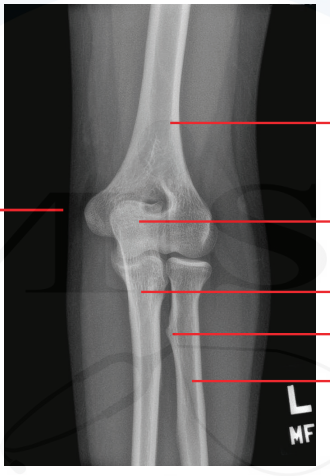

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FIG. X-ray of Elbow Joint

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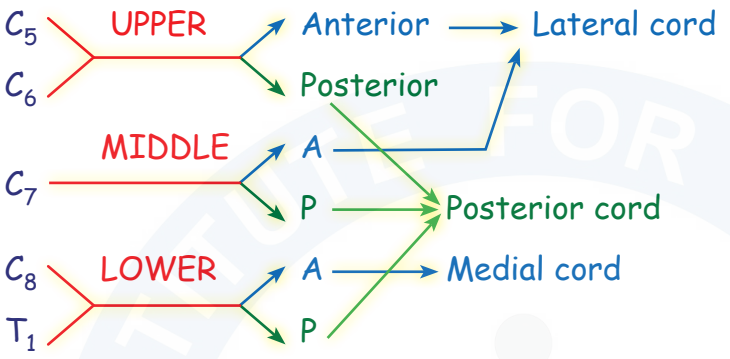
FIG. X-ray of wrist Joint & Hand

BRACHIAL PLEXUS & ITS LESIONS

1. Formation & Branches

❑ **Formation** → Ventral Rami of C_5 – T_1

Roots Trunk Divisions Cords

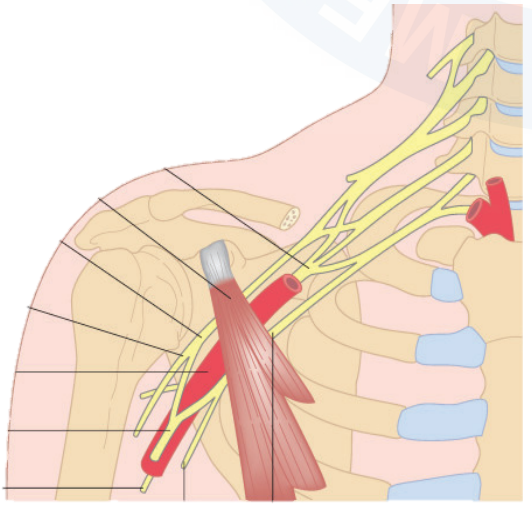


◀FIG. Brachial Plexus (Formation)

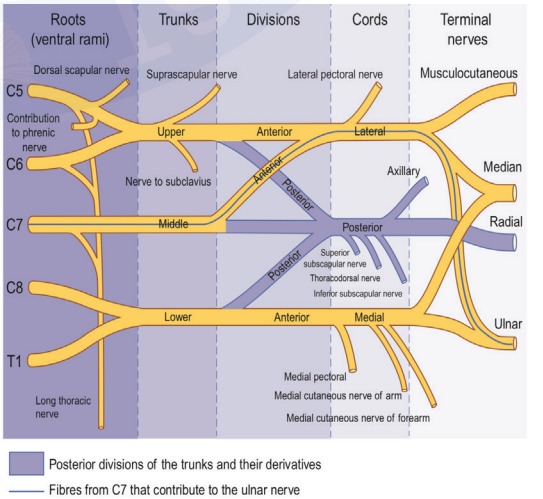
❑ **Components & Location**

Roots → Trunk → Divisions → Cords → Branches

Located in Neck (Posterior Triangle of Neck) | Behind Clavicle | Axilla



◀FIG. Brachial Plexus (Location)



◀FIG. Brachial Plexus

❑ Branches

Branches from Roots

1. Dorsal Scapular N
   - → Root Value - C_5
   - → .....
2. Long Thoracic N
   - → C_5 C_6 C_7
   - → Also called nerve of bell
   - → .....

Branches from Trunk

1. Suprascapular N
   - → C_5 C_6
   - → .....
2. Nerve to Subclavius
   - → C_5 C_6
   - → Subclavius

Lateral Cord Branches

1. Lateral Pectoral Nerve
   - → C_5 C_6 C_7
   - → .....
2. Musculocutaneous N
   - → C_5 C_6 C_7
   - → .....

3. Lateral Root of Median N
   - → C_5 C_6 C_7

Medial Cord Branches

1. Medial Pectoral N
   - → C_8 T_1
   - → Pectoralis major & P. minor
2. Medial Root of Median N
   - → C_8 T_1
3. Medial Cutaneous N of arm
   - → C_8 T_1
4. Medial Cutaneous N of forearm
   - → C_8 T_1

} Sensory Branches

5. Ulnar N → C_7 C_8 T_1

☞ Note

Medial root of Median N (C_8 T_1) → MEDIAN Nerve C_5 C_6 C_7 C_8 T_1

Lateral root of Median N (C_5 C_6 C_7) → MEDIAN Nerve C_5 C_6 C_7 C_8 T_1

Posterior Cord - Branches

1. Lower Subscapular Nerve
   - → C_5 C_6
   - → Subscapularis & Teres Major
2. Upper Subscapular N
   - → C_5 C_6
   - → Subscapularis

3. Nerve to Latissimus dorsii (Thoracodorsal N)
   - → C_6 C_7 C_8
   - → Latissimus Dorsii
4. Axillary N
   - → C_5 C_6
   - → .....
5. Radial N
   - → C_5 C_6 C_7 C_8 T_1

2. Erb's Palsy

- Injury of Upper Trunk
- Nerve roots involved → C5 C6
- Cause → Undue separation of head & shoulder
- Deformity is Policeman Tip deformity OR Waiter's Tip deformity OR Porter's Tip deformity

◀FIG. Policeman Tip Deformity

✦ Position of the Limb in the Deformity is

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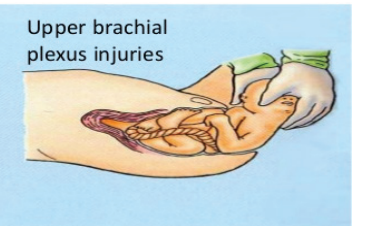


✦ Nerves involved

Axillary Nerve, Musculocutaneous Nerve, Suprascapular Nerve

✦ Muscles paralysed

Deltoid, Teres Minor, Biceps, Brachialis, Supraspinatus, Infraspinatus, Supinator

3. Klumpke's Palsy

- Injury of Lower Trunk
- Nerve roots involved → C8T1
- Cause - Hyperabduction of Arm
- Deformity → Claw Hand

  Hyperextension of Meta - carpophalangeal Joint &

  Flexion of Interphalangeal Joint (Proximal & distal)

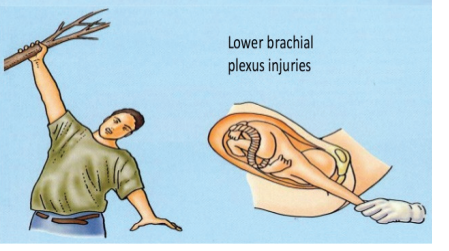
* Muscles paralysed - Intrinsic Muscles of Hand

◀FIG. Claw Hand (Anterior View)

◀FIG. Claw Hand (Lateral View)

* HORNER'S syndrome (If sympathetic fibres running through T1 are involved)

1. Ptosis (Drooping of upper Eyelid)
2. Miosis (Constriction of Pupil)
3. Anhydrosis (Absence of sweating on face)



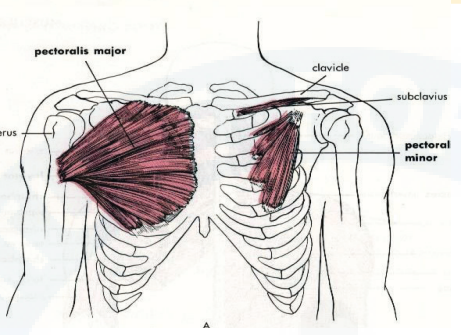
4. Enophthalmos (Sunken Eyeballs)

5. Loss of Ciliospinal Reflex

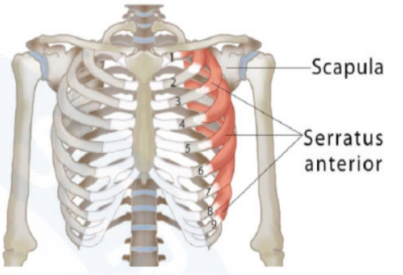
PECTORAL REGION

1. Pectoralis Major & Minor, Subclavius

1. Pectoralis Major
2. Pectoralis Minor
3. Subclavius
4. Serratus anterior



◀FIG. Pectoralis Major, Pectoralis Minor subclavius (Anterior View)



◀FIG. Serratus Anterior (Anterior View)

❑ **Pectoralis Major**

Origin → Clavicle, Sternum & Ribs

Insertion → Humerus

Action → .....

2. Serratus Anterior (Boxer's Muscle)

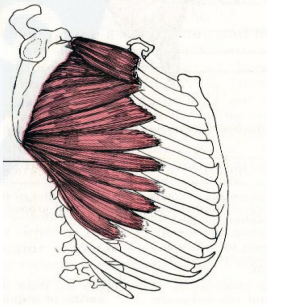
Origin → Upper 8 Ribs

Insertion → Scapula

Nerve supply → Long Thoracic Nerve

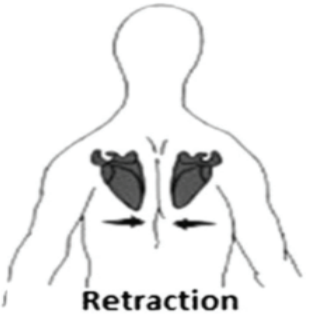
Action → .....

→ .....



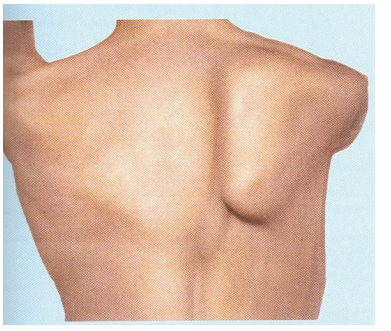
◀FIG. Serratus Anterior (Lateral View)

4. Protraction & Retraction of Scapula → Occurs at .....



3. Winging of Scapula

1. Due to Injury of Long Thoracic Nerve
2. Causes - Stab injury
   - During removal of breast carcinoma.
3. Paralysis of serratus anterior
4. Loss of protraction of scapula
5. Medial border & inferior angle of scapula becomes prominent



◀FIG. Winging of Scapula

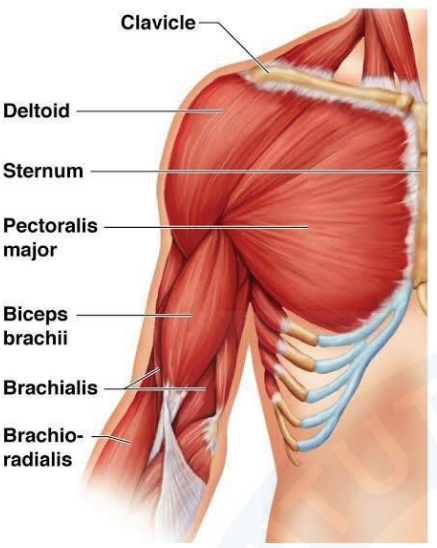
SHOULDER REGION

1. Deltoid

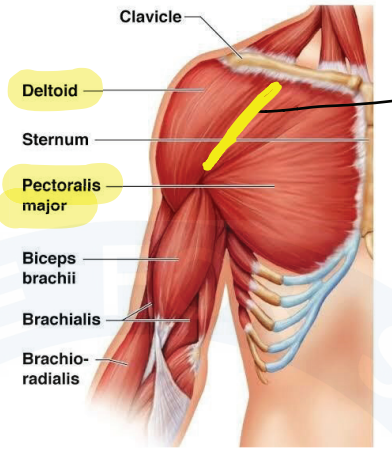
- Origin – .Clavicle & Scapula
- Insertion – Humerus (Deltoid Tuberosity)
- Nerve supply – Axillary Nerve
- Multipennate Muscle
- Action - Abduction at shoulder joint (15^0 - 90^0)

❑ **Abduction at shoulder Joint**

1. 0 – 15° (Initiation of abduction) → .....
2. 15° – 90° → .....
3. > 90° (overhead abduction) (90° – 180°) → .....



◀FIG. Deltoid (Anterior View)



◀FIG. Deltopectoral groove

2. Deltopectoral Groove / Triangle

- Between Deltoid & Pectoralis Major
- Content → Cephalic Vein
- Note - Deltopectoral Triangle - Between Deltoid, P. Major & Clavicle

❑ **Axillary N Injury**

Causes → • # of surgical Neck of Humerus
• dislocation of shoulder joint

1. Paralysis of Deltoid & Teres minor
2. Loss of ABDUCTION (15^0-90^0) at shoulder Joint
3. .....
4. .....

✦ **CLINICAL INTEGRATION**

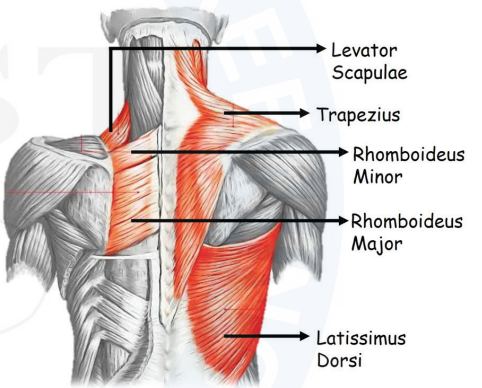
A Young boy met with accident and presented with anterior dislocation of shoulder. Abduction of shoulder is lost with loss of rounded contour of shoulder and sensory loss on upper & lateral aspect of arm. Which Nerve injury is there?

A. Median Nerve
B. Ulnar Nerve
C. Radial Nerve
D. Axillary Nerve

BACK REGION

❑ **Muscles of Back Region -**

1. Trapezius
2. Latissimus dorsi
3. Rhomboideus Major
4. Rhomboideus Minor
5. Levator scapulae



◀**FIG.** Muscles of Back Region (Posterior view)

1. Latissimus Dorsi

❑ **Latissimus Dorsii — also called Climber's Muscle**

- Action → Extension, Adduction & Medial Rotation at Shoulder Joint

2. Trapezius

✦ Nerve supply → .....

✦ Action →

- Upper fibres → Elevation of scapula
- Middle fibres → Retraction of scapula
- Lower fibres → Depression of scapula

✦ Injury / Cut / Incission at posterior triangle of Neck may lead to Injury of spinal accessory N → Trapezius paralyzed → 1 ....., 2 .....,

Sternocleidomastoid Paralyzed → .....

◀FIG. Dropped Shoulder

CLINICAL INTEGRATION

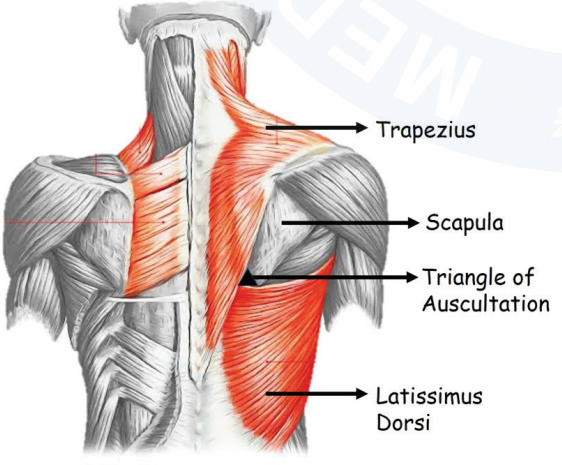
Q. A person was climbing the staircase, there was no light & suddenly he fell & a iron nail hit on his neck. After the injury, the patient is unable to shrug the shoulder & he is having difficulty in turning his head. What structure is most likely to be damaged?

a) Long thoracic nerve
b) Spinal Accessory nerve
c) Sternocleidomastoid
d) Deltoid

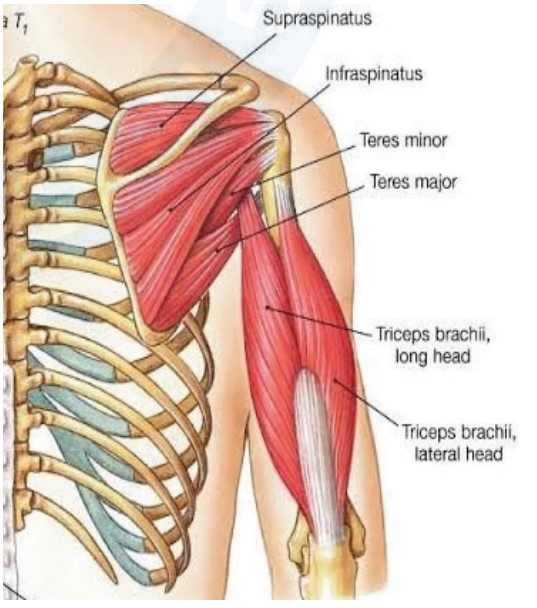
3. Triangle of Auscultation

❑ Boundaries

1. Trapezius, 2. Latissimus dorsi, 3.Scapula



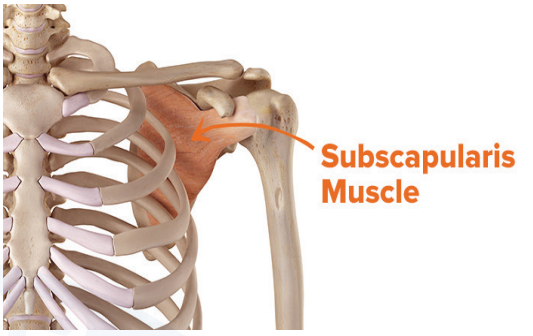
◀FIG. Triangle of Auscultation (Posterior View)



◀FIG. Muscles of scapular region (Posterior view)

MUSCLES OF SCAPULAR REGION

- 1. Supraspinatus
- 2. Infraspinatus
- 3. Teres Minor
- 4. T. Major
- 5. Subscapularis



◀FIG. Subscapularis (Anterior View)

ROTATOR CUFF

Rotator cuff or Musculotendinous cuff → (SITS)

Note : It is also called Dynamic stabilizer of shoulder Joint

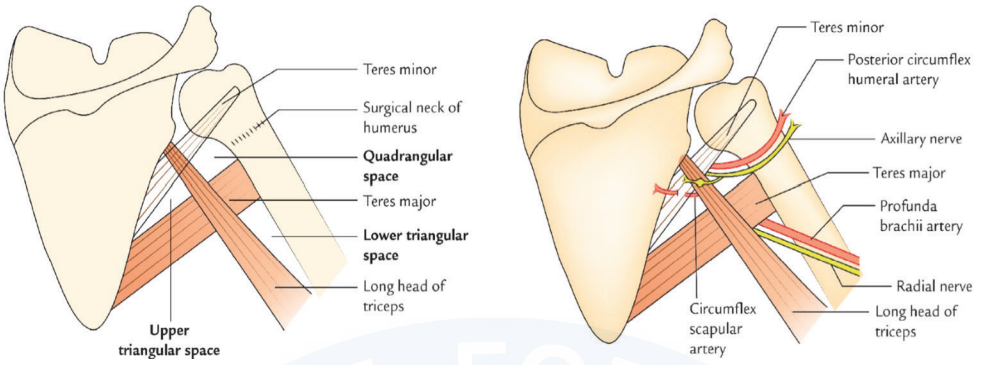
1. Supraspinatus → Most Commonly injured tendon of rotator cuff
2. Infraspinatus
3. Teres Minor
4. Subscapularis → Forgotten Muscle of Rotator cuff

INTERMUSCULAR SPACES

- **Quadrangular space**
  - Contents
    - Axillary Nerve &
    - Posterior Circumflex Humeral vessels

- **Lower Triangular space**
  - Contents
    - Radial Nerve &
    - Profunda Brachii vessels

- **Upper Triangular Space**
  - Contents
    - Circumflex Scapular Artery



◀FIG. Intermuscular spaces

ARM

1. Muscles of Anterior Compartment

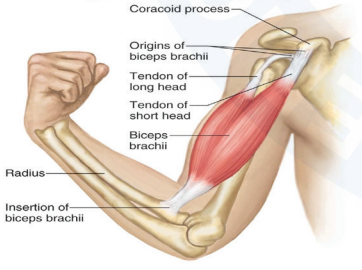
1. Biceps Brachii
2. Brachialis
3. Coracobrachialis

→ Nerve supply is Musculocutaneous N.

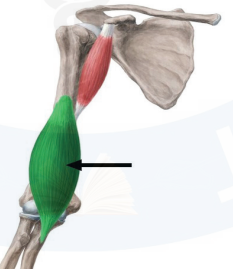
→ Action → flexion

❑ **Biceps**

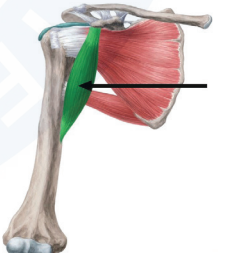
- Helps in Flexion & Supination
- Crosses both shoulder joint & elbow joint



◀FIG. Biceps brachii



◀FIG. Brachialis



◀FIG. Coracobrachialis

Note: Lateral Cutaneous nerve of forearm is a branch of musculo cutaneous nerve

❑ **Musculocutaneous nerve injury**

- .....
- .....

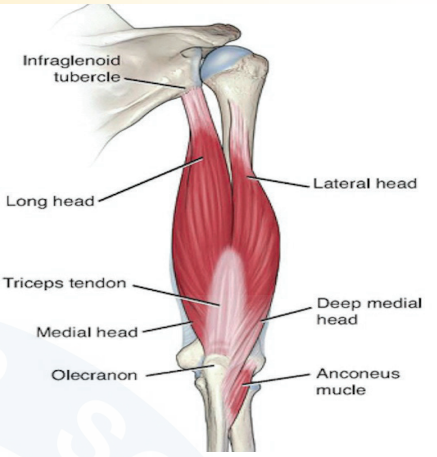
3. Muscles of Posterior Compartment

❑ TRICEPS

- N.S. Radial Nerve
- Action Extension

Arm → Anterior comp →
Posterior comp →

Forearm → Anterior comp →
Posterior comp →



◀FIG. Triceps (Posterior view)

FOREARM

1. Muscles of Anterior Compartment

❑ Superficial Group

1. Pronator Teres - Pronation
2. Flexor Carpi Radialis
3. Flexor Carpii Ulnaris
4. Palmaris Longus
5. Flexor digitorum superficialis

(2, 3, 4 - Flexion)

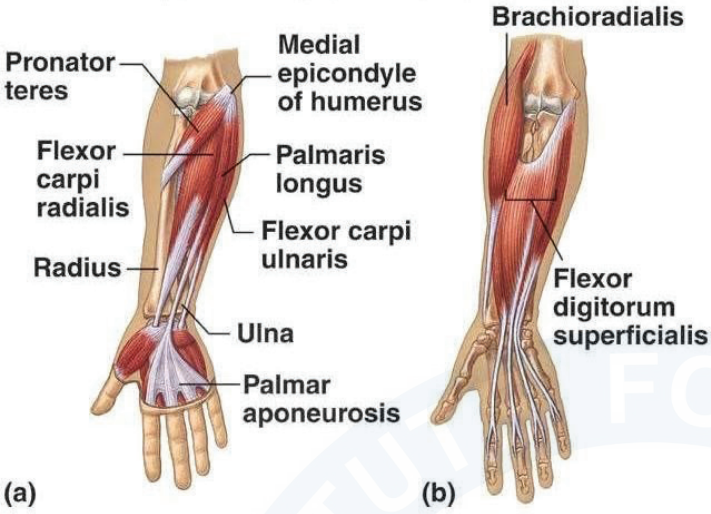
❑ Deep Group

1. Flexor digitorum profundus
2. Flexor pollicis longus

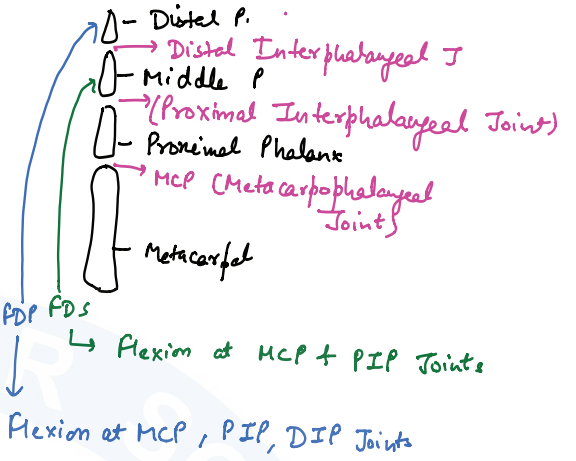
(1, 2 - Flexion)

3. Pronator Quadratus - Pronation

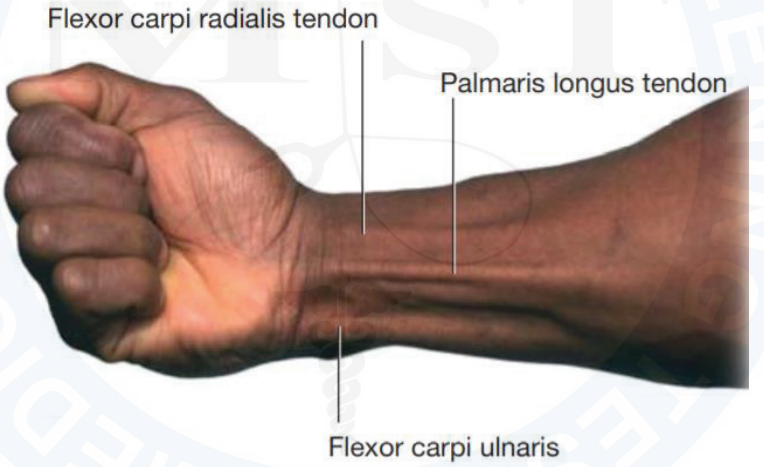
◀FIG. Muscles of Anterior Compartment of forearm (Deep Group)



◀FIG. Muscles of anterior Compartment of forearm (superficial Group)



◀FIG. Insertion of FDS (on middle Phalanx) & FDP on distal Phalanx



❑ Nerve supply

- All muscles of ant. comp. of. forearm Supplied by MEDIAN Nerve Except
  - 1. .....
  - 2. .....

❑ Other Points

- Radial Artery Pulsation .....
- Pronation & supination - Occurs at .....

- Pronation - Pronator Teres & Pronators Quadratus
- Supination - Supinator & Biceps Brachii
- Tennis Elbow → .....

  Golfer's Elbow → .....

  Student's Elbow → .....
- LABORER'S Nerve → Median Nerve

  MUSICIAN Nerve → Ulnar Nerve

2. Muscles of Posterior Compartment

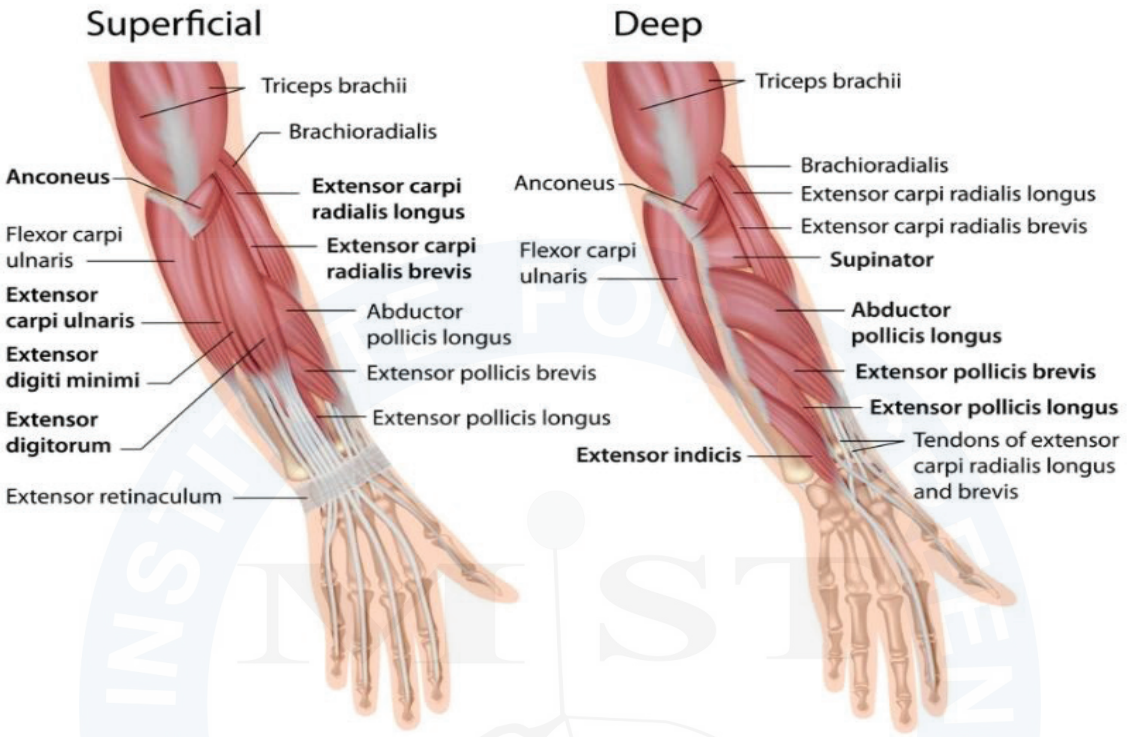
1. Extensor carpi Radialis Longus
2. Extensor carpi Radialis brevis
3. Extensor carpi ulnaris
4. Extensor digitorum
5. Extensor indicis
6. Extensor digiti minimi
7. Extensor Pollicis longus & brevis

Extension

8. Abductor Pollicis longus → Abduction of thumb
9. Brachioradialis → Flexion at Elbow joint
10. Anconeus → Screwing movement
11. Supinator → Supination

- All these muscles are supplied by Radial N

FIG. Brachioradialis (Anterior view)



◀**FIG.** Muscles of Posterior Compartment of forearm (Posterior view)

MUSCLES OF HAND

❑ **1. Thenar Group (4)**

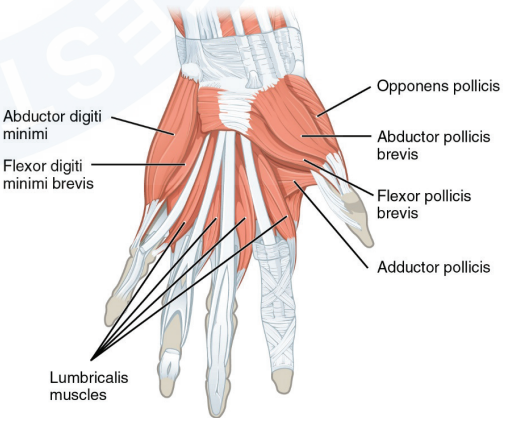
A. Flexor pollicis brevis
B. Abductor " "
C. Opponens " "
D. Adductor " " → Ulnar N

} Median N.

❑ **2. Hypothenar Group (4)**

A. Flexor digiti minimi
B. Abductor " "
C. Opponens " "
D. Palmaris Brevis

} Ulnar N



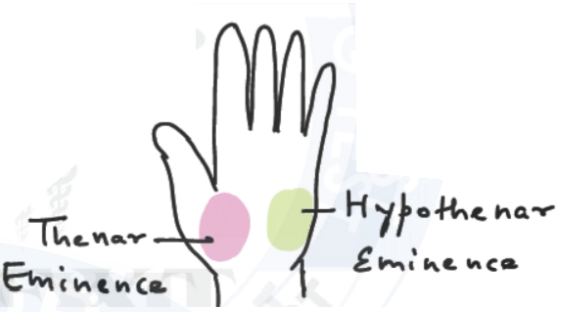
◀**FIG.** Muscles of Hand (including Lumbricals) (Anterior View)

❑ 3. Lumbricalis (4)

- 1st & 2nd lumbrical → Median N.
- 3rd & 4th lumbrical → Ulnar N.

❑ 4. Interossei

* Palmar I - (4)
* Dorsal I - (4)
} Ulnar N

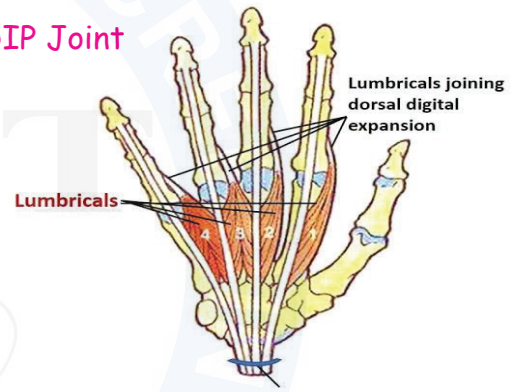


❑ Lumbricals

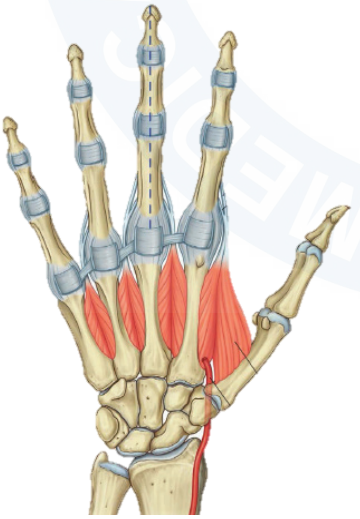
- Origin from tendon of FDP
- Action → Flexion at MCP & Extension & PIP & DIP Joint
  * (This action also done by both Interossei)

❑ Interossei

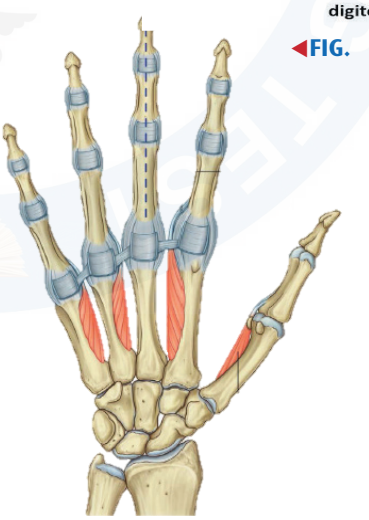
- Palmar I → Adduction of fingers
- Dorsal I → Abduction " "



◀FIG. Lumbricals



◀FIG. Dorsal Interossei (Anterior View)



◀FIG. Palmar Interossei (Anterior View)

✦ **Other Points**

1. Opposition Movement → Opposition of thumb (OPB - Main) & Little finger (ODM)
2. Unipennate → 1st & 2nd Lumbricals & All palmar Interossei
3. Bipennate → 3rd & 4th Lumbricals & All Dorsal Interossei
4. Palmar Interossei is not attached to central finger

ANATOMICAL SNUFF BOX

❑ **Boundaries**

- Medial Boundary → Extensor Pollicis Longus
- Lateral Boundary → Extensor Pollicis Brevis & Abductor Pollicis Longus

❑ **Floor**

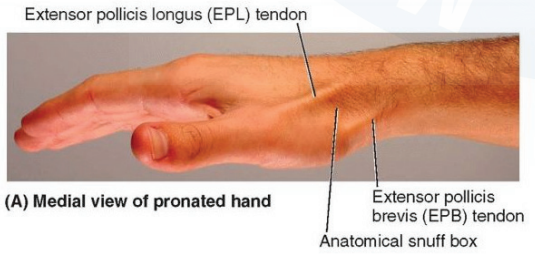
- Scaphoid & Trapezium

❑ **Roof**

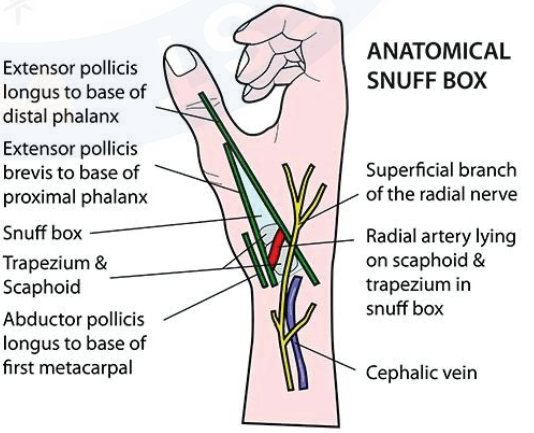
- Cephalic vein & Superficial branch of radial nerve

❑ **Content**

- Radial Artery



◀FIG. Anatomical Snuff Box

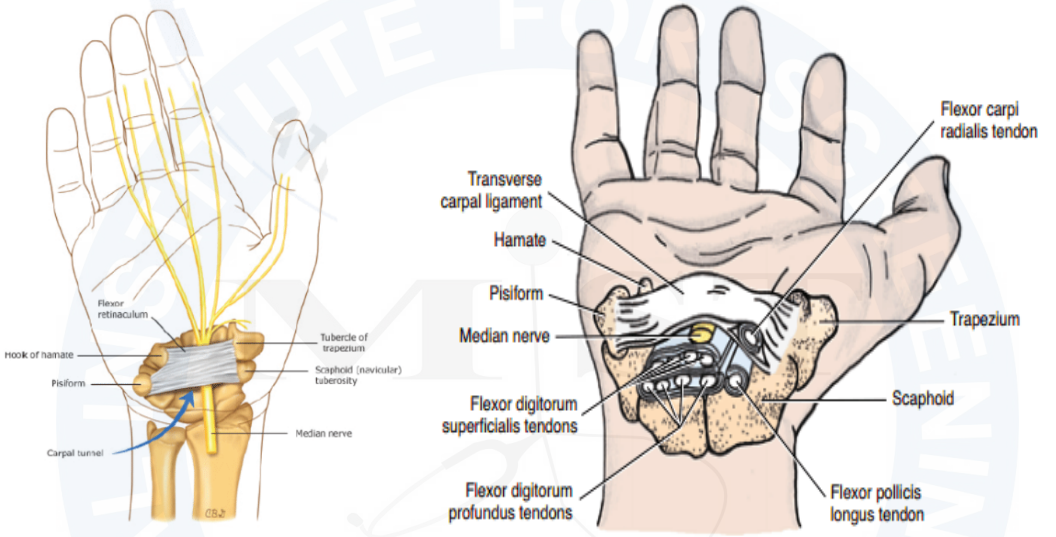


◀FIG. Anatomical Snuff Box

FLEXOR RETINACULUM

❑ **Structures passing deep to FR / passing through carpal tunnel**

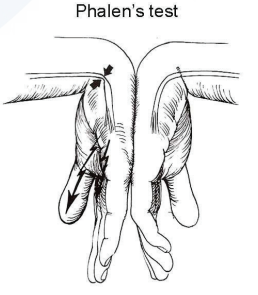
1. Median Nerve
2. Tendons of FDS (Flexor Digitorum Superficialis)
   FDP (Flexor Digitorum Profundus)
   FPL (Flexor Pollicis Longus)



◀**FIG.** Flexor Retinaculum

Carpal Tunnel Syndrome

- **Compression of Median Nerve**
- Complains Tingling, Numbness, Pin Prick Sensation and Paresthesia in the distribution of median N.
- Symptoms appear in night / increase in night
- Thenar wasting /reduced thenar eminence
- PHALEN Test Positive
- TINEL'S Sign



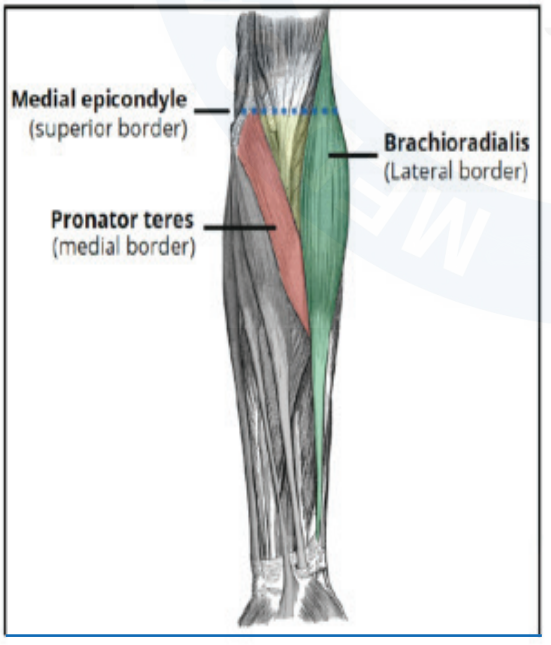
◀**FIG.** Phalen's Test

Note: **Cubital Tunnel Syndrome and guyon canal** syndrome compression of **Ulnar Nerve**

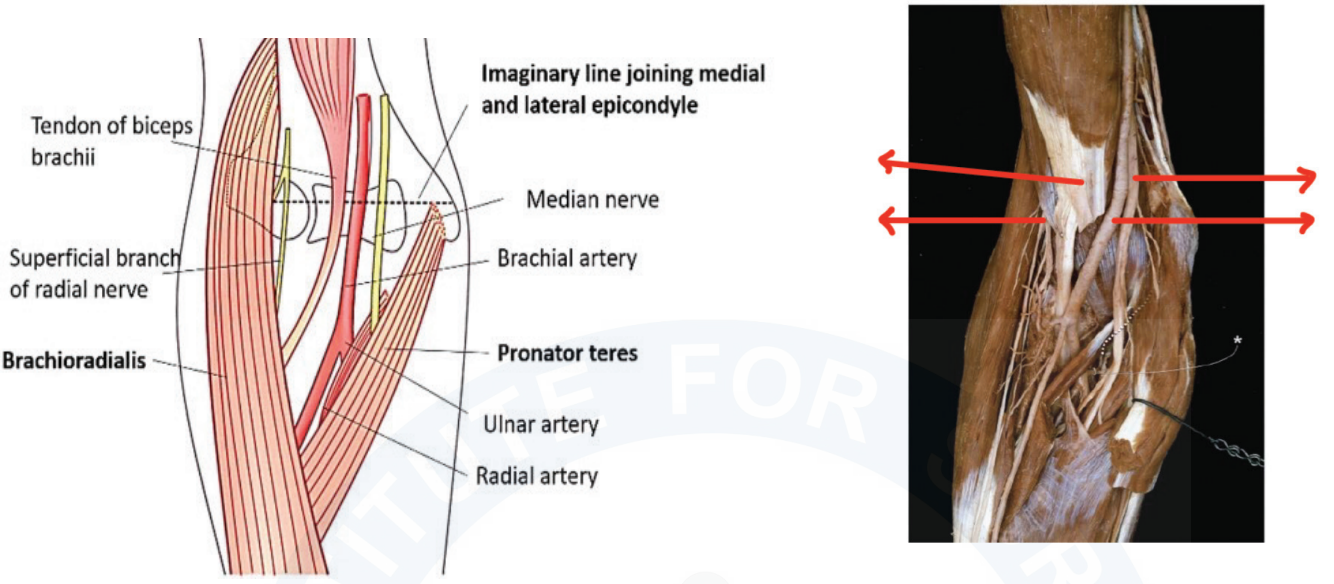
CUBITAL FOSSA

- **Boundaries**
  1. Medial Boundary - Prorator Teres
  2. Lateral Boundary - Brachioradialis
  3. Base → Imaginary line between Medial & Lateral Epicondyle of Humerus.
- **Roof** - Median Cubital Vein (MCV)
- **MCV** - Most commonly used for taking intravenous blood samples & for giving intravenous drugs
- **Floor** - Supinator & Brachialis
- **Contents (Medial to Lateral)**
  1. .....
  2. .....
  3. .....
  4. .....

- Note - Brachial Artery Pulsation is felt .....

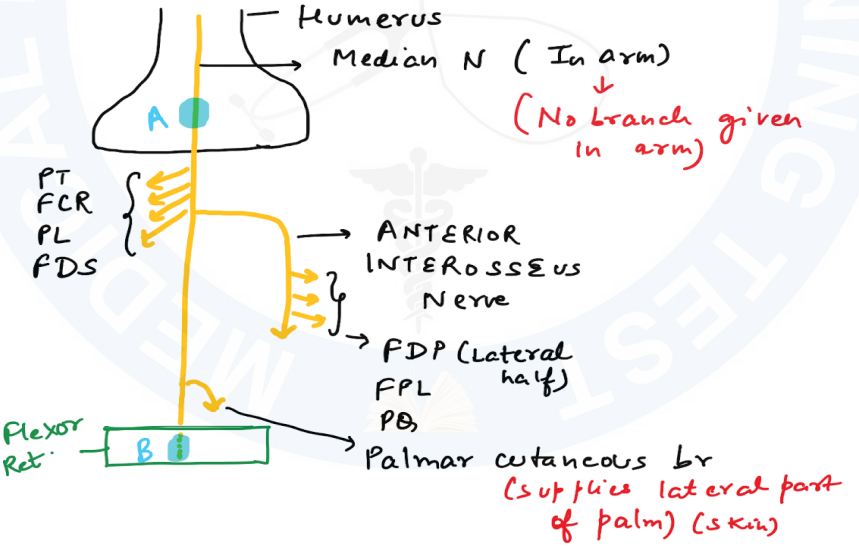


◀FIG Boundaries of Cubital fossa



◀**FIG.** Contents of cubital fossa

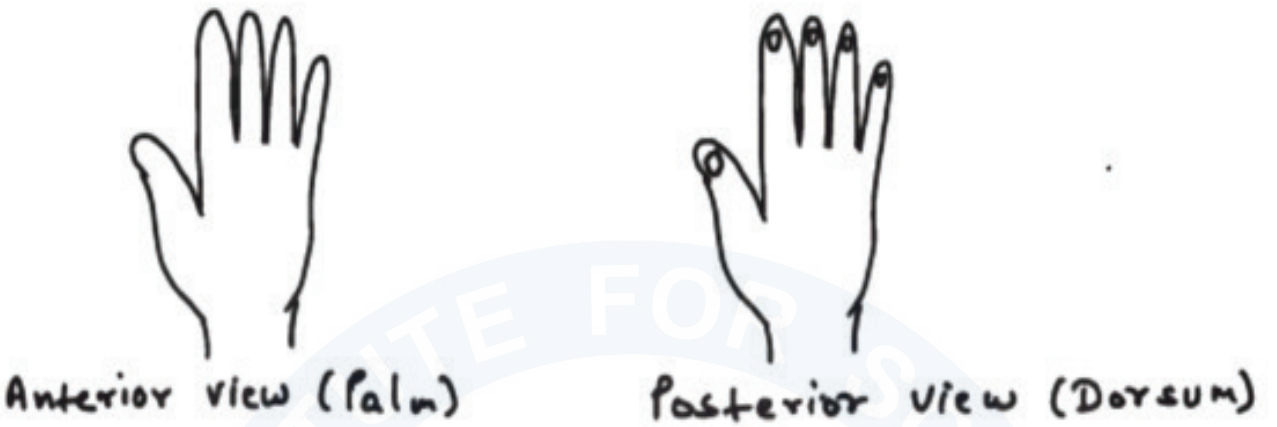
MEDIAN NERVE - COURSE, BRANCHES & LESIONS



◀**FIG.** Median Nerve (Course & Branches)

- Sensory supply
  - Skin of Lateral $3\frac{1}{2}$ fingers & adjacent palm
  - Nailbeds of $3\frac{1}{2}$ fingers (dorsum of hand)

❑ Sensory Supply (Skin) →

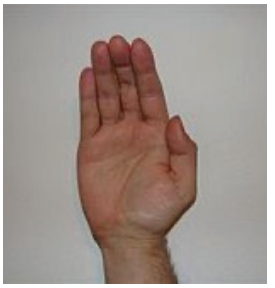


- Note -
  - Skin over anatomical snuff box → .....
  - Skin over thenar eminence → .....
  - Skin over hypothenar eminence → .....

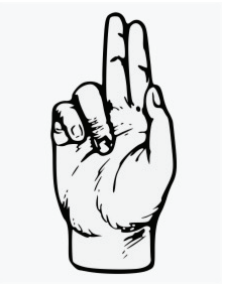
Lesions of Median Nerve

❑ Lesion A- At Elbow (eg. supracondylar # of Humerus)

- .....
- .....
- .....
- Thenar Atrophy / Thenar wasting / Reduced thenar Eminence
- Sensory loss (In distribution of the nerve)



◀FIG. Apethumb deformity

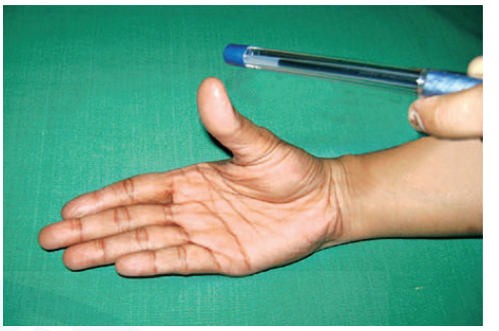


◀FIG. Benediction deformity

◀FIG. Pointing Index

❑ **Lesion B- At Wrist** (eg. wrist slash injury / suicidal cut)

- Ape thumb deformity
- Thenar atrophy
- sensory loss



◀**FIG.** Pen Test

❑ **Test**

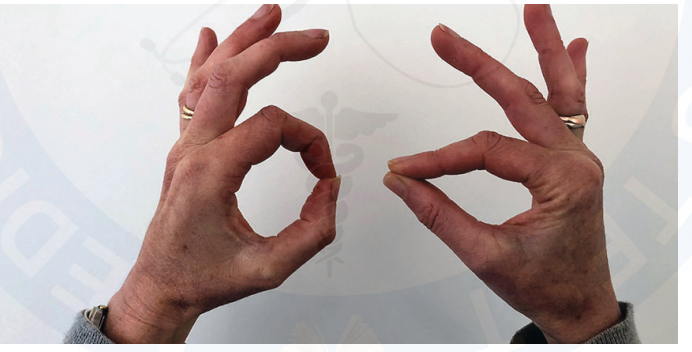
1. Pen test - To check abduction of thumb

CLINICAL INTEGRATION

Q. A 26 old male had an accident and had injuries in arm region. When he attempts to make a fist, the index finger and middle finger are unable to flex. On examination, pen test is positive. Which nerve is involved in this case?

a. Axillary nerve b. Radial nerve c. Median nerve d. Ulnar nerve

❑ **Ok Sign - For Anterior Interosseous Nerve Injury**



◀**FIG.** OK Sign

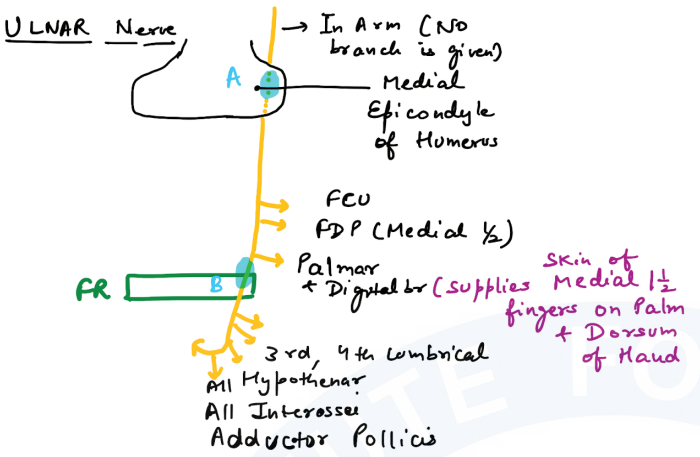
ULNAR NERVE - COURSE, BRANCHES & LESIONS

Sensory Supply → Medial 1½ fingers + adjacent hand (Both Palm and dorsum of Hand)

Lesions of ulnar Nerve

❑ **Lesion A- At Elbow** (eg Fracture dislocation of medial epicondyle of Humerus)

- Partial Claw Hand (due to involement of all Interossei & 3rd & 4th lumbricals) → (Clawing of medial two fingers)



◀FIG. Median Nerve (Course & Branches)

◀FIG. Partial/Ulnar Claw Hand

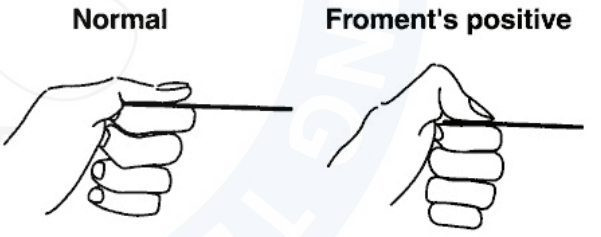
- Hypothenar atrophy / Hypothenar wasting / Reduced Hypothenar Eminence
- Sensory loss in the distribution of ulnar N.

❑ Lesion B- At Wrist

- ULNAR PARADOX - In low lesion, the severity of clawing will increase (due to sparing of FDP)
- Hypothenar atrophy
- Sensory loss

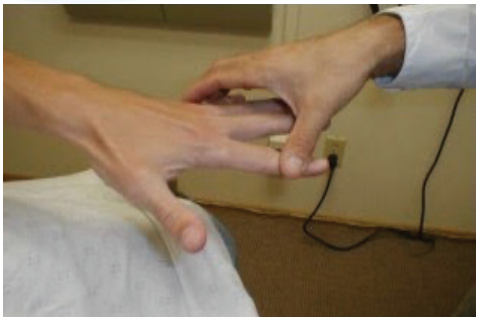
❑ Tests

1. Book fromment test → To check adduction of thumb
2. Card / Paper test → To check adduction of fingers
3. Egawa test → To check abduction of fingers



◀FIG. Book Fromments Test

◀FIG. Card Test



◀FIG. Egawa Test

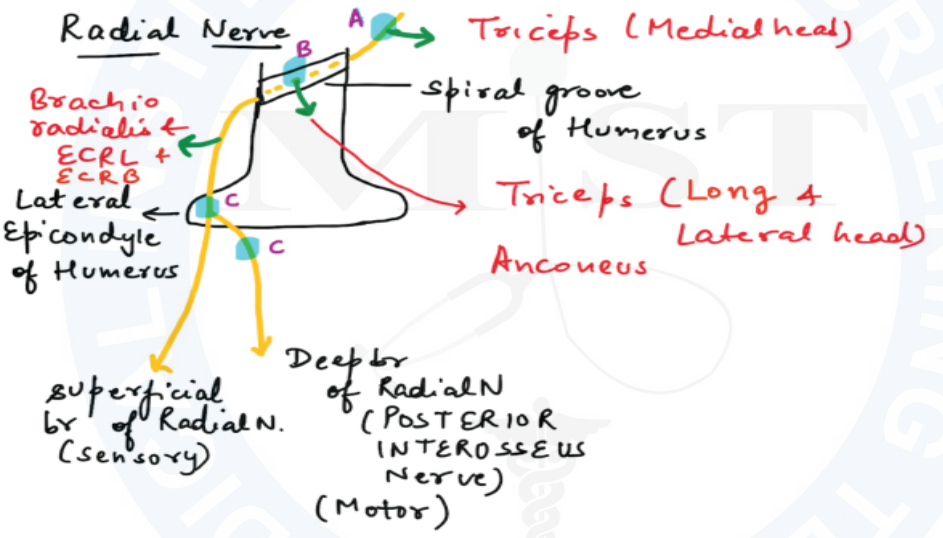
Note: **Complete Claw Hand** → Both Median N and Ulnar N.

CLINICAL INTEGRATION

A patient presents with hypothenar muscle wasting and loss of sensation on little finger and medial half of ring finger. which nerve is involved?

a. Axillary nerve
b. Radial nerve
c. median nerve
d. Ulnar nerve

RADIAL NERVE - COURSE, BRANCHES & LESIONS



◀FIG Radial Nerve - Course & Branches

ECRL - Extensor Carpi Radialis Longus
ECRB - Extensor Carpi Radialis Breivs

Sensory Supply → Supplies Skin Of Lateral $3\frac{1}{2}$ Fingersand Adjacent Hand (On Posterior Aspect) (Except Nailbeds)

Lesions of Radial Nerve

❑ **Lesion A - At axilla** (eg. Crutch Palsy)

- Loss of extension at Elbow
- Wrist drop
- Finger drop
- Sensory loss

◀FIG Crutch Palsy

◀FIG Saturday Night Palsy

❑ **Lesion B - At spiral groove** eg. # of shaft of Humerus, Saturday Night Palsy

- Wrist drop
- Finger drop
- Sensory loss

(Note - Extension of Elbow is weakened) (due to sparing of Triceps)

❑ **Lesion C (At Elbow)** → Triceps, Brachioradialis, ERCL Spared

At Lateral Epicondyle (# of L. E.) → Finger drop, Sensory loss

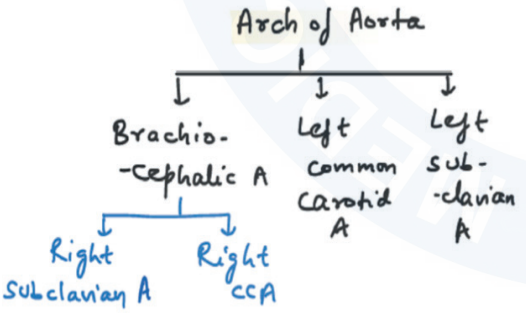
Posterior Interosseous Nerve Injury → Finger drop

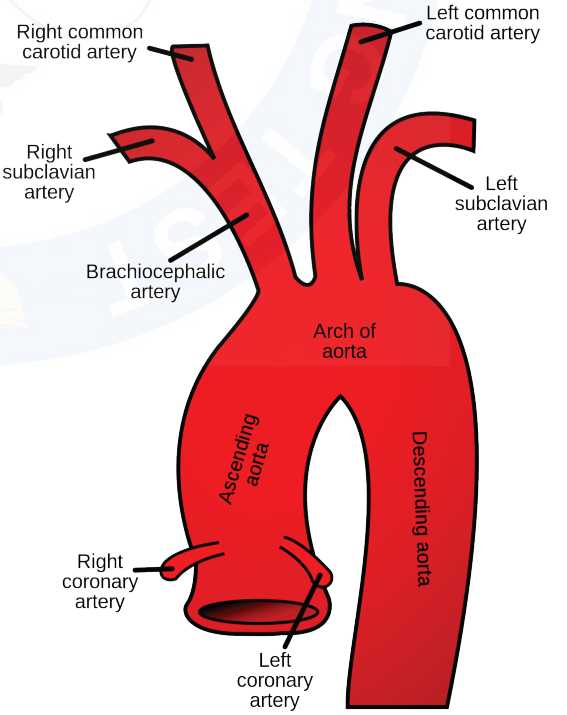
FIG. Wrist drop

ARTERIES OF UPPER LIMB

1. Complete overview

❑ **Arch of aorta & its branches**





Subclavian Artery (Neck)
↓
Axillary Artery (Axilla)
↓
Brachial Artery (Arm)
↓
Radial A Ulnar A (Forearm)

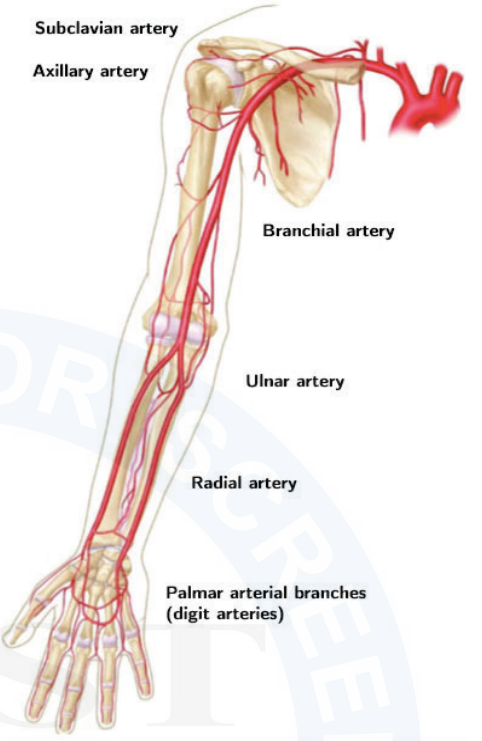
Radial A Ulnar A (Forearm)

Palmar or Arterial Arches (Hand)

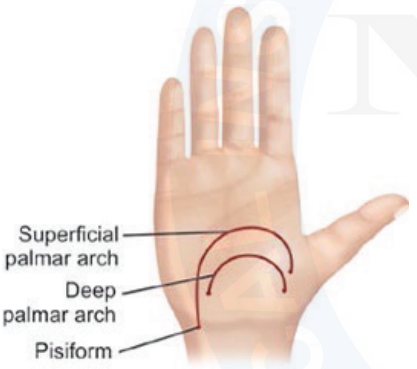
(Superficial & deep palmar arches)

Ulnar Artery (mainly) + Radial A

Radial A (mainly) + Ulnar A



◀FIG. Arteries of upper limb



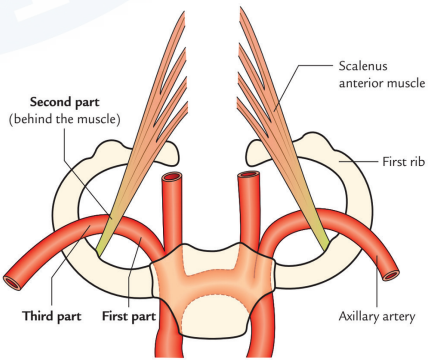
◀FIG. Palmar Arch

- superficial Palmar Arch (SPA)
  - * Ulnar Artery continues / Ends as SPA
  - * Located at the level of Distal border of Extended thumb
- Deep Palmar Arch → Located 1cm proximal to SPA

2. Subclavian Artery

❑ Subclavian Artery

- ✦ Extends from origin to outer border of 1st rib
- ✦ Continues as Axillary A
- ✦ divided into 3 parts by Scalenus Anterior



◀FIG. Subclavian Artery

❑ Branches

✦ 1st Part

- .....
- .....

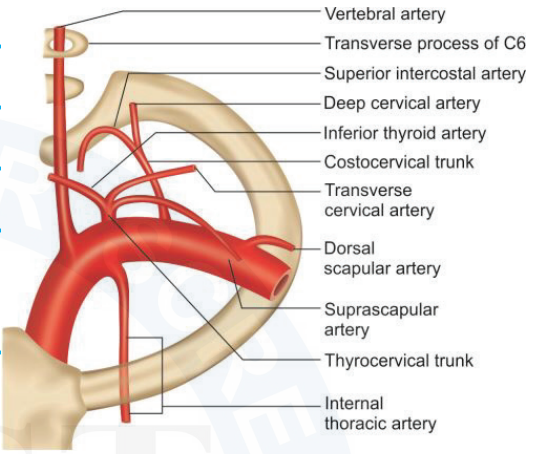
- .....
- .....
- .....
- .....

✦ 2nd Part

- .....

✦ 3rd Part

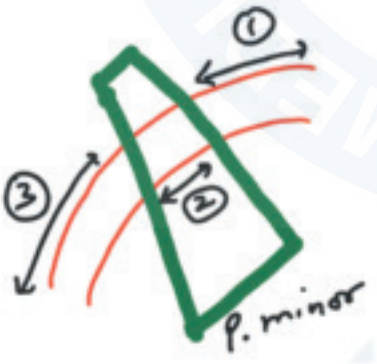
- .....

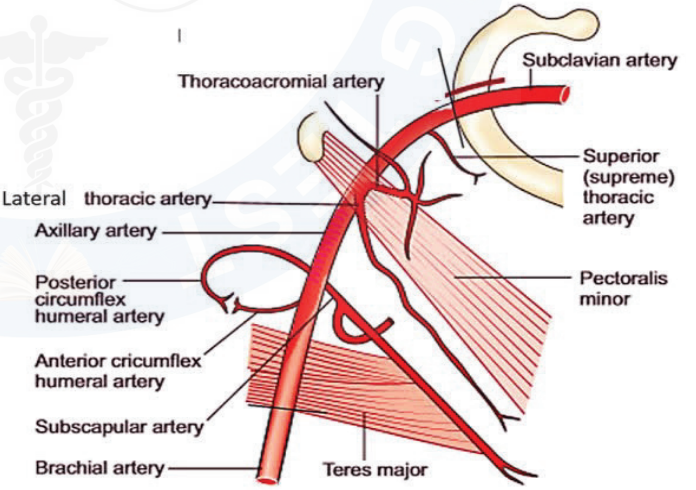


◀**FIG.** Subclavian Artery Branches

3. Axillary Artery

- Divided into 3 parts by Pectoralis Minor
- Extent - From outer border of 1st rib till lower border of teres major





◀**FIG.** Axillary Artery (Branches)

❑ Branches

Ist part → Superior Thoracic A

IInd part → Thoraco Acromial A

→ Lateral Thoracic A

IIIrd part → Subscapular A
→ Anterior Circumflex Humeral A
→ Posterior Circumflex Humeral A

Total - 6 branches

Note - Circumflex scapular A is a branch of Subscapular A.

4. Anastomosis around Scapula

✦ Between the branches of 1st part of subclavian A & 3rd part of Axillary A

❑ **Arteries participating in this anastomosis**

1. Suprascapular A
2. Transverse Cervical A
3. Subscapular A (Circumflex Scapular A)

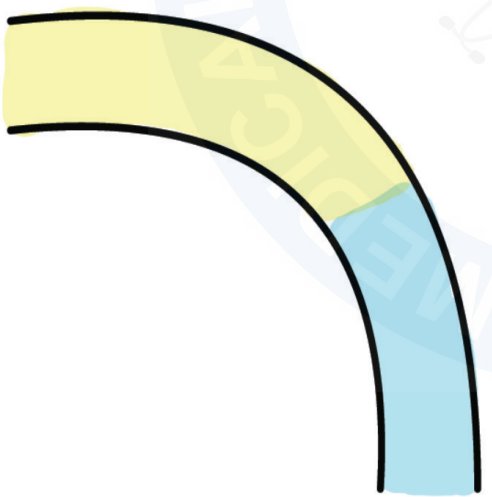


FIG. Anastomosis Around Scapula

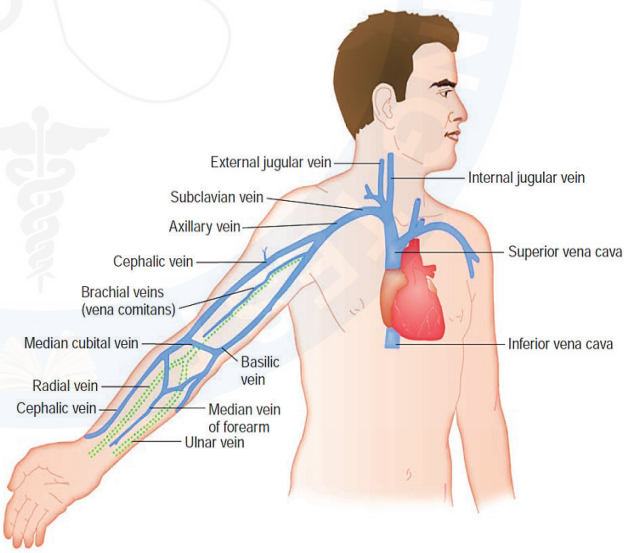
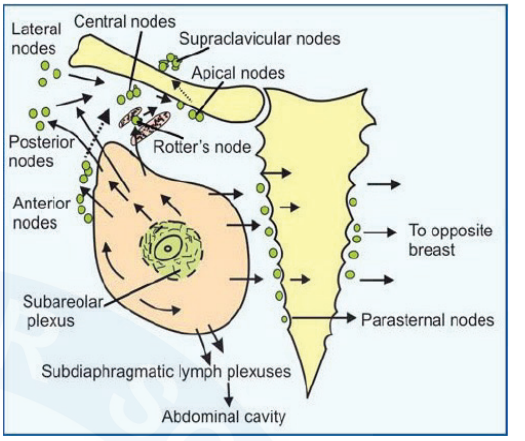


FIG. Veins of upper limb

BREAST

❑ **Lymphatic drainage of breast**

1. Axillary LN (75%) → Chief LN draining the breast is Anterior Axillary LN
2. Internal Mammary LN / Parasternal LN → 20%
3. Posterior Intercostal → 5% LN
4. Sub diaphragmatic & sub peritoneal lymphatic plexus.



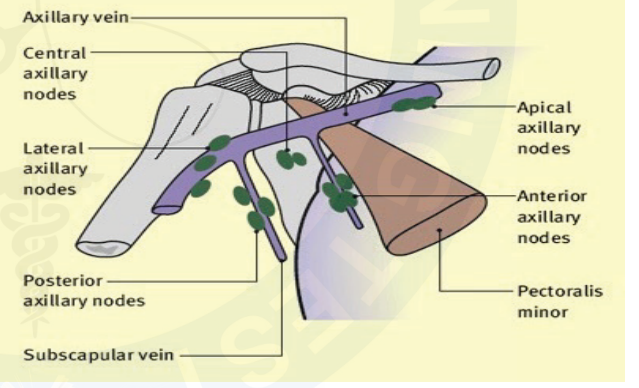
◀FIG. Breast (Lymphatic drainage)

❑ **Subareolar lymphatic plexus of Sappey** located deep to areola

AXILLARY LYMPH NODES

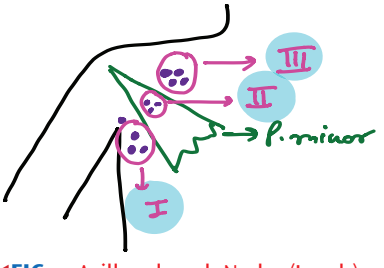
❑ **5 groups of axillary lymph nodes**

1. Anterior group
2. Posterior group
3. Lateral group
4. Central group
5. Apical group (Infraclavicular group)



◀FIG. Axillary Lymph Nodes

❑ **Levels of axillary LN** → According to Pectoralis minor.



◀FIG. Axillary lymph Nodes (Levels)

I → LN Lateral & below to P. minor

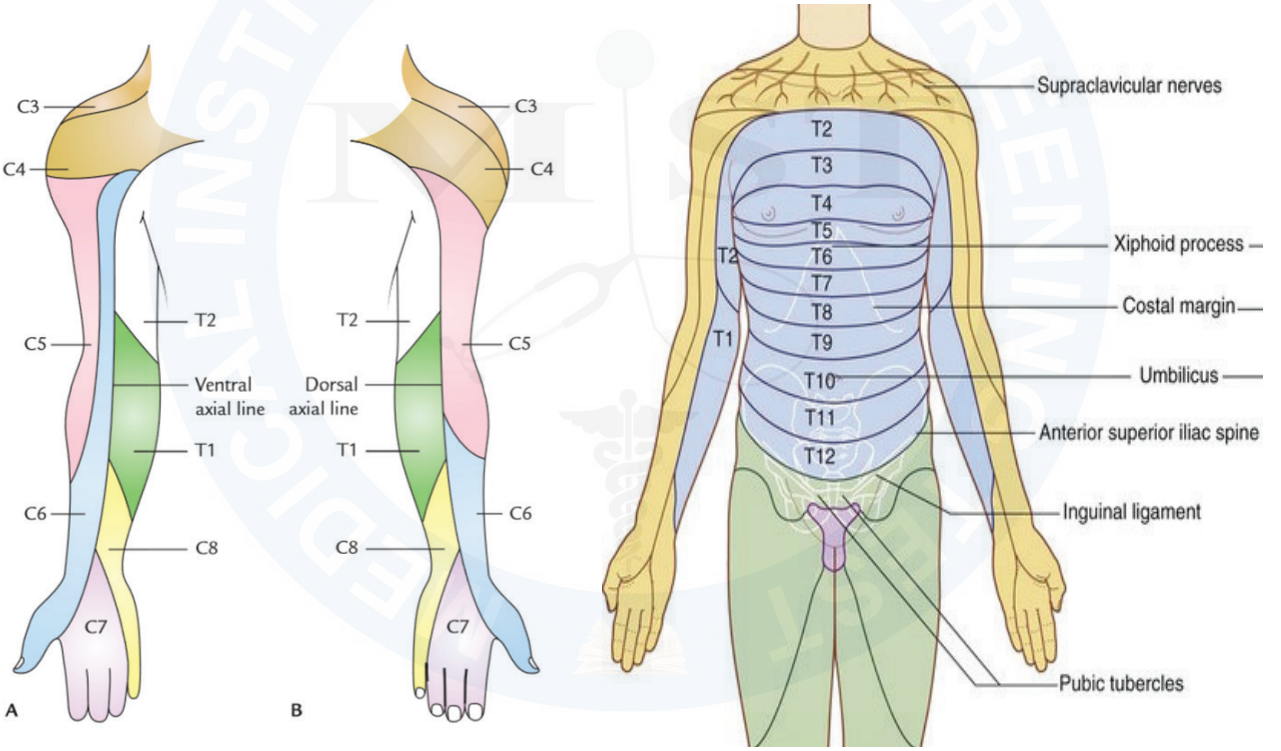
II → LN deep to P. minor

III → LN medial & above to P. minor

DERMATOMES OF UPPER LIMB

- Dermatome is an area of skin that is mainly supplied by a single spinal Nerve.
- Dermatome of
  - Thumb - C_6
  - Middle 3 fingers - C_7
  - Little finger - C_8

Note: Dermatome of Umbilicus is T_{10} Spinal Nerve
(Skin around umbilicus is supplied by T_{10})



◀**FIG.** Dermatomes of upper limb

- Note → Vertebral Level of Umbilicus
  1. L4
  2. Junction of L3 & L4

Other Points

1. Clavipectoral fascia - Structures piercing it
   (a) Lateral Pectoral Nerve (b) Cephalic Vein
   (c) Thoracoacromial Artery (d) Lymphatics from breast
2. • Angle of Humeral Torsion – 163° – 164°
   • Angle of femoral Torsion – 15°
3. • Biceps Jerk – C5C6
   • Triceps Jerk – C6C7C8
4. • Largest Bursa near shoulder Joint → Subacromial Bursa.
   • Which bursa communicates with synovial cavity of shoulder joint → Subscapular Bursa
5. Muscles
   • Multipennate – Deltoid, Subscapularis
   • Bipennate – 3rd & 4th Lumbrical, dorsal Interossei
   • Cruciate – Sternocleido mastoid, Masseter, adductor Magnus
   • Spiral – Pectoralis Major, Latissimus dorsi, Supinator.
6. Axilla -
   • Anterior wall - Pectoralis Major, P. minor, Subclavivs
   • Posterior wall - Subscapularis, L. dorsi, Ters Major
   • Medial wall - Serratus anterior & ribs
   • Laterral wall - Biceps & coracobrachialis
7. Coracoclavicular Ligament → Heips in weight transmission of upper limb to the body.

Workbook Questions

1. All of the following structures are passing deep to flexor retinaculum EXCEPT:
   (a) Flexor pollicis longus (b) Flexor digitorum superficialis
   (c) Palmaris longus (d) Flexor digitorum profundus

2. Root value to axillary nerve:
   (a) C5, C6 (b) C6, C7, C8
   (c) C5, C6, C7 (d) C5-T1

3. Which is the most commonly fractured carpal bone?
   (a) Hamate (b) Lunate (c) Scaphoid (d) Capitate

4. Which is the nerve pass behind medial epicondyle?
   (a) Ulnar (b) Radial
   (c) Median (d) Posterior interosseus

5. The medial boundary of the anatomical snuffbox is formed by the:
   (a) Ext. pollicis brevis (b) Ext Carpi radialis longus
   (c) Extensor pollicis longus (d) Ext Carpi radialis brevis

6. Winging of scapula is due to damage of:
   (a) Axillary nerve (b) Long thoracic nerve
   (c) Median nerve (d) Ulnar nerve

7. Median nerve supplies all EXCEPT:
   (a) Abductor pollicis brevis (b) Adductor pollicis brevis
   (c) Flexor pollicis brevis (d) Opponens pollicis

8. All are muscles of rotator cuff EXCEPT:
   (a) Teres major (b) Teres minor
   (c) Infraspinatus (d) Supraspinatus

9. Supination and pronation of upper limb occurs at which joint :
   (a) Wrist joint (b) Radioulnar joint
   (c) Elbow joint (d) Carpometacarpal joint

10. Inferior scapular angle is at which level?
    (a) T4 (b) T6
    (c) T7 (d) T2

1. C 2. A 3. C 4. A 5. C 6. B 7. B 8. A 9. B 10. C

JOINTS

Classification of Joints

- **Joints** - Fibrous
  Cartilaginous
  Synovial

- According to movement

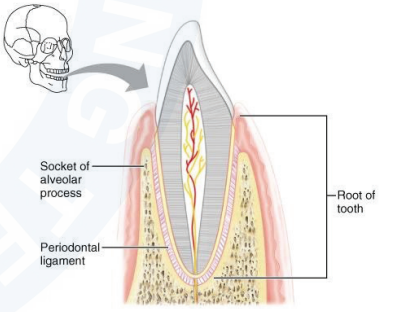
1) IMMOVABLE Joints (SYNARTHROSIS) → Eg. Fibrous J + primary cartilaginous J.

2) SLIGHTLY MOVABLE J. (AMPHIARTHROSIS) → Eg. secondary Cartilaginous

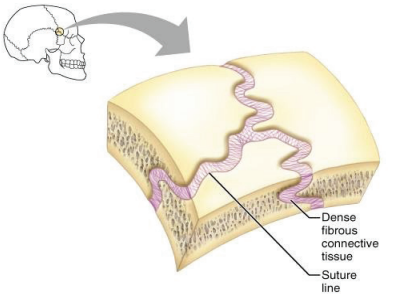
3) FREELY MOVABLE J. (DIARTHROSIS) → Eg. synovial J

Fibrous Joint

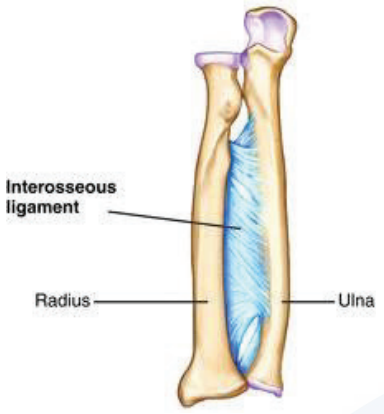
Sutures	Syndesmosis	Gomphosis
↓	↓	↓
Skull	• Middle Radioulnar Joint • Middle & Inferior Tibiofibular Joint	Tooth in alveolar socket (Bony)



◀FIG. 1.4 Fibrous Joint - Gomphosis



◀FIG. FIBROUS JOINT - SUTURES



◀**FIG. 1.3** Fibrous Joint - Syndesmosis

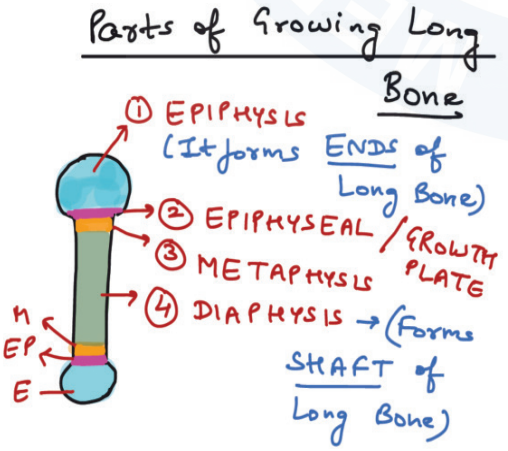
Cartilaginous J

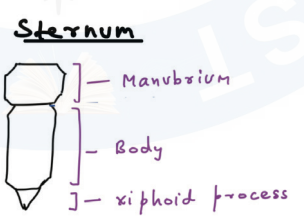
Primary C. J. (Synchondrosis)

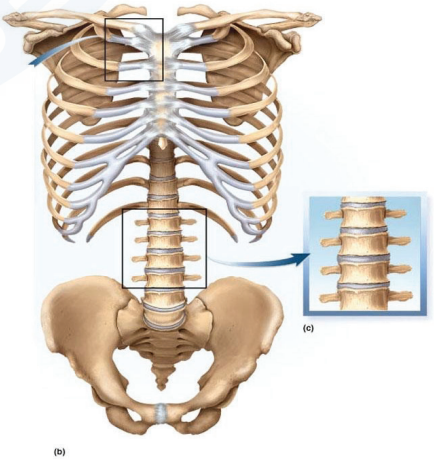
- Epiphyseal Plate (between epiphysis & diaphysis)
- Spheno-occipital joint

Secondary C. J. (Symphysis)

- Pubic Symphysis
- Manubriosternal J.
- Inter vertebral J.
- Xiphisternal J.







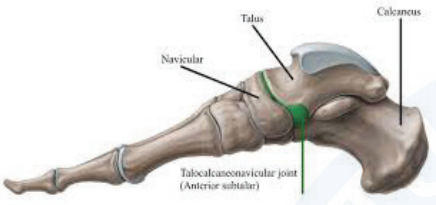
◀**FIG.** Secondary Cartilaginous Joint

❑ **For Reference**

Later on , Primary cartilaginous joint ossifies with age. This process is called synostosis

Synovial Joints

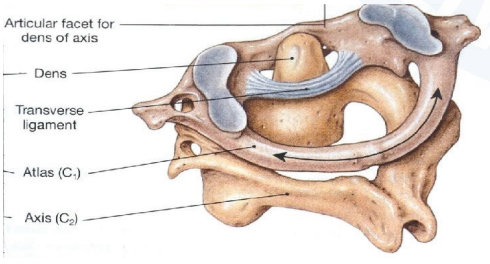
- **Ball & Socket J →**
  - Hip J
  - Shoulder J
  - Talocalcaneo - navicular J
  - B/w Incus & Stapes (Incudo - Stapedial Joint)



◀**FIG.** Talocalcaneonavicular Joint

◀**FIG.** Ear ossicles

- **Hinge J →**
  - Elbow J
  - Interphalangeal J
- **Pivot J →**
  - Superior & Inferior radioulnar J
  - Atlantoaxial J
    ("NO" Movement Joint)
    (It helps to look on right side & then on left side)

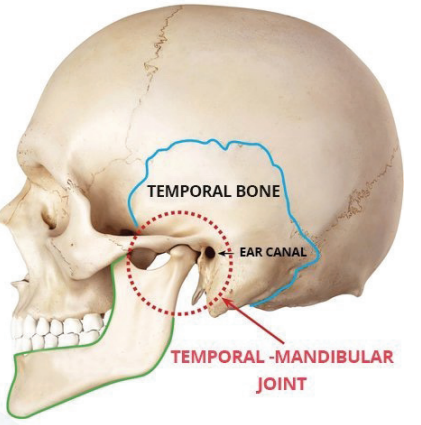


◀**FIG.** Atlantoaxial Joint

◀**FIG.** Atlanto occipital Joint

- **Ellipsoid J →**
  - Wrist J
  - Meta carpophalangeal J
  - Atlanto-occipital J

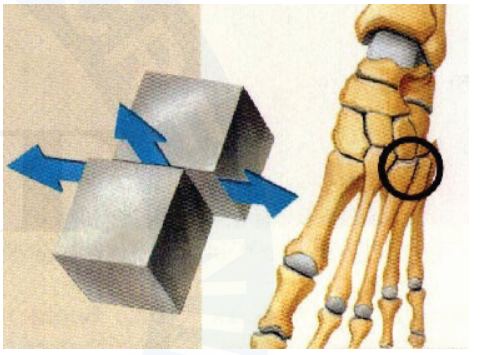
❑ **Saddle J →**
- Ist Carpometacarpal J.
- B/w Malleus & Incus - (Incudo - Malleolar Joint)
- Sternoclavicular J.
- Calcaneocuboid J.



◀**FIG.** Condylar Synovial Joint

❑ **Condylar J →**
- Knee J
- Temporo - mandibular J
- MCP Jt.

❑ **Plane/Gliding J →**
- Inter metacarpal J
- Inter carpal J
- Inter metatarsal J
- Inter tarsal J



◀**FIG.** Plane Synovial Joint

❑ **For Reference →**

MCP Joint

1) Ellipsoid (First Best ans)
2) Condylar

EPIPHYSIS

1. Pressure Epiphysis

- ✦ Upper End of Humerus
- ✦ Lower End of Radius
- ✦ Lower End of Femur
- ✦ Upper End of Tibia
- ✦ Upper End of Femur
- ✦ Lower End of Fibula

2. Traction Epiphysis

- Greater and Lesser Tubercle of Humerus.
- Greater and Lesser Trochanter of Femur
- Mastoid process

3. Atavastic Epiphysis

- Coracoid Process (Scapula)
- OS Trigonum (Posterior Tubercle of Talus)

4. Aberrant Epiphysis

- Head of 1st Metacarpal
- Base of 2nd to 5th Metacarpal

HYBRID/COMPOSITE MUSCLE

Workbook Questions

1.. Which is an example of pressure epiphysis?
- a. Head of first metacarpal
- b. Upper end of humerus
- c. Coracoid process
- d. Acromion process

Ans B

2.. All are hybrid muscles except?
- a. Pectoralis major
- b. Latissimus dorsi
- c. Flexor digitorum profundus
- d. Subscapularis

Ans B

3.. Pubic Symphysis is example of:
- (a) Fibrous joint
- (b) Primary cartilaginous joint
- (c) Secondary cartilaginous joint
- (d) Synovial joint

Ans C

4.. Temporomandibular joint is
- a. Ellipsoid joint
- b. Condylar joint
- c. Saddle joint
- d. Hinge joint

Ans B

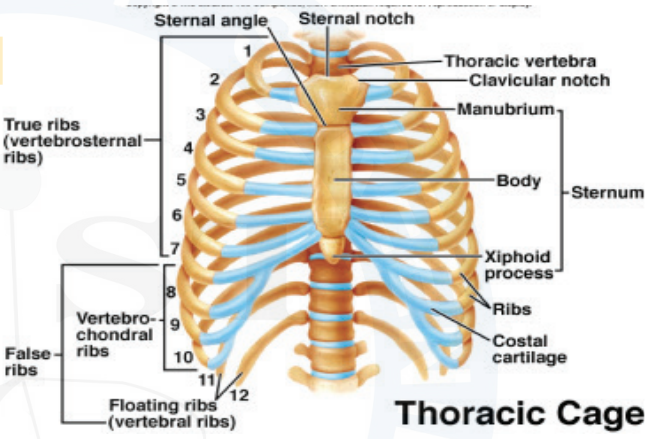
5.. "NO" movement joint is
- a. Atlantoaxial joint
- b. Atlanto-occipital joint
- c. Sternoclavicular joint
- d. Shoulder joint

Ans A

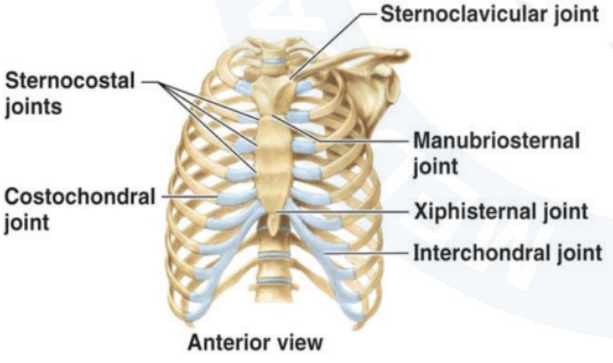
BONES OF THORACIC REGION

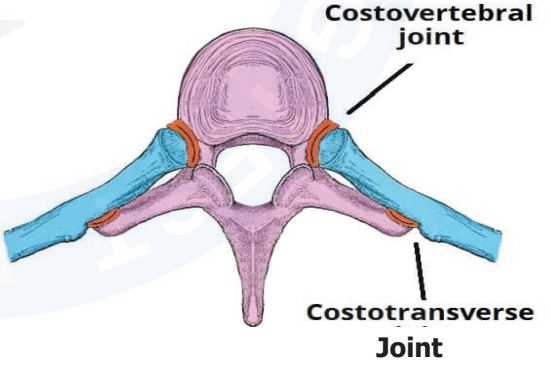
1. Ribs

- True Ribs - 1 to 7
- False Ribs - 8 to 12
- Vertebrosternal ribs - 1 to 7
- Vertebrochondral ribs - 8, 9, 10
- Vertebral ribs - 11, 12 (floating ribs)

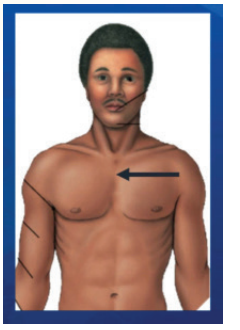


◀FIG. Ribs in thoracic Cage

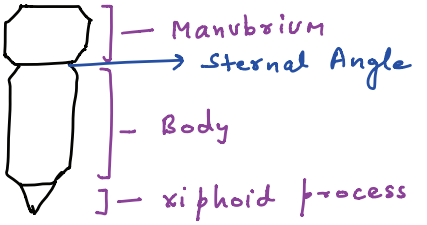




2. Sternum

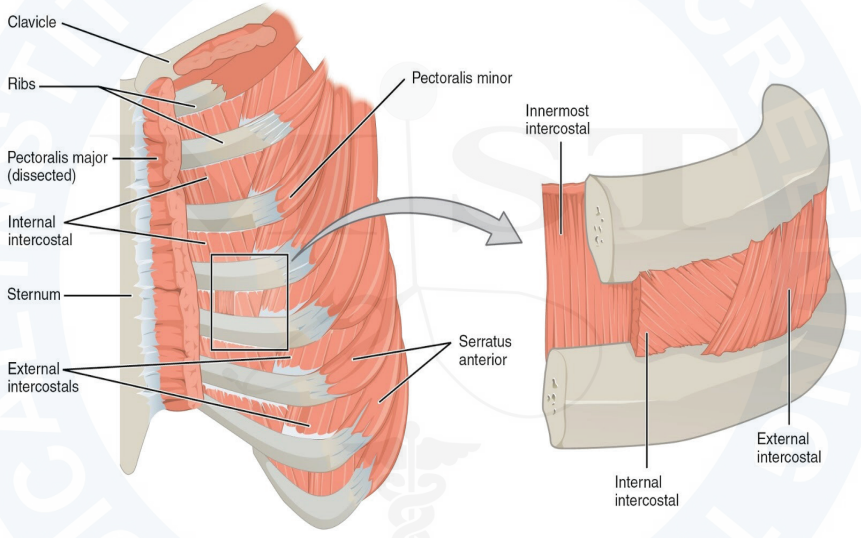


- **Sternal angle of Louis**
  - Between Manubrium & Body of sternum
  - Lies at level of Lower border of T_4



◀**FIG.** Sternum

INTERCOSTAL SPACE - MUSCLES & PLEURAL TAP



◀**FIG.** Intercostal space Showing Intercostal Muscles

- **Neuro vascular Bundle** → (Located at lower border of Rib)
  - Arrangement (above downwards)

V → Intercostal Vein
A → Intercostal Artery
N → Intercostal Nerve

- **Pleural Tap / Paracentesis Thoracis**
  - Needle is inserted along the .....

CLINICAL INTEGRATION

A needle has to be inserted by an intern. What is the most preferred site for this clinical procedure ?

a. Lower border of the upper rib
b. Upper border of the lower rib
c. Middle of the intercostal space
d. Lower border of the lower rib

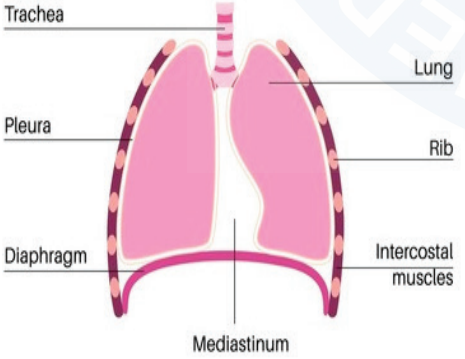
MEDIASTINUM

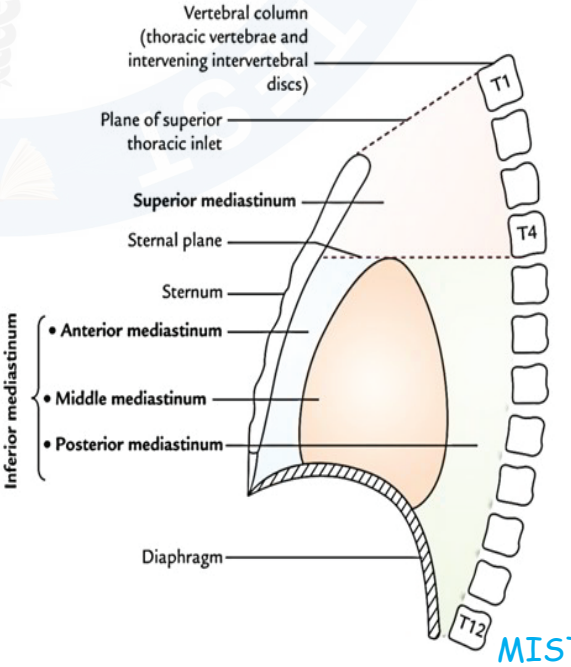
Mediastinum

↓

divide at the level of T_4 (Sternal angle)

- Superior Mediastinum
- Inferior Mediastinum
  - Anterior Mediastinum
  - Middle Mediastinum
  - Posterior Mediastinum

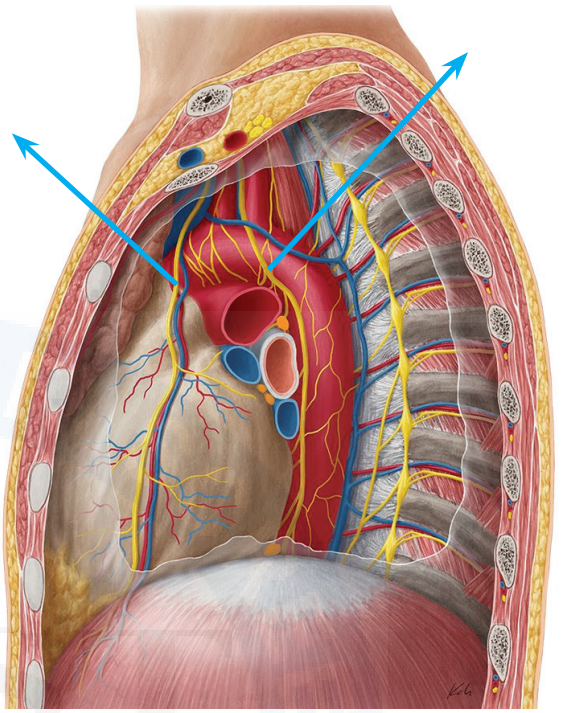




MIST

❑ Superior Mediastinum - Contents

1. .....
2. .....
3. .....
   .....
4. .....
   .....
5. .....
6. .....
7. .....
8. .....
9. .....
10. .....



Middle Mediastinum - Contents

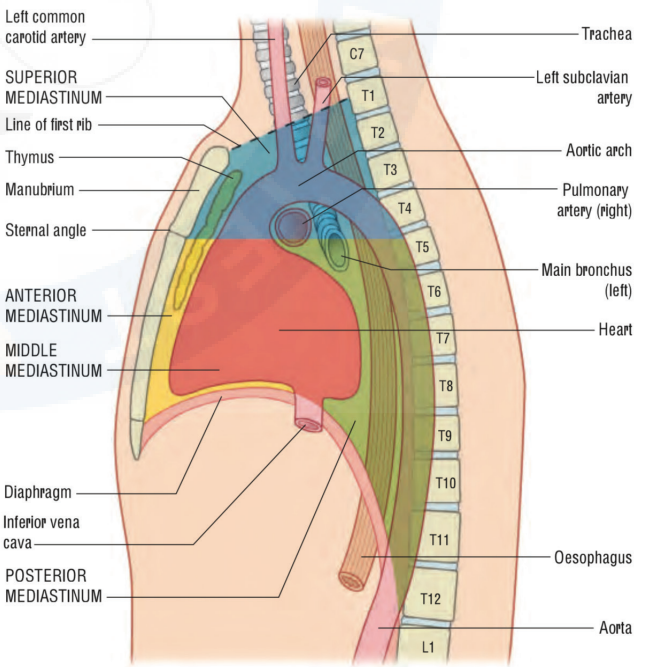
1. Heart
2. Phrenic Nerve

❑ Anterior Mediastinum - Contents

1. Lower part of thymus
2. Fibrofatty tissue
3. Sternopericardial ligament

❑ Posterior Mediastinum - Contents

1. .....
2. .....
3. .....
4. .....
5. .....



HEART

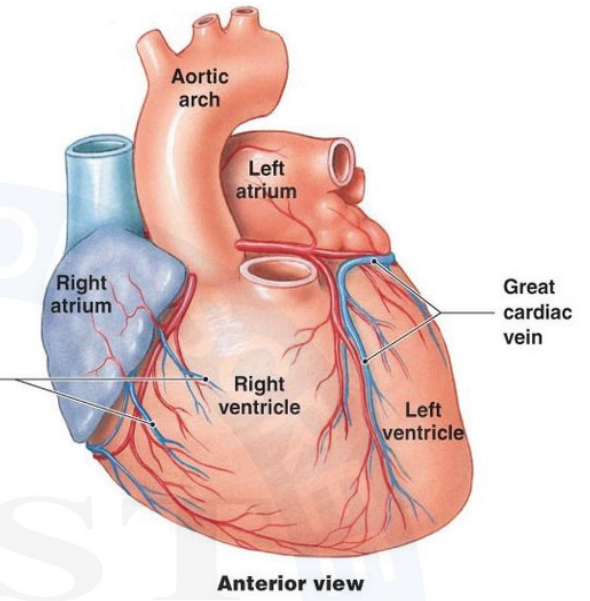
1. Parts, Surfaces & Grooves.

❑ Apex

- Formed by .....
- Location: Left 5th Intercostal space (9 cm from midline)- "APEX BEAT"

❑ Base .....

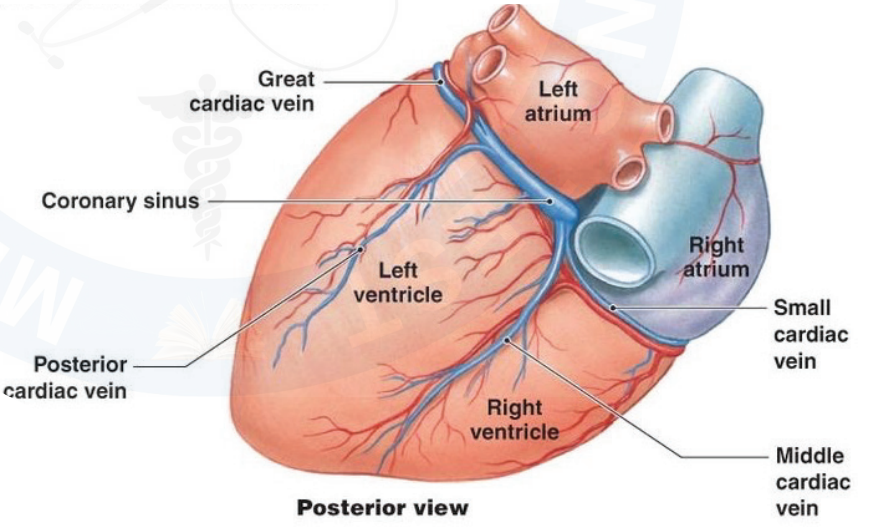
.....



◀FIG. Heart (Anterior view)

❑ 3 Surfaces

1. Anterior / Sternocostal surface
2. Inferior / Diaphragmatic surface
3. Left lateral surface



◀FIG. Heart (Posterior view)

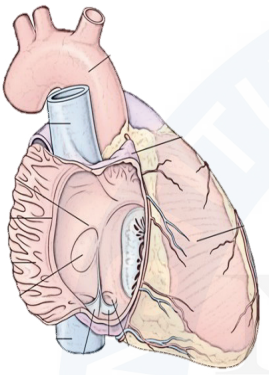
3. Interior

❑ Interior of Right Atrium

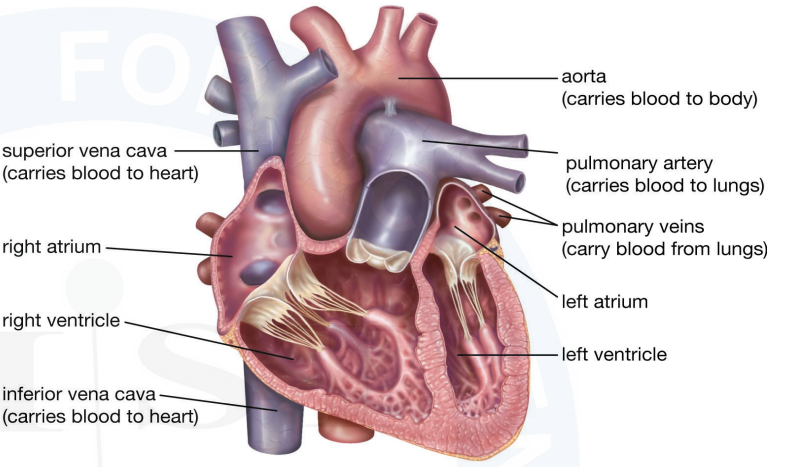
1. Fossa Ovalis
2. Annulus / Limbus Ovalis
3. Torus Aorticus

❑ Interior of Ventricle (RV & LV)

1. Chorda Tendinae
2. Papillary Muscle



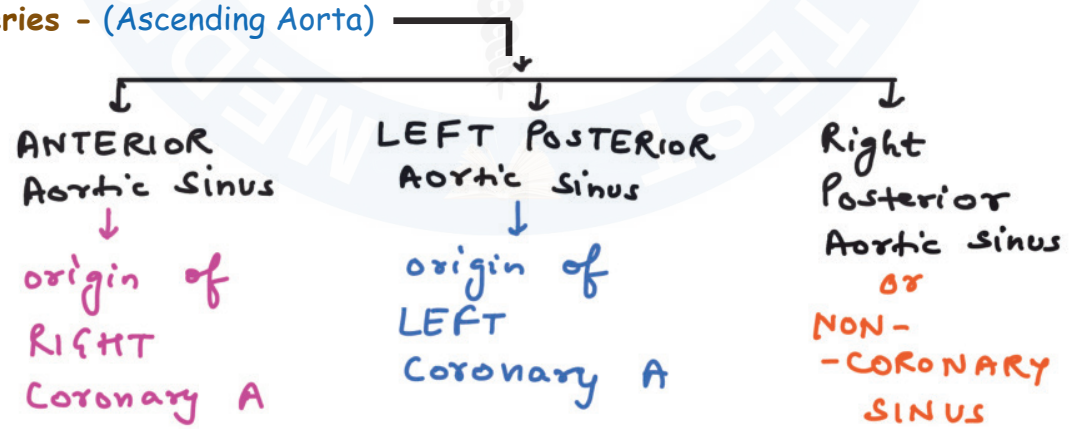
◀FIG. Interior of Right Atrium

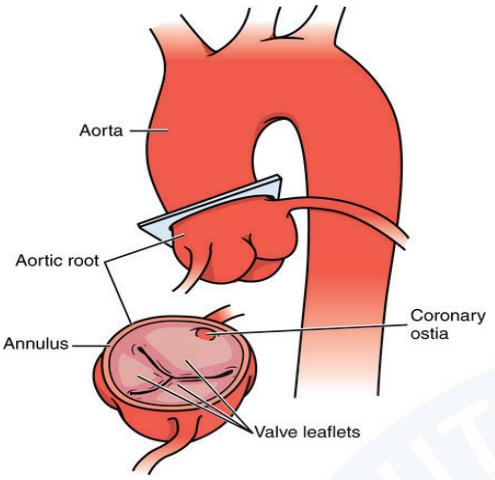


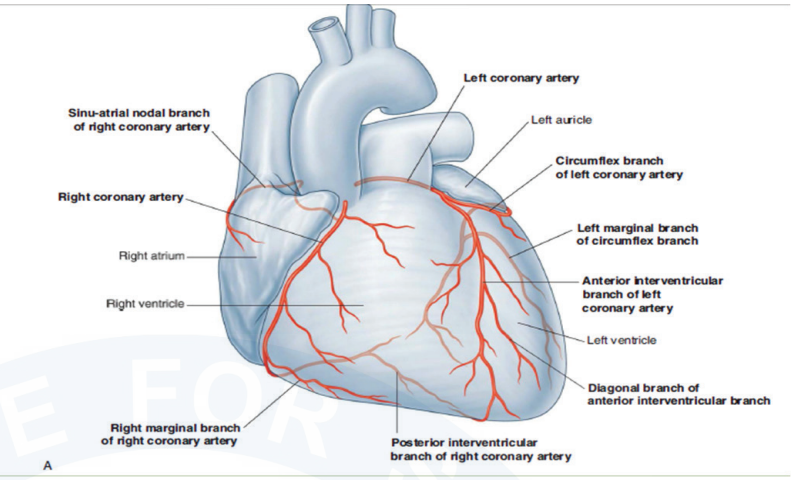
◀FIG. Interior of Ventricle

3. Arterial supply

❑ Arteries - (Ascending Aorta)







◀**FIG.** Arteries of Heart

❑ Right Coronary A - Branches

1. .....
2. Conus A → (also called Third Coronary A)
3. SA Nodal A & AV Nodal A
4. Marginal A

❑ Left Coronary A - Branches

1. .....
2. .....

Cardiac Dominance

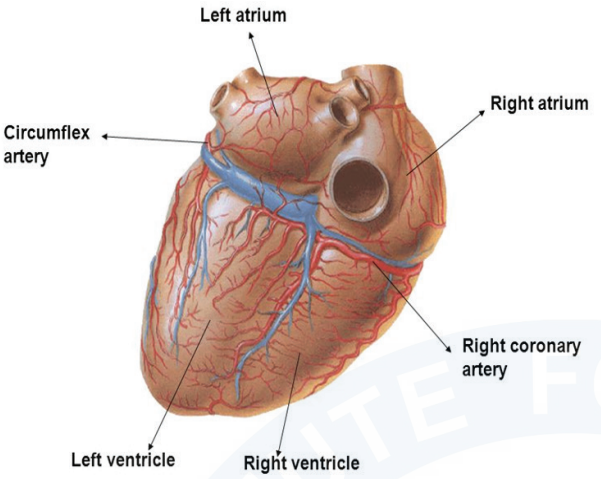
- According to the origin of Posterior Interventricular A (PIVA)
  - ✓ Right Cardiac Dominance → PIVA arises from RCA
  - ✓ Left Cardiac Dominance → PIVA arise from Left Circumflex Artery

5. Venous Drainage

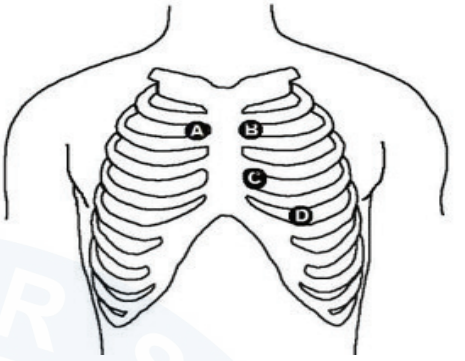
❑ Coronary Sinus

- Location → Coronary Sulcus
- All veins of Heart drain in to Coronary sinus Except Anterior Cardiac Vein
- Coronary sinus drains into Right Atrium

☞ Note - Anterior Cardiac vein drains directly in to Right Atrium



◀FIG. Coronary Sinus (Posterior View)



◀FIG. Auscltation Area

❑ Auscultation Area

(A) Aortic Area → Located at right 2nd Intercostal space

(B) Pulmonic Area → Left 2nd ICS

(C) Tricuspid Area → Left 4th ICS

(D) Mitral Area → Left 5th ICS

❑ Nerve Supply

- Sympathetic → T_1 to T_5 Spinal Nerves
- Parasympathetic → Vagus Nerve.

TRACHEA

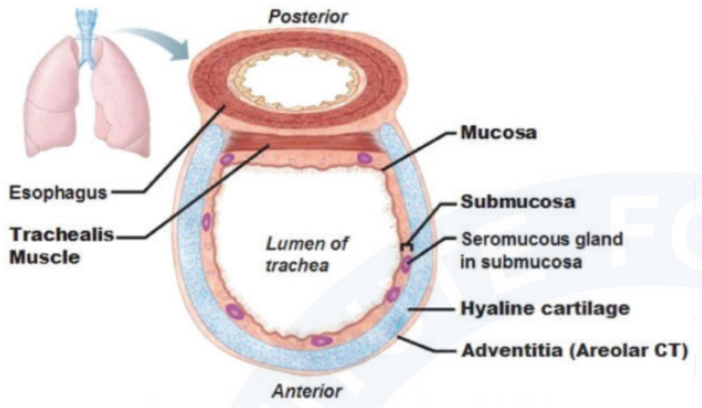
✓ 10 -12 cm length

✓ Extent → C_6 to T_6

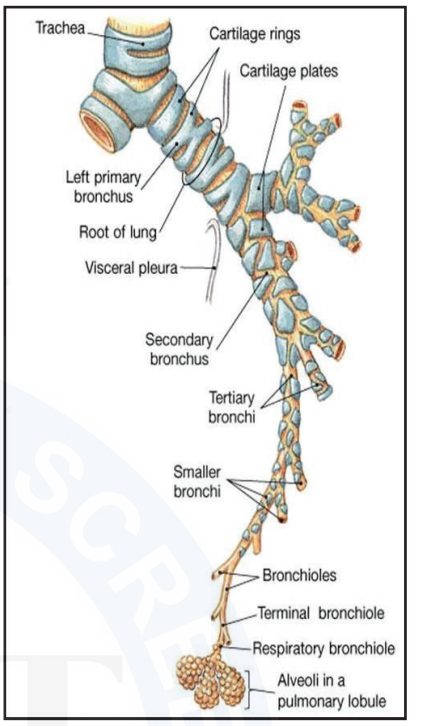
✓ Carina → T_6 (live person)

T_4 (Cadaver)

Tracheobronchial angle → Right -25⁰
→ Left -45⁰



◀FIG. Transverse Section of Trachea & Esophagus



◀FIG. Tracheobronchial Tree

❑ **Trachea**

↓

(Primary) Principal Bronchus (For Each Lung)

↓

Lobar Bronchus (For Each Lobe) (Secondary B.)

↓

Tertiary Bronchus (For Each Bronchopulmonary Segment) (Segmental B)

↓

Terminal Bronchiole

↓

Respiratory Bronchiole

↓

Alveolar duct

↓

Alveoli & alveolar sac

✦ Trachea & Bronchus - Hyaline cartilage present

✦ Alveoli → PNEUMOCYTES

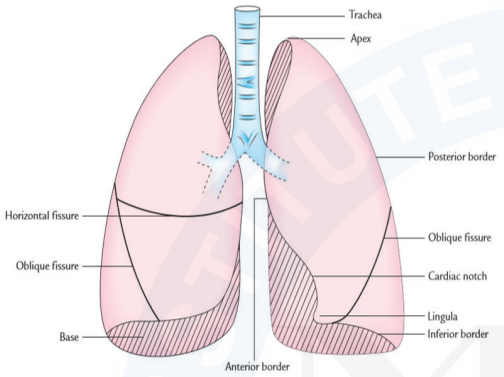
✦ Bronchioles → CLARA cells.

LUNG

1. Introduction

✦ Right Lung → (1) Upper Lobe
(2) Middle Lobe
(3) Lower Lobe

✦ Left Lung → (A) Upper Lobe
(B) Lower Lobe



◀FIG. Lung

2. Hilum

Medial aspect of Lung → Root of Lung / Hilum

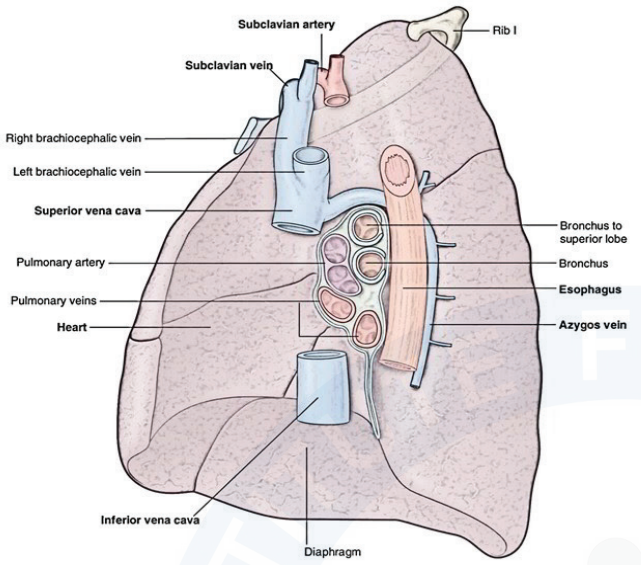
❑ **Right Lung (Above to Below)**

1. B - Bronchus (Eparterial) → Superior most structure
2. A - Artery (Pulmonary)
3. B - Bronchus (Hyparterial)
4. V - Vein (Pulmonary)

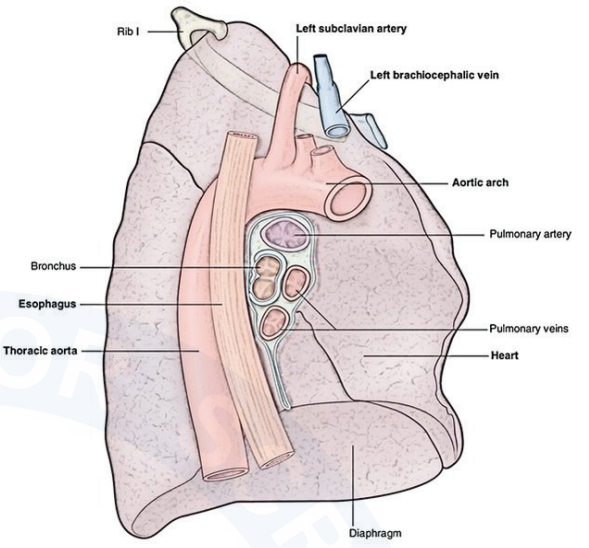
❑ **Left Lung (Above to Below)**

1. A - Artery (Pulmonary) → Superior most structure
2. B - Bronchus
3. V - Vein (Pulmonary)

☞ Note - Anterior to Posterior for both lungs → VAB



◀FIG. Hilum of Right Lung



◀FIG. Hilum of Left Lung

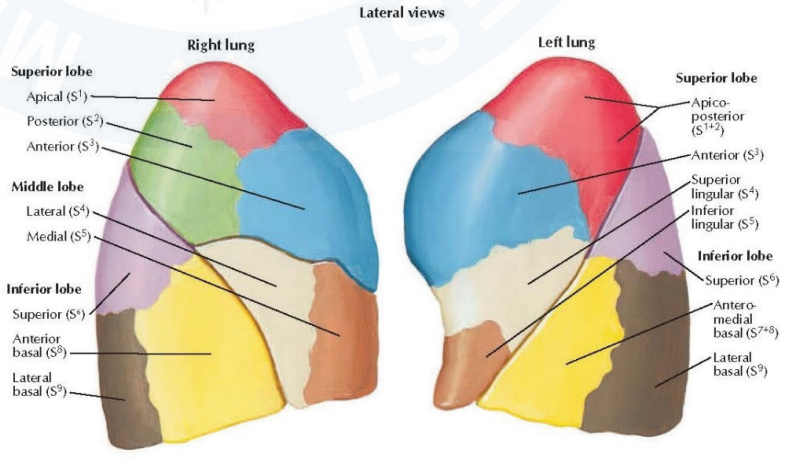
- Structure which arches over the hilum of Right Lung .....
- Structure which arches over the hilum of Left Lung.....

3. Broncho Pulmonary Segments

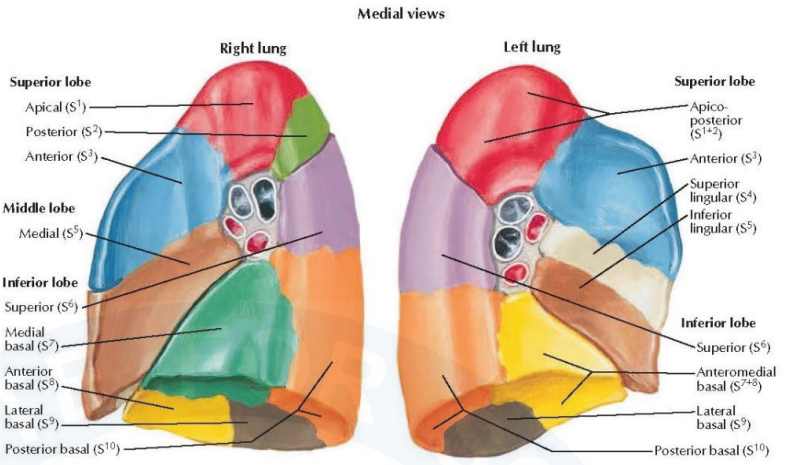
(10 in Right lung & 10 in Left lung)

❑ Right lung

- 1. Upper lobe
  - Apical
  - Anterior
  - Posterior
- 2. Middle lobe
  - Medial
  - Lateral



- 3. Lower lobe
  - Apical / superior
  - Anterior Basal
  - Posterior Basal
  - Medial Basal
  - Lateral Basal



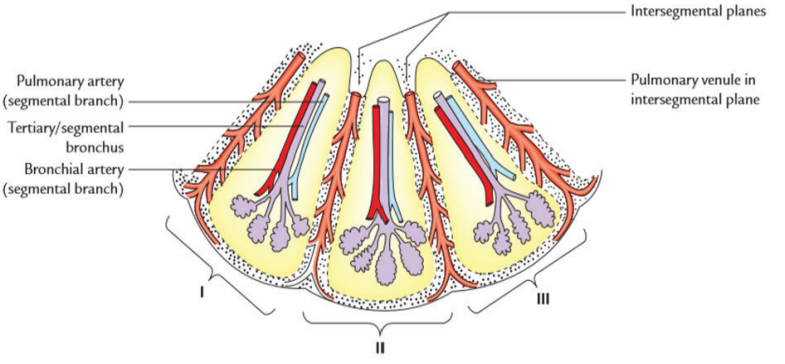
◀**FIG.** Bronchopulmonary Segment

Left lung

- 1. Upper lobe
  - Apical
  - Anterior
  - Posterior
  - Superior lingular
  - Inferior lingular
- 2. Lower lobe
  - Apical / superior
  - Anterior Basal
  - Posterior Basal
  - Medial Basal
  - Lateral Basal

Characteristics of Bronchopulmonary Segment (BPS)

- 1. Pyramidal in Shape
- 2. Each BPS has its own
  - Tertiary Bronchus
  - Pulmonary Artery
  - Nerves
  - Lymphatics



◀**FIG.** Bronchopulmonary Segment

- ✦ 3. Pulmonary Vein runs in inter segmental area
- ✦ 4. Segmental resection can be done

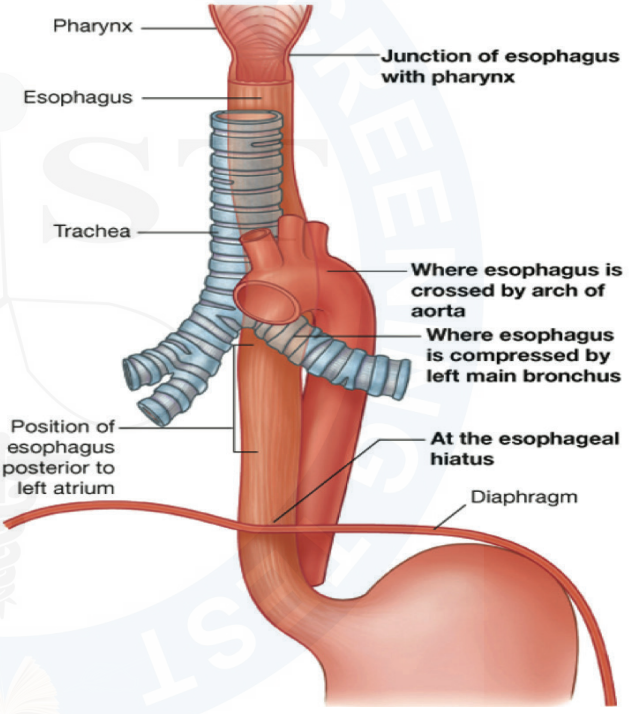
❑ **Foreign Body Aspiration (Aspiration Pneumonitis)**

⇨ In supine position, BPS most commonly involved is .....
.....

⇨ In standing / Erect position, BPS most commonly involved is .....
.....

ESOPHAGUS

- 25 cm length
- Extent - C_6 to T_{11}
- Esophageal Constrictions (4)



◀**FIG.** Esophageal Constrictions

❑ **CLINICAL INTEGRATION**

An endoscope is being passed by the clinician and he felt some resistance when the endoscope reached 25 cm from the incisor teeth. The endoscope has reached till which landmark?

(a) Arch of aorta
(b) Right bronchus
(c) Main bronchus
(d) Left bronchus

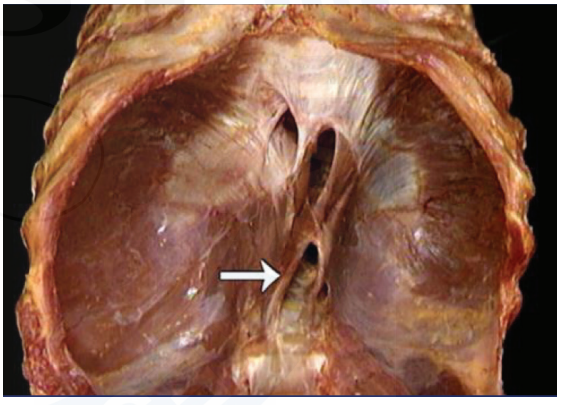
MIST

Location	Distance from Incissor Teeth	
	(cm)	(inch)
1. Pharyngo-esophageal Junction (cricopharyngeal Junction) (C_6)		6
2. When crossed by Arch of Aorta (T_4)		9
3. When crossed by Left Bronchus (T_6)		11
4. When it passes through diaphragm (T_{10})		15

DIAPHRAGM

→Supplied by Phrenic N (C3C4C5)

→Main Muscle of Inspiration



◀FIG. Esophageal Constrictions

❑ **Major openings**

✦ 1. Venacaval opening → T_8

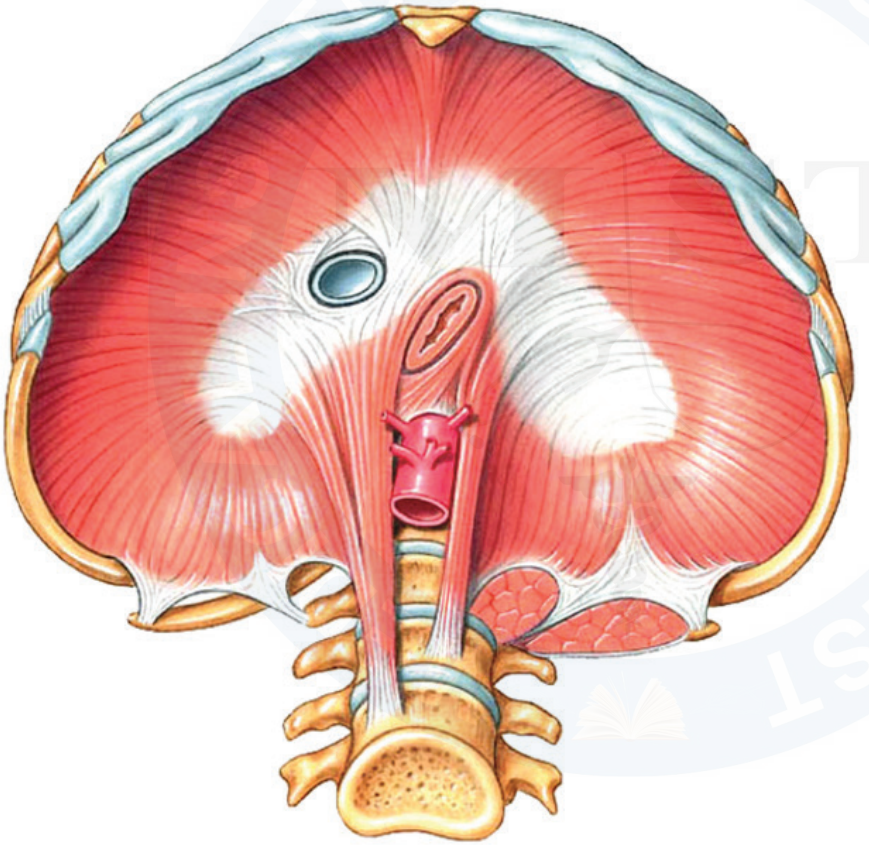
- located at Central tendon
- Structures passing through it
  - ✓ Inferior vena Cava
  - ✓ Right Phrenic Nerve

✦ 2. Esophageal opening → T_{10}

- located in muscular part
- Structures passing through it
  - ✓ Esophagus
  - ✓ Vagus Nerve
  - ✓ Left Gastric Artery (Esophageal branches)

✦ 3. Aortic opening → T_{12}

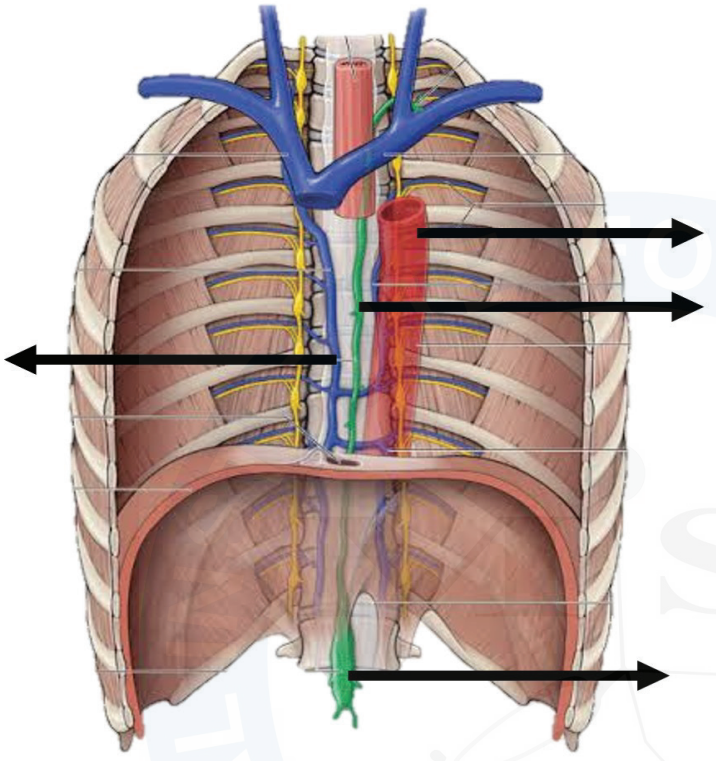
- located behind Median Arcuate Ligament
- Structures passing through it
  - ✓ .....
  - ✓ .....
  - ✓ .....



◀**FIG.** Diaphragm (Inferior View)

THORACIC DUCT

Drains at the Junction of .....



◀FIG. Thoracic duct

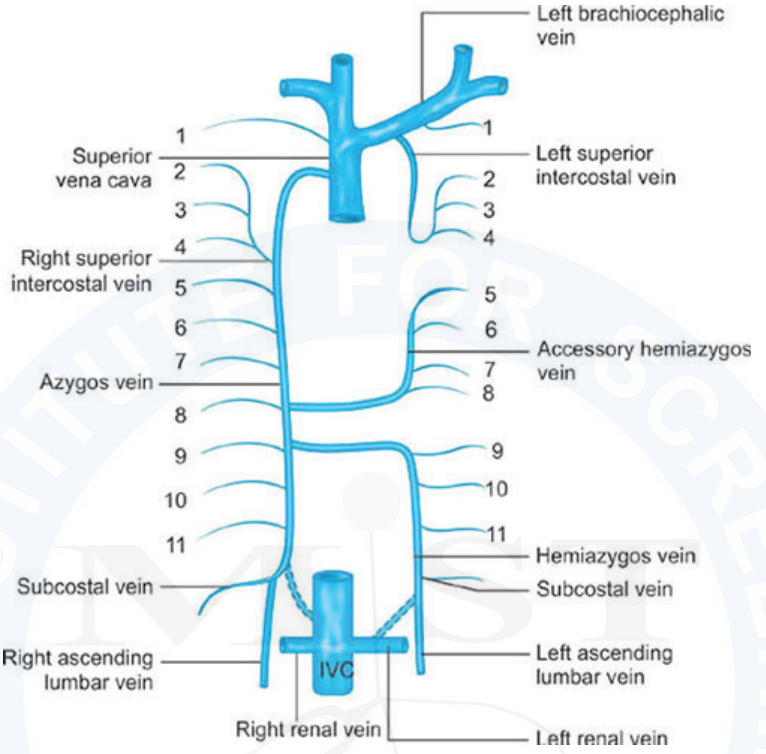
VENOUS DRAINAGE

1. Superior Venacava

- SVC is formed by union of .....
- Brachiocephalic vein is formed by union of .....
  .....

2. Azygos Venous System

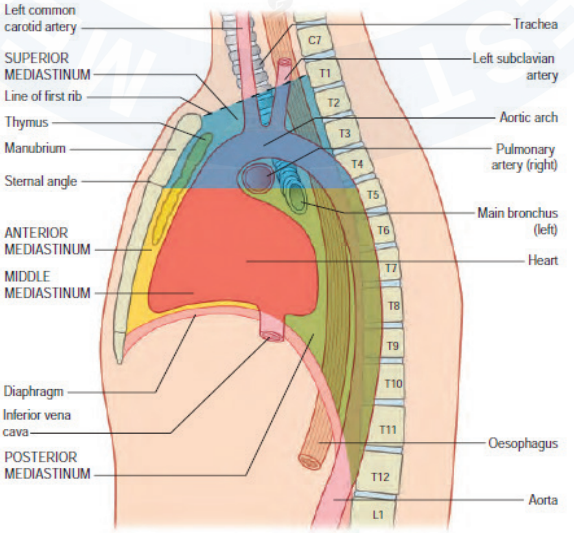
❑ **Azygos Vein** → Drains into Superior Vena Cava



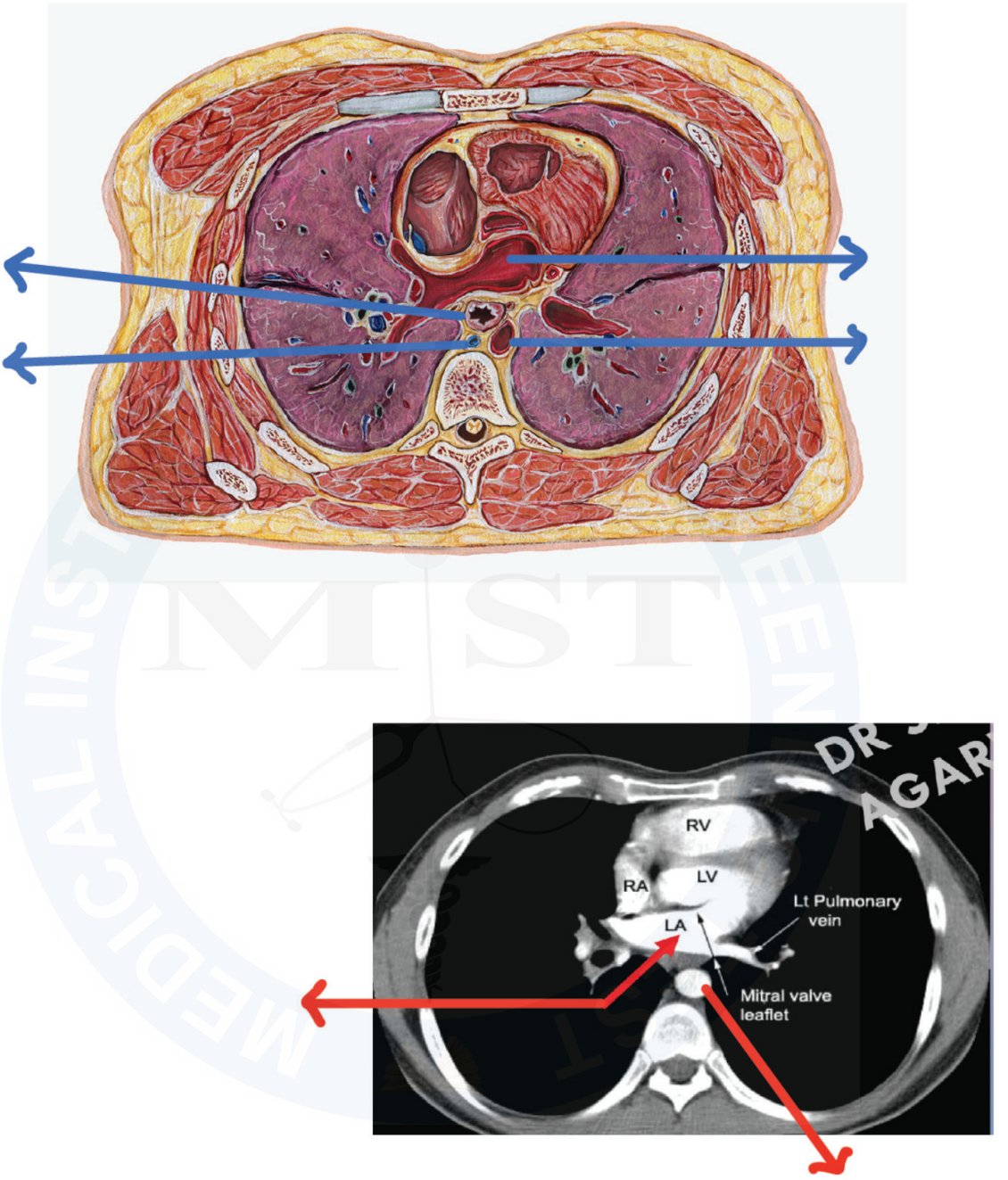
◀**FIG.** Azygos Venous System

Note: 1. Superior Intercostal Vein – is formed by union of 2nd, 3rd and 4th Posterior Intercostal Veins

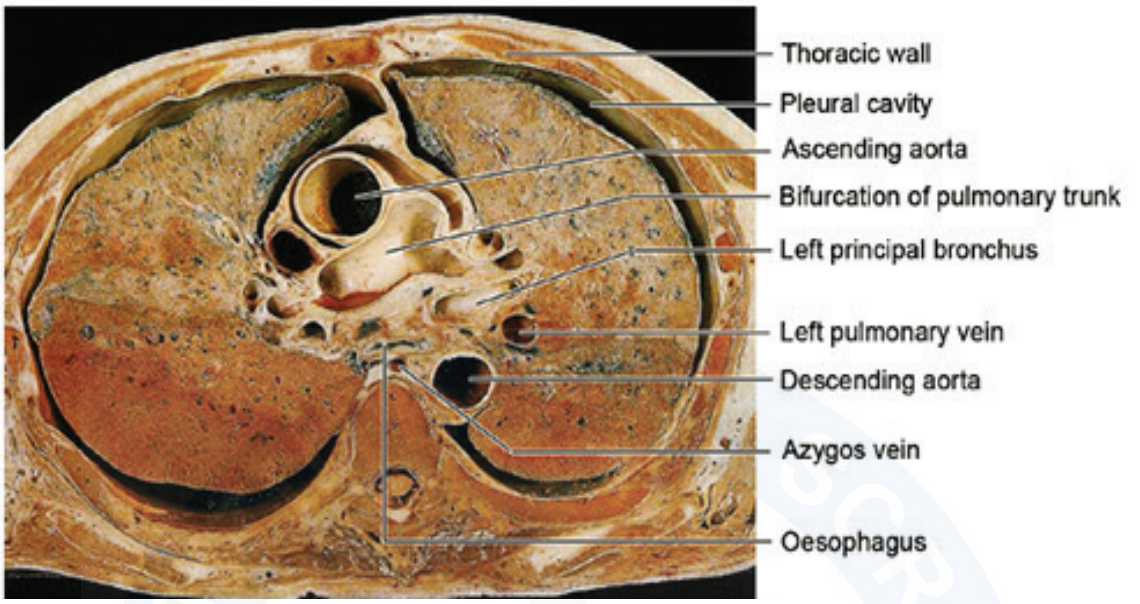
1. Subcostal Vein – 12th Posterior Intercostal Vein



◀**FIG.** Thorax (For Transverse Section)



◀**FIG.** Transverse Section 1 of Thorax



◀**FIG.** Transverse Section 2 of Thorax

Workbook Questions

1. Left anterior descending artery is a branch of

   (A) Right coronary artery (B) Left coronary artery

   (C) Ascending aorta (D) Coronary sinus

   Ans B

2. Right coronary artery arises from which sinus?

   (A) Anterior aortic sinus (B) Right posterior aortic sinus

   (C) Left posterior aortic sinus (D) From anterior and posterior aortic sinuses

   Ans A

3. Branches of arch of aorta are all except

   (A) Branchiocephalic trunk (B) Left common carotid

   (C) Left subclavian (D) Vertebral

   Ans D

4. Structures passing through aortic opening in diaphragm are all EXCEPT

(A)Aorta (B)Azygous vein (C)Thoracic duct (D) Vagus nerve

Ans D

5. Which of the following structure arches over the hilum of right lung:

(a) Azygos vein (b) Right thoracic duct (c) Aorta (d) Trachea

Ans A

6. Apex of the heart is formed by:

(a) RA (b) RV (c) LA (d) LV

Ans D

7. Which one is not a content of posterior mediastinum

(A) Oesophagus (B)Descending thoracic aorta

(C) Superior venacava (D) Vagus nerve

Ans C

8. Constrictions of esophagus are present at all sites except:

(a) At cricopharyngeal junction

(b) Where it is crossed by the arch of aorta

(c) Where it is crossed by azygos vein

(d) Where it is crossed by the left principal bronchus

Ans C

9. Needle for pleural tap is inserted along ?

(a) Upper border of rib (b) Middle border of rib

(c) Lower border of rib (d) Middle portion of intercostal space

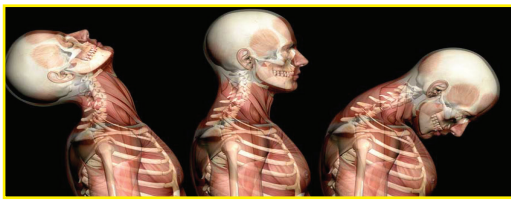
Ans A

10. Which of these is true about bronchopulmonary segment?

(A) Nonresectable (B) Spherical in shape

(C) Artery is intersegmental (D) Pulmonary veins are intersegmental

Ans D



HEAD & NECK

Carnial Nerves

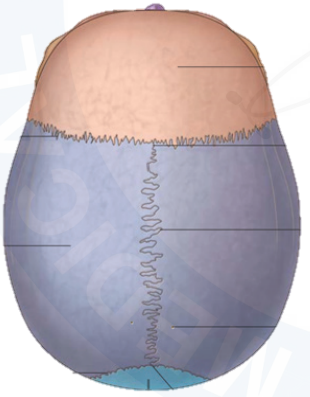
1. Olfactory N
2. Optic N
3. Occulomotor
4. Trochlear
5. Trigeminal
   - V1 – Ophthalmic N.
   - V2 – Maxillary N.
   - V3 – Mandibular N.
6. Abducent
7. Facial
8. Vestibulocochlear / Auditory
9. Glossopharyngeal
10. Vagus
11. Accessory (Spinal part and Cranial part)
12. Hypoglossal

❑ **Note:**
- Coronal section / Plane (along frontal plane) (divides the body into Anterior & posterior parts)
- Saggittal section / Plane (divides the body into right & left parts)
- Transverse or Horizontal section / Plane

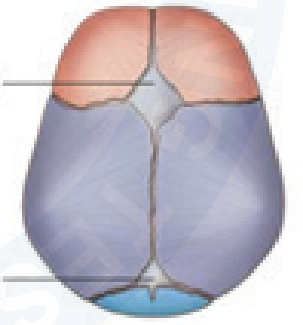
SKULL

Superior View (Skull)

- Total fontanelles → (6)
- **Bregma** (Anterior fontanelle) (Diamond Shaped)
  - Present between Frontal & Parietal Bones
  - Present between Coronal & Sagittal suture
  - Closes at .....
- **Lambda** (Posterior fontanelle)
  - Present b/w Parietal & occipital Bones
  - B/W Sagittal & Lambdoid sutures
  - closes at .....
  - Triangular in shape



◀ **FIG.** Norma verticalis (Adult Skull) (Superior view)



◀**FIG.** Norma verticalis (Fetal Skull) (Superior view)

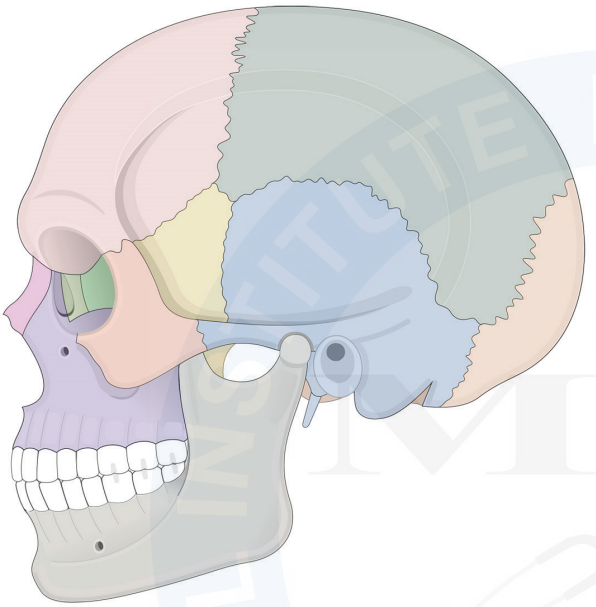
Lateral View (Skull)

- Pterion
  - 'H' shaped suture
  - Between: Frontal, Parietal, Temporal, sphenoid
  - Deep to it .....

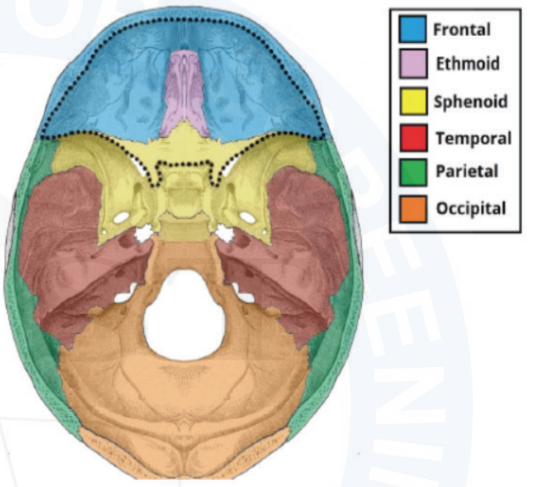
CLINICAL INTEGRATION

A patient suffered from Extradural Haemorrhage due to involvement of Middle Meningeal artery. What can be the location of injury on the skull ?

a. Inion b. Asterion c. Pterion d. Nasion



◀FIG. Lateral view of skull

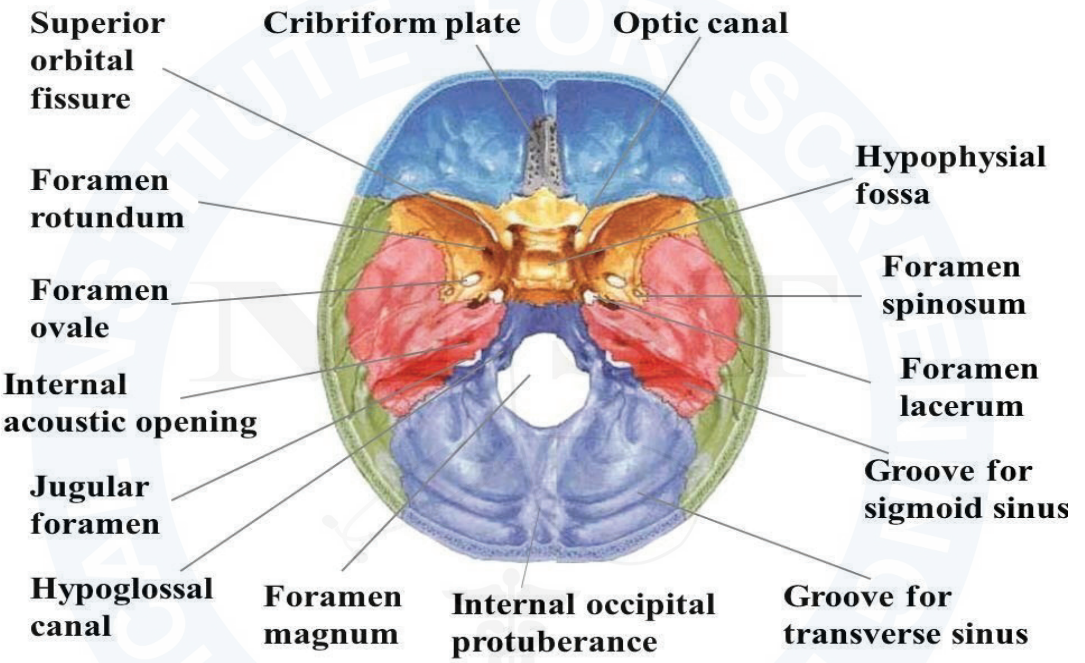


◀ FIG. Cranial fossa

Foramens of (Skull)

- 1. Cribriform Plate → .....
- 2. optic canal → .....
- 3. Superior Orbital Fissure (SOF) → .....
- 4. Foramen Rotandum: .....
- 5. Foramen ovale →
  - .....
  - .....
  - .....
  - .....
- 6. Foramen Spinosum: .....

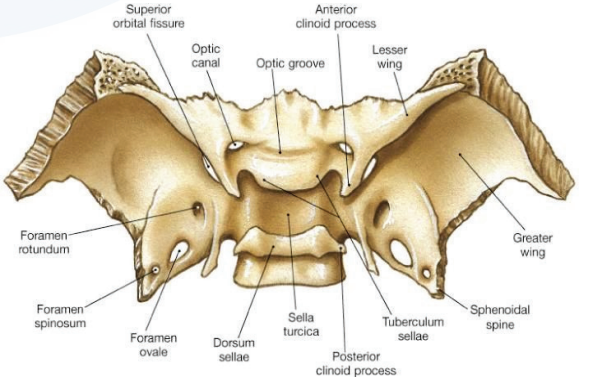
- **7. Internal Acoustic Meatus:** .....
- **8. Jugular foramen:** .....
- **9. Hypoglossal canal →** .....
- **10. Foramen magnum** Lower part of medulla & meninges, Vertebral Artery & Spinal Arteries (Anterior & Posterior), Membrana Tectoria, Cruciate Ligament, Apical Ligament of dens



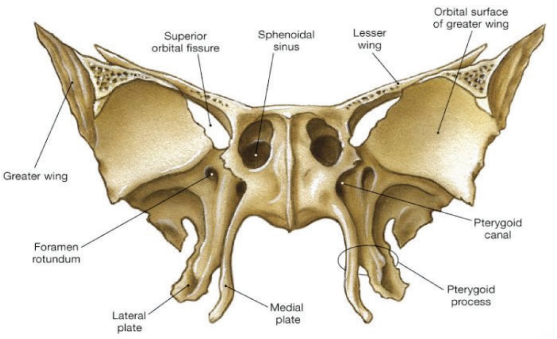
◀**FIG.** Cranial fossa

Sphenoid

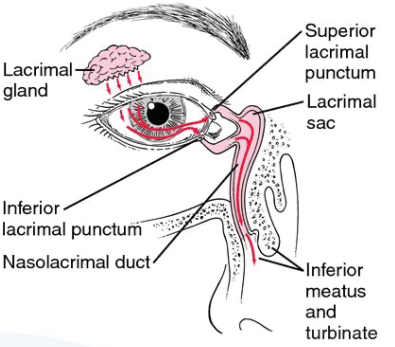
- **Parts.**
  1. Body
  2. Lesser wing
  3. Greater wing
  4. Pterygoid Process



◀**FIG.** Sphenoid (superior view)



◀FIG. Sphenoid (Anterior view)



◀FIG. Lacrimal Apparatus

Bony Orbit & Related Foramina

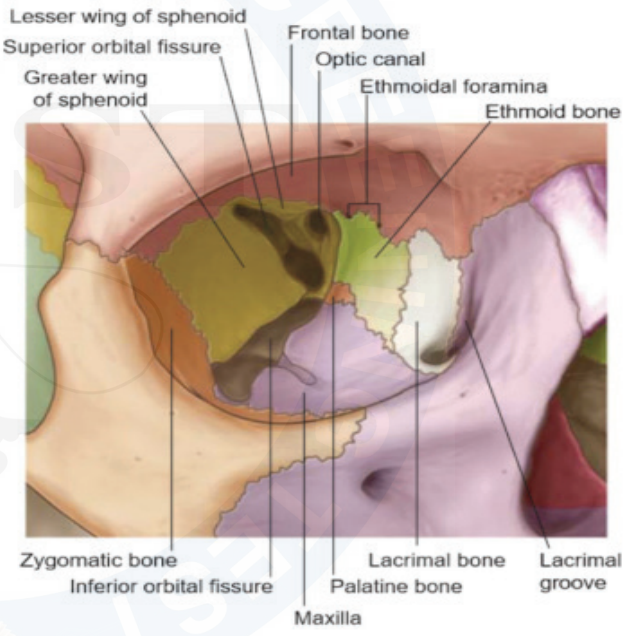
- Bony orbit → .....

❑ **Walls**

- Roof → .....
- Floor → .....

.....

- Medial wall
  - .....
  - .....
  - .....
  - .....
- Lateral wall .....



◀FIG Bony orbit

Foramens of Bony orbit

✦ **Inferior orbital fissure**

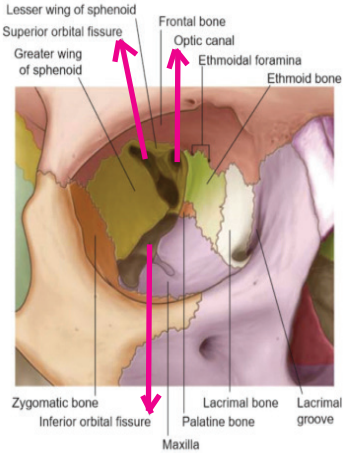
- ✓ Zygomatic Nerve
- ✓ Infraorbital Nerve

} (Branches of maxillary Nerve)

- ✓ Infraorbital vessels

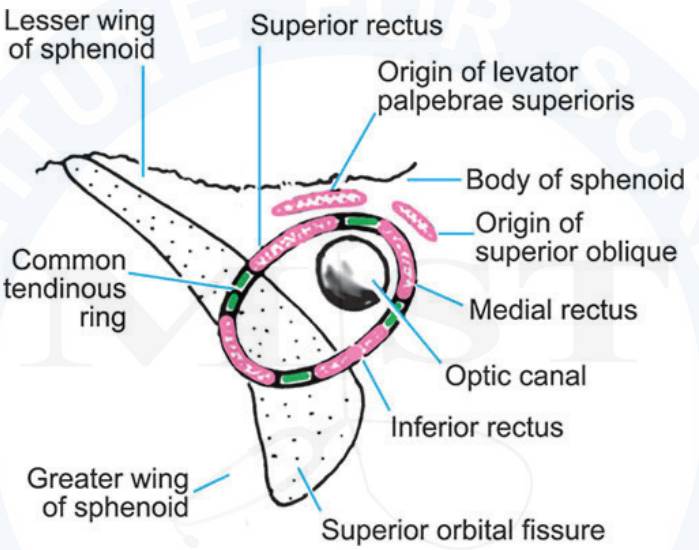
✦ **Superior orbital fissure**

1. Medial Compartment → .....



2. Lateral Compartment → .....
   - → .....
   - → .....

3. Intermediate Compartment
   - ✓ .....
   - ✓ .....
   - ✓ .....



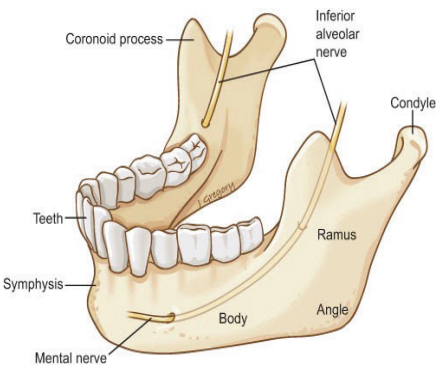
◀**FIG.** Superior orbital fissure

NOTE: LACRIMAL N, FRONTAL N & NASOCILIARY N ARE BRANCHES OF OPHTHALMINC N

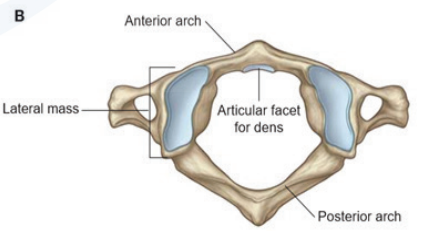
MANDIBLE

❑ Mandible

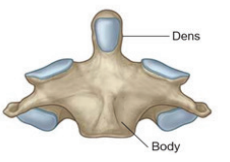
- Mandibular foramen → transmits .....
- Mental foramen → Mental Nerve + vessels



◀**FIG.** Mandible



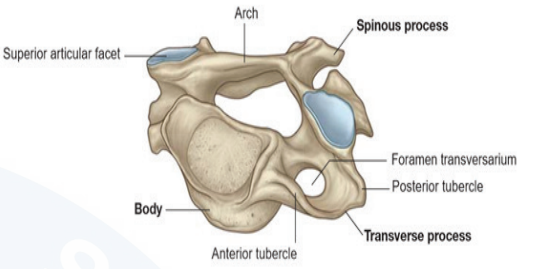
◀**FIG.** Atlas cervical vertebra



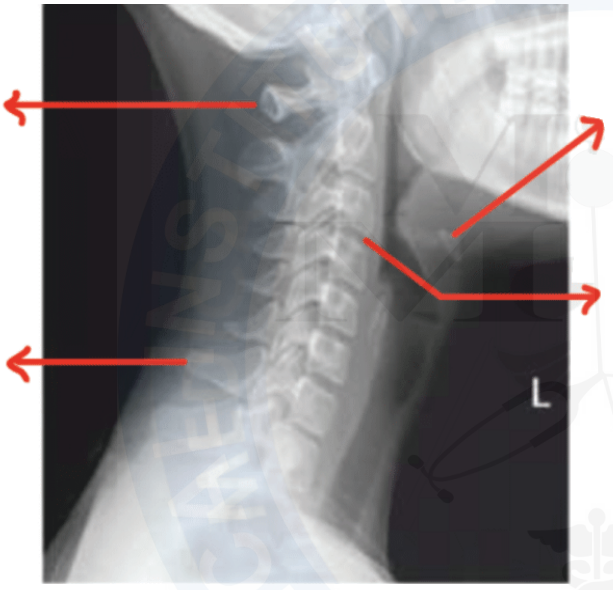
◀**FIG.** Axis cervical vertebra

CERVICAL VERTEBRA

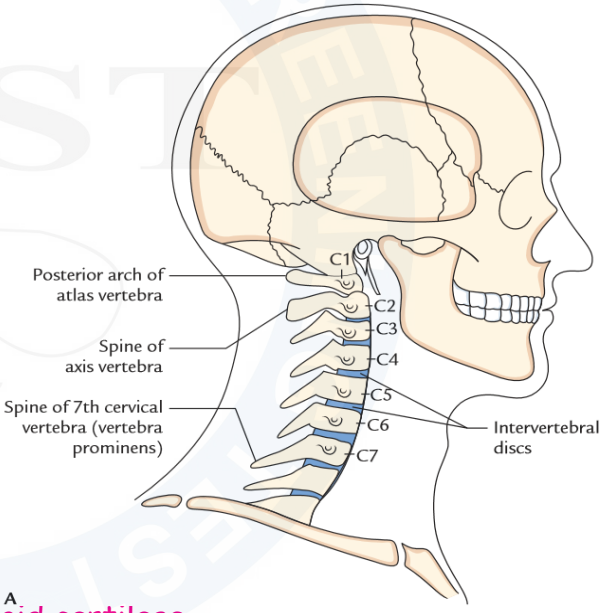
- **Characteristic feature** Presence of foramen Transversarium
  - Vertebral Artery passes through foramen transversarium of C_1 to C_6
    - C_1 - Atlas
    - C_2 - Axis
    - C_7 - Vertebra Prominens



◀FIG. Cervical vertebra (Foramen Transversarium)



◀FIG. X-Ray of Head & Neck (Lateral View)

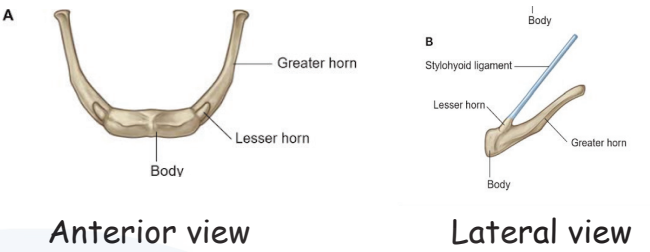


A.

- **Junction of C_3 & C_4**
  - Corresponds to upper border of thyroid cartilage
  - Bifurcation of common carotid artery
- **C_6**
  - Corresponds to lower border of cricoid cartilage
  - Pulsation of common carotid artery
    (Felt against carotid / chassaignac tubercle of C6)
  - Larynx & Pharynx Ends
  - Trachea & Esophagus Begins

HYOID BONE

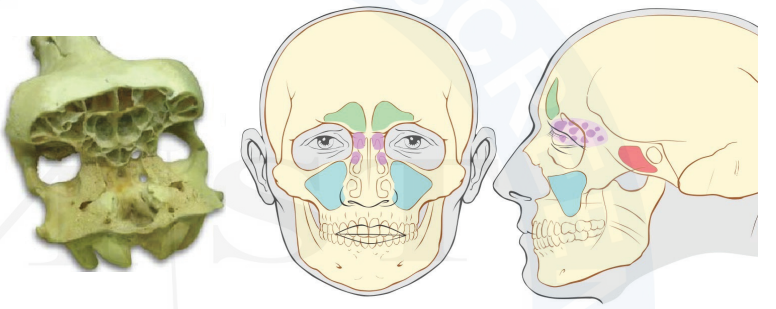
- **Hyoid Bone - Located at C_3 level**
  1. Body
  2. Greater cornu or horn
  3. Lesser cornu or horn



◀ **FIG.** Hyoid Bone

PNEUMATIC BONES

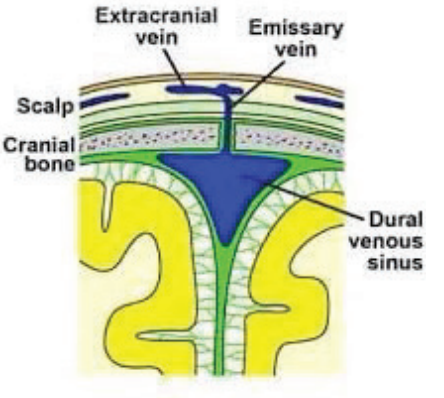
- ✓ .....
- ✓ .....
- ✓ .....
- ✓ .....
- ✓ .....



◀ **FIG.** Pneumatic Bone

SCALP - LAYERS, OCCIPITOFRONTALIS & DANGEROUS LAYER

- **Scalp - 5 layers (outer to inner) :** SCALP
  1. Skin
  2. Connective tissue (dense)
  3. Aponeurotic layer (Galea aponeurotica or Epicranial aponeurosis)
  4. Loose connective tissue
  5. Pericranium
- **4th layer** (Loose Connective Tissue layer): .....
  .....
- **Aponeurotic layer**
  - Fibrous Layer
  - It has Occipitofrontalis muscle



◀ **FIG.** Emissary Vein

❑ Black Eye -

◀FIG. Occipitofrontalis & Aponeurotic Layer

FACE

1. Nerve Supply

❑ Sensory Supply of Face

Skin of whole face is supplied by Trigeminal Nerve

.....

.....

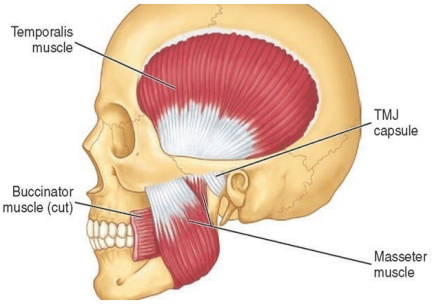
❑ Motor supply of face

✦ Muscle of whole face is supplied by Facial Nerve

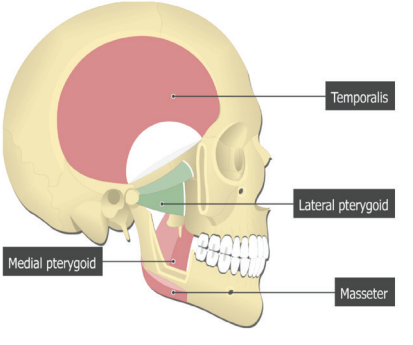
1. .....
2. .....

✦ Muscles of Mastication

1. Temporalis
2. Masseter
3. Medial Pterygoid
4. Lateral Pterygoid



◀FIG. Muscles of Mastication

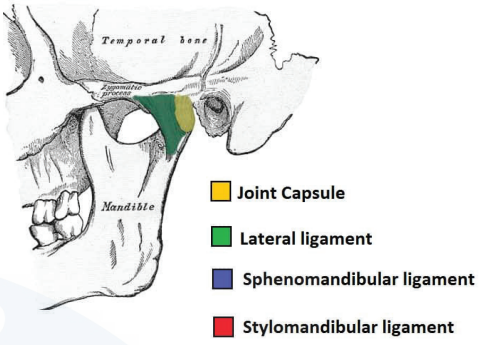


[Note : Buccinator → By Facial N]

TEMPOROMANDIBULAR JOINT

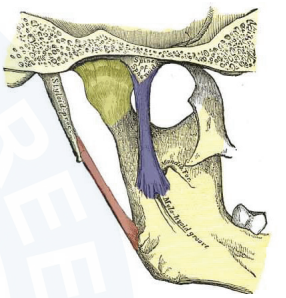
❑ Ligaments

1. Lateral Temporomandibular (Main)
2. Stylomandibular } (accessory ligament)
3. Sphenomandibular lig.



❑ Movements at TMJ

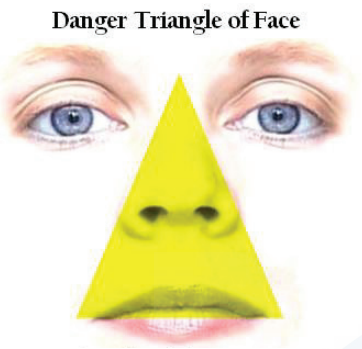
1. Depression of mandible → .....
   (opening of mouth)
2. Elevation of mandible → .....
   .....
3. Retraction of mandible → .....
4. Protraction of mandible → .....



◀**FIG.** Temporomandibular Joint (Ligaments) (Lateral view)

◀**FIG.** Movements of Mandible at TMJ

2. Dangerous Area of Face



Dangerous area (Triangle) of face

↓

Nose, Upper lip & adjacent area

↓

Drained by facial vein

↓

Deep facial vein

↓

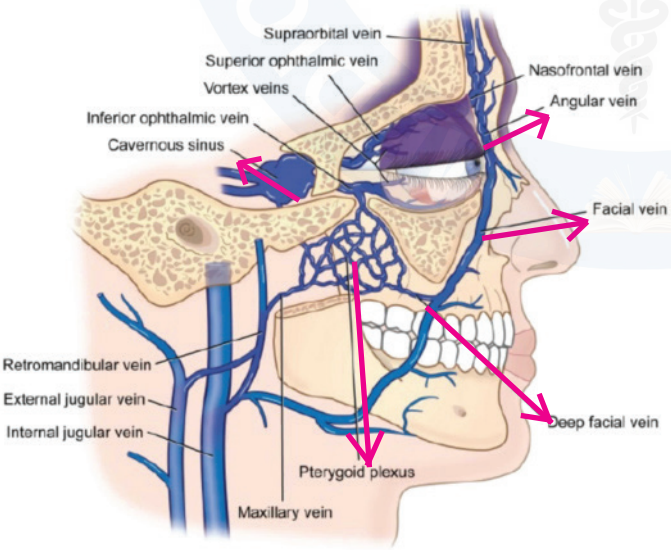
Pterygoid venous plexus

↓

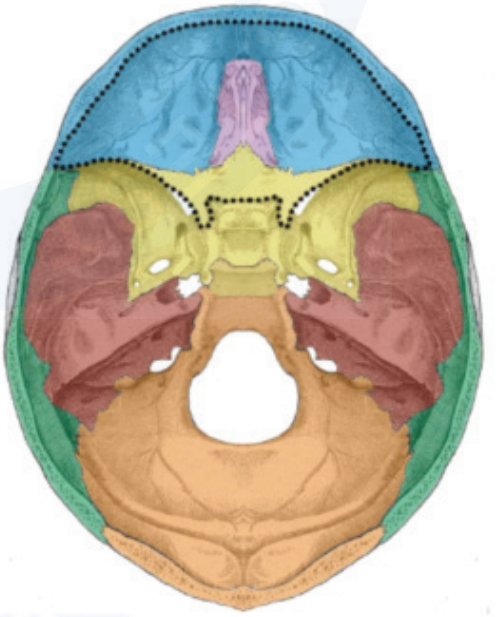
Emissary vein

↓

Cavernous sinus



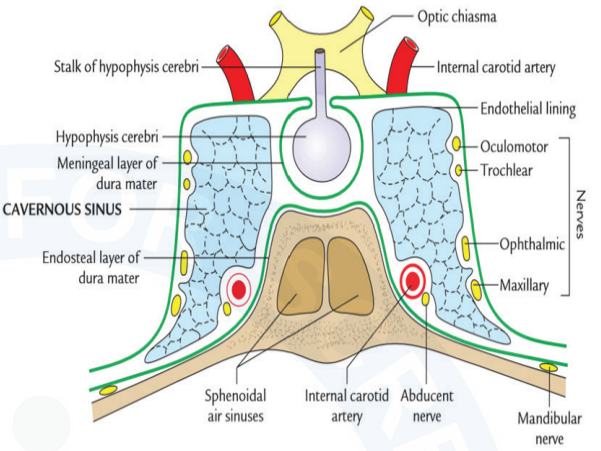
◀FIG. Venous connection Between facial vein & Cavernous Sinus



◀FIG. Cavernous Sinus Location

CAVERNOUS SINUS

- **Location:** Located on either side of body of sphenoid
- **Structures related to its Lateral wall**
  1. .....
  2. .....
  3. .....
- **Structures passing through CS**
  1. .....
  2. .....
- **Cavernous Sinus Thrombosis**
  - Most common cause
    - infection comes from dangerous area of face (via facial vein)
  - S/S
    1. Paralysis of all Extraocular Muscles (d/t involvement to CN 3, 4, 6) (Opthalmoplegia)
    2. Pain in forehead (d/t inv. of ophthalmic Nerve)
    3. Periorbital Edema and Exophthalmos (d/t venous congestion)



◀FIG. Cavernous Sinus

ARTERIAL SUPPLY OF HEAD & NECK

Common Carotid A, External Carotid A, Facial A

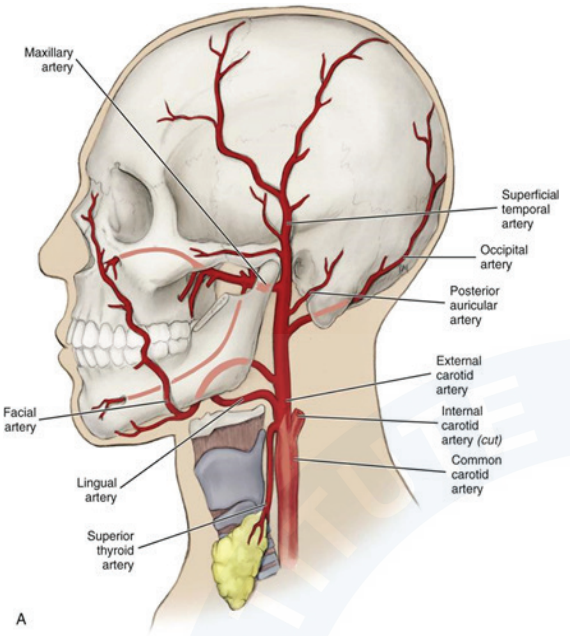
Common Carotid A

↓

Divides at Junction of C3 / C4 (Level of upper border of thyroid cartilage)

External Carotid A | **Internal Carotid A**

↓



◀**FIG.** External Carotid Artery Branches (Lateral view)

ICA

↓

No branch in Neck

↓

Pass through carotid canal & Branches in cranial cavity.

❑ ECA - Branches

Anterior branches

1. .....
2. .....
3. .....

Posterior branches

4. .....
5. .....

Medial branches

6. ..... → .....
   .....

Terminal branches

7. .....
8. .....

PARASYMPATHETIC GANGLIA

Ganglia	Cranial Nerve related to it	Structures supplied by it
1. Ciliary Ganglia		
2. Otic G.	.	
3. Pterygopalatine G.		
4. Submandibular G.		

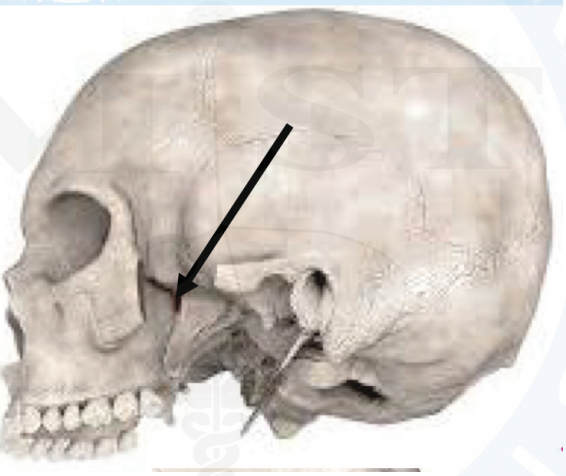


FIG. Location of Parasympathetic Ganglia

SALIVARY GLANDS

1. Parotid Gland

- Structures Passing through it

Sup 1. ..... (most superficial)

↓ 2. .....

deep 3. .....

- Parotid duct / STENSON'S duct

It Opens in oral cavity .....

.....

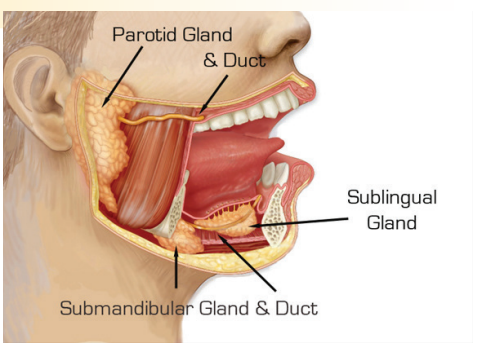


FIG. Ducts of Submandibular gland & Sublingual gland (Lateral view)

❑ **FREY'S syndrome / Gustatory Sweating**

- Due to surgery of Parotid Gland

↓

.....

.....

FIG. Frey's Syndrome

- Patient complains of redness, flushing, sweating & increased temperature on Lateral aspect of cheeks, while he is eating.

2. Submandibular & Sublingual Glands

❑ **Submandibular Gland -**

- Its duct is also known as Wharton's Duct
- It opens at Floor of oral Cavity (at sublingual Papilla)

❑ **Duct of Sublingual Gland**

- Opens at floor of oral cavity

FACIAL NERVE & ITS LESIONS

❑ **Facial Nerve Lesions**

☞ 1. UMN lesions of facial N. Palsy

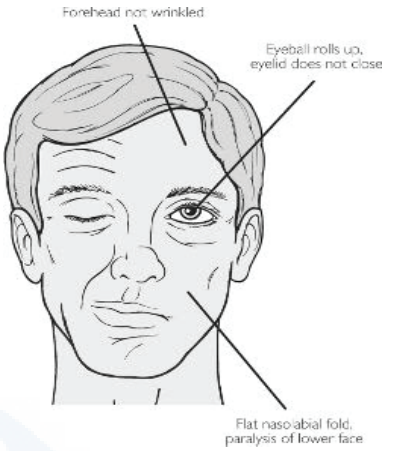
☞ 2. LMN lesion

.....

.....

☞ Bell's Palsy

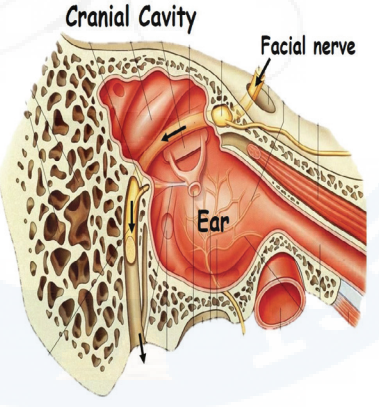
1. Loss of Transverse wrinkles on forehead
2. Inability to close eyes Lagophthalmos
3. Loss of Nasolabial fold
4. Dribbling of saliva
5. Accumulation of food
6. Angle of mouth tilted down on the same side of lesion

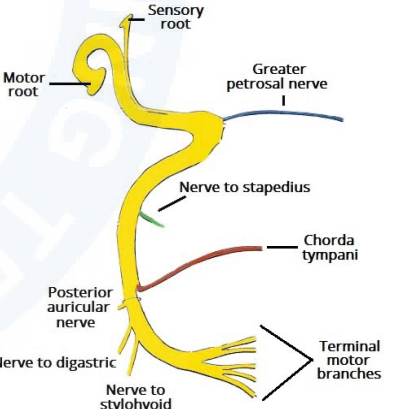


◀**FIG.** Bell's Palsy

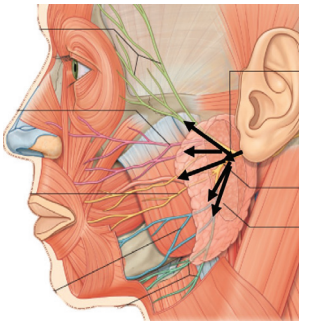
❑ Branches

1. .....
   .....
   .....
   .....
2. .....
3. .....
4. Posterior auricular N
5. Nerve to stylohyoid
6. Nerve to posterior belly of digastric
7. Terminal Branches "PES ANSERINUS" (Branching Pattern)





◀**FIG.** Facial Nerve (Branches)



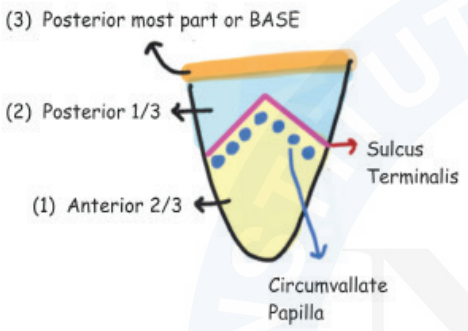
❑ CLINICAL INTEGRATION

A patient has loss of transverse wrinkles on left side of forehead. He can never close his left eye completely. He is having difficulty in eating food. What nerve is most likely to be damaged?

a. Greater Auricular nerve
b. Auriculotemporal nerve
c. Facial nerve
d. Vagus nerve

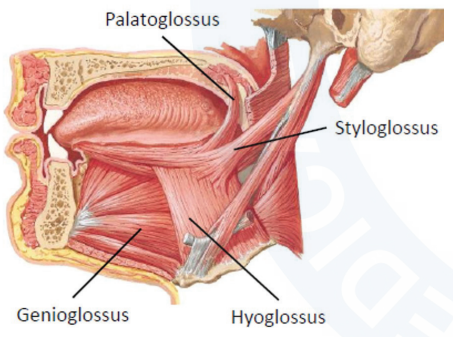
TONGUE & CRANIAL NERVE 12

Parts

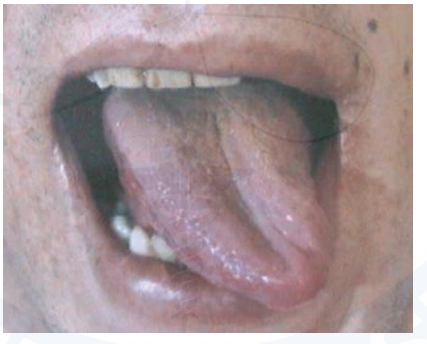


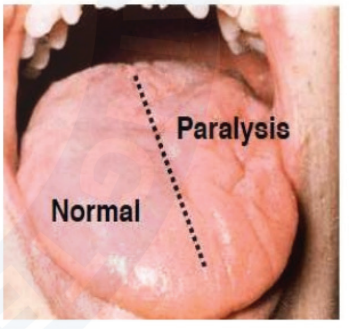
Papilla

1) Filiform → Most Numerous
→ Located throughout (every where)
2) Fungiform → Located at tip & sides of tongue
3) Circumvallate → Located anterior to sulcus terminalis
4) Foliate → Rudimentary in humans, Located at side of tongue near sulcus terminalis



◀FIG Extrinsic muscles of Tongue





◀FIG. Deviation of Tongue (Hypoglossal Nerve Palsy)

❑ Motor Nerve supply

All these muscles are supplied by Hypoglossal Nerve Except Palatoglossus (By Vagoaccessory complex)

❑ Sensory supply of Tongue

1. Anterior 2/3 part → General sensation → .....
   → Taste sensation → .....

Note : .....

.....

2. Posterior 1/3 part & Circumvallate Papilla → Both General & Taste sensation → Glossopharngeal nerve)
3. Posteriormost part → Both General & Taste sensation → Vagus Nerve (Internal Laryngeal nerve)

❑ **CLINICAL INTEGRATION**

A patient had the extraction of 3rd molar tooth & after this the patient was observed to have loss of sensation from anterior 2/3rd of the tongue & floor of the mouth. Which nerve is most likely to be damaged ?

a. Hypoglossal nerve
b. Lingual nerve
c. Accessory nerve
d. Inferior alveolar nerve

Geniglossus

- .....
- .....

❑ **Hypoglossal N. Injury**

1. Atrophy of tongue (on side of lesion)
2. On examination, when you ask patient to protrude the tongue, tongue is deviated to same side of lesion (due to unopposed Normal Genioglossus of the opposite side)

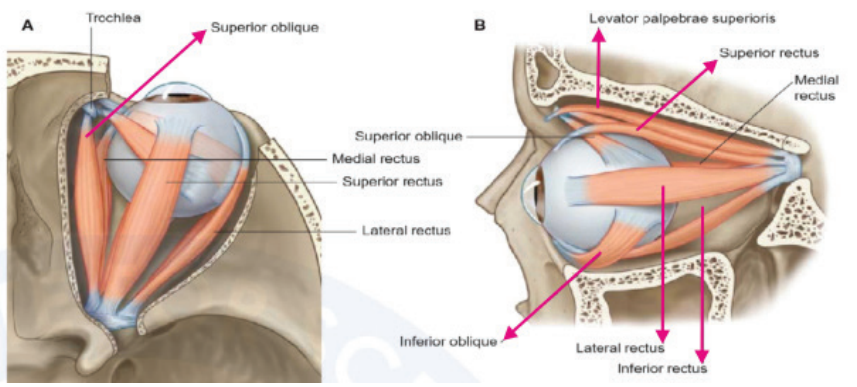
❑ **CLINICAL INTEGRATION**

A patient underwent surgery of submandibular gland during which injury to the hypoglossal nerve occurred. Which of the following will be found in the patient ?

a. Deviation of tongue to same side
b. Deviation of tongue to opposite side
c. Protrusion of tongue
d. Loss of general sensation in anterior 2/3rd of tongue

EXTRAOCULAR MUSCLES & CRANIAL NERVES 3, 4 & 6

1. Recti (4) -
   - Superior Rectus
   - Inferior Rectus
   - Lateral Rectus
   - Medial Rectus
2. Oblique (2)
   - Superior oblique
   - Inferior oblique



◀FIG. Extraocular Muscles (A- Superior view, B- Lateral view)

3. LPS -
   - Levator Palpabrae Superioris

❑ **Note**: All Extra ocular Muscles are attached to apex of orbit Except Inferior oblique.

❑ **Nerve supply**

All extra ocular muscles are supplied by

.....

.....

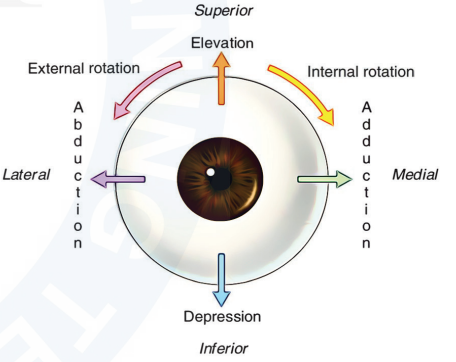
.....

Note - Ciliaris & sphincter pupillae are also supplied by Occulomotor N

❑ **Movements of eyeball**

- MR → .....
- LR → .....
- LPS → .....

1. Superior Rectus → Elevation + Intorsion + Adduction
2. Inferior Rectus → Depression + Extorsion + Adduction
3. Superior oblique → Depression + Intorsion + Abduction
4. Inferior oblique → Elevation + Extorsion + Abduction



◀FIG. Movements of Eyeball

❑ **Occulomotor Nerve lesion**

1. Ptosis (due to paralysis of LPS)
2. Fixed & Dilated pupil (d/t paralysis of sphincter Papillae & Ciliaris)
3. Down & Out Position of eyeball (d/t unopposed action of Lateral Rectus & superior oblique)
4. Diplopia

❑ **Abdvcent Nerve Lesion**

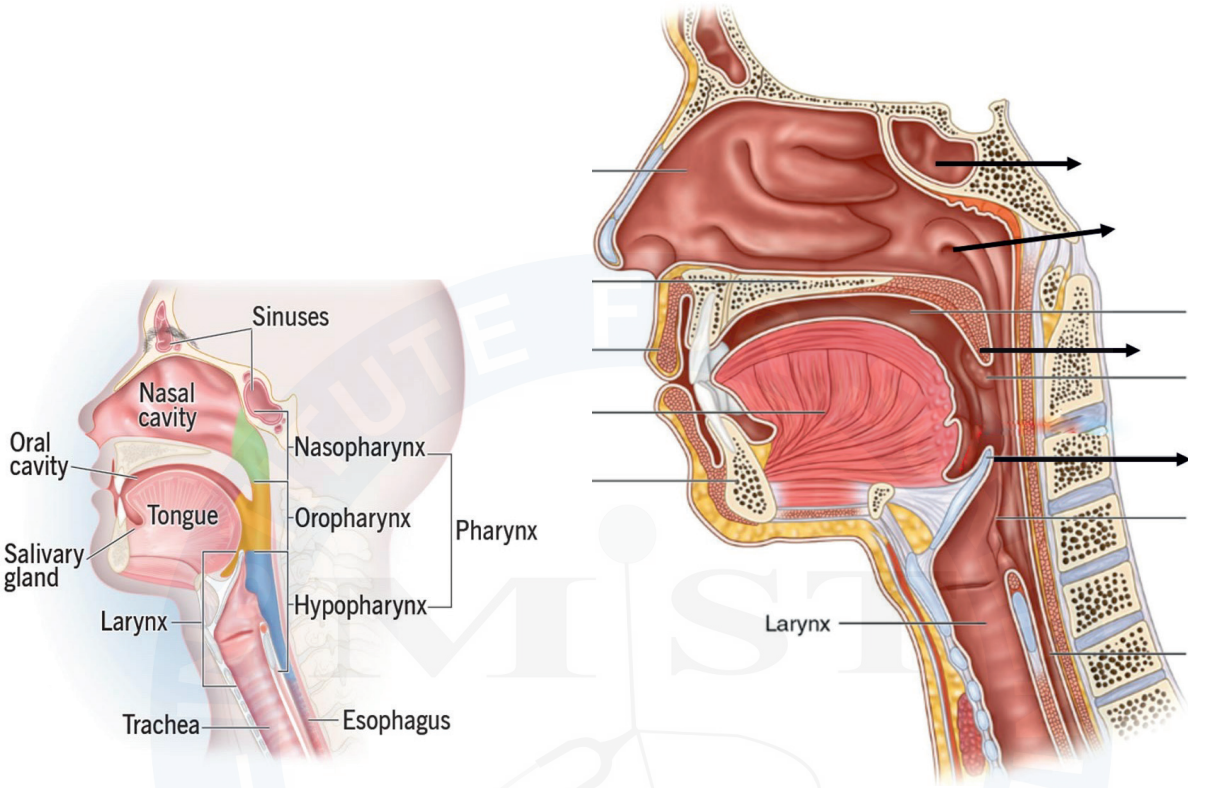
.....

❑ **Trochlear Nerve Lesion**

.....

.....

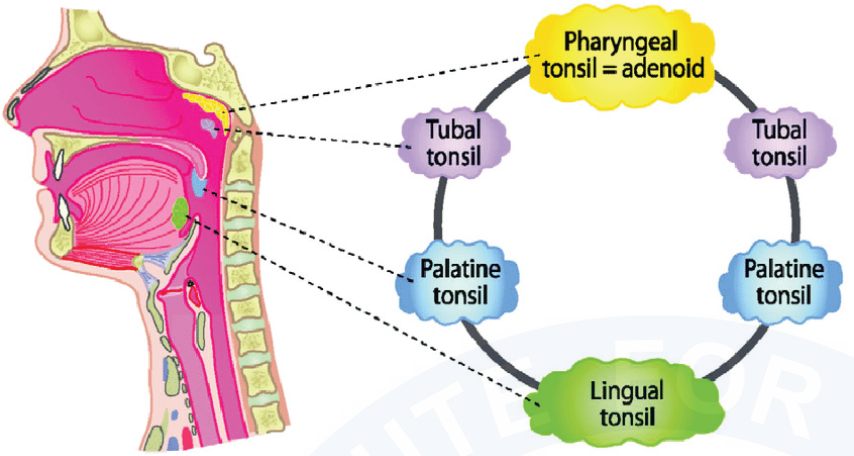
SAGGITAL SECTION OF H & N



◀**FIG.** Sagittal Section of H&N

LYMPHATIC DRAINAGE OF HEAD & NECK (INCLUDING WALDEYER RING)

- **Waldeyer's ring**
  - ✦ located at posterior aspect of nasal cavity & oral cavity.
  - ✦ Components
  1. Pharyngeal Tonsil
  2. Tubal Tonsil
  3. Palatine Tonsil
  4. Lingual Tonsil
- Lymph Node of Tongue → Jugulo-Omohyod LN
- Lymph Node of Tonsil → Jugulo - Digastric LN
- Tip of tongue → Submental LN
- Parotid Gland → Parotid LN or Preauricular LN



◀**FIG.** Waldeyeris Ring

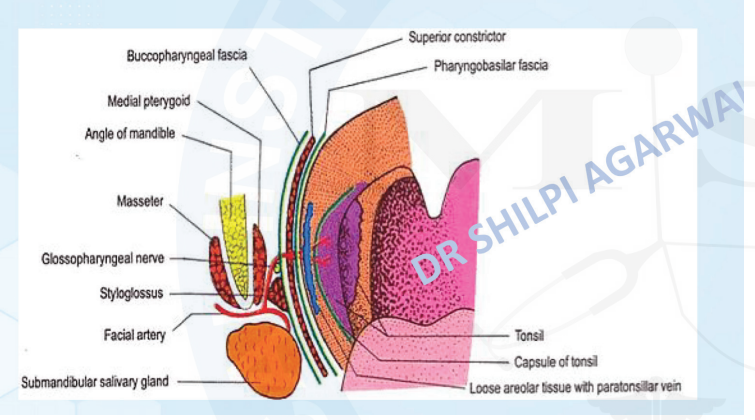
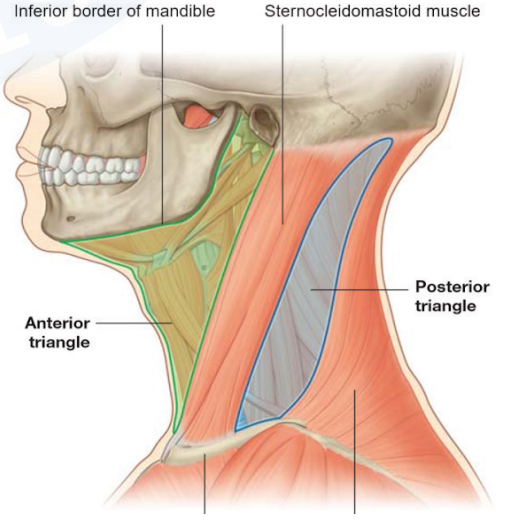


FIG. Palatine Tonsil

TRIANGLES OF NECK

1. Overview

- **Posterior Δ - Boundaries**
  1. Anteriorly → Sternocleidomastoid (SCM)
  2. Posteriorly → Trapezius
  3. Inferiorly / Base → Clavicle
- **Anterior Δ - Boundaries**
  1. Anteriorly → Imaginary midline of neck
  2. Posteriorly → SCM
  3. Superiorly / Base → Mandible (Lower Border)



◀**FIG.** Triangles of Neck (Lateral View)

CAROTID SHEATH

✓ derived from deep cervical fascia of Neck

❑ **Encloses**

1. Common Carotid Artery & Internal Carotid A
2. Internal Jugular vein
3. Vagus Nerve

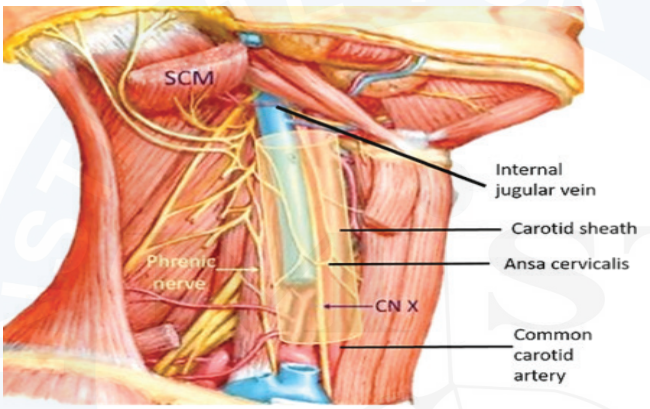
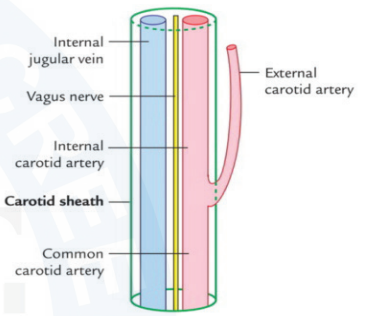


Fig. Carotid Sheath & Ansa Cervicalis (Lateral View)



◀**FIG.** Openings in Lateral wall

ANSA CERVICALIS

❑ **Ansa Cervicalis** Nerve loop in Neck

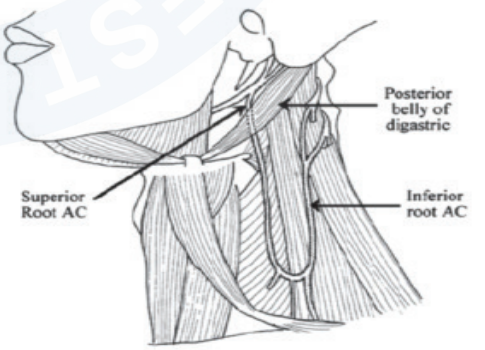
✓ C1 C2 C3

❑ **Muscles supplied by it**

✓ Sternohyoid

✓ Omohyoid (Superior & Inferior Belly)

✓ Sternothyroid



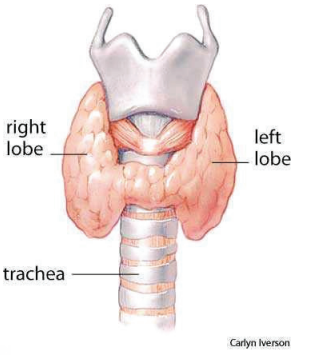
◀**FIG.** Ansa Cervicalis (Lateral View)

THYROID GLAND

❑ **Extent**

Lobe → From thyroid cartilage to 5th / 6th Tracheal ring → C_5 - T_1

Isthumus → opposite 2nd, 3rd & 4th tracheal rings → C_7

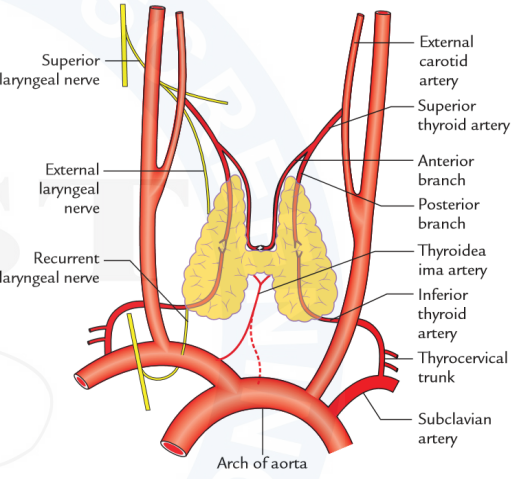


◀FIG. Thyroid Gland (Lateral View)

❑ **Arterial supply**

1. Superior thyroid A (ECA)
2. Inferior thyroid A (Subclavian A)
3. .....

.....



◀FIG. Thyroid Gland - Arteries

☞ Note - Parathyroid A - is a branch of Inferior Thyroid Artery.

PHARYNX

❑ **Extent** - Base of skull to C_6

❑ **Parts** - 1. Nasopharynx 2. Oropharynx 3. Laryngo pharynx

❑ **Nerve supply**

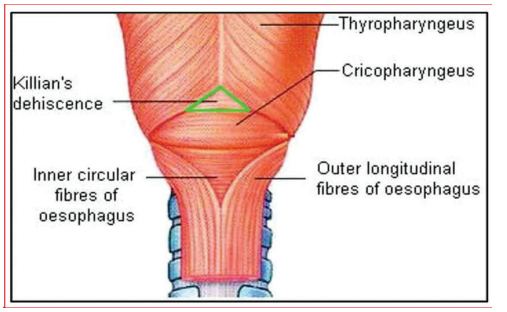
All these muscles are supplied by Vagoaccessory complex Except

1. Stylopharyngeus → By Glossopharyngeal N
2. Cricopharyngeus → By recurrent Laryngeal N.

❑ Killians Dehiscence

- Present in Inferior Constrictor
- B/W Thyropharyngeus & Cricopharyngeus

(Which are the two parts of inferior constrictor)



☞ Note - Eustachian tube opens in Nasopharynx

SOFT PALATE

❑ Nerve supply

All these muscles are supplied by Vagoaccessory Complex EXCEPT Tensor Veli Palatini (By Mandibular Nerve)

☞ opening of Auditory Tube is maintained by Tensor Veli Palatini & Levator veli Palatini

LARYNX

❑ Extent C3 – C6

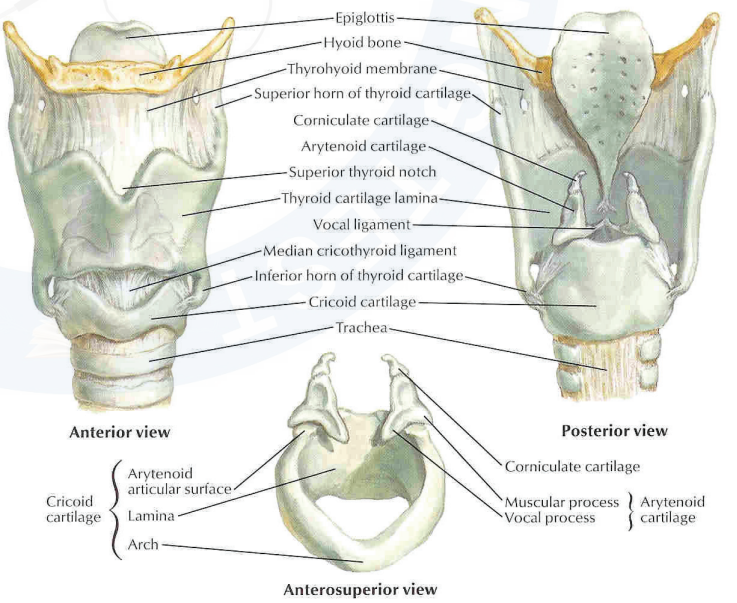
❑ Larynx Cartilages

(A) Unpaired

- ⇨ Thyroid cartilage
- ⇨ Cricoid cartilage
- ⇨ Epiglottis

(B) Paired

- ⇨ Arytenoid Cartilage
- ⇨ Corniculate C.
- ⇨ Cuneiform C.



❑ Nerve Supply - Motor

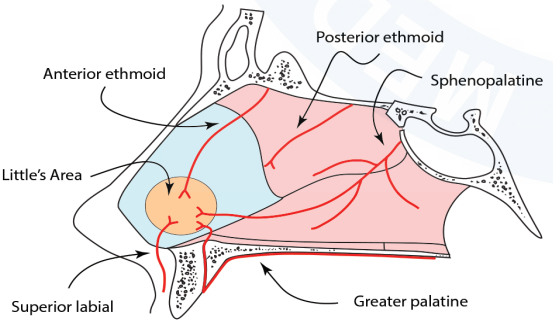
✦ All the muscles of larynx are supplied by Recurrent Laryngeal N. Except Cricothyroid (By External Laryngeal N)

- **Sensory supply**
  1. Mucosa above vocal cords → Internal Laryngeal N.
  2. Mucosa below vocal cords → RLN
- Posterior Cricoarytenoid → SAFETY Muscle of Larynx
  → Action → Abduction of Vocal Cord

NOSE

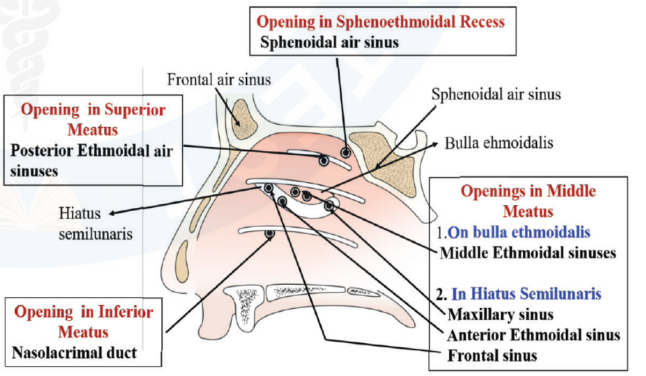
1. Nasal Septum

- **Little's Area**
  - antero inferior Quadrant of Nasal septum
  - vascular area.
  - site for epistaxis
  - KISSELBACH'S PLEXUS → Anastomosis of arteries
    1. Anterior Ethmoidal A
    2. Greater Palatine A
    3. Sphenopalatine A
    4. Superior Labial A



Nasal Septal Arterial Supply

◀FIG. Little's Area

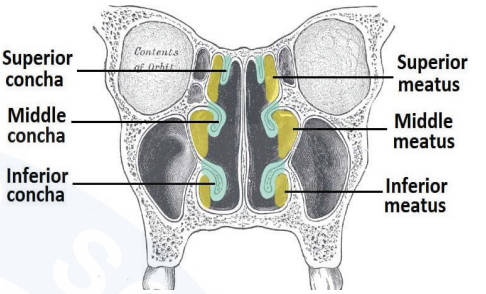


◀FIG. Lateral wall of nose

2. Lateral Wall

❑ Openings in Lateral wall of Nose

1. Sphenoethmoidal Recess → Sphenoid sinus drains here
2. Superior Meatus → Posterior Ethmoidal Sinus
3. Middle Meatus → Frontal sinus, Maxillary Sinus & Anterior & Middle Ethmoidal sinus
4. Inferior Meatus → Nasolacrimal Duct



ONE LINERS

1. Trimus - spasm of medial pterygoid
2. Cranial nerve

- Pure sensory - CN 1, 2, 8
- Pure motor - CN 3, 4, 6, 11, 12
- Mixed cranial - CN 5, 7, 9, 10
- Cranial which carry parasympathetic fibres - 3, 7, 9, 10
- They only CN which arise from dorsal (posterior) aspect of brainstem -Trochlear N.
- CN which undergo internal decussation -Trochlear N.
- CN involved in increased intracranial pressure-Abducent N.

3. Named Nerves

- Arnold's /Alderman's N -Auricular branch of vagus
- Jacobson N -Tympanic branch of glossopharyngeal nerve
- Vidian Nerve -Nerve to pterygoid canal
- N of wirsberg -sensory root of facial nerve
- Criminal Nerve of Grassi - Posterior Gastric branch of vagus N.

4. Nucleus ambiguous - CN. 9, 10, 11

- Nucleus tractus solitarius -CN 7, 9, 8

5. Origin of CN — Midbrain - 3, 4

— Pons - 5, 6, 7, 8

— Medulla - 9, 10, 11, 12

- At pontomedullary Junction → CN 6, 7, 8

6. Palatine Tonsil → Supplied by Glossopharyngeal N.

7. Lesser Petrosal Nerve → Branch of Glossopharyngeal N

8. Corneal Reflex → Afferent → CN5
   → Efferent → CN7

9. Light Reflex → Afferent → CN2
   → Efferent → CN3

Workbook Questions

1. Which of the following muscle is supplied by glossopharyngeal nerve :

   (a) Palatopharyngeus (b) Salpingopharyngeus

   (c) Stylopharyngeus (d) Superior constrictor of pharynx

   Ans C

2. Pterygopalatine ganglion supplies:

   a. Parotid gland b. Submandibular gland c. Lacrimal gland d. Sublingual

   Ans C

3. Skin over parotid region is supplied by:

   (a) Retroauricular nerve (b) Greater auricular nerve

   (c) Greater occipital nerve (d) Facial nerve

   Ans B

4. Parotid duct opens:

(a) Opposite to lower 1st molar tooth (b) Opposite to lower 2nd molar tooth

(c) In floor of mouth (d) In relation with upper 2nd molar tooth

Ans D

5. Depression of mandible is done by:

(a) Lateral pterygoid (b) Medial pterygoid (c) Temporalis (d) Masseter

Ans A

6. Which is the following structure passes through the lateral part of superior orbital fissure?

(a) Superior ophthalmic vein (b) Inferior ophthalmic vein

(c) Abducent nerve (d) Nasociliary nerve

Ans A

7. All of the following are pneumatic bones EXCEPT:

(a) Maxilla (b) Frontal (c) Mandible (d) Ethmoid

Ans C

8. 'Chassaignac's tubercle' is:

(a) Erb's point (b) Carotid tubercle on C6 vertebra

(c) Found on first rib (d) Medial condyle of humerus

Ans B

9. All of the following are branches of the external carotid artery except:

(a) Superior thyroid artery (b)Anterior Ethmoidal artery

(c) Occipital artery (d) Posterior auricular artery

Ans B

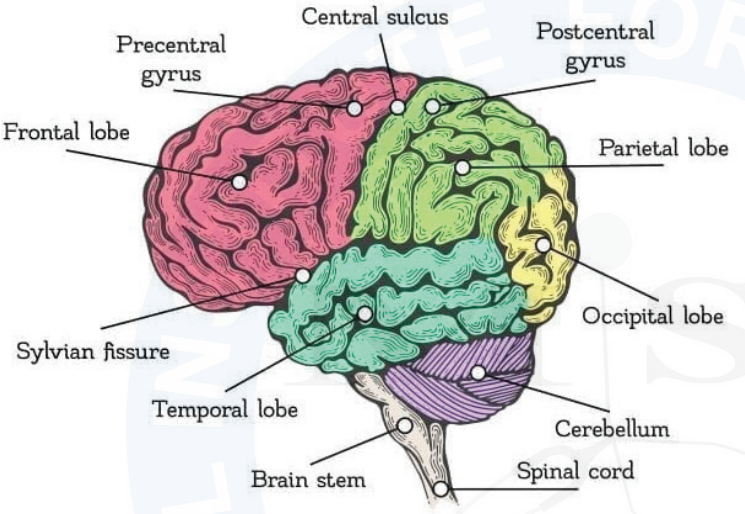
10. The superior oblique muscle is supplied by:

(a) 3rd cranial nerve (b) 4th cranial nerve (c) 5th cranial nerve

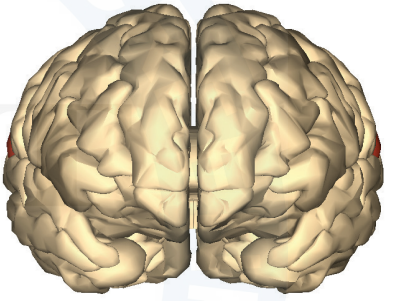
(d) 6th cranial nerve

Ans B

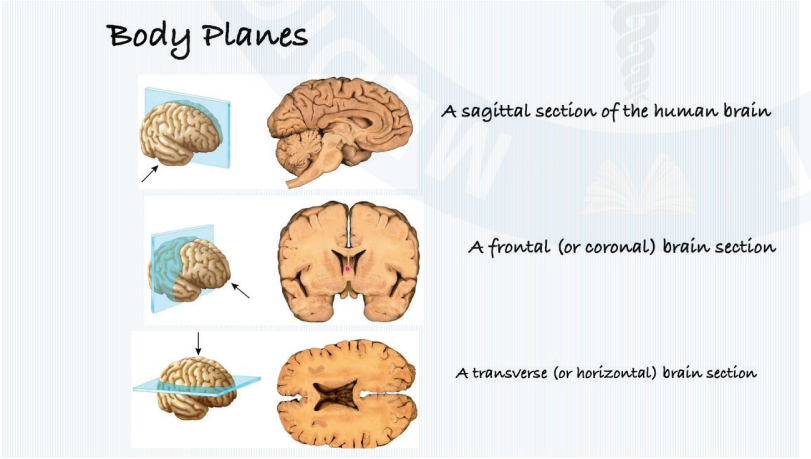
ORGANISATION OF NERVOUS SYSTEM & MENINGES



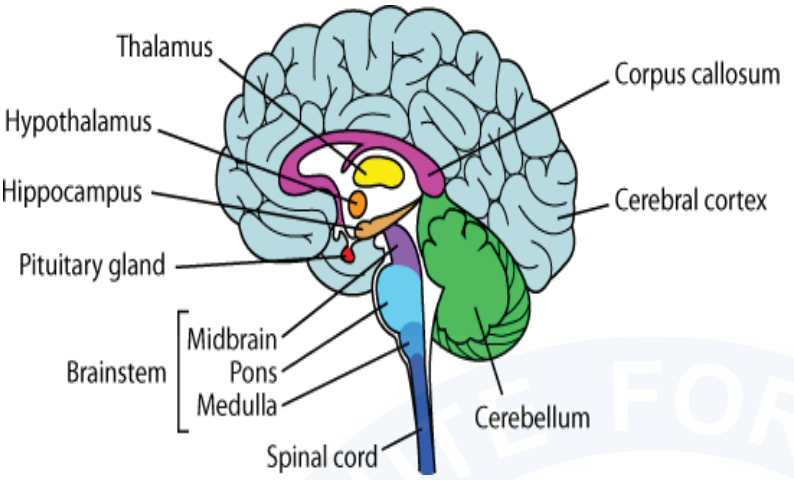
◀**FIG.** Brain (Lateral view)



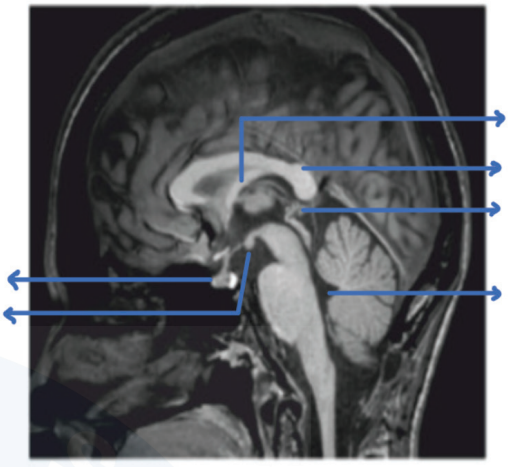
◀**FIG.** Brain (Anterior View)



◀**FIG.** Parts of Corpus Callosum (Sagittal Section)



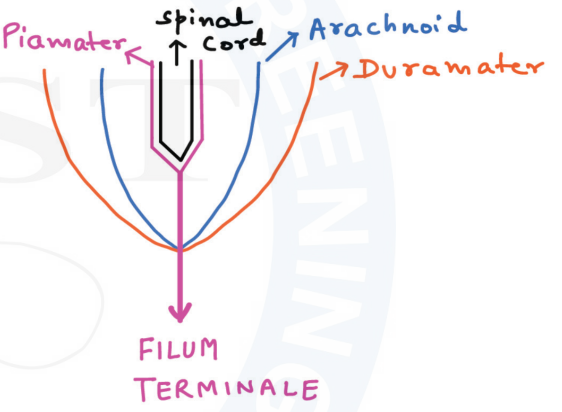
◀FIG. Brain (Sagittal section showing medial view)



◀FIG. Sagittal Section of Brain

❑ Meninges (Outer to inner)

1. Duramater
2. Arachnoid
3. Piamater



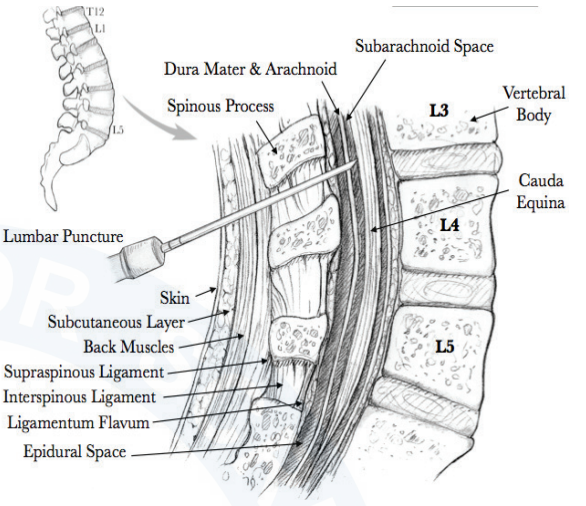
❑ Spaces

- ✦ 1. Subarachnoid Space
  - Between Arachnoid and Piamater
  - CSF (Cerebrospinal Fluid)
- ✦ 2. Subdural Space B/w duramater and arachnoid
- ✦ Epidural Space B/w Vertebra and duramater.
- ❖ .....
- ❖ .....
- ❖ .....
- ❖ .....
- ❖ .....
- ❖ .....

Lumbar Puncture

❖ Layers pierced by lumbar Puncture Needle

1. .....
2. .....
3. .....
4. .....
5. .....
6. .....
7. .....



◀FIG. Lumbar Puncture

❖ For therapeutic & diagnostic purpose

Note: .....

CEREBRUM

1, WHITE MATTER FIBRES

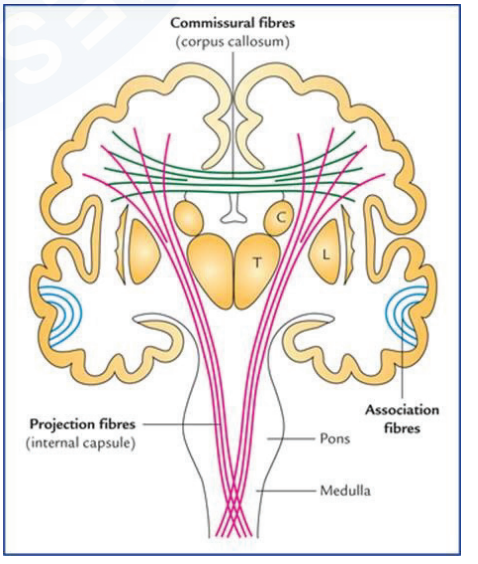
❑ **1. Association Fibres (in same cerebral hemisphere)**

- Longitudinal Fasciculus (Superior & Inferior)
- Uncinate Fasciculus
- Arcuate Fasciculus
- Cingulum

❑ **2. Commissural Fibres**

(Connect R and L Cerbral Hemispheres)

- Corpus Callosum (Largest Commisssural fibre)
- Anterior Commissure
- Posterior Commissure
- Hippocampal Commissure



◀FIG. White Matter Fibres of Cerebrum

- Habenular Commissure

❑ **3. Projection Fibres**

- Internal Capsule
- Corona Radiata
- Tracts

2. FUNCTIONAL AREAS

◀**FIG.** Functional areas of Cerebrum (Lateral view)

◀**FIG.** Functional areas of Cerebrum (Medial view)

✦ 1. Motor area
- Brodmanns Area No 4
- Located in Precentral gyrus

✦ 2. Sensory Area
- Area No. 3, 1, 2
- Postcentral gyrus
  - ❖ Paracentral Lobule → Represents Perineum & Lower Limb

✦ 3. Visual Area
- Area No. 17, 18, 19
- In occipital lobe

✦ 4. Auditory Area
- Area No. 41, 42
- Superior temporal gyrus

✦ 5. Wernicke's sensory Speech area
- Area No. 22, 39, 40

- Superior temporal gyrus
- For .....
- Lesion .....
- Patient is not able to understand but keeps speaking (Speaks meaningless words)

✦ 6. Broca's Motor Speech area →

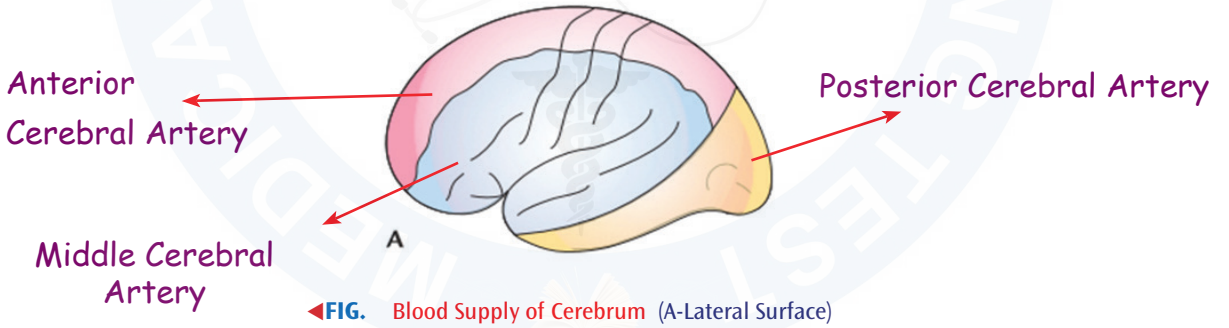
- Area No. 44, 45, Inferior frontal gyrus
- For .....
- Lesion .....
- Patient is able to understand but not able to speak / express (Speaks 2-3 words only)

❖ Arcuate fasciculus → Connects Brocas & Wernickes speech area

3. ARTERIAL SUPPLY

❖ Main artery of lateral surface → Middle Cerebral A

Note: Blood supply of visual area → Posterior Cerebral A



◀FIG. Blood Supply of Cerebrum (A-Lateral Surface)

DIENCEPHALON

❑ **Components**

1. Thalamus
2. Hypothalamus
3. Epithalamus → Pineal Body
4. Metathalamus
   - Lateral Geniculate Body → Optic Radiation
   - Medial Geniculate Body → Auditory radiation

BASAL GANGLIA

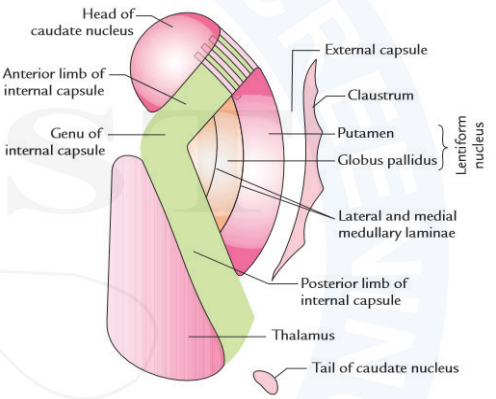
- **Components**
  1. Caudate
  2. Putamen
  3. Globus Pallidus
  4. Substantia Nigra
  5. Claustrum
  6. Amygdala
  7. Subthalamus

Putamen + Globus Pallidus = Lentiform Nucleus

Caudate + Lentiform Nucleus = Corpus Striatum

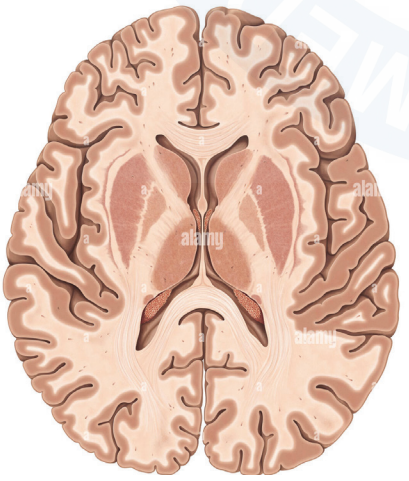
INTERNAL CAPSULE

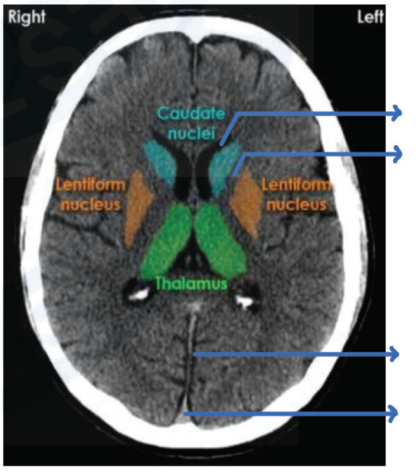
- **'V' shaped structure**
- **Relations**
  - Medially → Caudate, Thalamus
  - Laterally → Lentiform Nucleus



◀**FIG.** Internal Capsule (Relations)

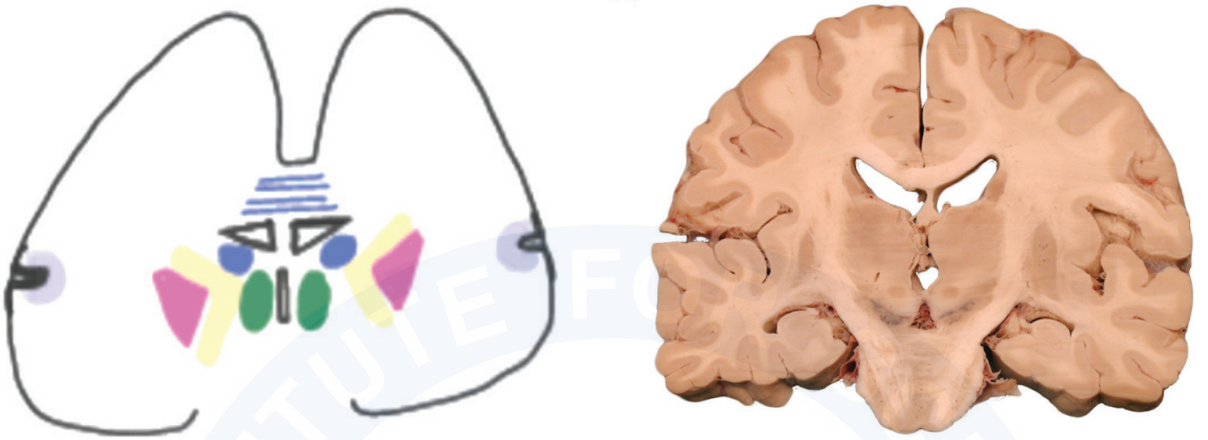
TRANSVERSE SECTION OF CEREBRUM





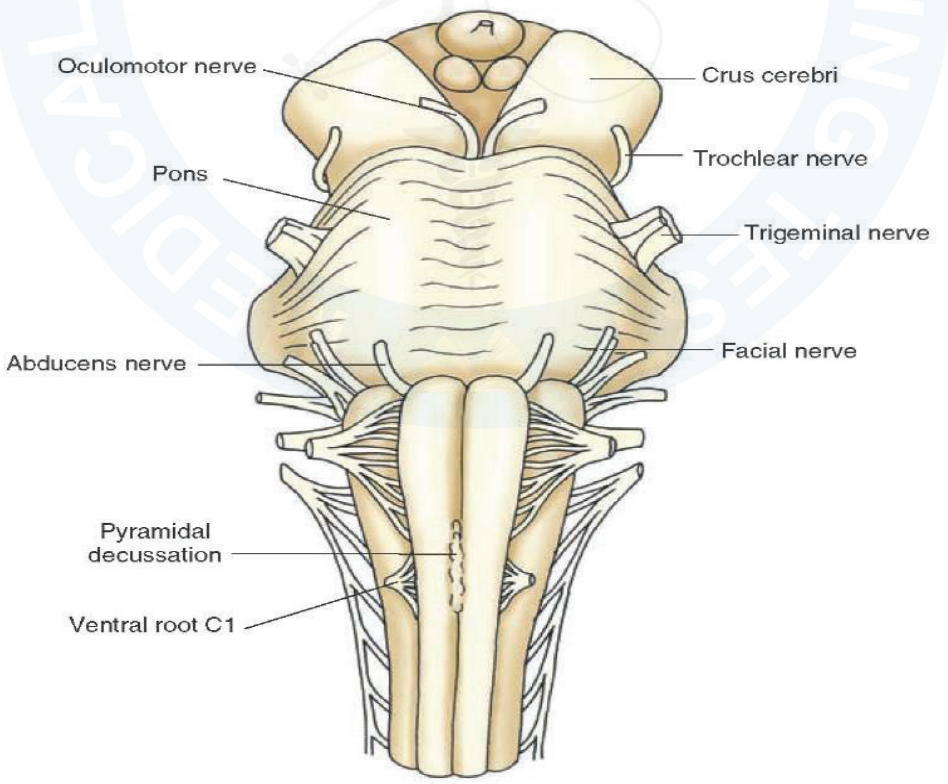
◀**FIG.** Transverse Section of Brain

CORONAL SECTION OF CEREBRUM

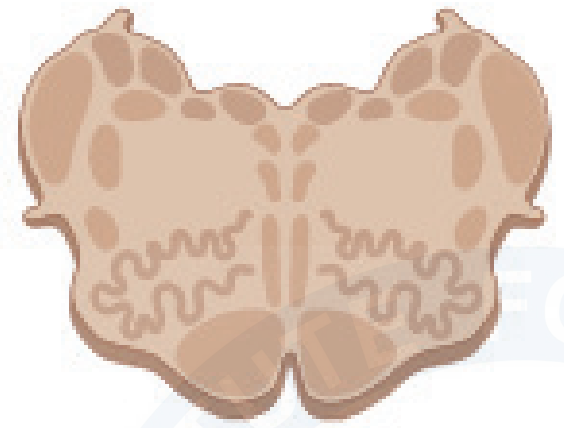


◀FIG. Coronal Section of Cerebrum

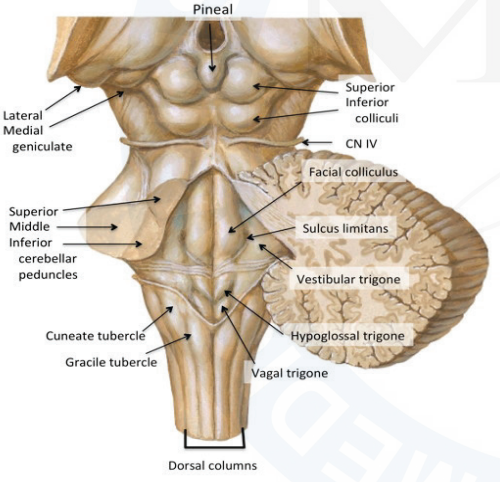
BRAINSTEM



◀FIG. Brainstem - Anterior view



◀FIG. Transverse Section at the Level of Medulla

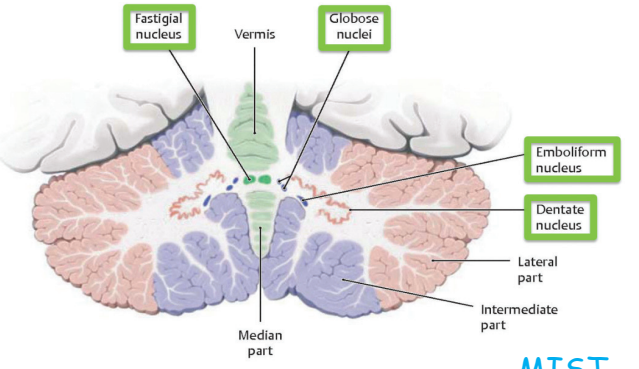


◀FIG. Brainstem - Posterior view

CEREBELLUM

Cerebellar Cortex

1. Purkinje Cell
2. Granule Cell
3. Golgi Cell
4. Stellate Cell
5. Basket Cell

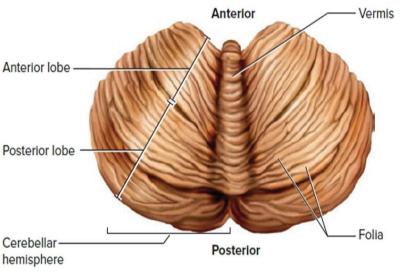


◀FIG. Deep Cerebellar Nuclei

MIST

❑ Deep Cerebellar Nuclei

1. .....
2. .....
3. .....
4. .....



◂FIG. Cerebellum

❑ Pathway

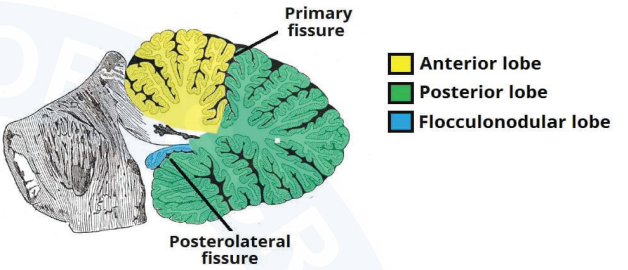
Afferent fibres → Cerebellar Cortex

↓

Purkinje cell

↓

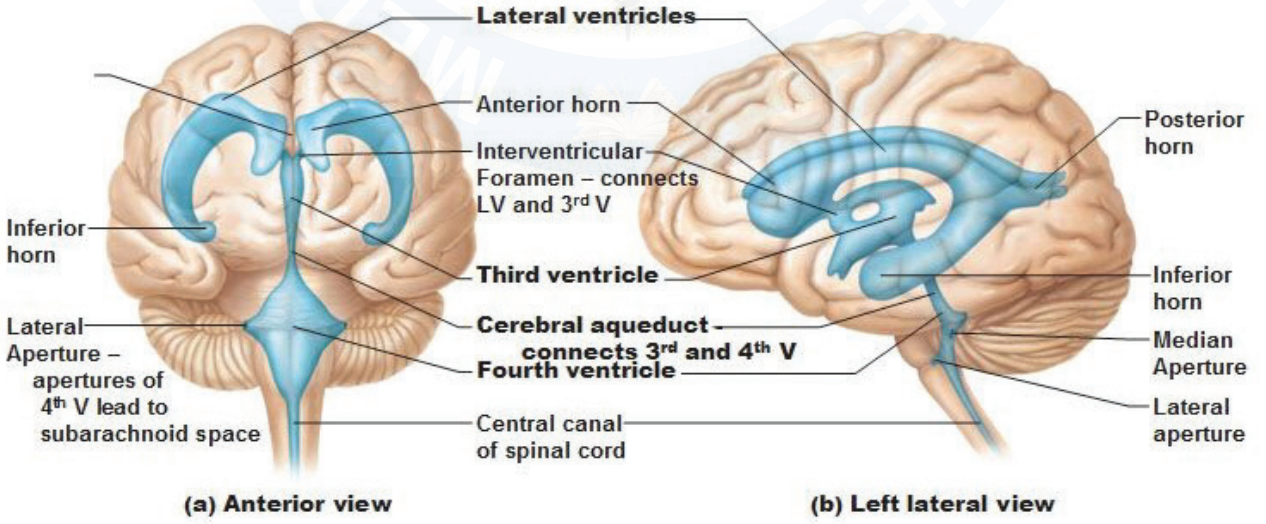
Efferent fibres ← Deep cerebellar Nuclei



VENTRICLES, CSF & HYDROCEPHALUS

❑ Ventricles of Brain

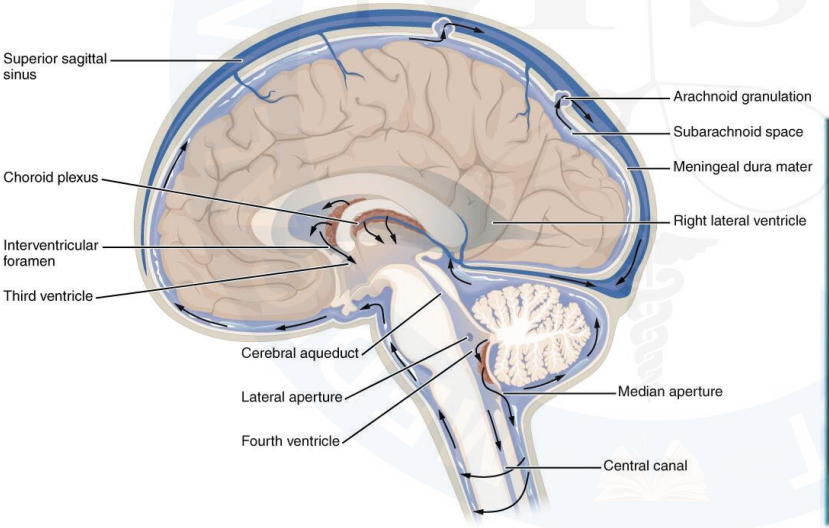
1. Right & Left Lateral Ventricles → located in R & L Cerebral Hemisphere
2. Third Ventricle → B/W Diencephalon (Thalamus)
3. Fourth ventricle → B/W Pons & Cerebellum



◂**FIG.** Ventricles of Brain

❑ Flow of Cerebrospinal fluid (CSF)

Formation of CSF → Choroid Plexus of Lateral Ventricle → Interventricular foramen of MONRO → 3rd Ventricle → Cerebral AQUE DUCT of SYLVIUS → 4th Ventricle → Foramen of MAGENDIE & Foramen of LUSCHKA → Subarachnoid Space → Absorption is by Arachnoid Granulations/Villi → Superior Sagittal Venous sinus



◀FIG. Flow of CSF

◀FIG. Hydrocephalus

HYDROCEPHALUS - (ACCUMULATION OF CSF)

❑ S/s

1. Disproportionate enlarged head
2. Anterior fontanelle → Enlarged & Tense & Bulging
3. Prominence / Bossing of forehead

MIST

ARTERIAL SUPPLY OF BRAIN

1. Vertebral Artery & Basilar Artery

❑ Branches of Vertebral A.

1. Posterior Inferior Cerebellar A (PICA)
2. Anterior spinal (ASA)
3. Posterior Spinal (PSA)

- ❖ Spinal cord → One ASA & Two PSA
- ❖ ASA - Supplies anterior 2/3 of spinal cord
- ❖ PSA - Supplies posterior 1/3 of spinal cord

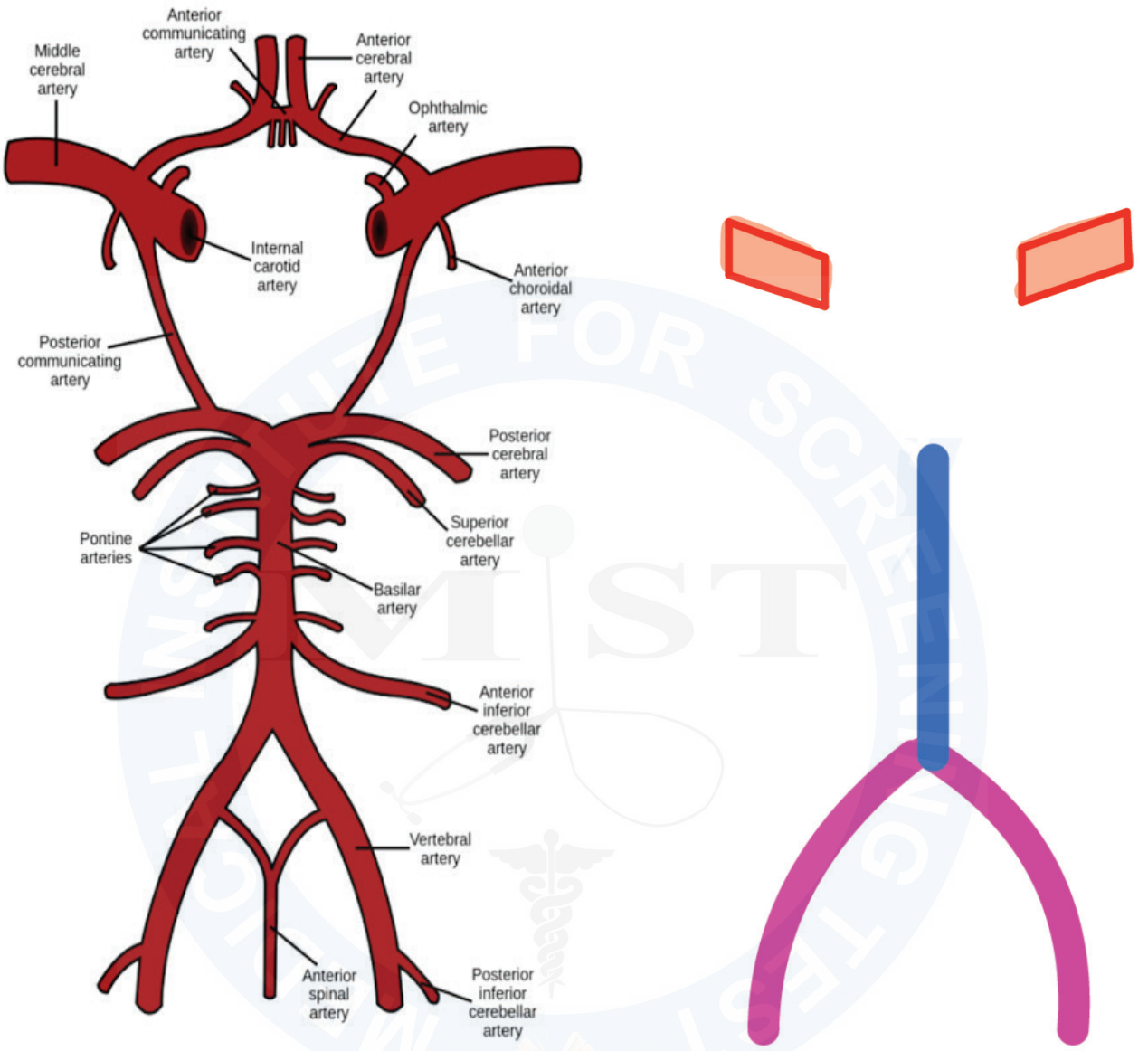
❑ Basilar Artery

- Formed by union of R & L Vertebral A
- ✦ Branches
  1. Superior Cerebellar A
  2. Labryinthine A
  3. Anterior Inferior Cerebellar A (AICA)
  4. Pontine A
  5. Posterior Cerebral A

2. Internal Carotid Artery & Circle of Willis

❑ Internal Carotid Artery

- ✦ Branches
  1. Anterior Cerebral A
  2. Middle Cerbral A
  3. Anterior Choroidal A
  4. Posterior Communicating A
  5. Ophthalmic A (gives a branch Central A of Retina)



◀**FIG.** Arterial Supply of Brain

❑ Circle of Willis

- Anastomosis is between the branches of ICA & Basilar A
- Located at inferior aspect of brain
- Formed by
  1. Anterior Cerebral A
  2. Anterior Communicating A
  3. Posterior Cerebral A
  4. Posterior Communicating A

MIST

3. Lateral & Medial Medullary Syndrome

❑ Lateral Medullary Syndrome or Wallenberg synd

- Due to the occlusion of .....

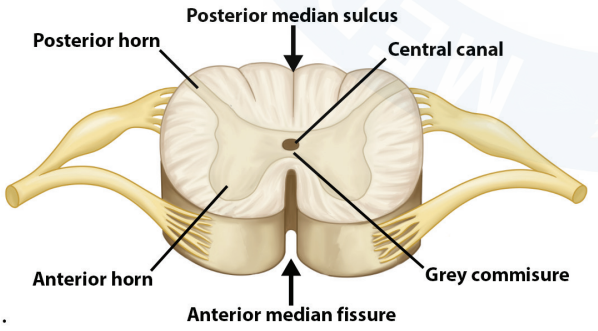
1. Nausea, Vomitting, dizziness, Giddiness
2. Difficulty in walking (Ataxia)
3. Difficulty in speaking & eating
4. Ipsilateral loss of taste sensation
5. Contralateral loss of Pain & Temperature

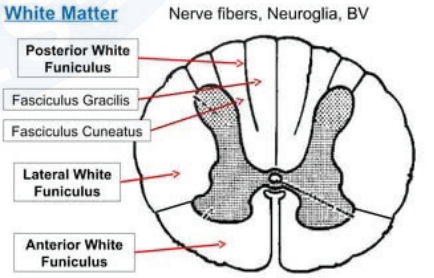
❑ Medial Medullary Syndrome

- Due to the occlusion .....

SPINAL CORD

1. Introduction





◀FIG. Transverse section of spinal cord

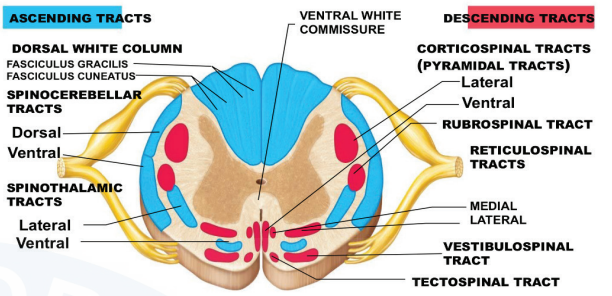
Note: Lateral Horn is present at T_1– L_2

2. Ascending Tracts (Dorsal column)

❑ Tracts

Ascending Tracts (Sensory)

Descending Tracts (Motor)



◀FIG. T. S. of Spinal cord showing ascending & Descending Tracts

Ascending Tracts

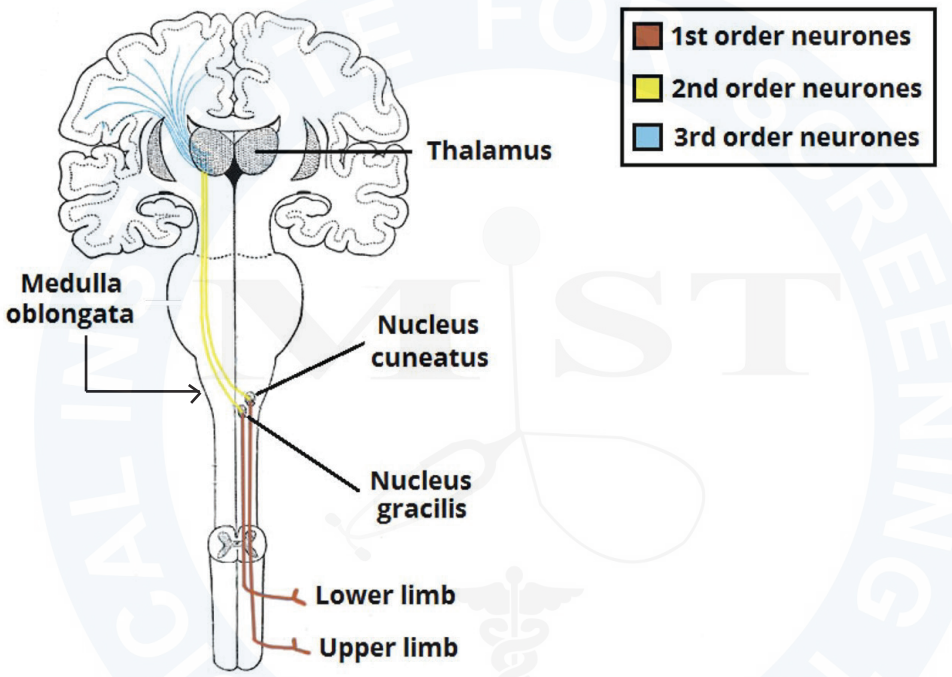
Periphery → 1st order Neuron → 2nd order Neuron → 3rd order Neuron → Cerebrum

❑ Dorsal Column (Ascending Tract)

- ✦ Sensations carried by it
  - ✓ .....
  - ✓ .....
  - ✓ .....
  - ✓ Tactile discrimination / Two Point Distrimination
  - ✓ Stereognosis
- ✦ Fasciculus Gracilis - Fibres from Lower Part of Body
- ✦ Fasciculus cuneatus – Fibres from Upper Part of Body

MEDICAL INSTITUTE FOR SCREENING TEST MIST

- 1st order Neuron – Dorsal Root Ganglia (DRG)
- 2nd order Neuron – Gracile Nucleus & Cuneate Nucleus
- 3rd order Neuron - Thalamus (VPL) (Ventral Posterolateral Nucleus)
- Crossing of fibres - at Medulla



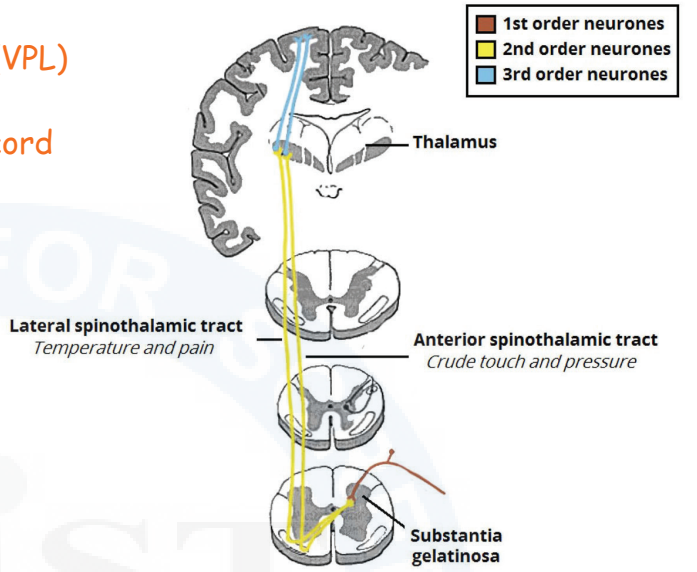
◀FIG. Dorsal Column

4. Ascending Tracts (Spinothalamic Tract)

❑ Spinothalamic Tract

- 1. Anterior S. T. → .....
- 2. Lateral S. T. → .....

- ✓ 1st order Neuron → DRG
- ✓ 2nd order Neuron → Substantia Gelatinosa
- ✓ 3rd order Neuron → Thalamus (VPL)
- ✓ Crossing of fibres → at spinal cord



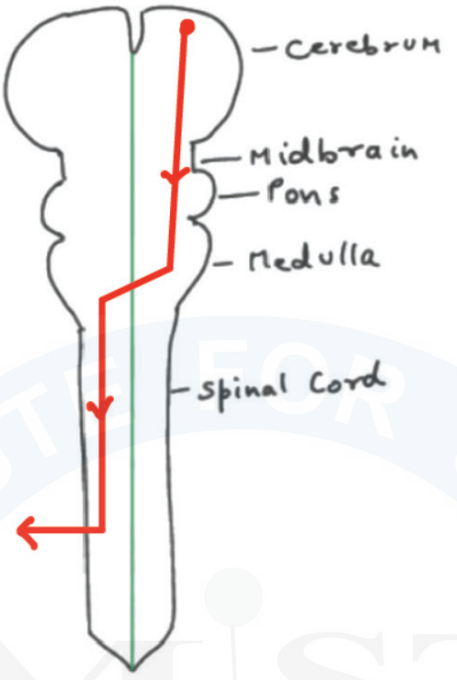
◀FIG. Spinothalamic Tract

5. Descending Tracts

Descending Tracts

Pyramidal Tract	Extrapyramidal Tract
or	↓ (for Automatic Involuntary movement)
Corticospinal Tract	
↓	1. Rubrospinal
• For Skilled Voluntary Movement	2. Olivospinal T.
• Crossing of Fibres occur at Medulla (Lower)	3. Vestibulospinal T.
	4. Reticulospinal T.
	5. Tectospinal T.

MIST



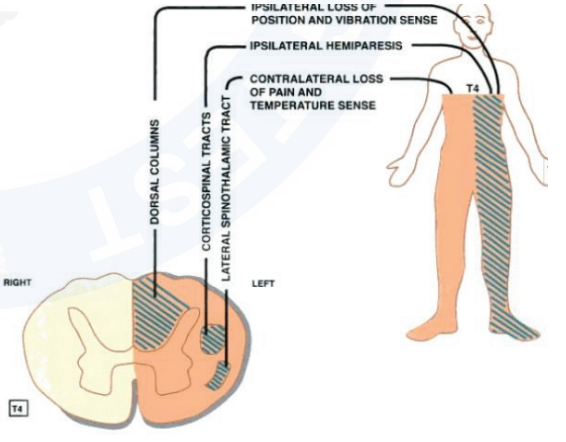
◀FIG. Corticospinal Tract

6. Brown Sequard Syndrome

❑ Brown Sequard Syndrome

✓ Hemisection of spinal Cord

1. Ipsilateral Hemiplegia
   (d/t involvement of corticospinal tract)
2. Ipsilateral Loss of fine touch & conscious propioception (d/t inv. of dorsal column)
3. Contralateral Loss of Pain & Temperature (d/t inv. of Lateral spinothalamictract)



Other Points.

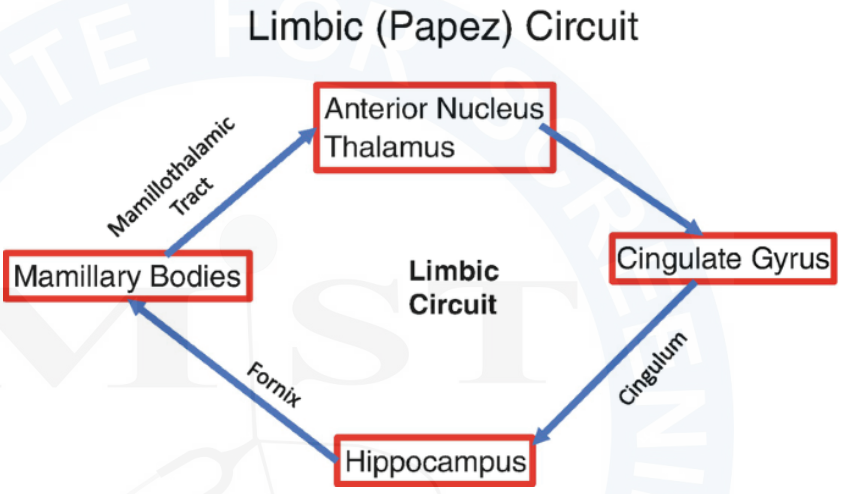
Note: Weber's syndrome (Midbrain)

1. Ipsilateral CN 3 involvement
2. Contralateral Hemiplegia

LIMBIC SYSTEM

✓ Components

1. Hippocampus
2. Fornix
3. Mamillary Body
4. Cingulate Gyrus



Workbook Questions

1. Blood supply of spinal cord is by how many spinal arteries?

   (a) 1 anterior and 2 posterior
   (b) 2 anterior and 1 posterior
   (c) 2 anterior and 2 posterior
   (c) 1 anterior and 1 posterior

   Ans A

2. Subarachnoid space extends up to :

   (a) Lower border L1
   (b) Upper border L1
   (c) Upper border of S2
   (d) Lower border of S2

   Ans D

3. Which of the following cranial nerve carries parasympathetic fibres?
(a) CN 1 (b) CN 2 (c) CN 3 (d) CN 4
Ans C

4. Corticospinal tract (pyramidal tract) cross at what level
(a) Midbrain (b) Pons (c) Upper medulla (d) Lower medulla
Ans D

5. During lumbar puncture, after piercing which structure the needle will enter in epidural space?
A. Ligamentum flavum B. Interspinous ligament
C. Supraspinous ligament D. Ligamentum nuchae
Ans A

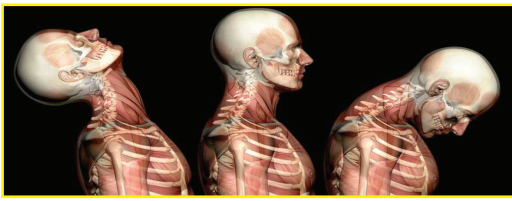
6. Posterior inferior cerebellar artery a branch of
(a) Internal carotid (b) External carotid (c) Vertebral (d) Basilar
Ans C

7. Which is not a part of basal ganglia?
(a) Caudate nucleus (b) Thalamus (c) Lentiform nucleus (d) Globus pallidus
Ans B

8. Which of the following cranial nerve is a pure sensory nerve?
(a) CN 3 (b) CN 5 (c) CN 8 (d) CN 9
Ans C

9. The spinal cord in infants ends at the level of:
(a) L1 (b) L2 (c) L3 (d) L4
Ans C

10. Which of the following areas of the brain is not a primary sen sory cortex?
(a) 1 (b) 2 (c) 3 (d) 4
Ans D



LOWER LIMB

BONES OF LOWER LIMB

1. Hip Bone & Bony Pelvis

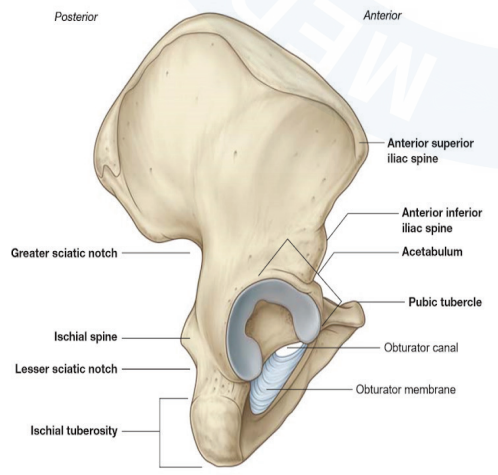
- Highest Point of Iliac Crest →
- PSIS →
- Inguinal Ligament →

✦ Midinguinal Point

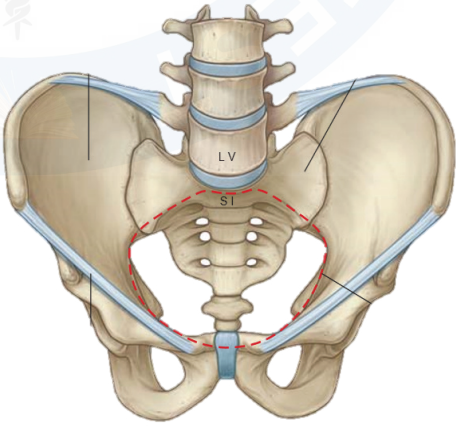
- .....
- .....

✦ ASIS (Anterior Superior Iliac spine)

- .....

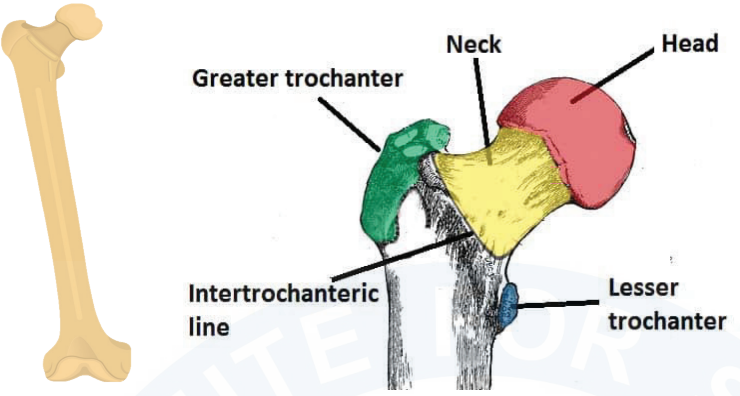


◀FIG. Hip Bone (Lateral View)



◀FIG. Bony Pelvis (Anterior View)

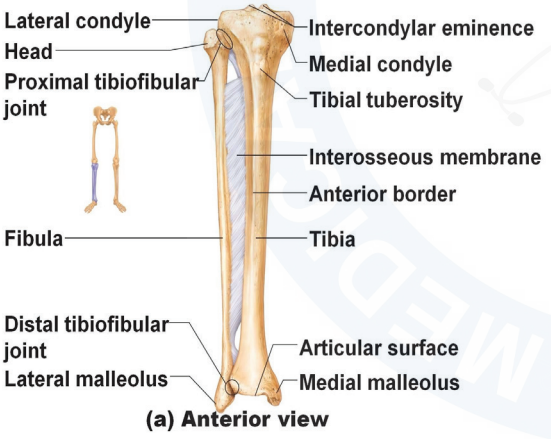
2. Femur



◂**FIG.** Femur

3. Tibia & Fibula

- Tibia - Medial Malleolus
- Fibula - Lateral Malleolus

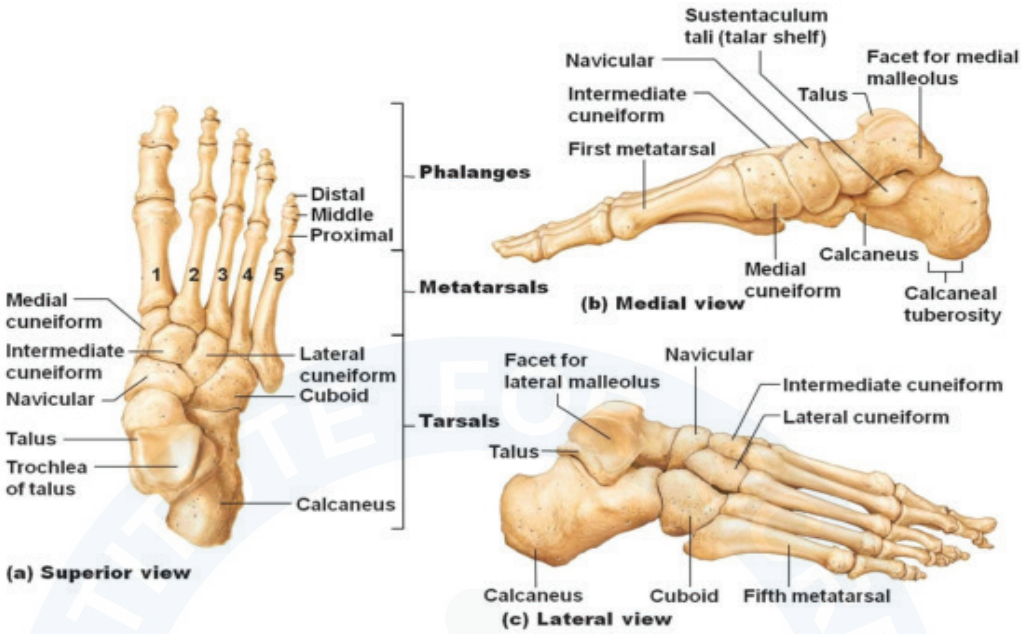


◂**FIG.** Tibia & Fibula

◂**FIG.** Tarsal Bones

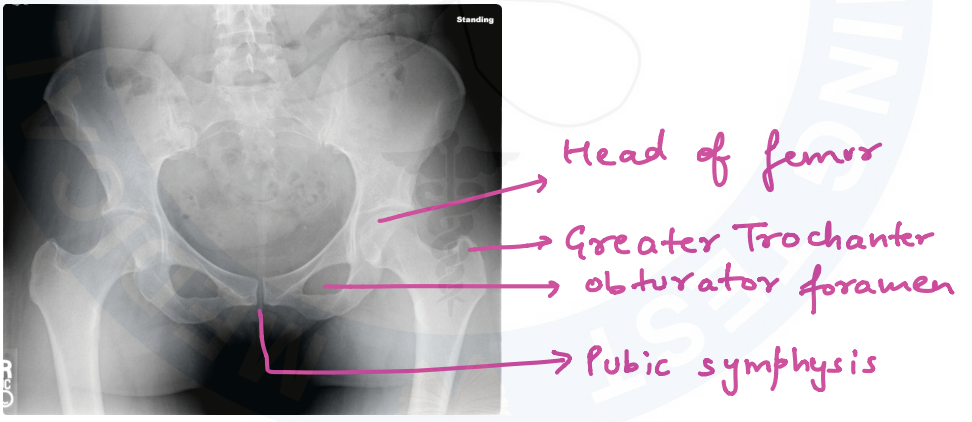
4. Tarsal bones, Metatarsals & Phalanges

- Talus – It has No muscle attachment
- Talocalcaneo – Navicular Joint – spring ligament
- Ankle Joint – Deltoid lig
- Hip Joint – Iliofemoral lig of Bigelow (strongest lig of Body)

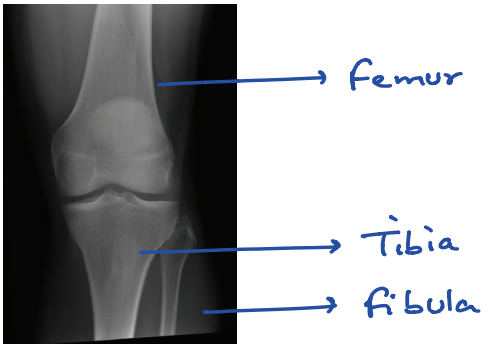


◀**FIG.** Bones of foot

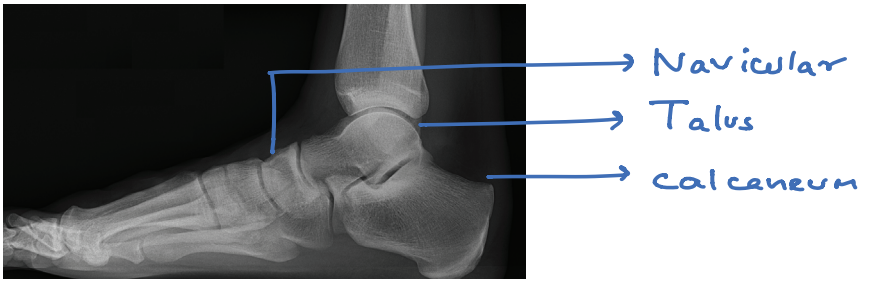
RADIOLOGICAL ANATOMY OF LOWER LIMB



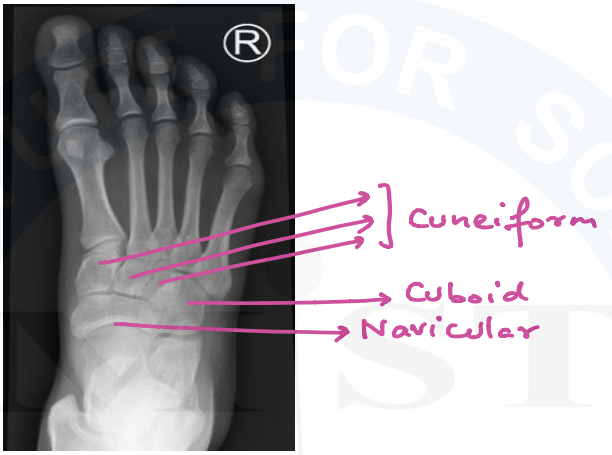
◀**FIG.** X ray Bony Pelvis & Hip Joint



◀**FIG.** X ray Knee Joint



◀FIG. X ray Foot (View from Medial side)



◀FIG. X ray Foot (superior view)

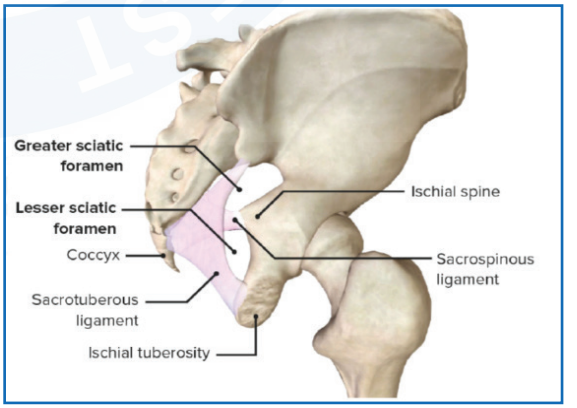
GREATER & LESSER SCIATIC FORAMEN

❑ Structures passing through Greater sciatic foramen

1. Piriformis
2. Superior Gluteal Nerve & vessels
3. Inferior Gluteal Nerve & Vessels
4. Sciatic Nerve
5. Posterior Cutaneous Nerve of thigh
6. **PIN** – Pudendal Nerve

   Internal Pudendal Vessels

   Nerve to obturator Internus



❑ **Structures passing through Lesser sciatic foramen**

1. **PIN**
2. Obturator Internus Tendon

ANTERIOR COMPARTMENT OF THIGH

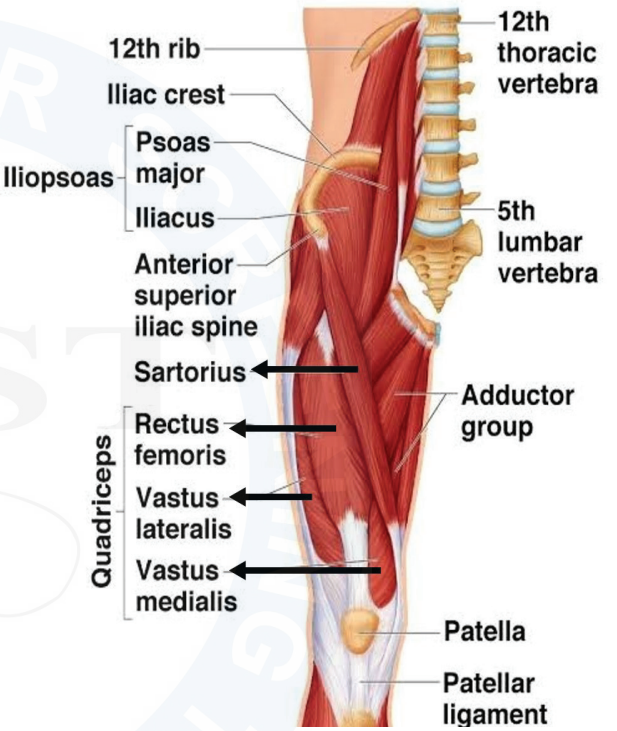
❑ **Muscles of anterior compartment of thigh**

1. Sartorius
2. Quadriceps Femoris
   - Vastus medialis
   - V. Lateralis
   - V. Intermedius
   - Rectus femoris

⇨ Nerve of this comp → .....

.....

.....



◀**FIG.** Muscles of anterior compartment of thigh

❑ **Quadriceps**

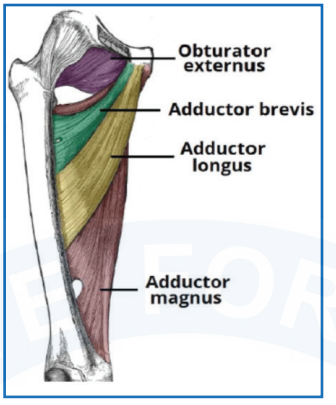
✦ Common action → .....

❑ **Knee Jerk** → Tap on Patellar ligament / P. tendon → Quadriceps activated & Extension occurs at knee joint

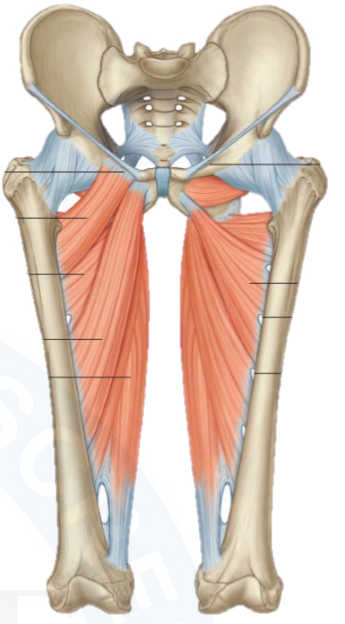
❑ **Sartorius** → Tailor's Muscle
→ Longest Muscle of body
→ Action → Hip Joint →
→ Knee Joint →

MEDIAL COMPARTMENT OF THIGH

1. Adductors-
   - A. longus
   - A. magnus
   - A. brevis
2. Gracilis
3. Obturator Externus
4. Pectineus



- Nerve of this Comp .....
  - ✦ Common action of all muscles - .....



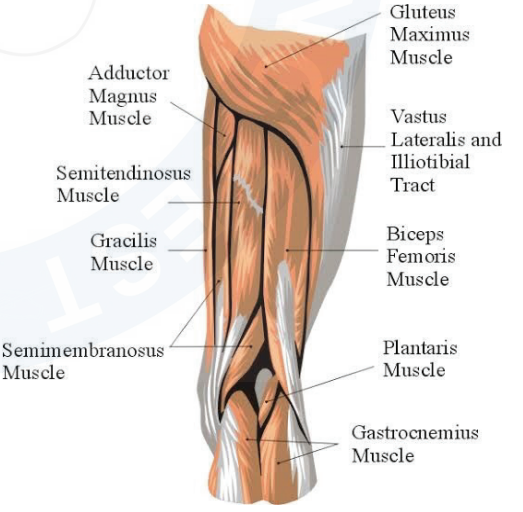
◀FIG. Muscles of Medial Compartment of thigh

POSTERIOR COMPARTMENT OF THIGH - MUSCLES

- ✦ Nerve of this Comp → .....
- ✦ Muscles
  1. Hamstrings
  2. Short Head of Biceps Femoris

❑ **Hamstrings**

1. Long Head of Biceps femoris
2. Semi membranous
3. Semitendinosus
4. Ischial Head of adductor Magnus



◀FIG. Posterior Compartment of Thigh

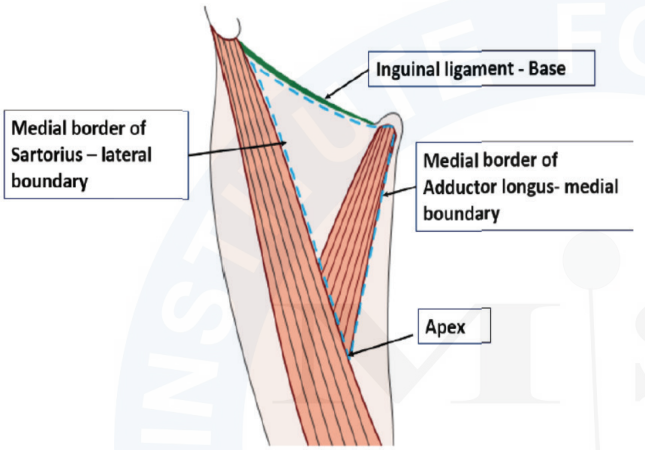
❑ **Common Action** .....

(For Reference → Hamstrings are supplied by Tibial part of sciatic nerve

1. Femoral Triangle

❑ Boundaries

1. Medial Boundary → .....
2. Lateral Boundary → .....
3. Base / Superior Boundary → .....



◀FIG. Boundaries of Femoral Triangle

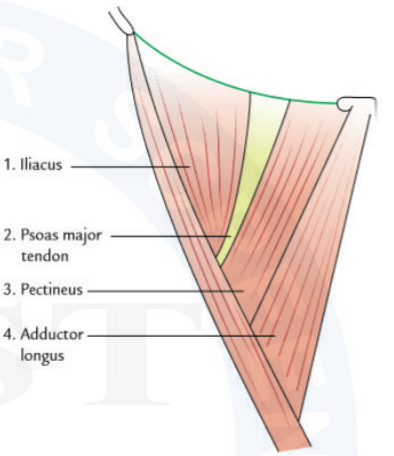
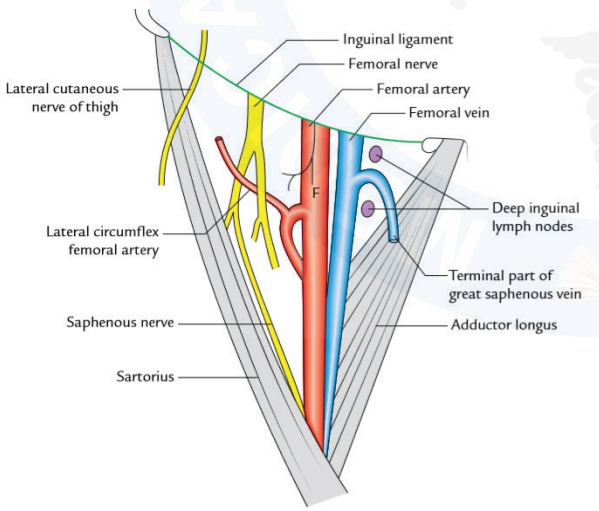


Fig. Floor of Femoral Triangle



◀FIG. Contents of Femoral Triangle

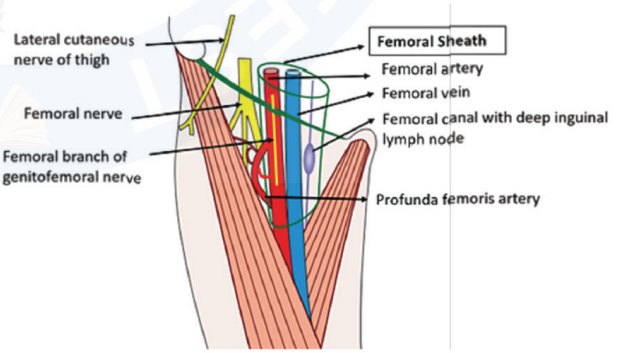


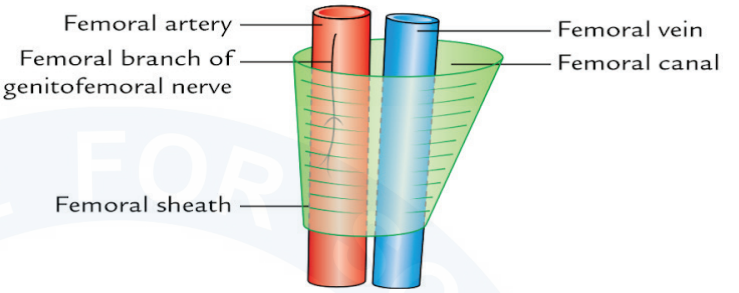
Fig. Femoral Sheath

❑ **Floor -** .....

1. **Adductor Longus**
2. **Pectineus**
3. **Psoas Major**
4. **Iliacus**

❑ **Contents -**

1. Deep Inguinal Lymph Node
2. Femoral Vein
3. Femoral Artery.
4. Femoral Nerve
5. Lateral Cutaneous nerve of thigh
6. Genito femoral N (Femoral Branch)



◀FIG. Femoral Sheath

2. Femoral Sheath

- Encloses upper part of femoral vessels

3 compartments

Medial Comp / Femoral Canal	Intermediate Comp	Lateral Comp
↓	↓	↓
Lymph Node of Cloquet or Rosenmuller	Femoral Vein	Femoral Artery & Genitofemoral Nerve

Note-

- **Genitofemoral N** → (Nerve of Cremastric Reflex)
  - Genitofemoral N → Femoral br → Supplies skin of upper & medial aspect of Thigh (or skin over femoral triangle)
    ↓
    Genital br → Supplies Cremaster Muscle
- **Cremastric Reflex**
  - Afferent → .....
  - Efferent → .....
- **Lateral Cutaneous N. of thigh** → Its compression leads to Meralgia Paresthetica → Tingling at lateral aspect of thigh

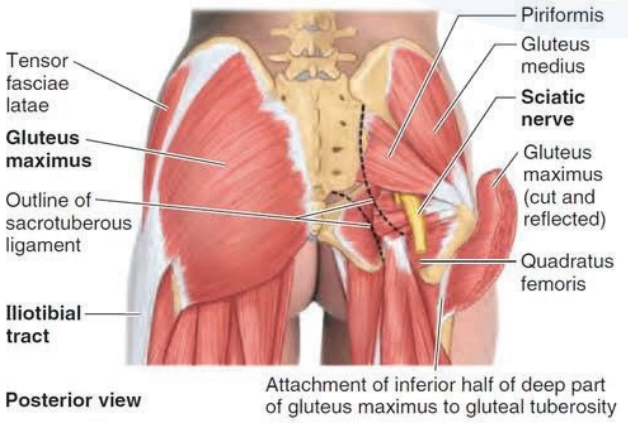
ILIOTIBIAL TRACT

- Located on lateral aspect of thigh
- Thickening of Deep fascia of thigh (Fascia Lata
- Gives attachment to
  1. Gluteus Maximus
  2. Tensor Fascia Lata
  3. Vastus Lateralis

◀FIG. Iliotibial Tract Anterior View & Lateral View

GLUTEAL REGION

1. Muscles



◀FIG Gluteal Region

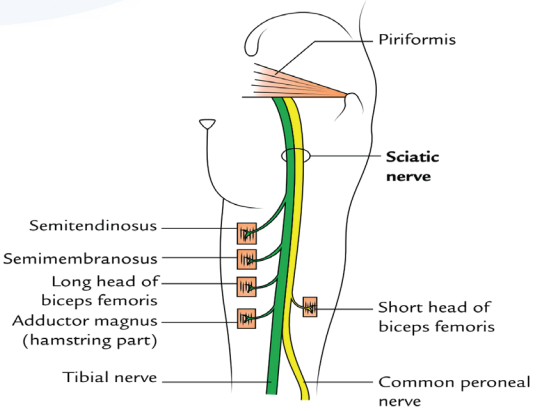


FIG. Sciatic Nerve & its divisions

- Safest Quadrant for Intramuscular Injection is upper & outer quadrant

❑ **G. maximus**

- N. S. → .....
- Action → .....
  → .....

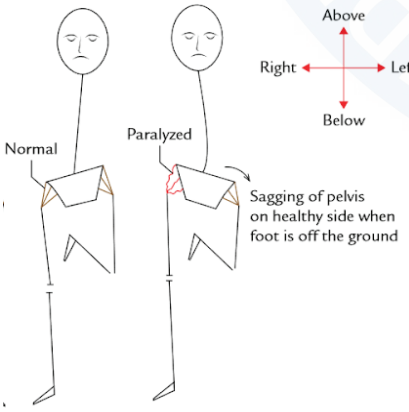
❑ **G. medius & G. minimus**

- N. S. → .....
- Action → .....

2. Superior Gluteal Nerve Injury

1. Glu teus Medius & Gluteus Minimus paralysed
2. Pelvis sags/falls to the opposite side
3. Trendelenberg sign is positive
4. - Lurching Gait (If Unilateral)
   - Waddling gait (If Bilateral)

◀FIG. Trendelenberg Sign



◀FIG. Trendelenberg Sign

❑ **CLINICAL INTEGRATION**

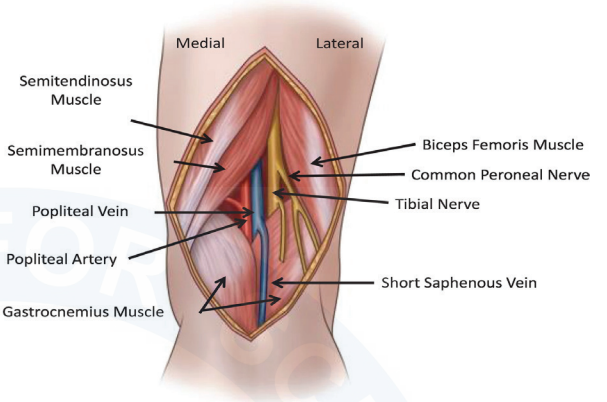
During examination, when a patient was asked to stand on left foot, then drop of right HIP/Side was observed. Which muscles are paralysed in this cas ?

a. Right gluteus medius & gluteus minimus
b. Left gluteus medius & gluteus minimus
c. Right gluteus maximus
d. Left gluteus maximus

POPLITEAL FOSSA

❑ **Contents**

1. Popliteal Artery
2. Popliteal Vein
3. Sciatic Nerve
4. Tibial N.
5. Common Peroneal N.
6. Popliteal fat & lymph node



◀**FIG.** Popliteal fossa

KNEE JOINT

❑ **Menisci** .....

❑ **Cruciate Lig →**

.....

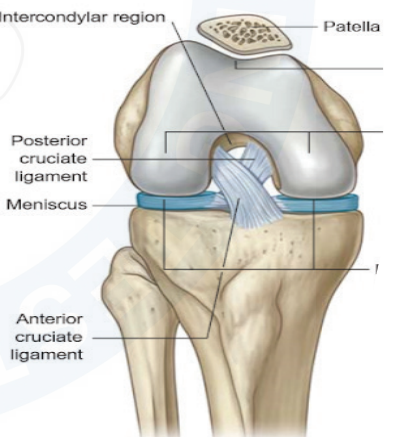
.....

❑ **Anterior Cruciate Lig →**

.....

.....

.....



◀**FIG.** Knee Joint

Locking	Unlocking
1. Occurs at End of Extension	1. Occurs at beginning of flexion
2. Medial Rotation of femur	2. Lateral rotation of femur
3.	3. .....

SCIATIC NERVE, TIBIAL NERVE & COMMON PERONEAL NERVE

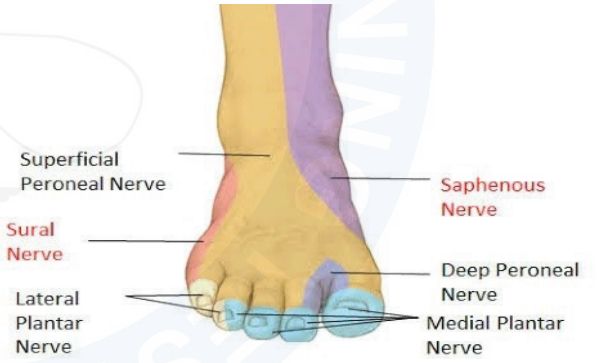
- **Sensory Supply (Skin)**
- **Deep Peroneal Nerve →**

  .....
- **Superficial Peroneal Nerve →**

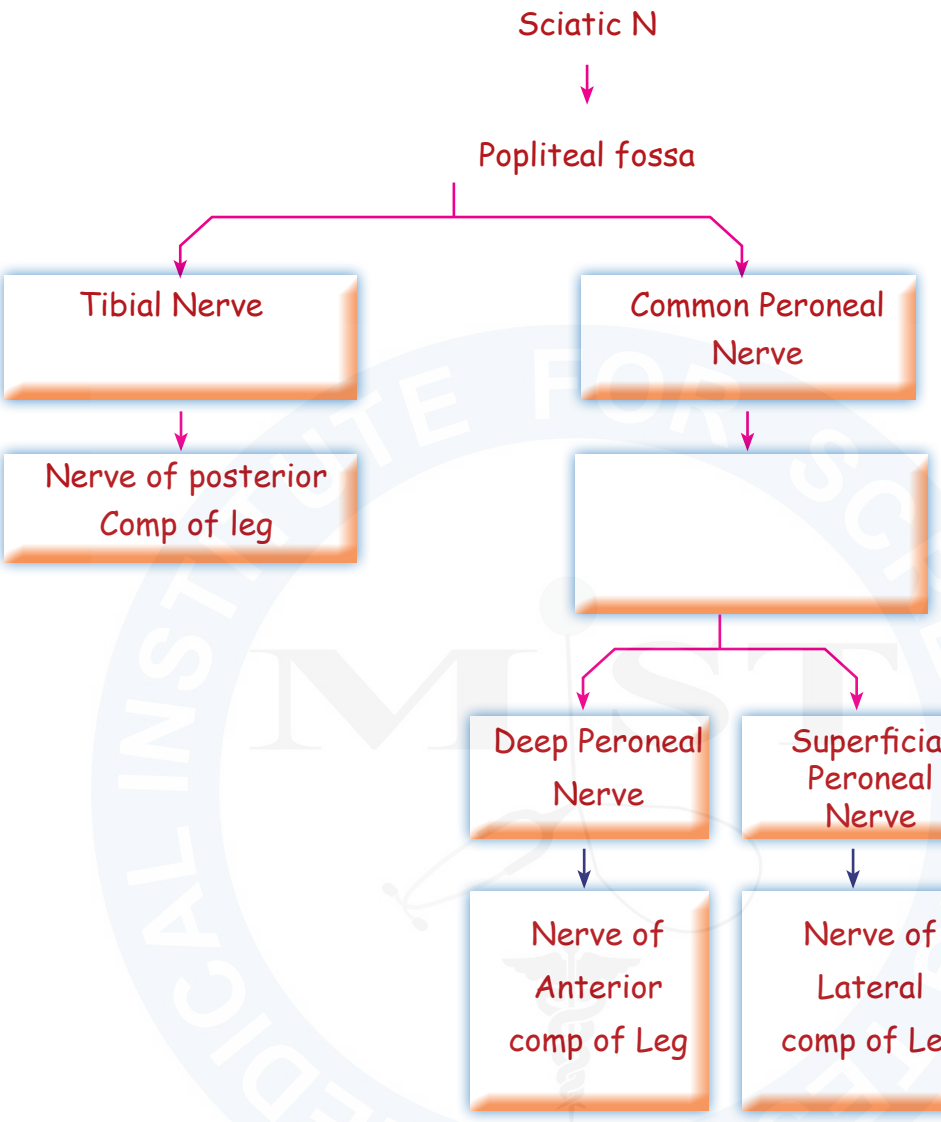
  .....

  .....

  .....
- **Tibial Nerve →** .....

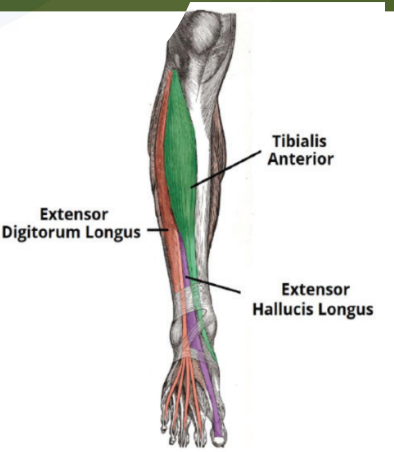


◀FIG. Sensory Supply of foot



MUSCLES OF ANTERIOR COMPARTMENT OF LEG

- **Nerve of Anterior Compartment of Leg - Deep Peroneal Nerue**
- **Muscles**
  1. Tibialis Anterior
  2. Extensor Hallucis Longus
  3. Extensor Digitorum longus
  4. Peroneus Tertius
- **Common Action** Dorsiflexion of Foot



◀**FIG.** Anterior Compartment

MUSCLES OF POSTERIOR COMPARTMENT OF LEG

❑ **Nerve of Posterior Compartment of Leg - Tibial Nerve**

❑ **Muscles**

✦ (A) Superficial Group

- Soleus
- Gastrocnemius
- Plantaris

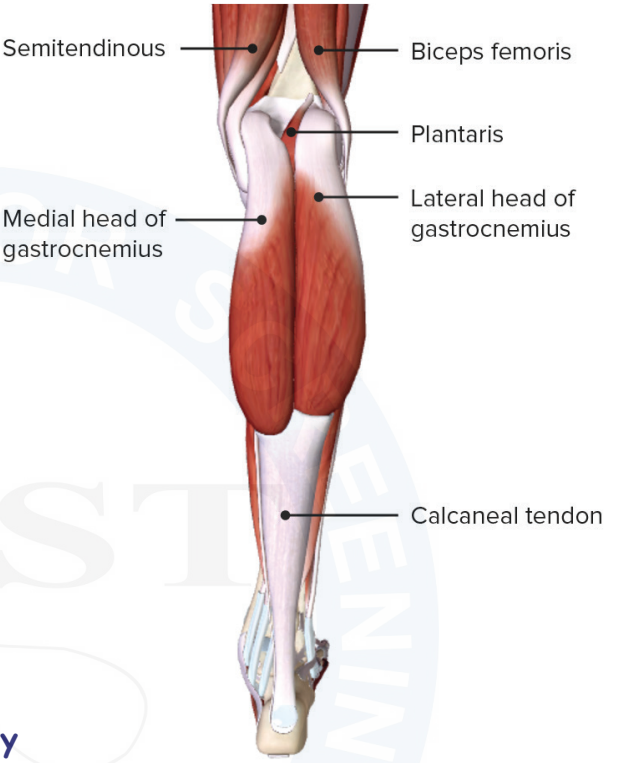
✦ (B) Deep Group →

- Tibialis Posterior
- Flexor Hallucis Longus
- Flexor digitorum longus
- Popliteus

❑ **Action**

Plantar Flexion of Foot

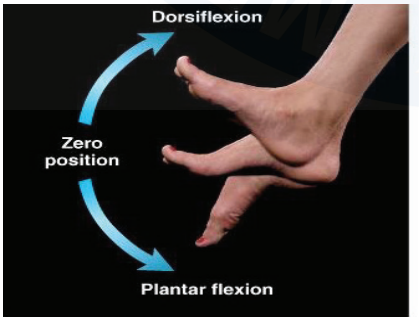
Note- Soleus → Peripheral Heart of Body



◀FIG. Posterior Compartment of LEG

DORSIFLEXION & PLANTAR FLEXION OF FOOT

❑ **Occurs at** .....



Note: Most Stable position of foot is → Dorsiflexion

MUSCLES OF LATERAL COMPARTMENT OF LEG

- **Lateral Compartment of Leg**
  - Nerve of this Comp is Superficial Peroneal Nerve
- **Muscles**
  - 1. Peroneus Longus
  - 2. P. Brevis
- **Common action**
  - Eversion of foot

INVERSION & EVERSION OF FOOT

- **Occurs at →** .....
- **Eversion →** Peroneus longus & Peroneus Brevis
- **Inversion →** Tibialis Anterior & Tibialis Posterior

(b) Inversion

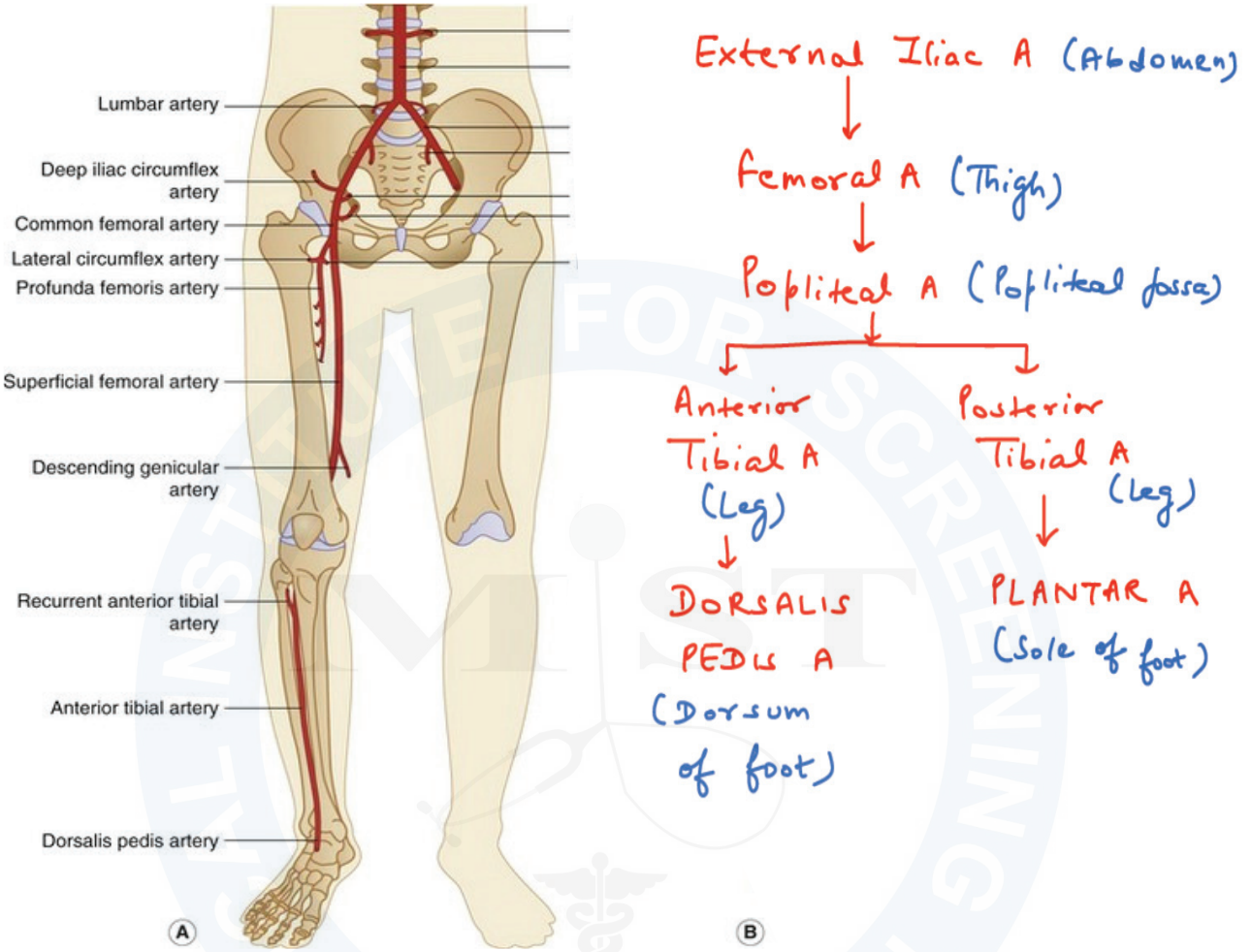
(c) Eversion

FOOT DROP

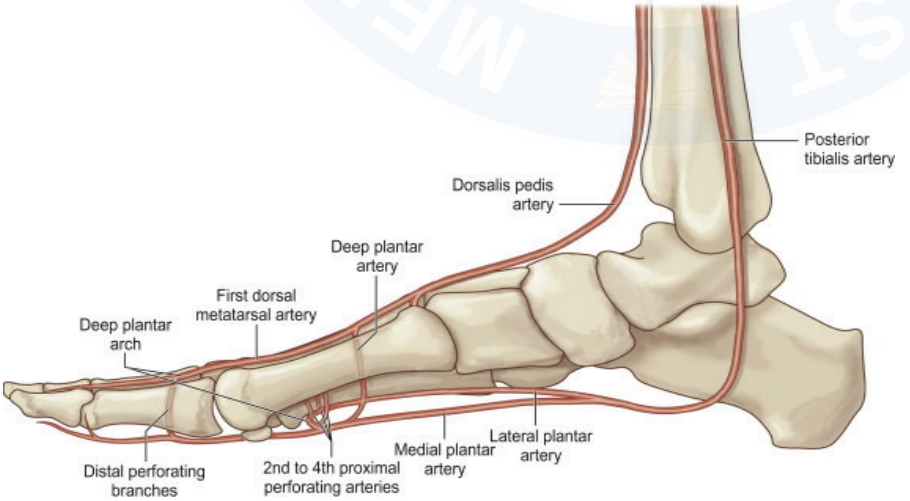
- **Foot Drop** 1. Injury of .....
  - → Loss of Dorsiflexion
  - → Loss of Eversion
  - → Sensory Loss

2. Injury of .....
  - → Loss of Dorsiflexion
  - → Sensory Loss at 1st webspace only

ARTERIES OF LOWER LIMB



◀**FIG.** Arteries of Lower limb (A- Anterior view, B- Posterior view)



◀**FIG.** Arteries of Lower limb (Foot) (Lateral view)

MIST

Dorsalis pedis artery pulse

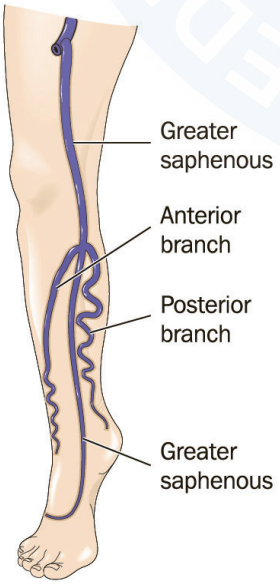
Posterior tibial artery pulse

VENOUS DRAINAGE OF LOWER LIMB

- **Superficial veins** → Great Saphenous Vein & Small Saphenous vein
- **Deep veins** → Femoral V, Popliteal V & Tibial Veins
- **Perforators** → Has Valves
  - → Allow flow of blood in one direction
  - → From superficial V to deep V
- **Varicose veins**
  - ✓ Incompetency of valves of Perforator
  - ✓ Enlarged tortous Veins (superficial veins)
  - ✓ Bluish black in color

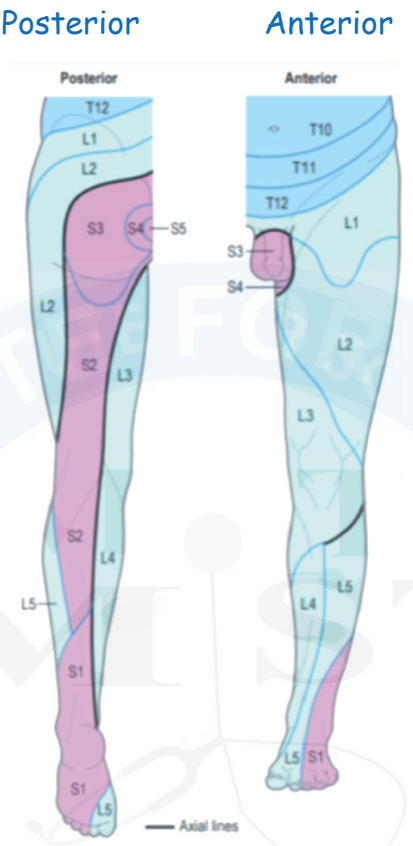
  Note: Great saphenous vein is related to saphenous Nerve

◀**FIG.** Varicose Veius

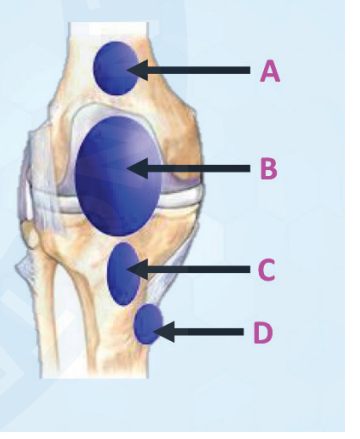


◀**FIG.** Great Saphenous Vein

DERMATOMES OF LOWER LIMB



◀FIG. Dermatormes of Lower limb



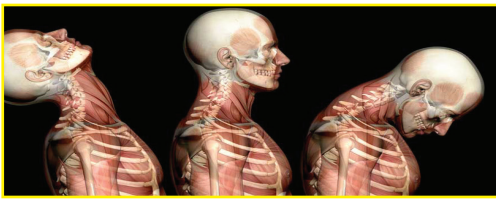
◀FIG. Bursa around knee joint

One Liners

1. **HOUSEMAID'S KNEE** → PREPATELLAR BURSITIS
2. **CLERGYMAN KNEE** → INFRAPATELLAR BURSITIS
3. **TARSAL TUNNEL SYNDROME** → DUE TO COMPRESSION OF TIBIAL NERVE
4. **ADDUCTOR CANAL**
   - MEDIAL BOUNDARY → ADDUCTOR LONGUS
   - LATERAL BOUNDARY → VASTUS MEDIALIS
5. MUSCLE WHICH HAS ATTACHMENT TO CAPSULE OF HIP JOINT → RECTUS FEMORIS
6. CAPSULE OF KNEE JOINT IS SUPPLIED BY GENICULAR BRANCH OF OBTURATOR N.
7. GUY ROPES / PES ANSERINUS → 3 MUSCLES TOGETHER → SARTORIUS, GRACILIS AND SEMITENDINOSUS

Workbook Questions

1. Locking of the knee joint is caused by the contraction of:
   (a) Popliteus (b) Biceps femoris (c) Semimembranosus (d) Quadriceps femoris
   Ans D
2. Structure passing through both greater and lesser sciatic foramen are all except:
   (a) Pudendal nerve (b) Internal pudendal vein
   (c) Nerve to obturator internus (d) Tendon of obturator internus
   Ans D
3. Which is true about origin of Femoral nerve root value?
   (a) Anterior division of L2, L3, L4 ventral rami
   (b) Anterior division of L1, L2, L3 ventral rami
   (c) Posterior division of L2, L3, L4 ventral rami
   (d) Posterior division of L1, L2, L3 ventral rami
   Ans C
4. Skin of First web space of foot is supplied by:
   (a) Common peroneal nerve (b) Superficial peroneal nerve
   (c) Deep peroneal nerve (d) Sural nerve
   Ans C
5. Which of the following muscles is an evertor of foot?
   (a) Peroneus longus (b) Tibialis anterior
   (c) Extensor hallicus longus (d) Extensor digitorum longus
   Ans A
6. Gluteus medius is supplied by:
   (a) Superior gluteal nerve (b) Inferior gluteal nerve
   (c) Nerve to obturator internus (d) Nerve to quadratus femoris
   Ans A
7. In injury to the neck of fibula, which of the following nerve is usually injured?
   (a) Superficial peroneal nerve (b) Deep peroneal nerve
   (c) Common peroneal nerve (d) Tibial nerve
   Ans C
8. Nerve root of pudendal nerve is:
   (a) S¬1S2S3 (b) S¬2S3S4 (c) S¬3S4 (d)S¬2S3
   Ans B
9. Tibialis posterior muscle is supplied by which nerve?
   (a) Posterior tibial nerve (b) Deep peroneal nerve
   (c) Femoral nerve (d) Sural nerve
   Ans A
10. Femoral triangle base is formed by:
    (a) Medial border of Sartorius (b) Inguinal ligament
    (c) Medial border of adductor longus (d) Iliacus
    Ans B



ABDOMEN & PELVIS

ABDOMEN - PLANES & QUADRANTS

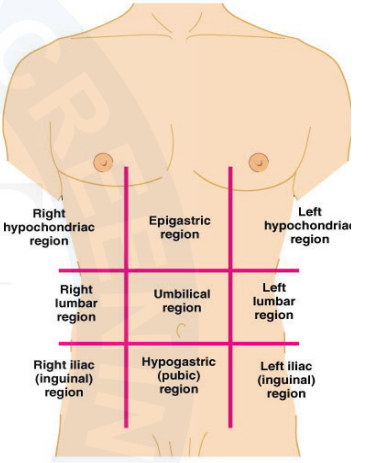
Abdominal Planes & Quadrants

2 Vertical Planes

Midclavicular Line / Plane (from midpoint of clavicle till midinguinal point)

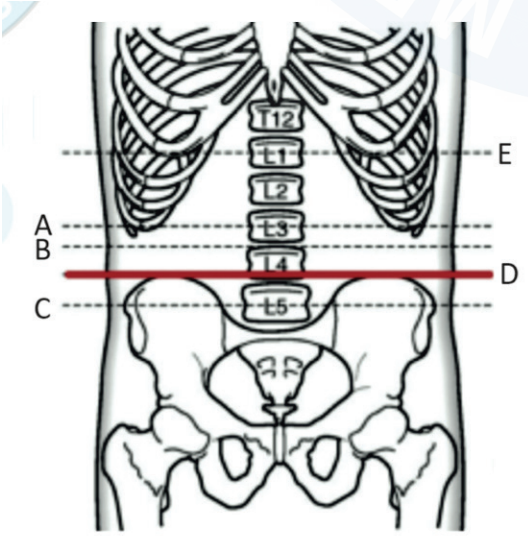
2 Horizontal Planes

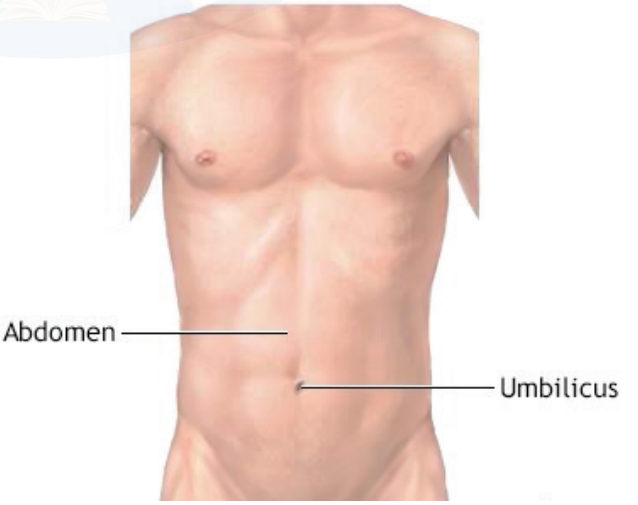
* Subcostal Plane (L_3- upper border)
* Transtubercular Plane (L_5 - upper border)



◀FIG. Abdominal Quadrants

- **Trans-pyloric Plane of Addison** - Lower border of L_1
- **Supracristal Plane** - Lower borderof L4
  (Present at the level of highest point of Iliac crest)

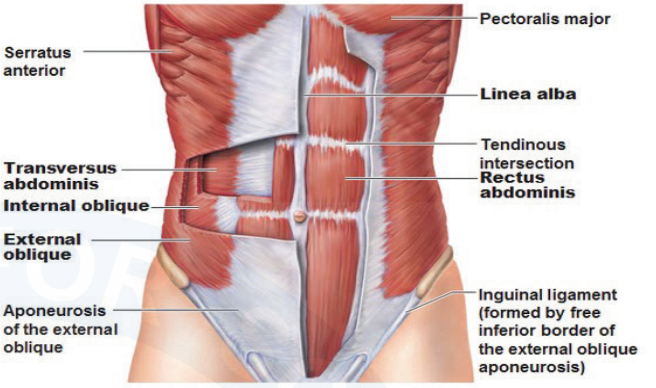




ANTERIOR ABDOMINAL WALL

1. Layers (Outer to inner)

1. Skin
2. Superficial foscia
3. External oblique
4. Internal oblique
5. Transversus abdominis
6. Fascia Transversalis
7. Extraperitoneal fat
8. Peritoneum (Parietal layer)

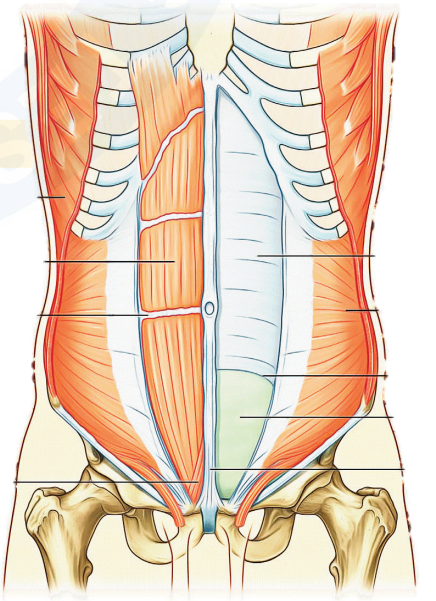


◀FIG. Muscles of Anterior Abdominal Wall

- Neurovascular bundle is present B/W Internal oblique & Transversus abdominis
- **Linea ALBA** → Common Insertion of EO, IO & TA
  → Extends from xiphoid process to pubic symphysis
- **Linea SEMILUNARIS** → Corresponds to Lateral border of Rectus Abdominis
- **INGUINAL Ligament** →

3. Rectus Sheath

- **Rectus Sheath**
  - Formed by aponeurosis of EO, IO, TA
- **Contents of Rectus Sheath**
  1. Rectus abdominis
  2. Pyramidalis - (supplied by subcostal nerve)
  3. Superior & Inferior Epigastric vessels
  4. Lower 6 thoracic nerves

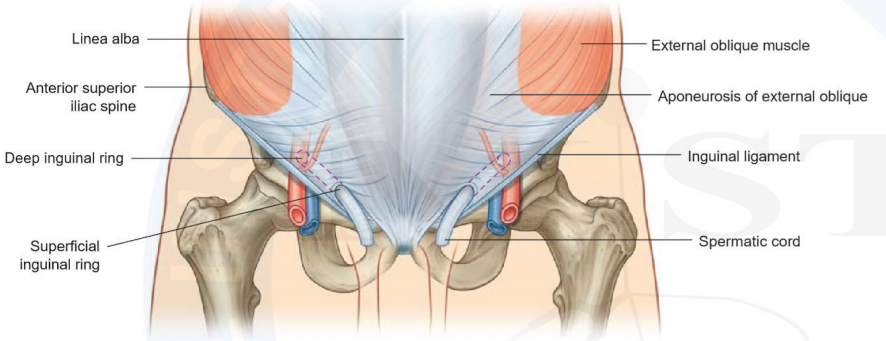


◀FIG. Rectus Sheath

INGUINAL REGION

1. Superficial & Deep Inguinal Ring

- Superficial Inguinal Ring
  - ✓.....
  - ✓Located above pubic tubercle
- Deep Inguinal Ring
  - ✓.....
  - ✓Located 1.25cm above Mid-inguinal point



◀FIG. Inguinal Region

2. Inguinal Canal

❑ **Inguinal Canal Boundaries**

- Anterior Wall →
- Posterior Wall →
- Roof →
- Floor →

❑ **Inguinal Canal** → Length - 4 cm

- Structure passing through it
  1. Supermatic Cord (Males) / Round Ligament of uterus (Females)
  2. Ilioinguinal Nerve

❑ Conjoint Tendon

→ Formed by Aponeurasis of .....

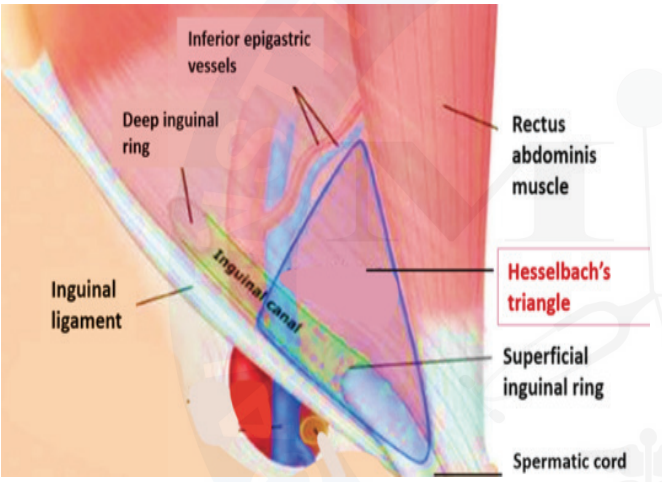
→ Related to ..... Inguinal Ring

→ Gives small contribution to ..... wall of Inguinal canal

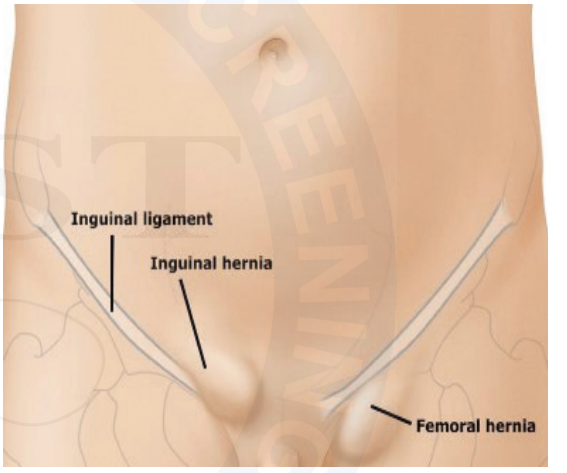
3. Hesselbach's Triangle

✦ Boundaries

1. Medial Boundary →
2. Lateral Boundary →
3. Inferior Boundary →



◀FIG. Hesselbach's Triangle



◀FIG. Inguinal Hernia & Femoral Hernia

4. Inguinal Hernia & Femoral Hernia

Inguinal Hernia	Femoral Hernia
✓ Swelling is above & Medial Pubic Tubercle	✓ swelling is below & lateral to pubic Tubercle
✓ Intestinal loop pass through Inguinal ring / canal	✓ pass through femoral ring and then femoral Canal

❑ **CLINICAL INTEGRATION**

A PATIENT HAS PRESENTED WITH HERNIATION OF INTESTINAL LOOP WHICH PASSES ALONG THE SPERMATIC CORD & REACHES TILL SCROTUM ?

A. FEMORAL HERNIA
B. DIRECT INGUINAL HERNIA
C. INDIRECT INGUINAL HERNIA
D. PARAUMBILICAL HERNIA

PERITONEAL CAVITY

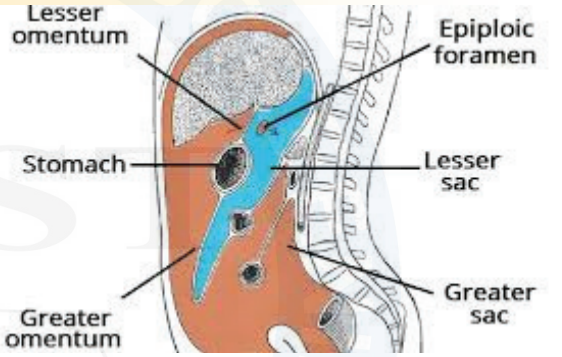
Sacs & Epiploic foramen

❑ **Peritoneal Cavity**

- Lesser sac / Omental Bursa
- Greater sac

❑ **Epiploic Foramen**

- Communication b/w greater sac & lesser sac.



◀FIG. Peritoneal Cavity (sacs)

ARTERIES OF ABDOMEN

1. Abdominal Aorta

❑ **Abdominal Aorta**

↓

Terminates as Common Iliac A (R & L)

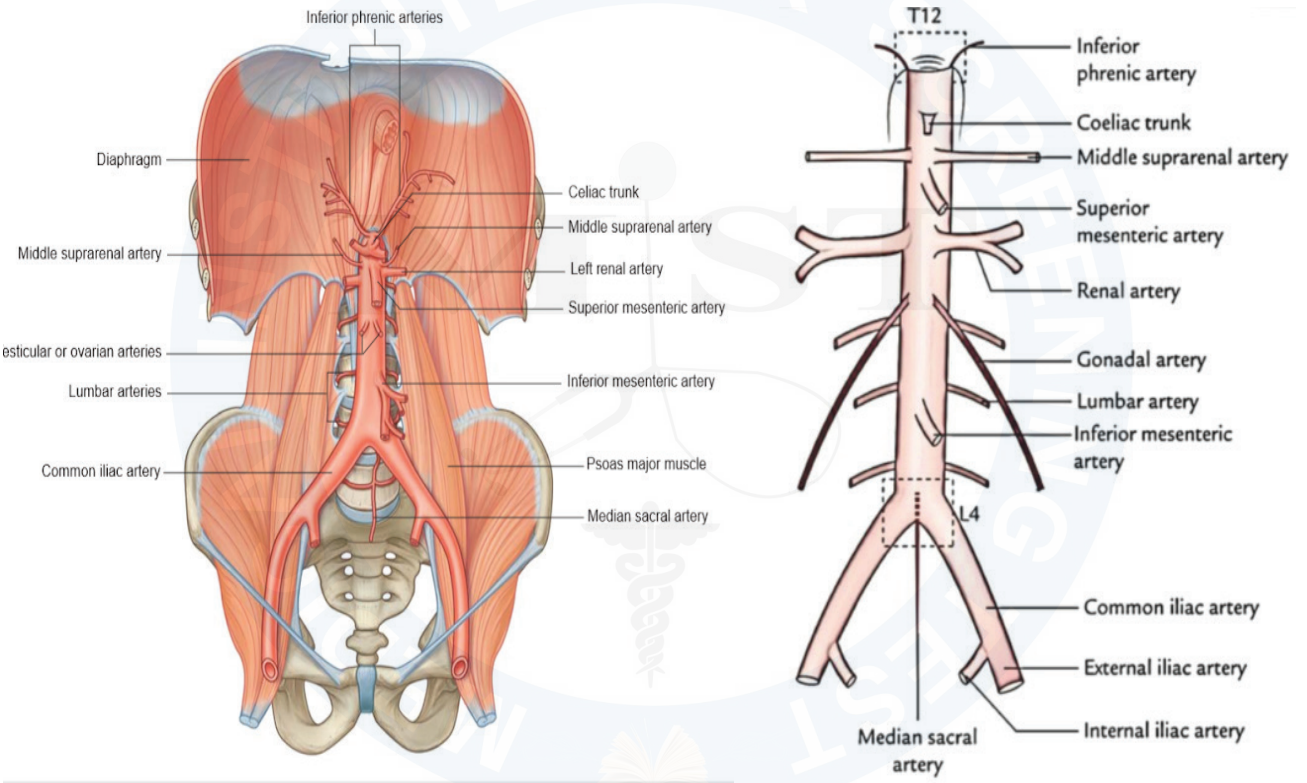
External Iliac A Internal Iliac A

Note: Inferior Epigastric A is branch of External Iliac A

❑ Branches of Abd Aorta

✦ Anterior Branches

1. Celiac Trunk → .....
2. Superior Mesenteric A → .....
3. Inferior Mesenteric A → .....



◀**FIG.** Abdominal Aorta

✦ Lateral Branches

4. R & L Renal A
5. R & L Middle Suprarenal A
6. R & L Inferior Phrenic A
7. R & L Gonadal A (Testicular A/ovarian A)

- Posterior Branches
  8. Median Sacral A
  9. Lumbar A
- Terminal Branches
  10. R & L common Iliac A

2. Celiac Trunk

Celiac Trunk

- 3 branches
  1. Left Gastric A
  2. Splenic A
  3. Common Hepatic A

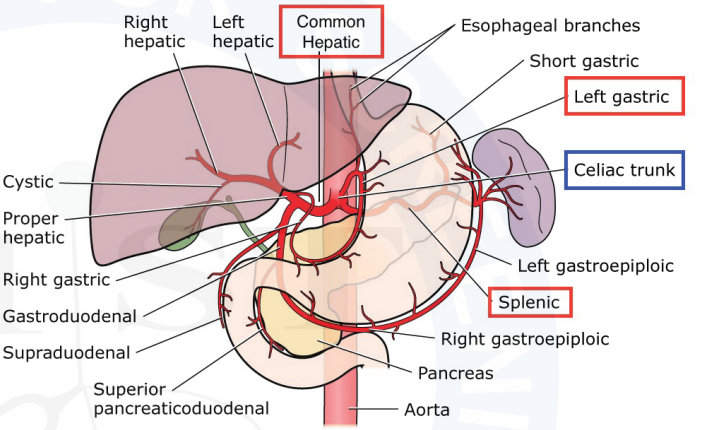


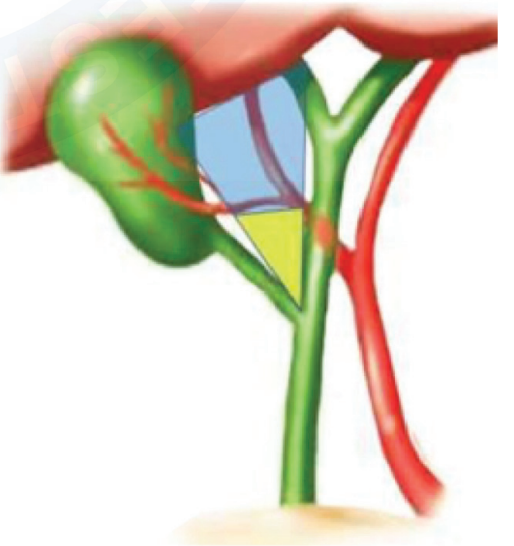
FIG. Celiac Trunk

Splenic A

1. Short Gastric A (Vasa Brevia)
2. Left Gastroepiploic A
3. Pancreatic A (Arteria Pancreatica Magna)
4. Hilar Branches

Common Hepatic A.

1. Right Gastric A
2. Gastro Duodenal A
   - Right Gastroepiploic A
   - Superior Pancreatico duodenal A
3. Hepatic Artery Proper



Note: Cystic A is a branch of Right Hepatic A

3. Superior Mesenteric Artery

❑ Superior Mesenteric A - Branches

1. Inferior Pancreatico duodenal A
2. Jejunal A
3. Ileal A
4. Ileocolic A → Gives a branch Appendicular A
5. Right colic A
6. Middle Colic A

4. Inferior Mesenteric Artery

❑ Inferior Mesenteric A - Branches

1. Left Colic A
2. Sigmoidal A.
3. Superior Rectal A

5. Watershed Area

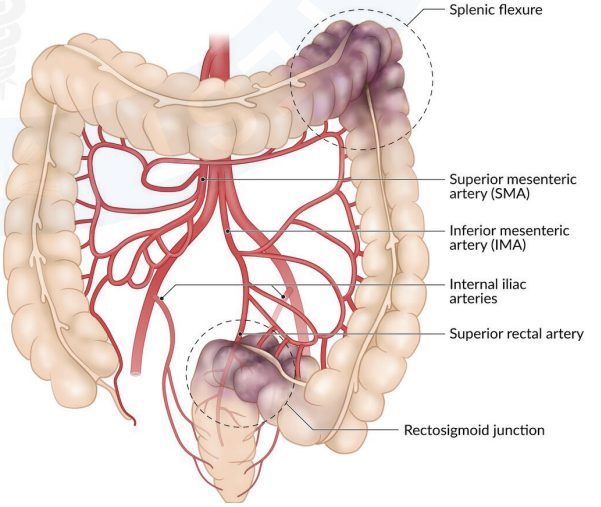
❑ Watershed Area

✦ 1. Griffith's Point

- .....
- .....

✦ 2. Sudeck's Point

- .....
- .....
- .....
- .....



◀**FIG.** Watershed Area

6. Internal Iliac Artery

❑ Internal Iliac A - Branches

Anterior Division

- ✓ Superior & Inferior Vesical A
- ✓ Middle Rectal A
- ✓ Uterine A
- ✓ Vaginal A
- ✓ Internal Pudendal A.
- ✓ Obturator A
- ✓ Inferior Gluteal A

Posterior Division

- ✓ Superior Gluteal A
- ✓ Lateral sacral A
- ✓ Iliolumbar A

PORTAL VEIN & PORTOCAVAL ANASTOMOSIS

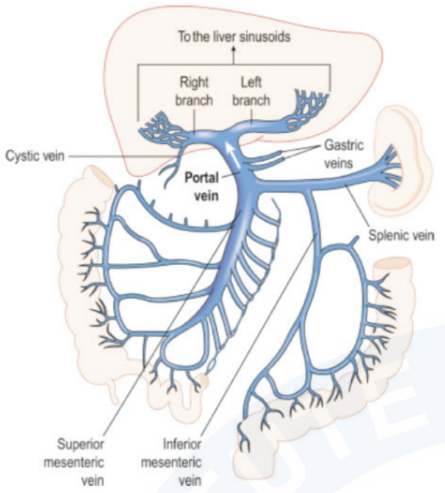
❑ Portal vein

- Formation → By union of splenic vein & superior mesenteric vein

❑ Portocaval anastomosis

(Between tributaries of portal vein & systemic vena cava system)

1. At Lower End of Esophagus
   - ✦ If Enlargement of this anastomosis- .....
2. At Umbilicus
   - ✦ ..... (If Enlargement of this anastomosis)
3. At Lower part of rectum & anal canal
   - ✦ ..... (If Enlargement of this anastomosis)

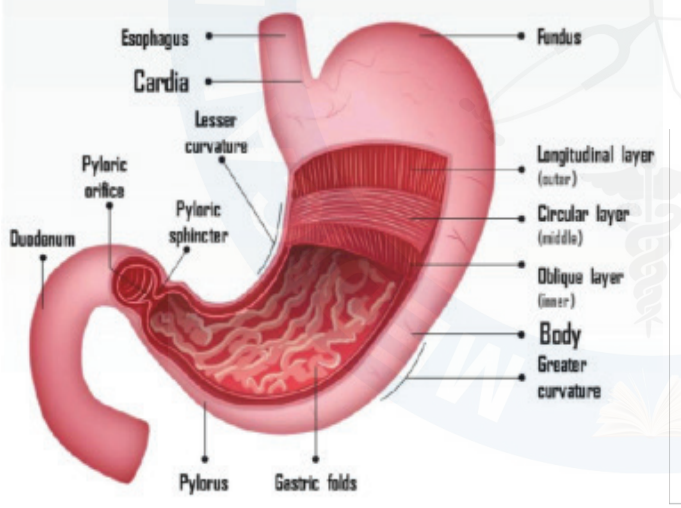


◀FIG. Portal vein

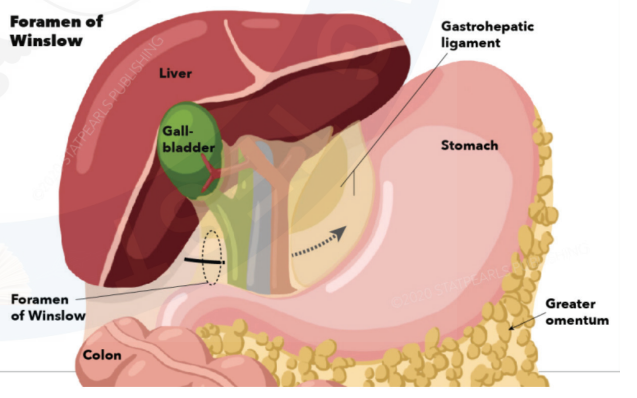
◀FIG. Caput Medusae

NOTE: IVC FORMATION ⟶ .....

STOMACH



◀FIG. Stomach



❑ Visceral - Relations

✦ Stomach Bed

- Structures related to posterior surface of stomach & they are separated from stomach by lesser sac

1. Diaphragm
2. Splenic Vessels
3. Suprarenal Gland (Left)
4. Splenic Flexure of colon
5. Kidney (Left)
6. Pancreas
7. Mesocolon (Transverse)

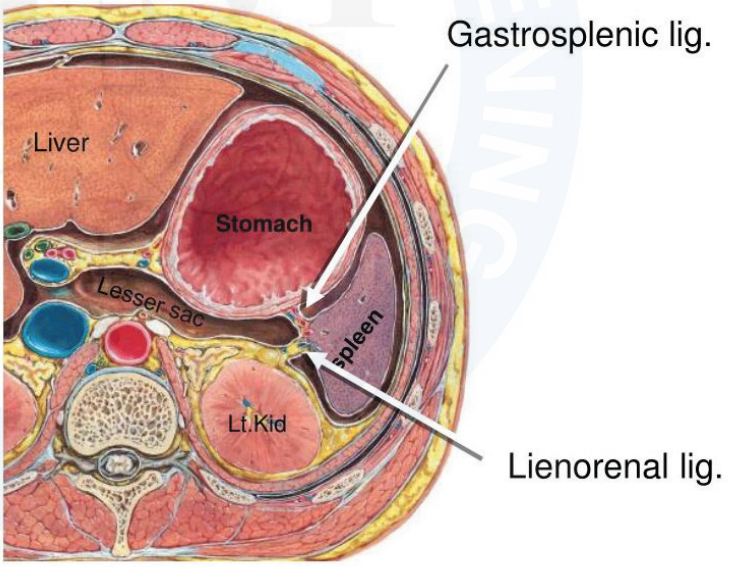
- .....

SPLEEN

❑ Relations

Inner surface (visceral)

- Stomach
- Left Kidney
- Splenic flexure of colon
- Pancreas (Tail)

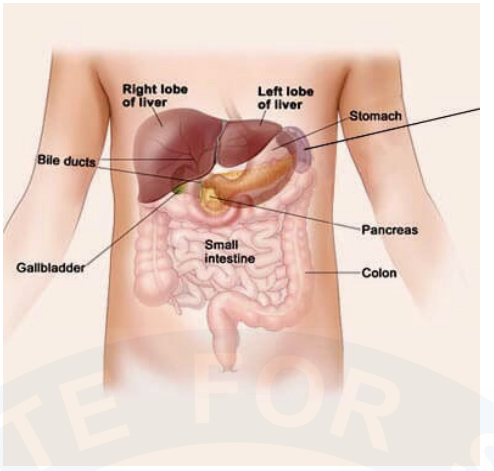


◀FIG. Portal vein

❑ Peritoneal ligaments

1. Gastrosplenic lig (Extends from stomach to spleen)
2. Splenorenal lig / Lienorenal lig (Extends from spleen to Left kidney) → Contents
   - A. .....
   - B. .....

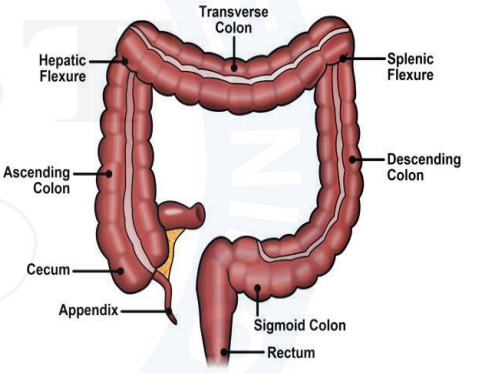
✦ Spleen .....



◀FIG Spleen

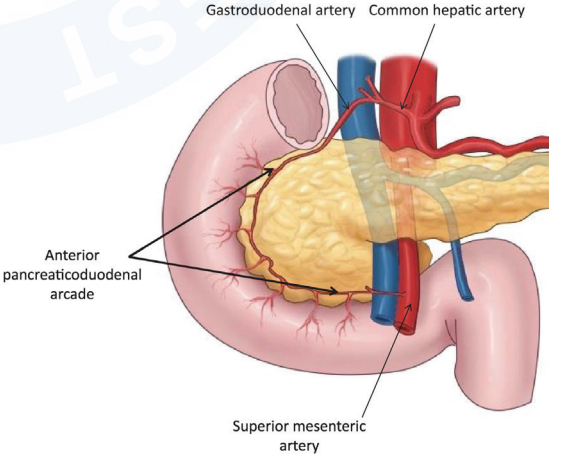
LARGE INTESTINE

- Longest Part → Transverse colon
- Appendix → Most common position
  → Retrocaecal
- Length – Small Intestine → 6 m
  - Large Intestine → 1.5 m
  - Rectum → 10 – 12 cm
  - Anal canal – 4 cm



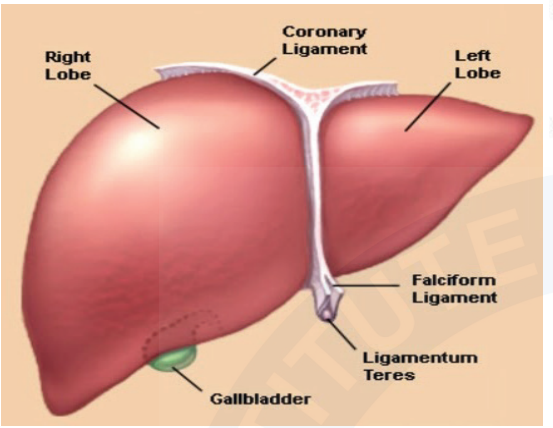
SMALL INTESTINE

- **Duodenum – 25 cm length**
  - 'C' shaped
  - Parts - 4 →(3rd Part is longest)
  - Extent - L1 L2 L3 vertebral levels
  - Brunner's Gland
- **Ileum → Peyer's Patch**
- **Liver → Space of Disse**

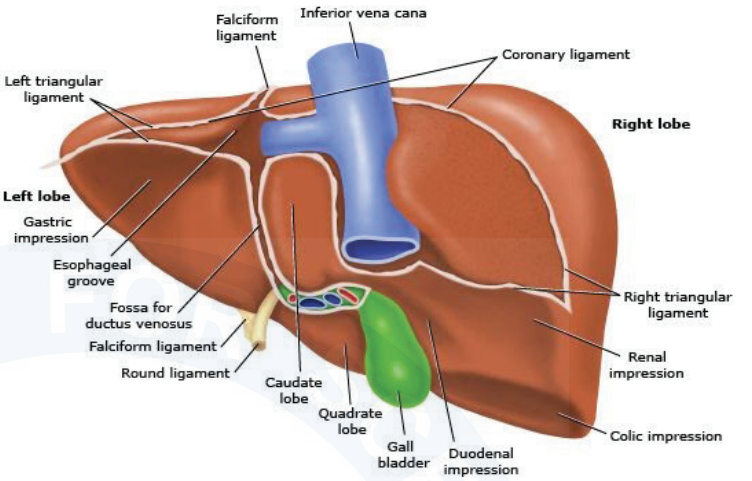


◀FIG. Duodenum

LIVER



◀FIG. Liver (Anterior view)

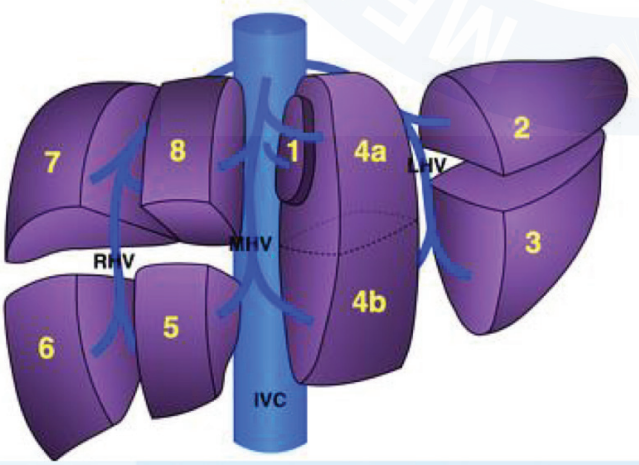


◀FIG. Liver (Posterior view)

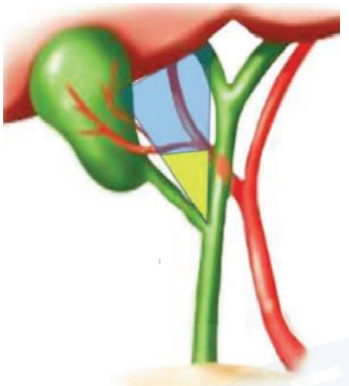
❑ Couinaud's Classification

On Basis of branching pattern of .....

- ✓ .....
- ✓ .....
- ✓ .....
- ✓ .....



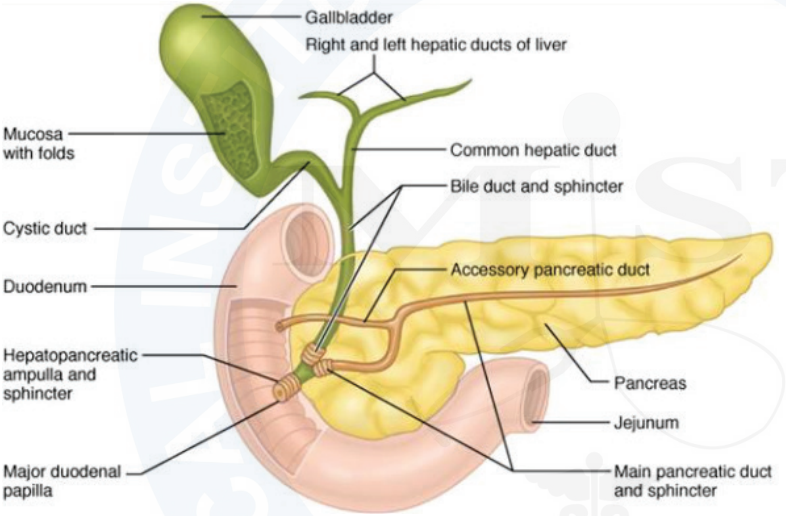
◀FIG Liver - Couinaud's Classification



- Calot triangle- boundaries-

1. .....
2. .....
3. .....

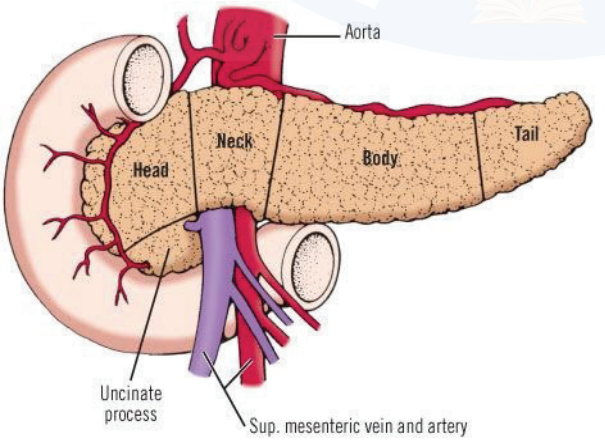
EXTRA BILIARY APPARATUS



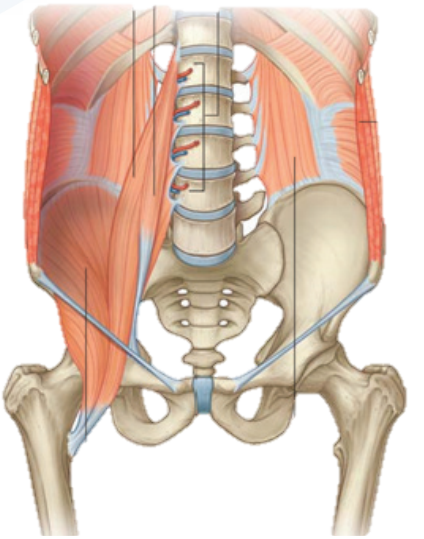
◀**FIG.** Extrabiliary apparatus

- Common Bile duct & Pancreatic duct open at Major Duodenal Papilla in 2nd part of Duodenum

PANCREAS



◀**FIG** Pancreas



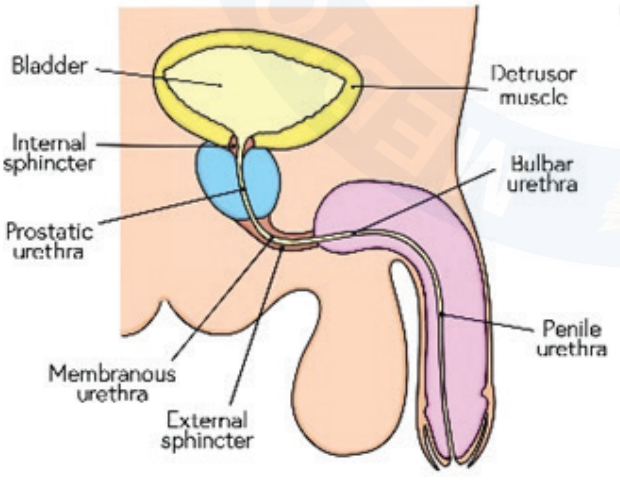
KIDNEY

- Extent - .....
- Malphigian Corpuscle
- Duct of Bellini
- Posterior relation → Psoas Major, Quadratus Lumborum & Transversus obdominis (Medial to Lateral)

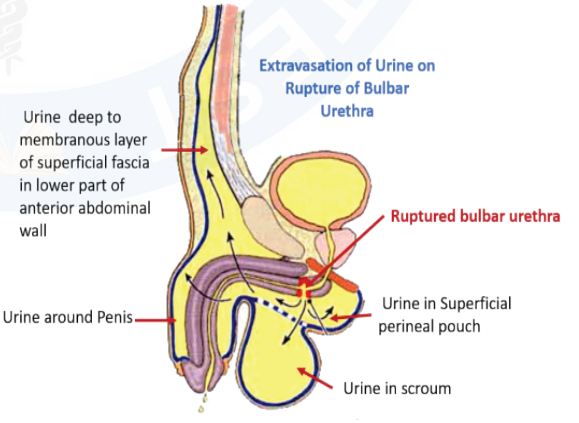
MALE URETHRA & URETHRAL INJURIES

- Length- 20 cm
- Parts -

1. Prostatic part (passes through the prostate) - widest
2. Membranous part - narrowest
3. Spongy or penile part - Bulbar urethra is the initial portion passing through bulb of penis



◀FIG. Male Urethra



◀FIG. Bulbar Urethra Rupture

❑ Urethral Injuries/ Rupture of the urethra-

✦ Bulbar urethra rupture - Urine extravasates (superficial extravasation) into the

- .....
- .....
- .....
- .....

✦ Membranous urethra rupture - Urine extravasates (deep extravasation) into the

- Deep Perineal pouch
- Around Prostate & Around urinary bladder

❑ CLINICAL INTEGRATION

A PERSON HAD A ROAD TRAFFIC ACCIDENT & REPORTED IN EMERGENCY. HE SUFFERED FROM INJURY TO BULBAR URETHRA. WHAT CAN BE THE LOCATION OF COLLECTION OF THE URINE DUE TO THIS INJURY?

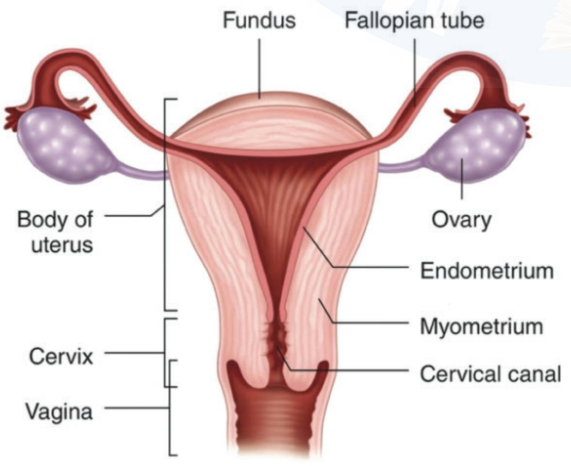
A. DEEP PERINEAL POUCH

B. PERITONEAL CAVITY

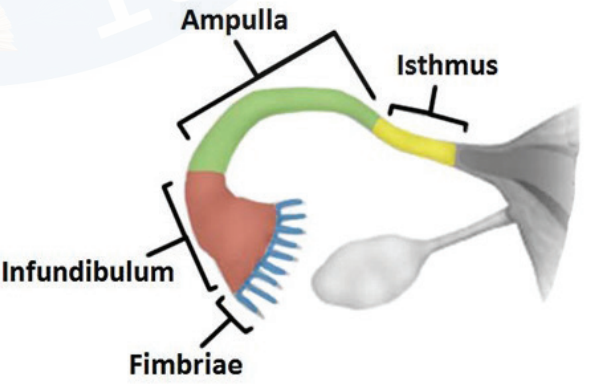
C. ANTERIOR ABDOMINAL WALL

D. ISCHIORECTAL FOSSA

FEMALE REPRODUCTIVE SYSTEM



◀FIG. Female Reproductive System

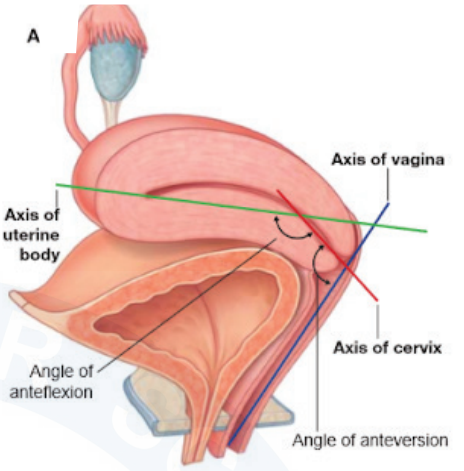


◀FIG. Fallopian Tube

1. Uterus

❑ Uterine axis

1. Angle of Anteflexion
   - ✓ .....
   - ✓ B/W axis of body & axis of cervix
2. Angle of Anteversion
   - ✓ .....
   - ✓ B/W axis of cervix & axis of vagina



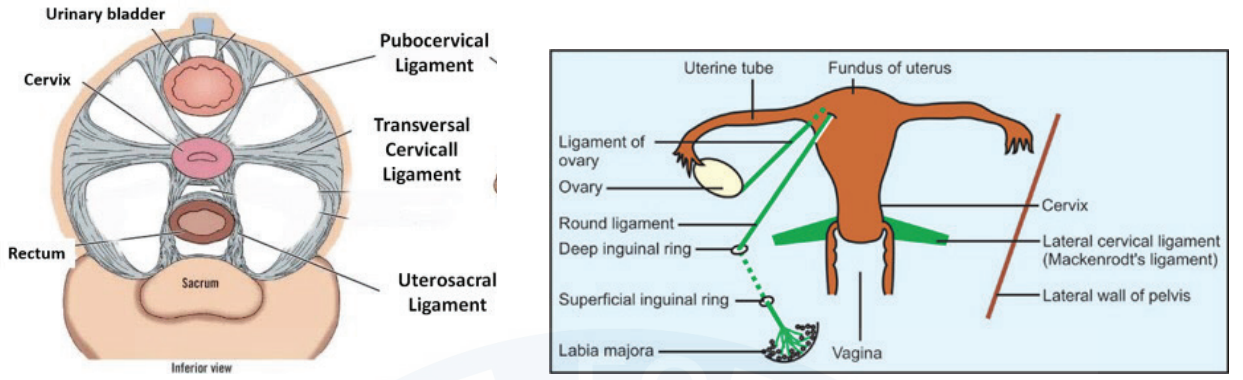
◀FIG. Angle of Anteflexion & Anteversion

❑ Arterial supply

Primary Supports of Uterus

- ✦ (A) Muscular supports -
  - ✓Levator ANI
  - ✓Perineal Body
- ✦ (B) Ligaments
  1. Mackenrodt's / Cardinal /Transverse Cervical Lig
  2. Uterosacral Lig
  3. Pubocervical Lig
  4. Round Lig of uterus
- ✦ (C) Uterine Angles

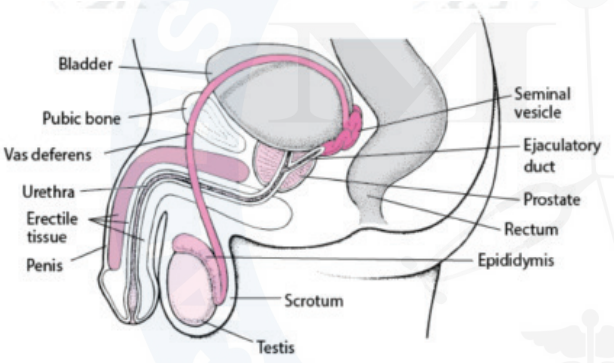
☞ [Note - Broad ligament does not provide support to uterus]



◀**FIG.** Uterus - Ligaments

MALE REPRODUCTIVE SYSTEM

1. Overview

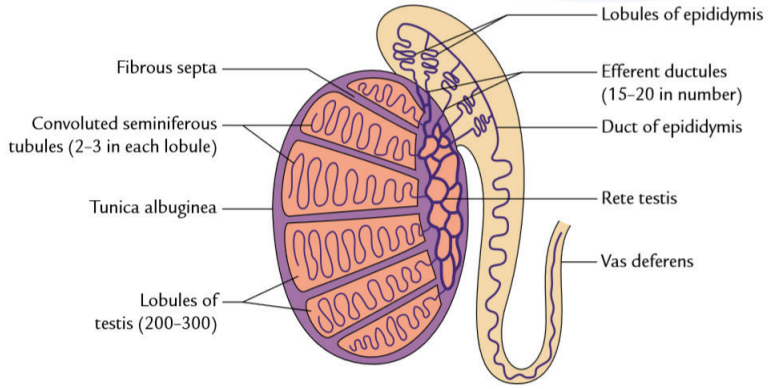


◀**FIG.** Male Reproductive System

2. Testis

❑ Testis

✦ Flow of Sperm



◀ **FIG.** Testis

Seminiferous Tubule
↓
Straight Tubule
↓
Rete Testis
↓
Efferent Ductules
↓
Epididymis

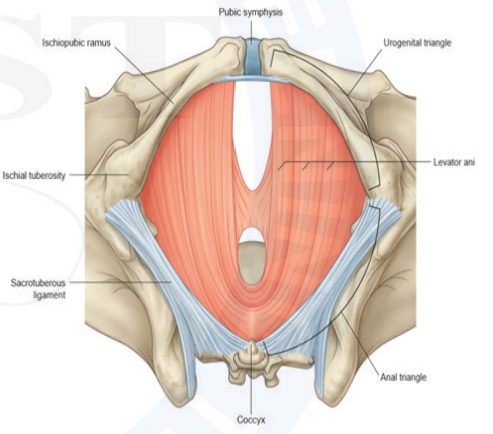
MIST

- **Descent of Testis-**
  1. 3rd month IUL → Iliac fossa
  2. 4th - 6th month IUL → Deep Inguinal Ring
  3. 7th month IUL → Inguinal Canal
  4. 8th month IUL → Superficial Ing. Ring
  5. 9th month IUl → Scrotum
- **Undescended Testis** → Arrest in descent
- **Ectopic Testis** → Abnormal location

PERINEUM

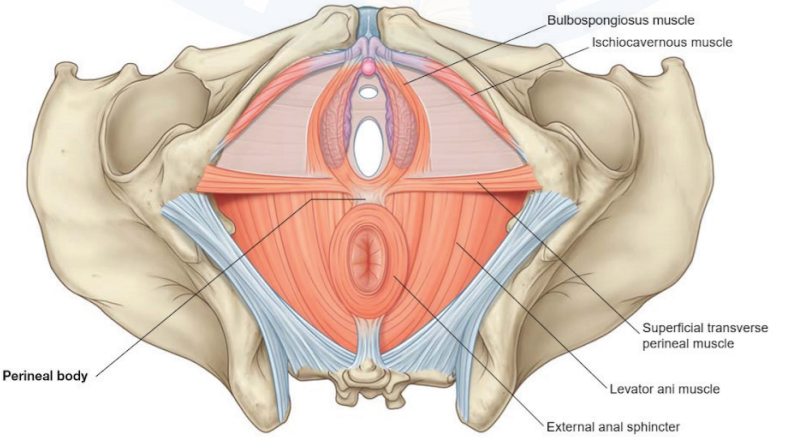
1. Pelvic Diaphragm

- **Pelvic Diaphragm**
  1. Levator Ani
  2. Ischiococcygeus / coccygeus
- **Levator Ani**
  - ✦ Components
    - ✓ Iliococcygeus
    - ✓ Pubococcygeus
    - ✓ Puborectalis



◀FIG. Levator ani

2. Perineal Body



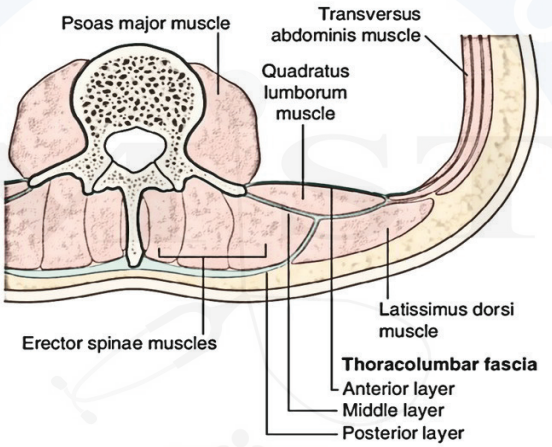
◀FIG. Perineal Body & Muscles of Perineum

❑ **Muscles attached to it (10)**

1. Bulbospongiosous (R & L)
2. Superficial Transverse Perineii (R & L)
3. Deep Transverse Perincii (R & L)
4. Levator Ani (R & L)
5. External anal sphincter
6. Longitudinal muscle of anal canal

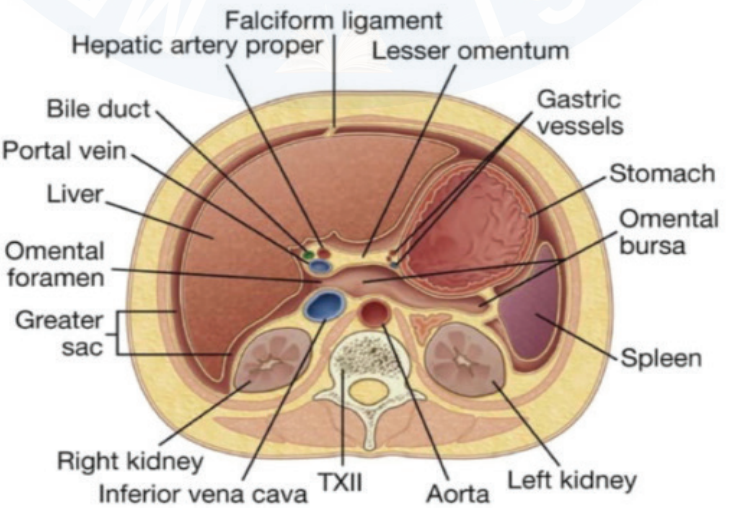
Note: Ischiocavernosus is not attached to perineal Body

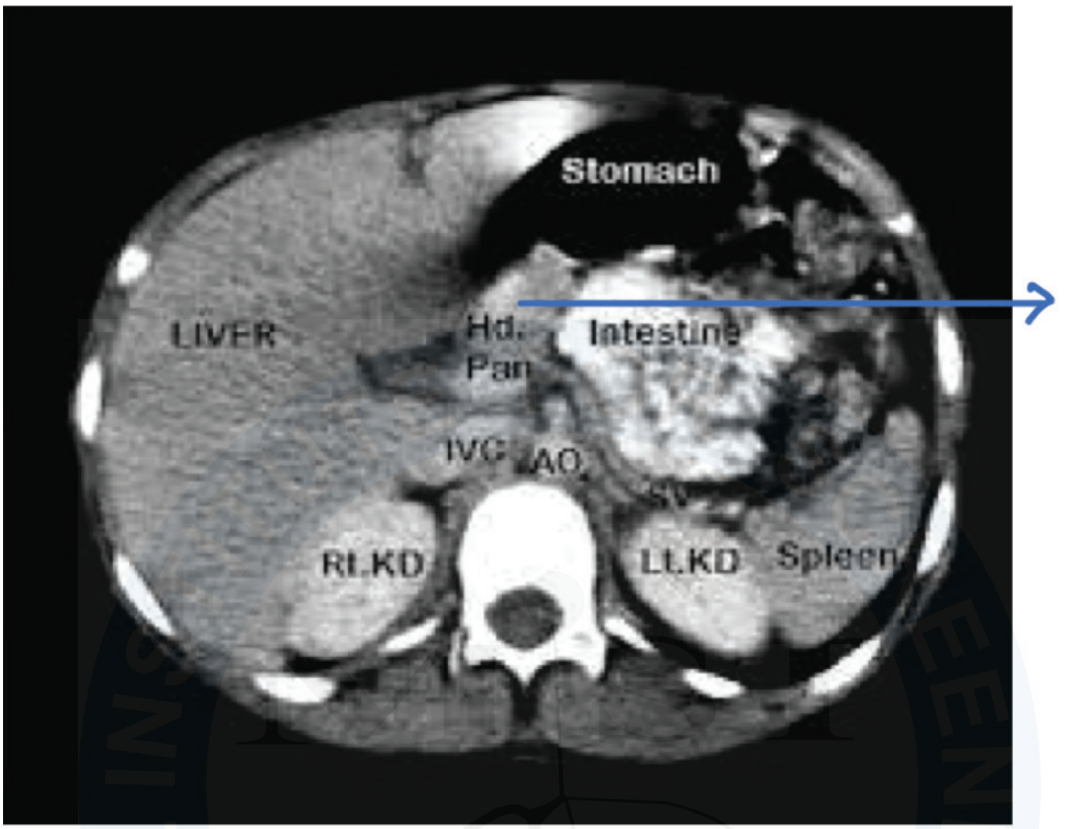
THORACOLUMBAR FASCIA



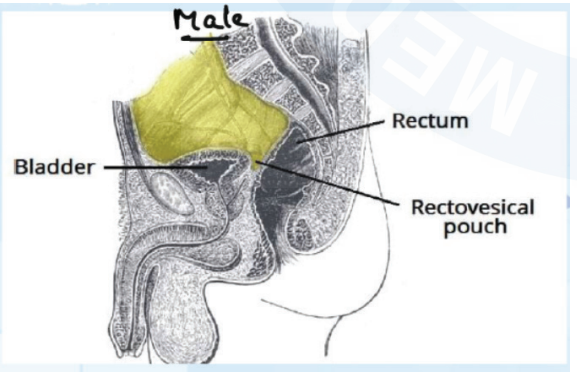
◀**FIG.** Thoracolumbar Fascia

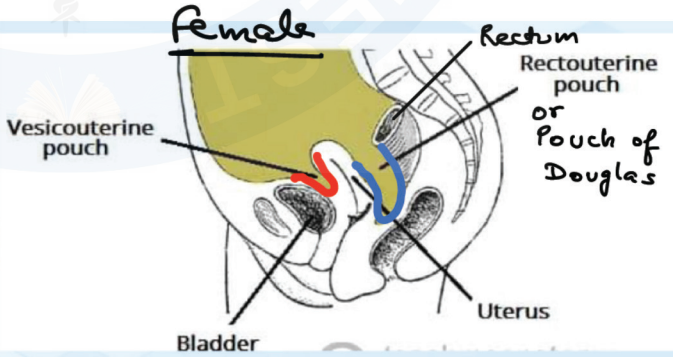
TRANSVERSE SECTION OF ABDOMEN





❑ Sagittal Section of Pelvis





Workbook Questions

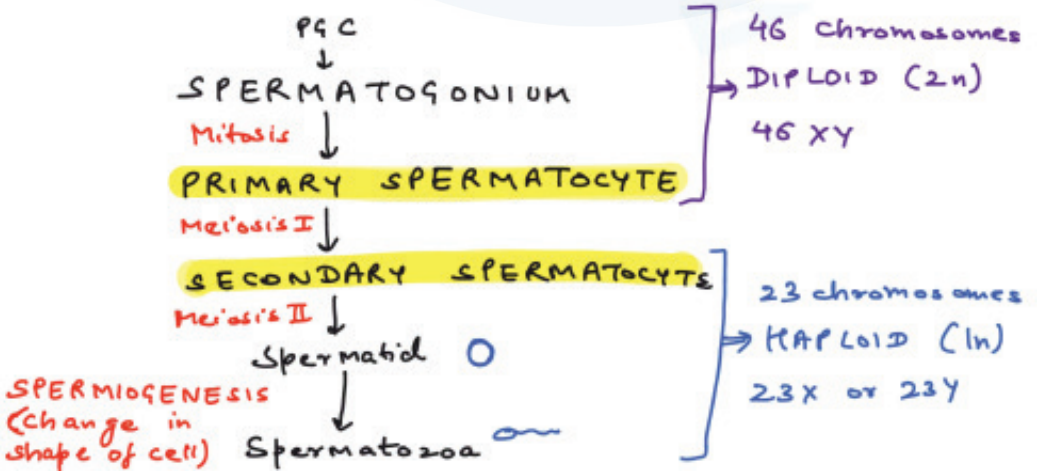
1. Which of the following is not a branch of internal iliac artery?
   (A) Superior rectal artery (B) Middle rectal artery
   (C) Superior gluteal artery (D) Inferior Vesical artery
   Ans A
2. Left gastroepiploic artery is a branch of
   (a) Gastroduodenal artery (b) Common hepatic artery
   (c) Left Gastric artery (d) Splenic artery
   Ans D
3. The Couinaud's segmental nomenclature is based on the position of the :
   (a) Hepatic veins and portal vein (b) Hepatic veins and biliary ducts
   (c) Portal vein and biliary ducts (d) Portal vein and hepatic artery
   Ans A
4. Supports of the uterus are all except
   (a) Uterosacral ligament (b) Broad Ligament
   (c) Mackenrodts' ligament (d) Levator ani
   Ans B
5. Deep inguinal ring is a defect in :
   (a) External oblique muscle aponeurosis
   (b) Internal oblique muscle aponeurosis
   (c) Transversus abdominis muscle aponeurosis
   (d) Transversalis fascia
   Ans D
6. Structures forming perineal body are all EXCEPT:
   (a) Levator ani muscle (b) Bulbospongiosus
   (c) Deep transverse perinei (d) Ischiococcygeus
   Ans D
7. Cremasteric muscle nerve supply:
   (a) Ilioinguinal nerve (b) Iliohypogastric nerve
   (c) Genital branch of genitofemoral nerve (d) Femoral nerve
   Ans C
8. Posterior superior iliac spine is at which level?
   (a) S1 (b) S2 (c) L1 (d) L2
   Ans B
9. Epiploic foramen is bounded by A/E
   (a) Caudate Lobe (b) Duodenum (c) Lesser Omentum (d) Greater Omentum
   Ans D
10. All structures are present within splenorenal Ligament A/E
    A. Tail of Pancreas B. Head of pancreas C. Splenic Artery D. Splenic Vein
    Ans B

PHASES OF DEVELOPMENT

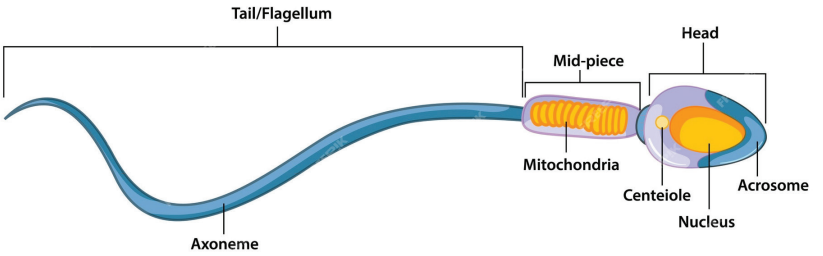
- **3 Phases of Development → (From Day of fertilization)**
  1. Germinal (ovular or pre embryonic) - Initial 2 wks IUL
  2. Embryonic Phase - 3rd to 8th wk IUL
  3. Fetal Phase - From 9th wk till birth
- PGC - Primordial Germ Cell → formed by EPIBLAST / (Yolk Sac)

SPERMATOGENESIS

- **Spermatogenesis**
  - Location → (Formation) Seminiferous Tubule (Testis)
  - Duration → 74 days
  - Sperms are stored in Epididymis
  - Sperms gain motility in Epididymis
- **Steps**



MIST

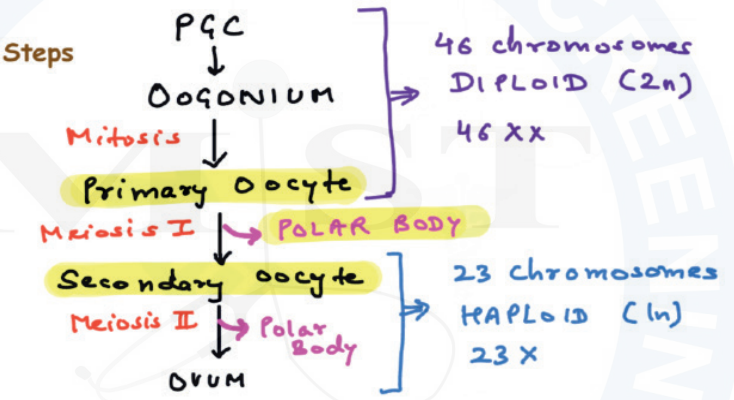


◀**FIG.** Spermatozoa

OOGENESIS

Location → ovary & fallopian Tube

❑ Steps



Primary oocyte (formed before birth)

↓

Primary oocyte is arrested at Prophase I

↓

This Arrest is released .....

↓

Secondary oocyte formed

↓

Secondary oocyte is arrested at Metaphase II

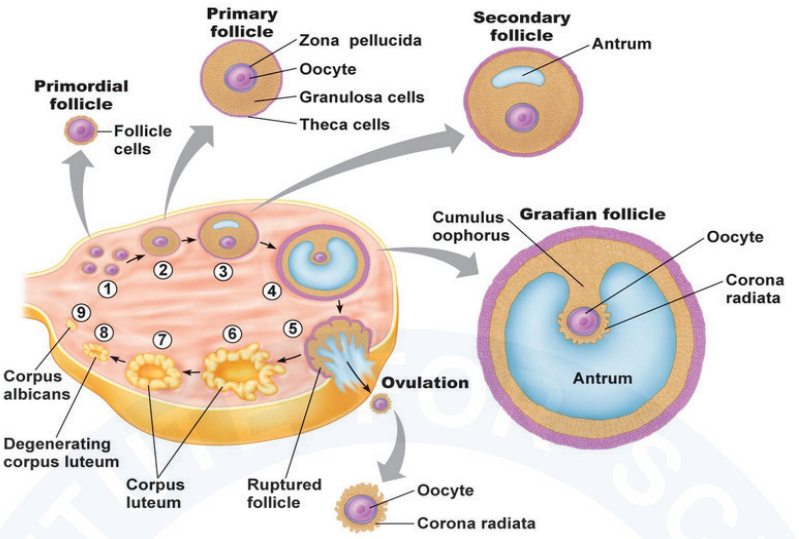
↓

This arrest is released → If fertilization does not occur Sec. oocyte degenerates

.....

↓

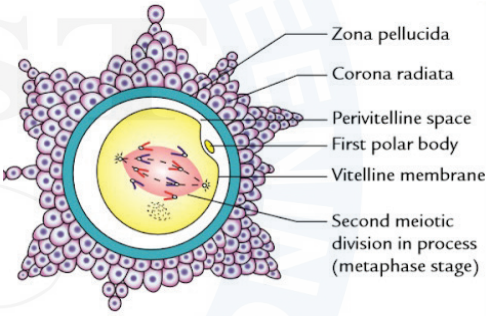
Ovum



◀FIG. oogenesis & Maturation of follicles

❑ **Primary oocyte (ovary)**

1. Primordial follicle
2. Primary follicle
3. Secondary follicle
4. Graffian (mature) follicle - This undergoes ovulation



◀FIG. Secondary oocyte

Note: Corpus Luteum → Yellow is color & located in ovary

- Corpus Albicans → whitish fibrous scar like tissue & located in ovary

FERTILIZATION

❑ **Location**

✓ Ampulla of Fallopian Tube

❑ **Capacitation**

✓ Location - Female Genital Tract (Fallopian Tube)

✓ Duration - 7 hrs

✓ Only the capacitated sperms can pass through corona radiata

❑ **Acrosome reaction**

✓ Acrosomal cap releases lysozymes to pass through zona pellucida

MIST

❑ Cortical Reaction

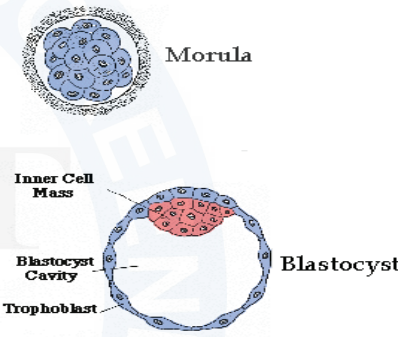
- ✓ As the head of sperm comes in contact with the oocyte surface, it results in release of Lysosomal enzymes from cortical Granules
- ✓ Due to this zona pellucida becomes impermeable to other sperms (Zona Reaction)
- ✓ This prevents Polyspermy

DEVELOPMENTAL PERIOD - WEEK 1

Morula, Blastocyst & Implantation

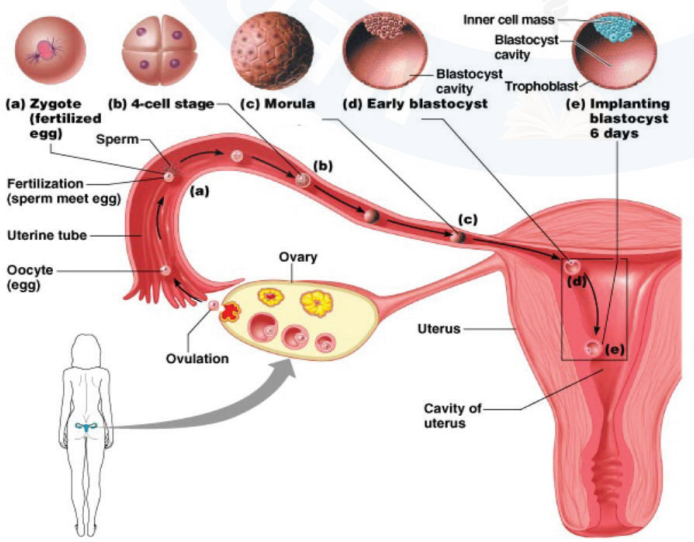
❑ Ist week of development

- Morula - 16 cell stage, formed on _____
- Morula enters Uterine cavity by _____
- Blastocyst → formed at $4\frac{1}{2}$ day IUL
  - ↳ It undergo implantation by _____



❑ Blastocyst → Has Blastocyst cavity

- outer cell mass or Trophoblast
- Inner cell mass → also called embryoblast



◀FIG. Week 1 of Development

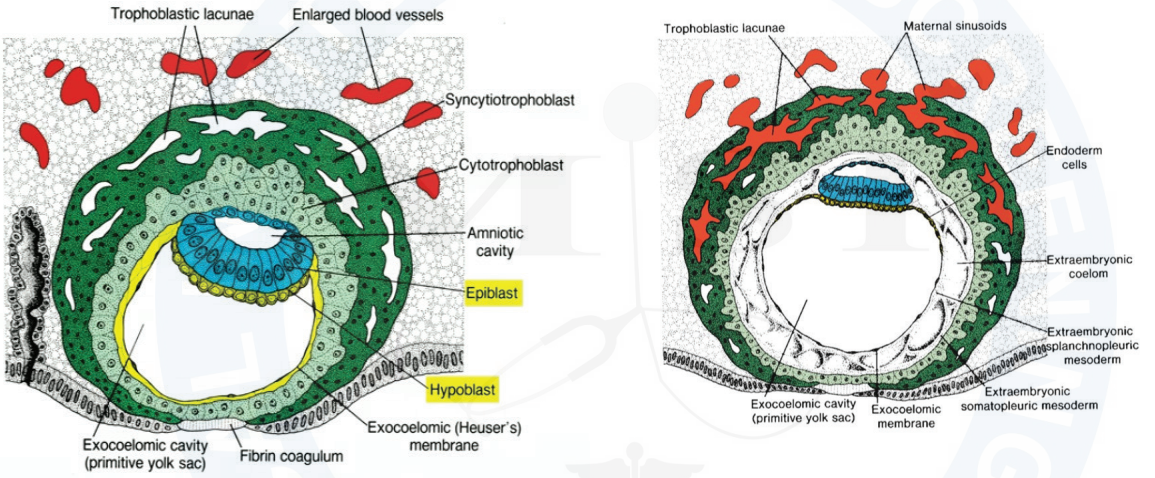
❑ Implantation

- Interstitial Implantation
- Location → Posterior wall of uterine cavity (Near fundus)

DEVELOPMENTAL PERIOD - WEEK 2

❑ 2nd week of development

1. Inner cell mass → Epiblast & Hypoblast
2. Trophoblast → Cyto - trophoblast
   → Syncytio - trophoblast
3. 2 cavities → Amniotic cavity
   → Yolk sac
4. Extraembryonic mesoderm
5. Primary chorionic villi



◀FIG. Week 2 of Development

DEVELOPMENTAL PERIOD WEEK 3

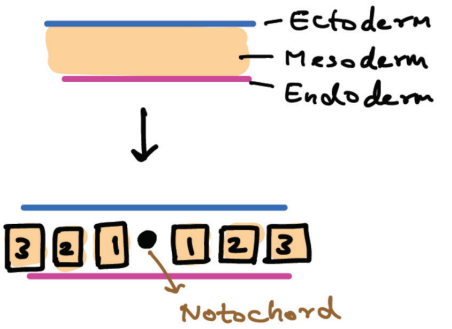
❑ 3rd week of development

1. Prochordal Plate → derived form Hypoblast
2. Primitive streak → derived form Epiblast
3. Gastrulation → Formation of 3 germ layers
   → Derived from Epiblast
   → First layer to be formed is Endoderm
   → 1. Endoderm, 2. Mesoderm 3. Ectoderm
4. Notochord → Derived from Epiblast
   → Helps in formation of Neural Tube
   → Adult Remnant → _____

◀FIG. SIRENOMELIA (Defect in Gastrulation)

MIST

5. Secondary & Tertiary Chorionic villi
6. Formation of Neural tube begins.
7. Intraembryonic mesoderm - divisions



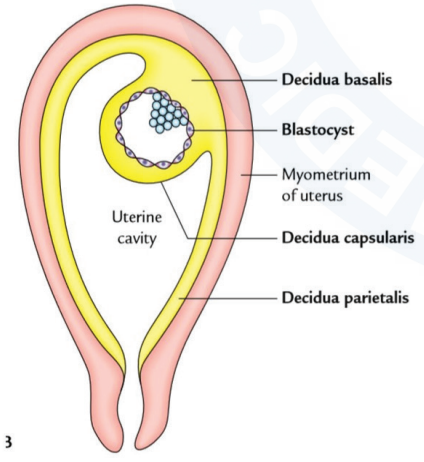
- Para-axial Mesoderm
- Intermediate Mesoderm (Kidney & Gonads)
- Lateral Plate Mesoderm

Para.axial Mesoderm Or Somites

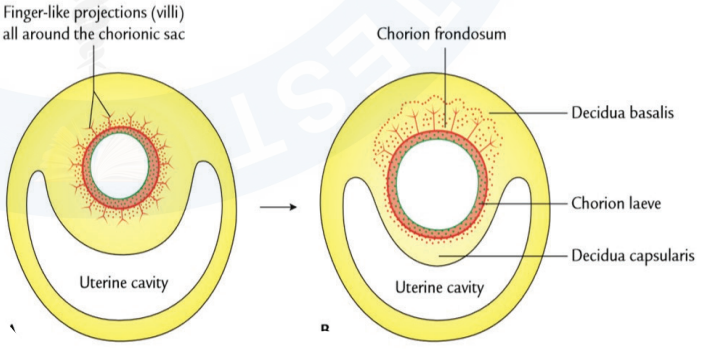
- Dermatome (Dermis of skin)
- Myotome (Muscles)
- Sclerotome (Ribs & Vertebra)

PLACENTA DEVELOPMENT

1. Decidua & Chorion



◀FIG. Decidua



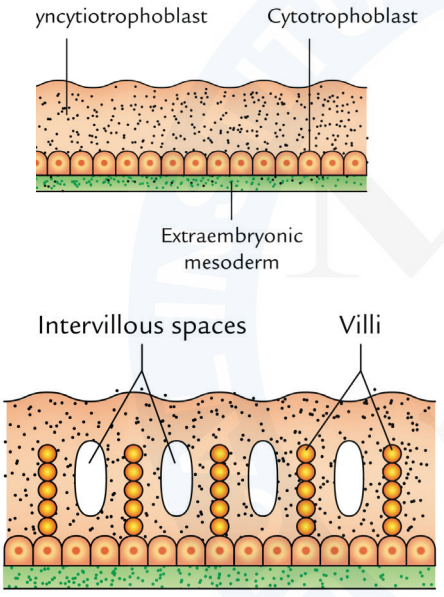
◀FIG. Chorion

❑ **Placenta.**

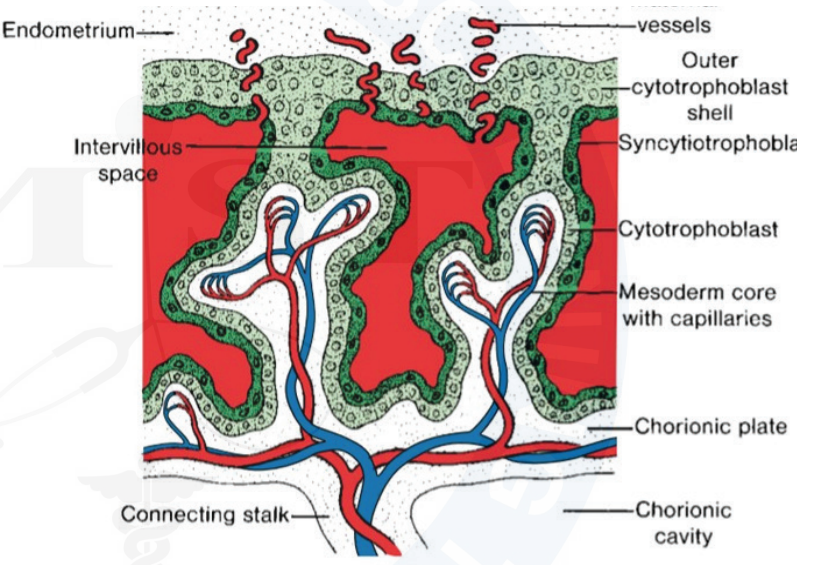
2 Chorionic Villi

❑ **Chorionic villi → finger like projections.**

1. Primary villi → Components - .....
2. Secondary villi → .....
3. Tertiary villi → .....



◀FIG. Primary villi



◀FIG. Tertiary villi

3. Placental Barrier

❑ **Placental Barrier**

- Layers B/W Maternal Blood & Fetal blood

❑ **Components**

1. .....
2. .....
3. .....
4. .....

❑ **At 20 weeks of pregnancy, Placental Barrier is formed by**

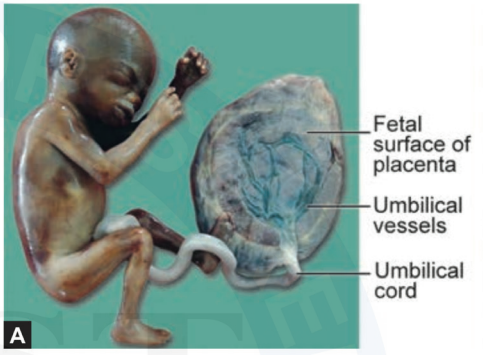
1. ST
2. Endothelium of fetal Vessels

UMBILICAL CORD

❑ **Umbilical Cord**

✦ Contents

1. Two Umbilical Artery (R & L)
2. One Umbilical Vein (LEFT)
3. Vitello intestinal duct
4. Allantois
5. Wharton's Jelly



✦ Note:-

1. Umbilical vein carries oxygenated blood from Placenta to fetus.
2. Umbilical Artery carries deoxygenated blood from fetus to Placenta.

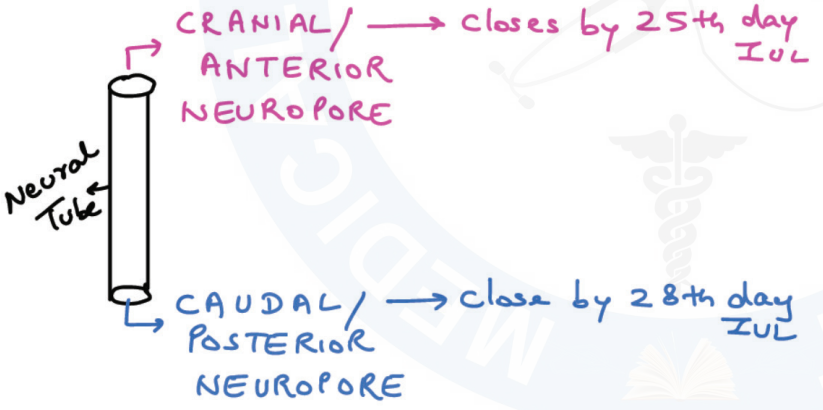
Embryological Structure	Adult Remnant
1. Umbilical Vein	Ligamentum Teres Hepatis
2. Umbilical Artery	• Proximal Part → Superior Vesical Artery • Distal Part → Medial Umbilical Ligament
3. Ductus Venosus	Ligamentum Vensum
4. Ductus arteriosus	Ligamentum Arteriosum

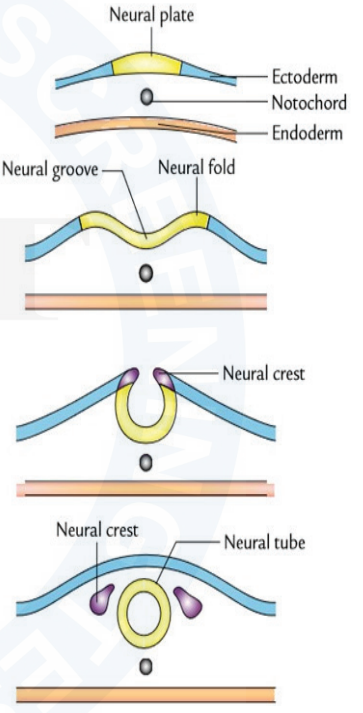
DEVELOPMENT OF CNS

1. Neural Tube formation & Congenital Defects

❑ **Neural tube -** Ectoderm

- ✓ Formation begins in .....
- ✓ Fusion of Neural Tube begins at .....
- ✓ Formation is completed by .....

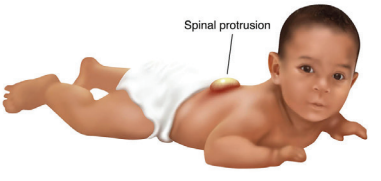




◀FIG. Neural Tube formation

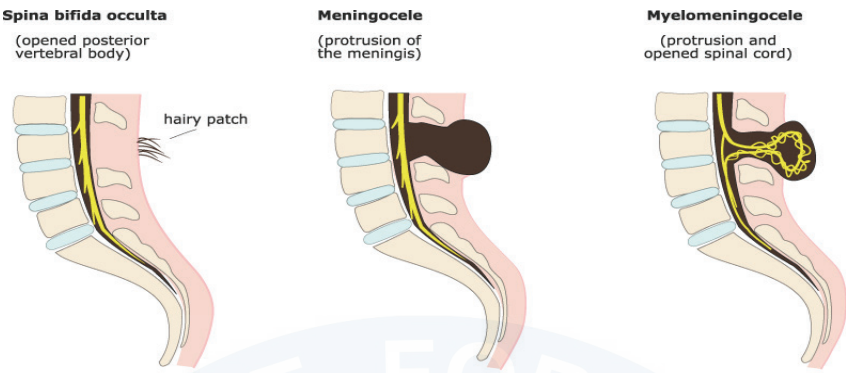
❑ **Neural tube defects-**

◀FIG. Anencephaly



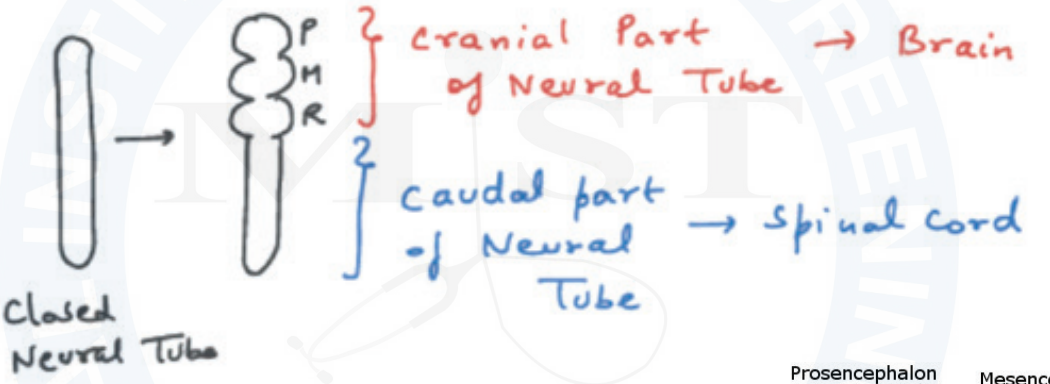
◀FIG. Spina Bifida

◀FIG. Craniorachishisis

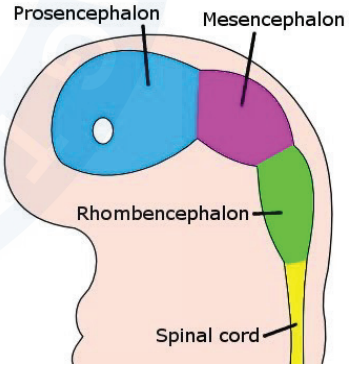


◀FIG Spina Bifida

2. Neural Tube Divisions



◀FIG Neural Tube Divisions



P Prosencephalon
- Telencephalon → Cerebrum
- Diencephalon → Part of Diencephalon

M Mesencaphalon → Midbrain

R Rhombencephalon
- Metencephlon → Pons & Cerebellum
- Myelencephalon → Medulla

3. Neural Crest Cells

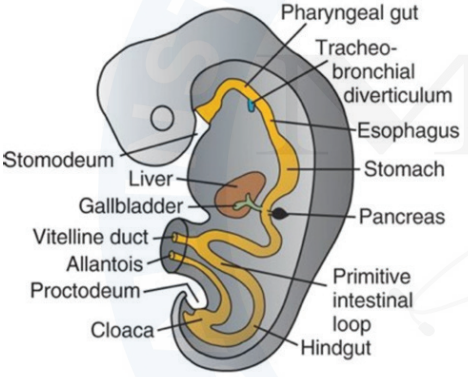
❑ Neural Crest cells Derivatives

1. Schwann cells
2. Odontoblast
3. Melanoblast
4. Adrenal Medulla
5. Chromaffin Tissue
6. Leptomeninges
7. Parafollicular 'C' cells of Thyroid

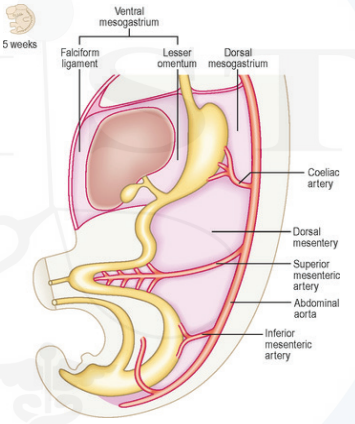
DEVELOPMENT OF GIT

1. Overview

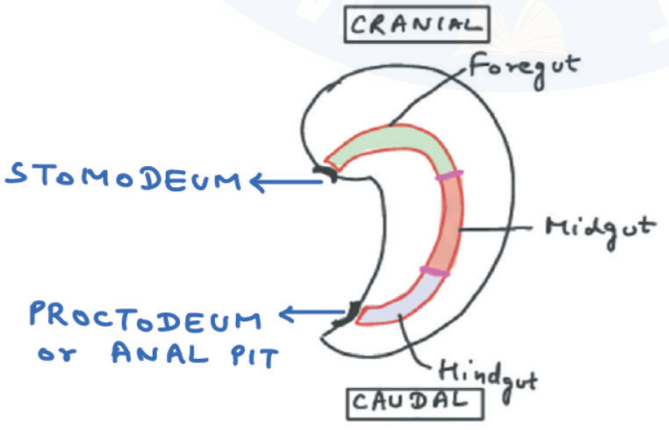
❑ Gut tube (Endodermal)



◀FIG Gut tube



◀FIG. Gut tube (Arteries)



◀FIG. Gut tube (schematic diagram)

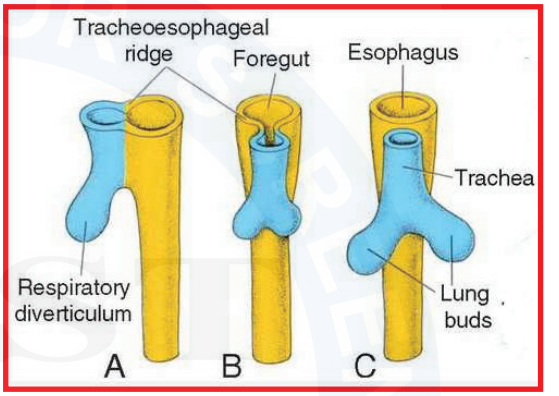
- Artery of foregut → Celiac Trunk.
- Artery of Midgut → Superior Mesenteric A.
- Artery of hindgut → Inferior Mesenteric A.

2. Foregut (Derivatives & Respiratory Bud)

❑ Foregut Derivatives

1. Pharynx
2. Esophagus
3. Stomach
4. Duodenum

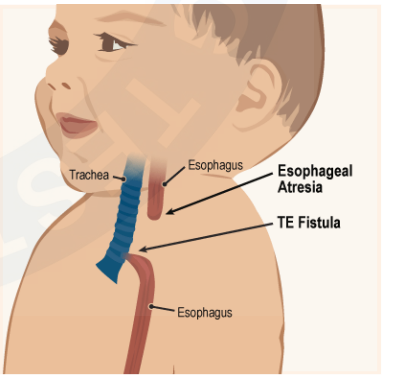
   (till major duodenal papilla)
5. Trachea & Lung
6. Liver & Gall bladder
7. Pancreas.



◀FIG. Respiratory Bud

❑ Respiratory Bud / Diverticulum

- ✓ Forms Trachea & Long
- ✓ Tracheo - Esophageal Septum Separates Trachea & Esophagus



◀FIG Tracheo-esophageal fistula

❑ Tracheo-esophageal fistula

- Defect in formation of Tracheo-esophageal septum

3. Foregut (Liver & Gall Bladder)

Hepatic Bud → Liver & Gall Bladder

4. Foregut (Pancreas)

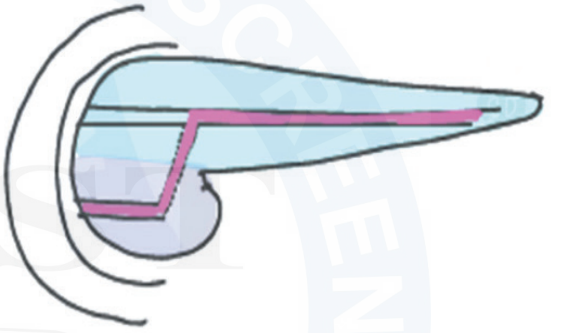
Pancreatic Buds

- **Ventral P. B** → Forms Lower part of Head & uncinate process
- **Dorsal P. B** → Rest of Pancreas

❑ Main Pancreatic duct (Duct of wirsung)

1. Ventral P. duct
2. Communication between two ducts
3. Distal part of Dorsal P. duct.

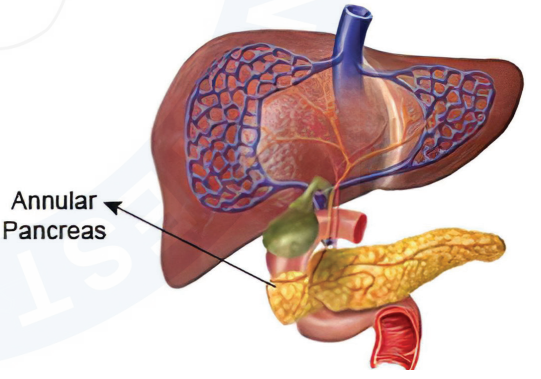
☞ Opens at Major Duodenal Papilla



◀FIG. Formation of Main Pancreatic duct (colored portion)

❑ Congenital Anomalies of Pancreas

- ❖ Most common congenital anomaly is .....
  .....
- ❖ Annular Pancreas

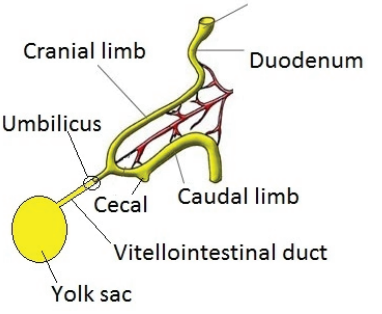


◀FIG Annular Pancreas

5. Midgut (Derivatives & Vitellointestinal Duct)

❑ Midgut Derivatives

1. Duodenum (below major duodenal papilla)
2. Ileum
3. Jejunum
4. Caecum
5. Appendix
6. Ascending colon
7. Right 2/3 of Transverse colon

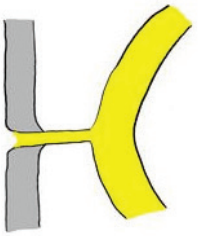


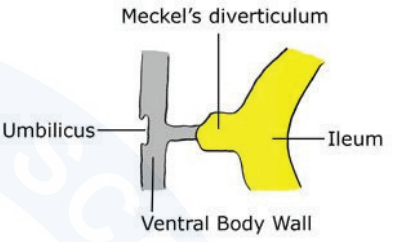
◀FIG. Vitellointestinal duct

❑ **Vitellointestinal duct / vitelline duct**

❑ **Congenital anomalies**

1. Vitellointestinal duct (VT duct)
   completely patent
   ↓
   .....
2. Proximal part of VI duct patent
   ↓
   Meckel's Diverticulum





◀FIG. Vitellointestinal duct Congenital anomalies

6. Midgut Rotation

✓ Physiological Hernia of midgut loop → occurs in .....

✓ Reduction of Physiological Hernia of midgut (Return of midgut loop) → occurs at .....

✓ Total rotation → .....

✓ Axis of rotation → .....

❑ **Omphalocele / Exomphalos**

- ✓ Non Return of midgut loop
- ✓ Intestinal loops present outside at level of umbilicus
- ✓ covered by amniotic membrane

◀FIG. Omphalocele

◀FIG. Gastrochisis

☞ Note -

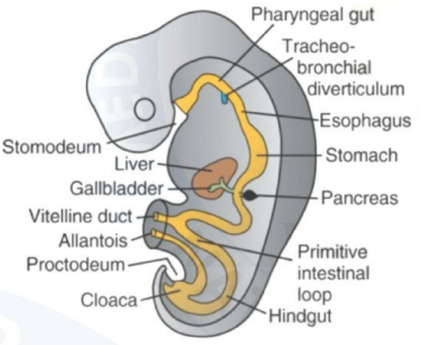
❑ **Gastrochisis**

- ✓ D/t Defect in fusion (formation) of anterior abdominal wall
- ✓ No amniotic membrane covering it
- ✓ Not at the level of umbilicus

7. Hindgut (Cloaca)

❑ Hindgut Derivatives

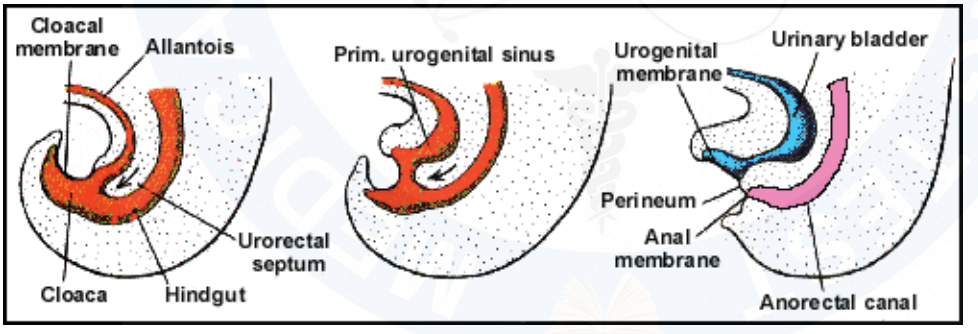
1. Left 1/3 of Transverse colon
2. Descending colon
3. Sigmoid colon
4. Rectum
5. Upper part of anal canal
6. Urinary Bladder



◀FIG. Gut tube

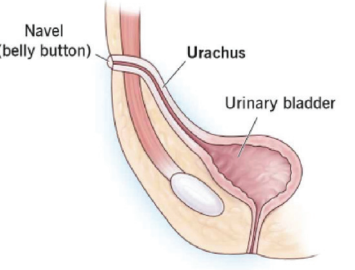
CLOACA (Lower Part of Hindgut)

- Urogenital Sinus → U. Bladder (Except trigone)
- Primitive Rectum → Rectum & Anal Canal (Upper Part)



◀FIG. Hindgut - Cloaca (Its division)

8. Hindgut (Allantois)

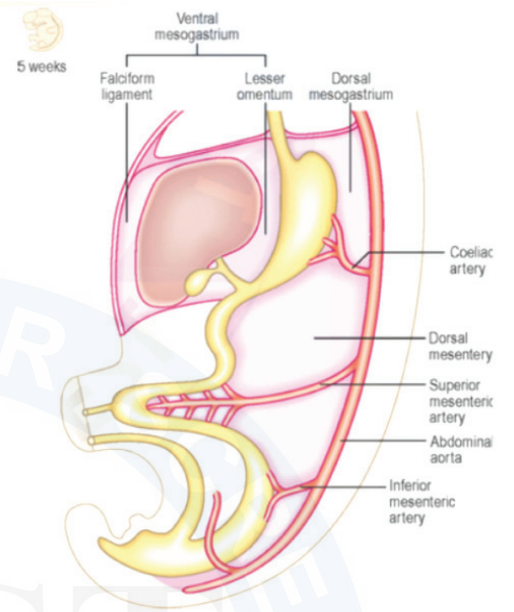


❑ Allantois

- ✓ B/W Urinary Bladder & Umbilicus
- ✓ Adult Remnant → _____
- ✓ If Allantois is completely Patent → _____

9. Mesentry

- ❑ **Ventral Mesentry / Mesogastrium**
  - ✦ Derivatives
    1. Lesser Omentum
    2. Falciform Ligament
    3. Coronary lig
    4. Triangular lig
- ❑ **Dorsal Mesogastrium / Mesentry**
  - ✦ Derivatives
    1. Greater Omentum.
    2. Gastrophrenic lig
    3. Gas trosplenic lig
    4. Splenorenal lig



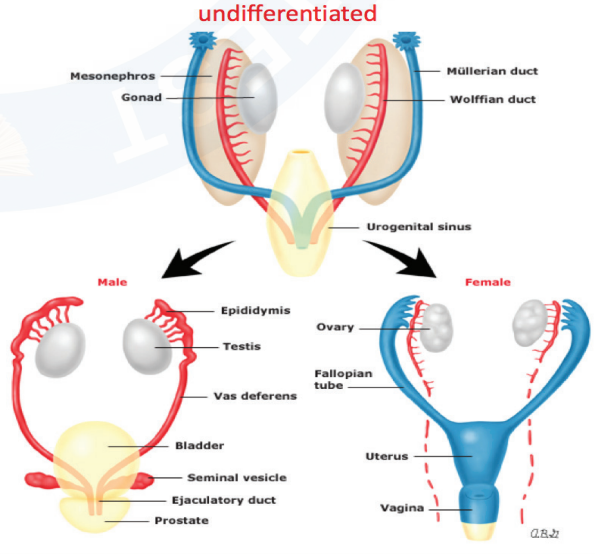
◀**FIG.** Gut tube (Arteries)

DEVELOPMENT OF GENITOURINARY SYSTEM

1. Mesonepharic ducts

Mesonephric duct (Wolffian duct) (Mesodermal)

- ❑ **Derivatives**
  - ✦ (A) In both males & females
    - ✓ Trigone of Urinary Blodder
    - ✓ Ureteric Bud (Ureter, Renal Pelvis & Major & Minor calyx)
  - ✦ (B) In Males -
    - ✓ Epididymis
    - ✓ Vas deferens
    - ✓ Seminal Vesicle
    - ✓ Ejaculatory Duct
    - ✓ Prostate



◀**FIG.** Development of Genitourinary System

- (C) In females -

  (Adult Remnant)

  - ✓ Epoophoron
  - ✓ Parophoron
  - ✓ Gartner's Duct

2. Paramesonephric ducts

Paramesonephric / Mullerian duct (Mesodermal)

- **Derivatives**
  - (A) In females
    - ✓ Uterus
    - ✓ Cervix
    - ✓ Fallopian Tube
    - ✓ Upper Part of Vagina
  - (B) In males (Adult Remnant)
    - ✓ Prostatic Utricle
    - ✓ Appendix of Testis

☞ Note - Lower part of vagina derived from Sinovaginal Bulb (Urogenital Sinus)

3. Homologous Structure

- **(Homologous organs / structure)**

Embryological structure	Females	Males
1. Gonadal / Genital Ridge	Ovary	Testis
2. Genital swellings	Labia Majora	Scrotum
3. Genital Tubercle	Clitoris	Penis
4. Genital folds	Labia Minora	Penile Urethra

DEVELOPMENT OF URINARY SYSTEM

❑ Urinary System

1. Urinary Bladder → Urogenital sinus (cloaca)

   Its Trigone → Mesonephric duct

2. Ureteric Bud → Mesonephric duct

3. Kidney → .....

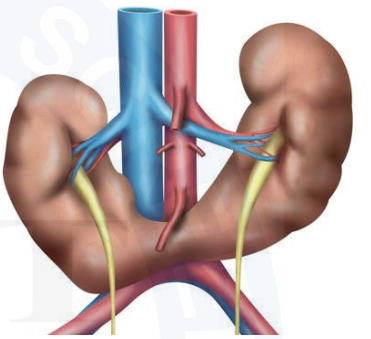
❑ Horse shoe kidney

⇨ Lower parts are fused

⇨ Ascent of Lower fused part is arrested by.....

.....

⇨ Lower fused part lies at the level .....



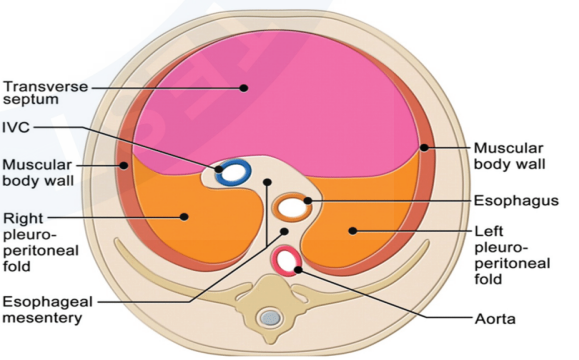
◀FIG. Horse Shoe Kidney

DEVELOPMENT OF DIAPHRAGM

❑ Development of Diaphragm

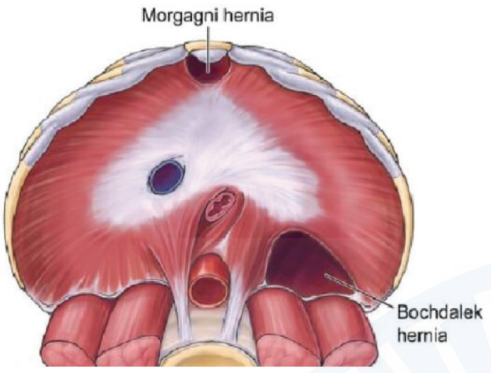
1. Septum Transversum
2. Pleuro-Peritoneal Membrane
3. Lateral Body wall Mesoderm
4. Dorsal Mesentry of Esophagus

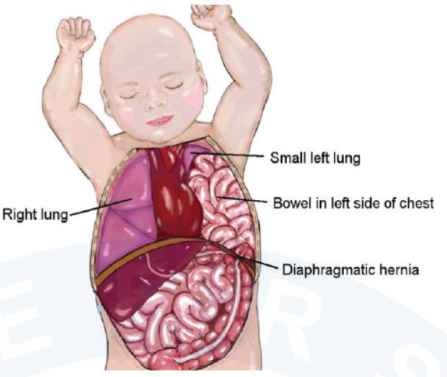
✓ Muscles → Cervical somites / myotomes

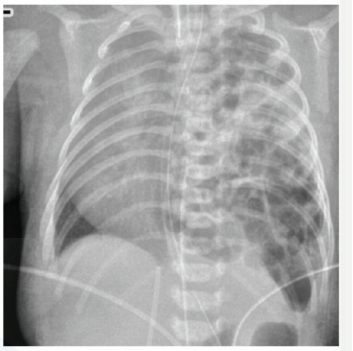


◀FIG. Development of Diaphragm

BOCHDALEK CONGENITAL DIAPHRAGMATIC HERNIA







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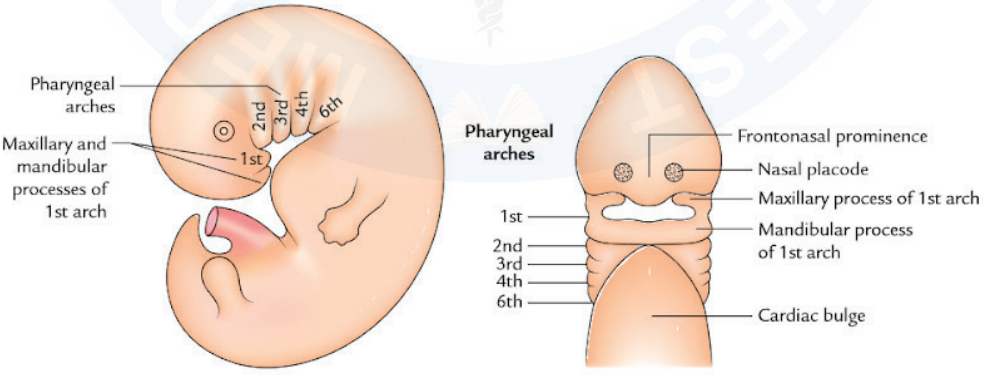
DEVELOPMENT OF HEAD & NECK

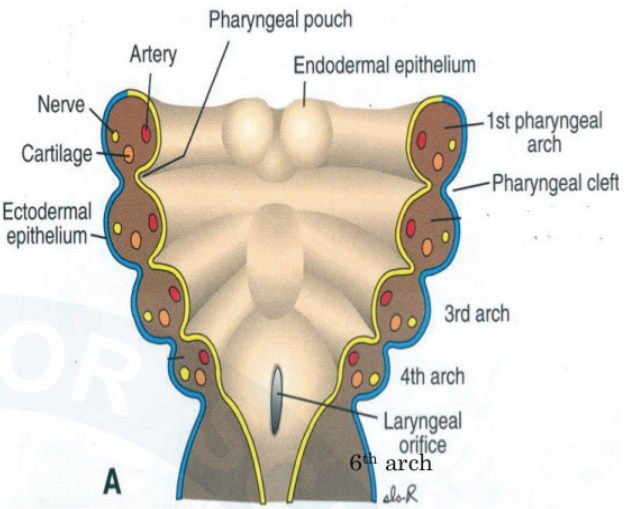
1. Pharyngeal Arches



◀**FIG.** Pharyngeal Arch

❑ Pharyngeal apparatus (Branchial)

1. Pharyngeal arch → Mesoderm
2. P. pouch → Endoderm
3. P. cleft → Ectoderm.

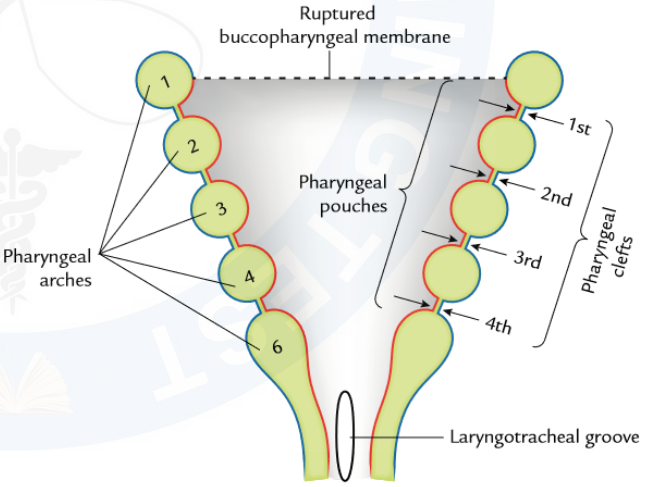


◀FIG. Pharyngeal apparatus (Arch, Pouch & Cleft)

❑ Pharyngeal Arch -

✦ 1st Pharyngeal Arch - Derivatives

- ✓ Malleus
- ✓ Mandible
- ✓ Maxilla
- ✓ Muscles of Mastication
- ✓ Mylohyoid
- ✓ Mandibular Nerve (Nerve of 1st Arch)
- ✓ Maxillary Artery (Artery of 1st Arch)
- ✓ Meckel's Cartilage
- ✓ Anterior Ligament of Malleus
- ✓ Anterior Belly of Digastric
- ✓ Spine of Sphenoid
- ✓ Sphenomandibular Ligament
- ✓ Tensor Tympani
- ✓ Tensor Veli Palatini
- ✓ Incus



✦ 2nd Pharyngeal arch - derivatives

- ✓ Reichert's Cartilage
- ✓ Styloid Process
- ✓ Sty lohyoid ligament
- ✓ Sty lohyoid Muscle
- ✓ Stapes
- ✓ Stapedius
- ✓ Smaller Cornu &

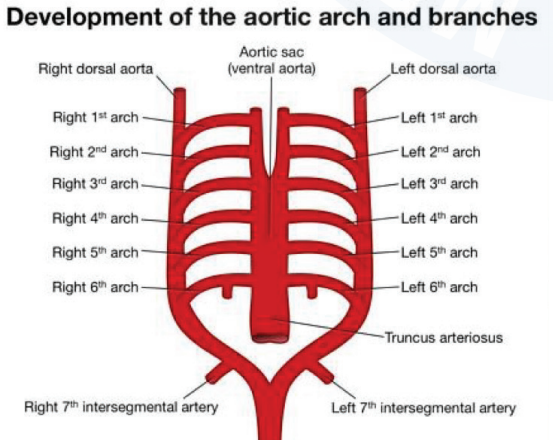
- ✓ Superior part of Body of Hyoid
- ✓ Stapedial Artery & Hyoid A (Artery of 2nd arch)
- ✓ Posterior Belly of digastric
- ✓ Facial Muscles
- ✓ FAcial Nerve (Nerve of 2nd Arch)

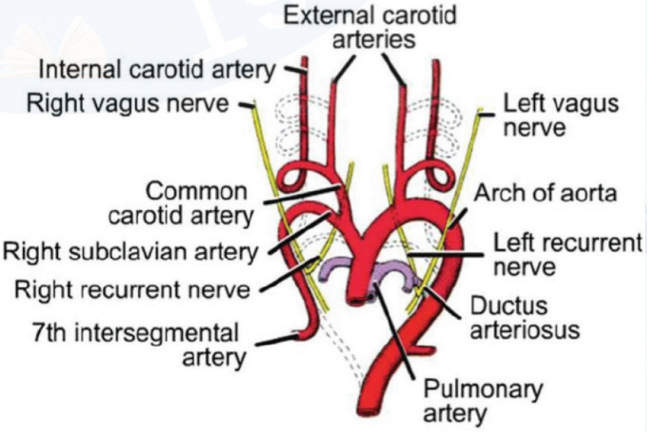
✦ 3rd arch - derivatives

- ✓ Rest of Hyoid
- ✓ Stylo Pharyngeus
- ✓ Nerve of 3rd arch → Glossopharyngeal N.
- ✓ Artery of 3rd arch → Carotid Arteries

✦ 4th and 6th arch - derivatives

- ✓ 4th & 6th arch → Muscles of Pharynx & Larynx
  → Cartilages of Larynx
- ✓ Nerve of 4th arch → Superior Laryngeal Nerve
- ✓ Artery of 4th arch → Left → Arch of Aorta
  → Right → Right Subclavian A
- ✓ Nerve of 6th arch → Recurrent Laryngeal N
- ✓ Artery of 6th arch → R & L Pulmonary Artery
  (on left side, Ductus Arteriosus is also formed)

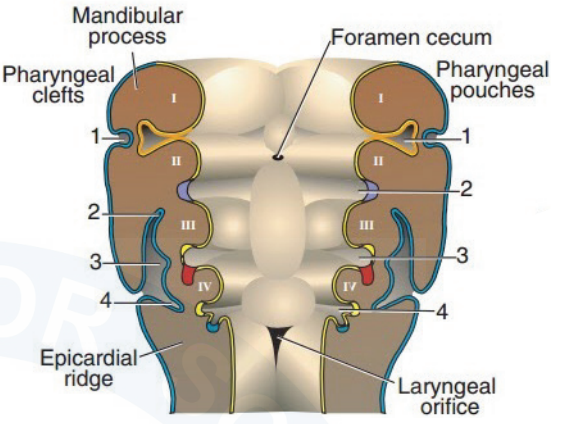




2. Pharyngeal Pouches

❑ Endodermal Pouch - derivatives

- ✦ 1st Pouch
  - ✓ Eustachian Tube
  - ✓ Middle Ear Cavity
  - ✓ Mastoid Antrum
- ✦ 2nd Pouch
  - ✓ Palatine Tonsil
  - ✓ Tonsillar Fossa
- ✦ 3rd Pouch
  - ✓ .....
  - ✓ .....
- ✦ 4th Pouch
  - ✓ Superior Parathyroid Gland



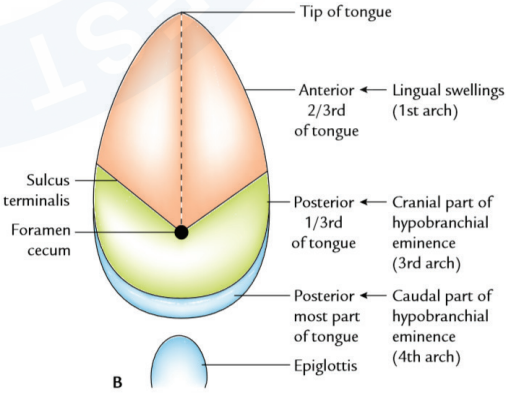
◀FIG. Endodermal Pouch & Ectodermal Cleft

☞ Note :

Branchial Cyst → Remnant of .....

3. Tongue

1. Formed by 1st 3rd & 4th arch
2. Lingual swellings, Tubercular Impar & Hypobranchial Eminence (Copula of His)

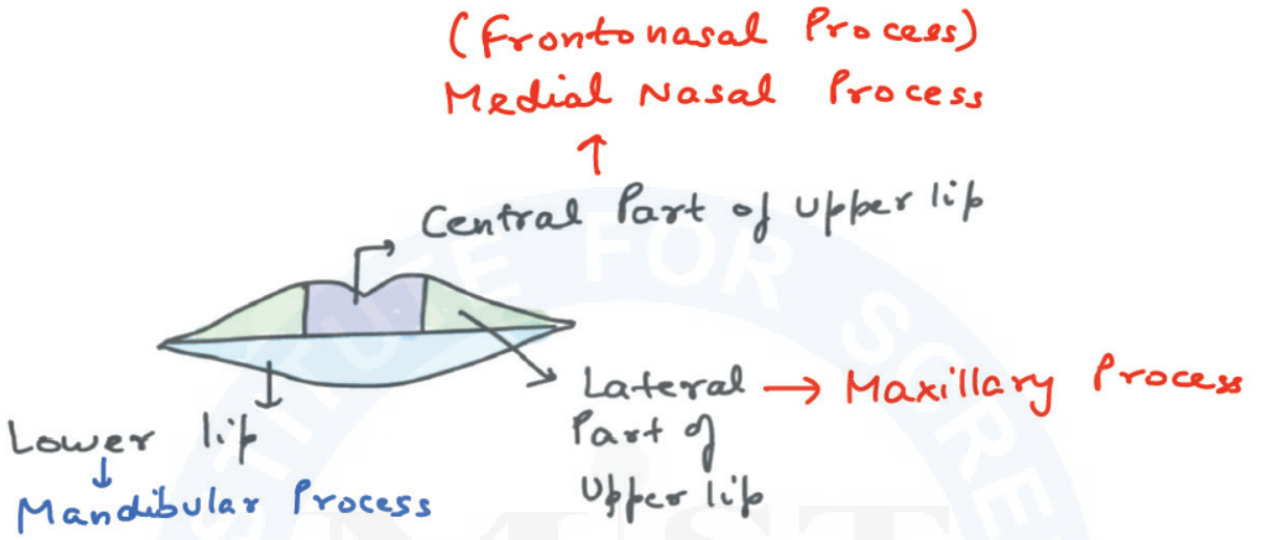


◀FIG. Tongue - development

☞ Muscles of tongue are derived from Occipital Somites / Myotomes

4. Lips

❑ Lips



◀FIG. Development of Lips

❑ Cleft lip

- Non fusion of medial nasal process & maxillary process

◀FIG. Cleft lip (A- Unilateral, B- Bilateral)

◀FIG. Preauricular Sinus / PIT (Defect in fusion of 1st & 2nd arch)

One Liners

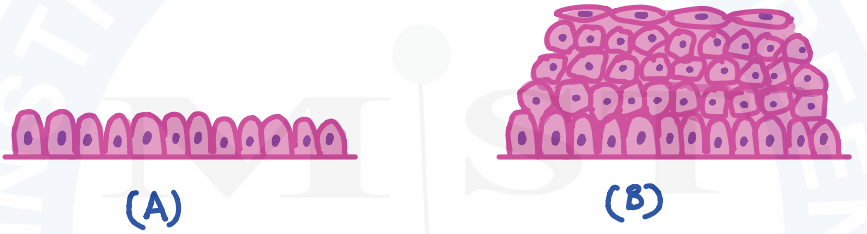
1. SUPERIOR VENA CAVA → ANTERIOR CARDINAL VEINS
2. TYMPANIC MEMBRANE FORMED BY ALL THREE GERM LAYERS
3. EAR PINNA – BY MESODERM OF 1ST & 2ND ARCH
4. THYROID GLAND – BY THYROGLOSSAL DUCT
5. PITUTARY GLAND – BY RATHKE'S POUCH
6. FOSSA OVALIS → IT IS ADULT REMNANT OF FORAMEN OVALE
   → FLOOR OF FOSSA OVALIS IS FORMED BY SEPTUM PRIMUM
7. ANNULUS OVALIS → ADULT REMNANT OF SEPTUM SECUNDUM
8. PARAFO LLICULAR 'C' CELLS OF THYROID GLAND FORMED BY → NEURAL CREST CELLS > ULTIMOBRANCHIAL BODY

Workbook Questions

1. Morula reaches uterus on.
   (A) 3rd day after fertilization (B) 4th day after fertilization
   (C) 6th day after fertilization (D) 8th day after fertilization
   Ans B
2. Testis are developed from.
   (A) Genital ridge (B) Genital tubercle
   (C) Wolffian duct (D) Gubernaculum
   Ans A
3. Tongue develops from all except
   (a) Lingual swellings (b) Tuberculum impar
   (c) Hypobranchial eminence (d) Frontonasal process
   Ans D
4. Which is not a derivative of dorsal mesogastrium?
   (A) Greater omentum (B) Gastrosplenic ligament
   (C) Gastro-phrenic ligament (D) Coronary ligament
   Ans D
5. Failure of closure of cranial neuropore leads to
   (a) Rachichisis (b) Spina bifida
   (c) Anencephaly (d) Meningocele
   Ans C
6. Which of the following structure is derived from umbilical vein?
   (a) Ligamentum arteriosum (b) Medial umbilical ligament
   (c) Ligamentum venosum (d) Ligamentum teres
   Ans D
7. Which of the following is true about horseshoe kidney?
   (a) Fused at upper pole, lies in pelvis (b) Fused at lower pole, lies in pelvis
   (c) Fused at upper pole, lies in front of L1
   (d) Fused at lower pole, lies in front of L4
   Ans D
8. Outer cell mass gives rise to:
   (a) Embryo proper (b) Trophoblast (c) Yolk sac (d) None
   Ans B
9. Maxillary artery develops from:
   (a) 1st (b) 3rd arch (c) 4th arch (d) 5th arch
   Ans A
10. Urachus fistula is a remnant of?
    (a) Yold sac (b) Allantois (c) Chorion (d) Amnion
    Ans B

EPITHELIUM

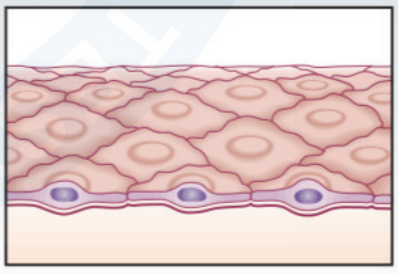
- **Epithelium** → Simple (single layer of cells)
  → Stratified (Multiple layers of cells).



◀**FIG.** (A) Simple Epithelium (B) Stratified Epithelium

Simple Squamous Epithelium

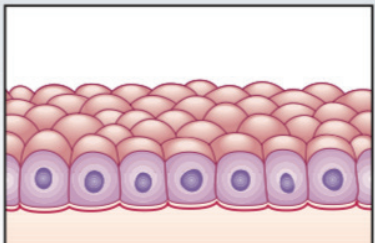
- **Flat cells**
  - ✦ Location -
    - Alveoli of Lung
    - Endothelium of blood vessels
    - Endocardium
    - Mesothelium (of Pleura, pericardium & peritoneum)



◀**FIG.** Simple Squamous Epithelium

Simple Cuboidal Epithelium

- Height, Length & Breadth of cell is equal
- Nucleus is central in position & round in shape
  - ✦ Location -
    - Thyroid Gland
    - Germinal Epithelium of ovary
    - PCT of Kidney



◀**FIG.** Simple Cuboidal Epithelium

Simple Columnar Epithelium

- Height of cell is more than width
- Nucleus - oval in shape
  - ✦ Location -
    - Stomach
    - Large Intestine

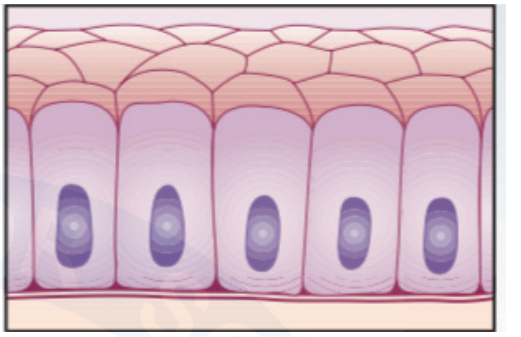


FIG. Simple Columnar Epithelium

Simple Columnar Epithelium with cilia

- ✦ Location -
  - Uterus
  - Fallopian Tube
  - Ventricles of Brain
  - Central canal of spinal cord

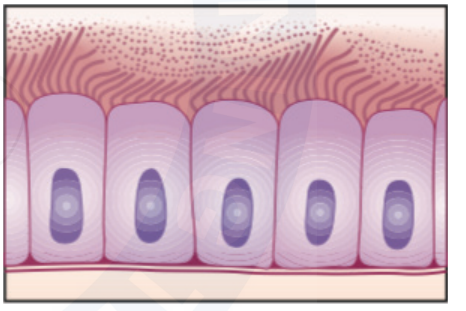


FIG. Simple Columnar Epithelium with cilia

Simple Columnar Epithelium with microvili

- ✦ Location -
  - Gall Bladder → (Brush Border Appearance)
  - Small Intestine

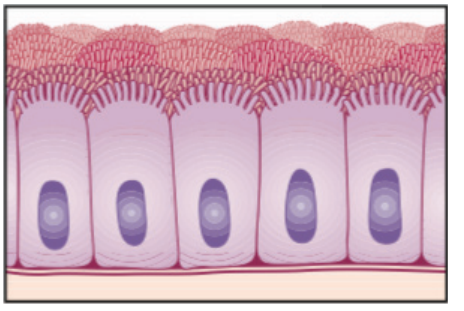
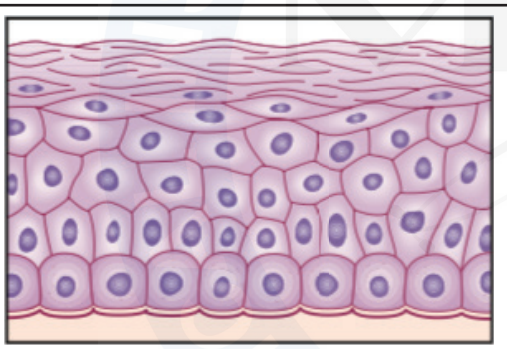
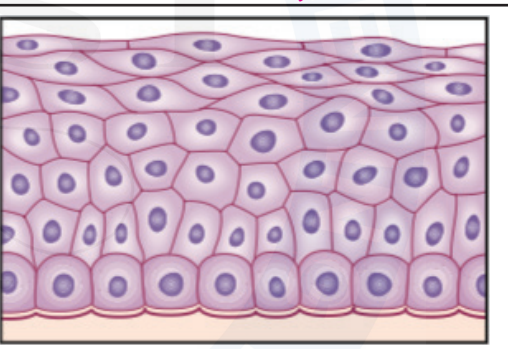
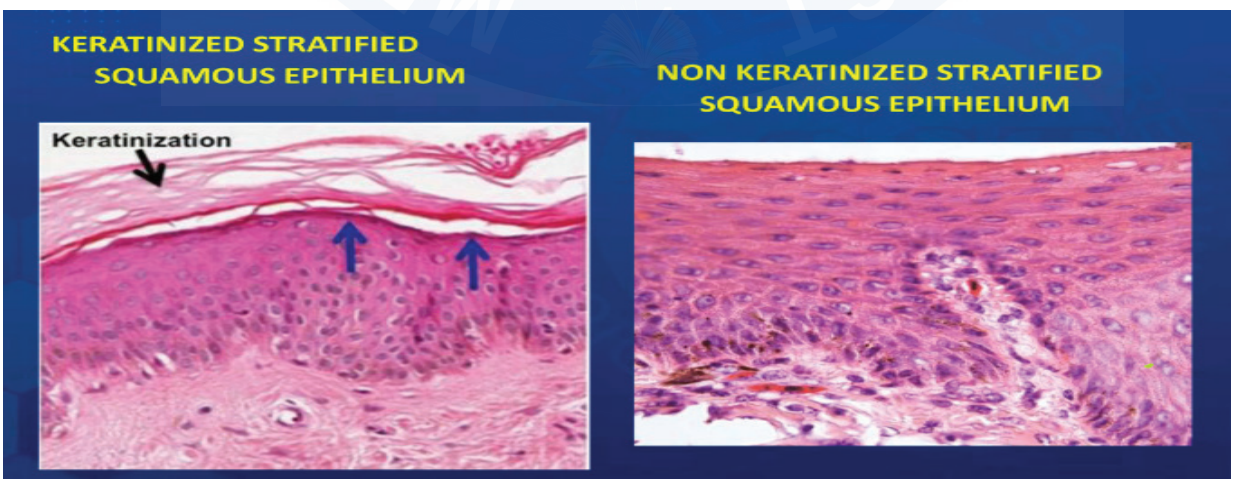


FIG. Simple Columnar Epithelium with microvili

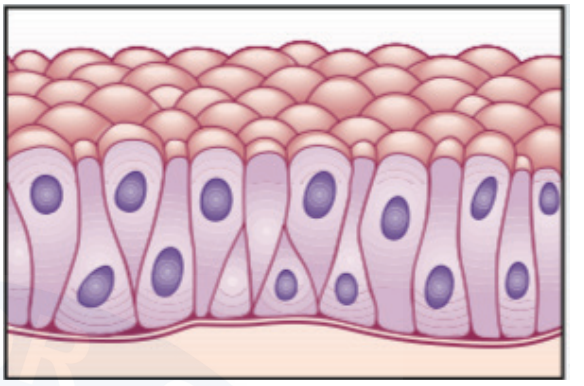
Stratified Squamous Epithelium

❑ Keratinized (Keratin layer is present)	❑ Non-Keratinized
✦ Location -	✦ Location -
• Skin (Thick & Thin)	• Tongue • Tonsil • Epiglottis (oral surface) • Oesophagus • Cornea • Vagina • Anal canal (Below Pectinate line)
	
◀FIG. Keratinized Stratified squamous Epithelium	◀FIG. Non-Keratinized Stratified squamous Epithelium



Pseudostratified Ciliated Columnar Epithelium

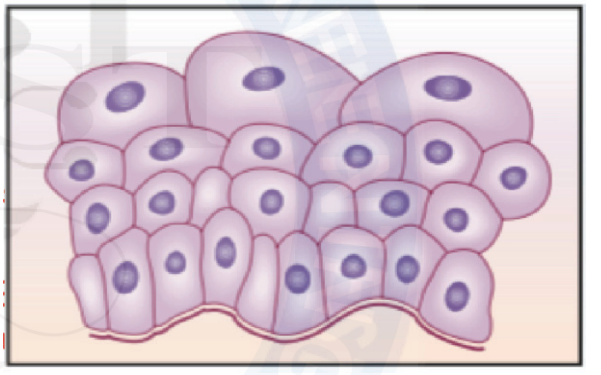
- Respiratory Epithelium
  - ✦ Location -
    - Trachea
    - Bronchus
    - Epiglottis (Laryngeal surface)



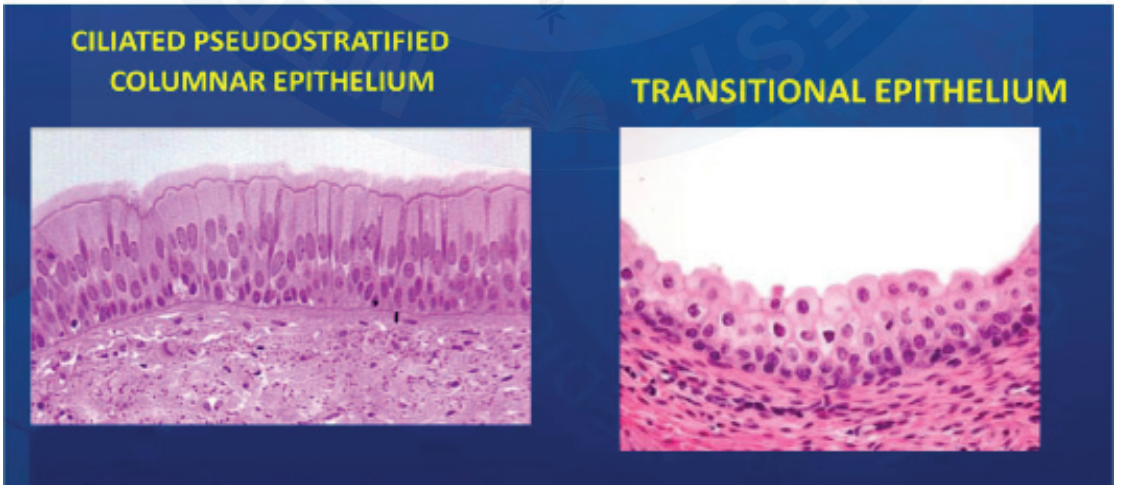
◀FIG. Pseudo stratified Ciliated Columnar Epithelium

Transitional Epithelium (Urothelium)

- ✦ Location -
  - Renal calyces & Pelvis
  - Ureter
  - Urinary Bladder



◀FIG. Transitional Epithelium



Cartilage

1. ✦ Hyaline C - (Most Common type)

   Location - Trachea, Bronchus,

   Nasal C, Thyroid C.,

   Cricoid C, Base of Arytenoid

2. Costal C, Articular C, Epiphyseal Plate

   ✦ Elastic - Epiglottis

   - Ear Pinna, Eustachian Tube, External Acoustic Canal

3. - Apex of Arytenoid, Corniculate, Cuneiform

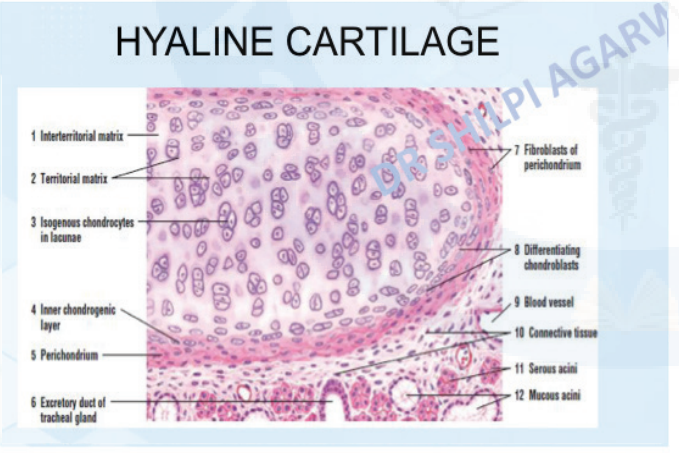
   ✦ Fibrocartilage

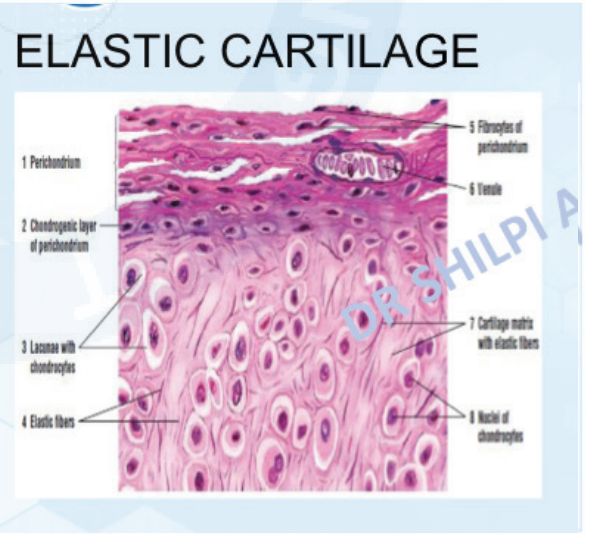
   - Intervertebral Disc

   Mensici of Knee Joint

   ✦ Which cartilage is not covered by Perichondrium → Fibrocartilage

   ✦ Which cartilage can ossify with age → Hyaline C.





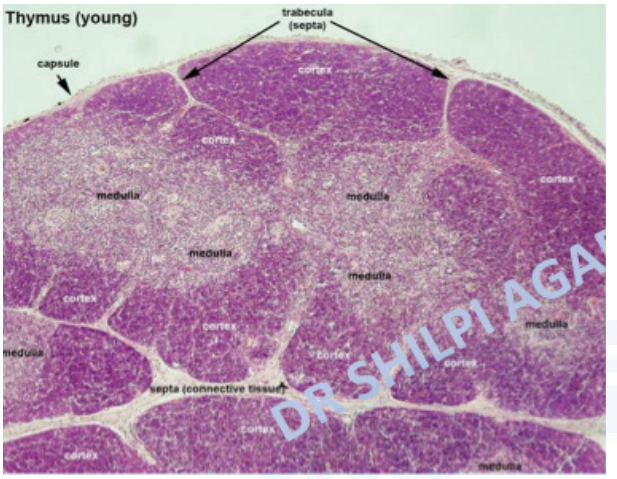


FIG. THYMUS

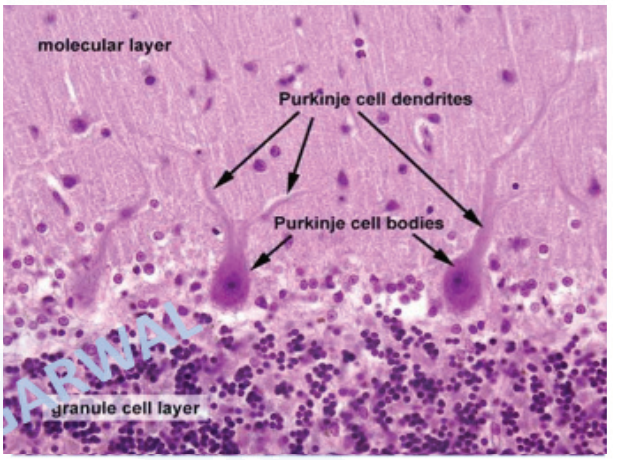


FIG. CEREBELLUM

Workbook Questions

1. Trachea is lined by?
   (a) Simple columnar
   (b) Pseudostratified columnar
   (c) Simple cuboidal
   (d) Stratified squamous, nonkeratinized
   Ans B

2. Transitional epithelium is seen in:
   (a) Gallbladder
   (b) Urinary bladder
   (c) Thyroid gland
   (d) Stomach
   Ans B

3. Esophagus is lined by?
   (a) Simple columnar
   (b) Pseudostratified columnar
   (c) Simple cuboidal
   (d) Stratified squamous, nonkeratinized
   Ans D

4. All are lined by simple columnar epithelium EXCEPT ?
   (a) Gallbladder
   (b) Small intestine
   (c) Thyroid gland
   (d) Stomach
   Ans C

5. Thyroid gland is lined by?
   (a) Simple columnar
   (b) Pseudostratified columnar
   (c) Simple cuboidal
   (d) Stratified squamous, nonkeratinized
   Ans C

Workbook Sheet

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MIST

Workbook Sheet

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